

July 10, 2026

Prof. Roland Rathelot

Editor

The Economic Journal

Subject: Revision of “Boots in the Beat: Effects of Beat Policing on Victimization and Crime Perception”

Dear Prof. Rathelot,

Thank you for inviting us to revise and resubmit our manuscript. We carefully read your comments and the comments from the three referees. We are very grateful for this detailed and constructive feedback, which helped us improve the paper substantially.

Below we provide a point-by-point response document with detailed answers to each comment. We also indicate how the manuscript has been revised in response.

Sincerely,

Andrés Barrios Fernández on behalf of the team

Editor's comments

Editor Comment 1

Dear Doctor Barrios-Fernandez,

Thank you for submitting your fascinating paper to The Economic Journal.

I have received three reviews from experts who are familiar with the topic of your paper, the literature, and the methods you use. While all reviewers find much to like in your paper, they disagree in their recommendations. R3 and R1 recommend a revise and resubmit (R1 emphasising that they expect a major revision), while R2 recommends rejecting the paper. After reading the paper and the feedback from the reviewers, I believe that there could be a path to publication, provided that you manage to address the main points raised by the reviewers—which will not be easy. If you are ready to take up the challenge, I am pleased to offer you an opportunity to revise and resubmit your paper.

1. Framing, contribution, and clarity. I very much side with R1 and R2 on this. The exposition of the policy is not clear. The term “specialisation”, which appears in the title and the first sentence of the abstract, does not help. We could think of three ways policing reforms may affect their effectiveness: (i) police size, (ii) police methods or strategies (i.e., how they interact with the public), and (iii) police allocation across areas. The paper is not sufficiently clear on what exactly the situation was pre-reform, what the reform entailed (and how it interacted with other potentially contemporaneous reforms), what you estimate, and how we should interpret the results. This is linked to your contribution and to how you can disentangle mechanisms. This is an aspect that all reviewers comment on, but I find R1 and R2 particularly relevant on this point.

Response to Editor Comment 1:

Thank you very much for this comment. We agree with you and the referees that the paper needed a clearer and more detailed account of *Plan Cuadrante* and of how it reshaped police work. To address this point, we contacted Carabineros de Chile and held multiple meetings with its Research Department. These interactions gave us access to documents on the design and implementation of the program, as well as evaluation reports produced by the Chilean Budget Office (DIPRES). We also reviewed once more the academic literature on policing approaches both in economics—following some of the referees’ advice—and in criminology.

Based on the interactions with the Chilean police and on our revision of the specialized literature, we first clarify the terminology used to describe *Plan Cuadrante*. Rather than referring to it as a “geographic specialisation” reform, we now describe the program as a reform that shifted policing in Chile toward a beat policing model. In police practice and in the criminology literature, beat policing refers to permanently assigning officers to a fixed, smaller territorial area—rather than rotating officers across a large catchment area—so that they can build local knowledge, strengthen links with the local community, and be more effective in preventing crime (Cordner, 1995; Skogan and Hartnett, 1997). Although this policing approach has been widely discussed in the criminology literature, we still lack rigorous evidence on its impacts on crime outcomes and police effectiveness. To the best of our knowledge, this is the first paper providing causal evidence on the effects of this policing approach on victimization, crime perception, and trust in the police. In addition to adjusting the terminology used to describe *Plan Cuadrante*, we reorganized the description of the program and the discussion of mechanisms around the framework you suggested, distinguishing

three key elements of the police production function: resources, deployment of resources across territory, and policing strategy. Using this structure, we expanded Section 2 with additional details on the design, implementation, and roll-out of *Plan Cuadrante*, added institutional detail, and revised the mechanisms discussion. We also updated the title—the new title reads “Boots in the Beat: Effects of Beat Policing on Victimization and Crime Perception”—, abstract, introduction, and conclusions to keep the framing consistent throughout the manuscript.

Building on the framework you suggested, we now clarify that although in some cases the introduction of *Plan Cuadrante* was accompanied by an increase in police resources, the reform’s core objective was not to increase police resources but to make police work more effective by moving to a beat policing model. In particular, *Plan Cuadrante* permanently assigned officers in the municipality to new, well-defined, smaller territorial units—quadrants—so that they could build local knowledge, engage with residents, and adapt their practices to local crime patterns and community dynamics through this local specialization. Thus, the implementation of *Plan Cuadrante* changed the three elements of the police production function we described: it increased police resources in some municipalities, it changed the deployment of resources by defining a new and smaller operational unit—i.e., quadrants—and permanently assigning officers to them, and changed police strategy by enabling officers to accumulate local knowledge and adapt their practices to local crime patterns and community dynamics. We show in the paper that although in some municipalities the implementation of *Plan Cuadrante* involved an increase in resources to make it feasible to move to a beat policing model, the increase in resources alone is not enough to explain the increase in public safety and police effectiveness we observe. We are not claiming that police resources do not matter, but rather that the change in policing approach was key to achieving the decline in victimization and crime perception and the increase in trust in police we document in the paper.

Based on our detailed review of policing approaches in the literature, we now also discuss similarities and differences between beat policing and other policing approaches, such as hot-spot policing or community-based policing. This is useful both to clarify the nature of the reform we study and to highlight the contributions of our paper.

Below, we provide the edited sections in the updated manuscript:

Abstract: the new abstract reads as follows:

This paper provides causal evidence on the effects of beat policing on public security and police effectiveness. Using rich administrative and survey data from Chile, we study *Plan Cuadrante*, a reform that shifted policing toward a beat-policing model by subdividing urban municipalities into smaller geographic units—i.e., quadrants—and permanently assigning officers to these quadrants rather than rotating officers across the municipality. Exploiting the staggered implementation of the program across municipalities, we find large improvements in both public safety and security perceptions. Among surveyed households, twelve-month victimization rates declined by 10 percentage points (36%), and administrative records show a 10% decline in crimes reported to the police. Arrests increased by 14% even as crime declined. The reform also improved security perceptions. The share of households perceiving increased criminal activity fell by 15 percentage points (36%), and, consistent with this pattern, household investment in private security measures declined by 7.7 percentage points (37%). In some municipalities, implementation of *Plan Cuadrante* was accompanied by significant increases in police resources. However, our mechanism

analyses indicate that our findings are not driven by resource increases alone, as they remain sizeable in municipalities with resource increases similar to those in control municipalities. The beat-policing components of the reform—i.e., permanent officer assignment and greater local knowledge—appear to be key to reducing crime and improving security perceptions.

Keywords: police strategy, beat policing, territorial deployment.

Introduction: the new introduction reads as follows:

Crime imposes significant social costs, straining the criminal justice system and increasing both public and private security expenditures. Given these stakes, crime consistently ranks among the public’s top concerns and features prominently in political debate (Cafferata and Scartascini, 2021; Gallup, 2023). Police forces play a central role in controlling crime and maintaining public safety. While there is extensive evidence on the effects of police resources on crime (see, for instance, Chalfin and McCrary, 2017; Levitt, 1997; Di Tella and Schargrodsy, 2004), we know much less about how the deployment of those resources, and the strategy changes enabled by different territorial assignments, shape police effectiveness. Studies of public service provision show that decentralization and local specialization can improve performance by aligning operations with local needs (see, for instance, Besley and Coate, 2003). This suggests that policing models that foster local knowledge and allow officers to adapt practices to local conditions may be more effective. Beat policing reflects these insights. Rather than rotating officers across large areas, it permanently assigns each officer to a smaller territorial unit, enabling the accumulation of local knowledge and adaptation to local crime patterns and community dynamics (Cordner, 1995; Skogan and Hartnett, 1997). This model has become increasingly common across police departments worldwide (National Academies of Sciences, Engineering, and Medicine, 2018; Ministerio del Interior y Seguridad Pública, 2014). Yet despite its widespread adoption, rigorous causal evidence on its effects on public safety and security perceptions remains scarce.

This paper combines rich administrative and survey data from Chile to provide causal evidence that beat policing can significantly improve public security and police effectiveness along multiple dimensions. We exploit quasi-random variation from the staggered adoption of a large reform, *Plan Cuadrante*, which introduced a beat-policing model in urban areas by dividing police operational areas—typically municipalities—into smaller zones—quadrants—and permanently assigning officers to them. This permanent assignment was designed to help officers accumulate local knowledge of territorial features, crime patterns, and communities, and to adapt their practices accordingly. Using a difference-in-differences strategy, we show that adopting municipalities experienced significant declines in victimization, reported crime, and perceived insecurity.

Plan Cuadrante was implemented in municipalities with at least 70% urban population and rolled out between 1998 and 2013 across the 150 municipalities that met this condition. Its central objective was to reduce victimization and insecurity perceptions. Rather than changing police hierarchy or operational doctrine, the reform reorganized frontline deployment by subdividing police operational areas—typically municipalities—into quadrants and permanently assigning officers to each of them. On average, municipalities were divided into seven quadrants, with size and boundaries defined by local police authorities based on road networks,

geographic features, and available resources.

To examine the effects of *Plan Cuadrante*, we use household-level data from the National Urban Security Survey (ENUSC), which reports crime victimization, security perceptions, trust in police, and private security investment for roughly 25,000 urban households across 101 municipalities each year. We complement these data with administrative records on arrests and crimes reported to the police. To leverage staggered adoption, we employ a difference-in-differences strategy comparing early and late adopters using the procedure developed in Callaway and Sant’Anna (2021). Analyses based on survey data focus on 46 urban municipalities that adopted the program between 2005 and 2012 and one never-treated municipality; together, they represent over one-quarter of Chile’s population. Analyses based on administrative data use a broader sample of 96 municipalities that adopted between 2003 and 2013 and 196 never-treated municipalities, representing over half of Chile’s population.

We find that households in adopting municipalities become 10 percentage points (36%) less likely to experience crime in the previous 12 months, with effects appearing within one year and growing over time. Consistent with this finding, we document a 10% decline in crimes reported to the police. These estimates may understate the reform’s true impact because not all crimes are reported and, by bringing police closer to communities, the program could have increased reporting rates. Nevertheless, the alignment between survey-based and administrative estimates provides reassuring evidence of the reform’s effectiveness.

Beyond reducing crime, the reform also reduced insecurity perceptions and private investment in security. This is noteworthy because actual crime and crime perceptions can diverge, potentially leading to suboptimal security expenditures. For example, Ajzenman et al. (2023a) show that rising immigration in Chile between 2000 and 2016 did not increase crime but did increase insecurity perceptions.

A potential identification concern is that crime reductions in adopting municipalities could reflect displacement to neighboring areas rather than genuine effectiveness gains. Hot-spot policing, which concentrates police resources in high-crime locations, has been shown to generate this type of spatial displacement (Blattman et al., 2021a). In our setting, spillovers are less likely because municipalities are large geographic areas and beat policing reorganized deployment across the full municipal territory rather than concentrating resources in selected locations. Nevertheless, we directly assess spatial spillovers through three complementary exercises. We find no evidence that *Plan Cuadrante* displaced crime to neighboring municipalities, diverted police resources from control to treated areas, or generated spillovers in victimization and insecurity outcomes in the ENUSC sample. These results support the interpretation that improvements in treated municipalities reflect genuine effectiveness gains rather than crime displacement.

To improve public safety and security perceptions, police work must strengthen prevention and crime resolution. Both dimensions can be affected by police resources, territorial deployment, and policing strategy. In *Plan Cuadrante*, the core beat-policing components were the reorganization of deployment and the operational strategy adjustments enabled by that reorganization. At the same time, some municipalities experienced sizeable increases in police resources, which raises the question of whether resource expansion alone can explain the effects we estimate.

During our study period, public expenditure on security was rising, and most municipalities experienced increases in police resources. Police resources were not the central component

of *Plan Cuadrante*, but adopting the model required a minimum operational capacity, which in some municipalities implied increases in officers and patrol vehicles above those observed elsewhere. Thus, part of the effects we document could reflect resource increases. However, changes in police resources alone do not explain our findings. Municipalities that adopted *Plan Cuadrante* but experienced resource increases similar to those in control municipalities still show large and statistically significant improvements in public safety outcomes. Moreover, for most outcomes, these effects are close in magnitude to those in municipalities where resources increased more substantially. In addition, variation in police resources unrelated to *Plan Cuadrante* adoption does not seem to generate comparable gains. Taken together, these patterns indicate that while resource increases helped enable implementation and may explain part of the effects in some municipalities, the beat-policing components of the reform were central drivers of the overall gains.

Against this background, the core institutional change introduced by *Plan Cuadrante* was on the deployment margin. The reform permanently assigned officers to smaller, fixed quadrants while leaving national operational guidelines unchanged across treated and control municipalities throughout our study period (Carabineros de Chile, 2018; Winchester et al., 2007). The Chilean police is a highly hierarchical institution with centrally defined doctrine and limited officer discretion, so adoption did not involve a top-down redesign of policing strategy. Instead, any strategic adjustments likely emerged from the new deployment itself. By remaining in the same territory, officers accumulated local knowledge of the area and its community and could adapt day-to-day practices—patrol routes and prevention activities—to local conditions. In this sense, *Plan Cuadrante* changed police work primarily through deployment, with strategy evolving endogenously within a common institutional framework.

The evidence supports this interpretation. Following the adoption of *Plan Cuadrante*, households report higher police presence in their neighborhoods, a pattern that is consistent with both resource expansion and deployment changes. However, our broader results suggest effects beyond patrolling intensity alone. While earlier work finds that police presence can reduce crime while officers are present but may be attenuated by displacement across places or time windows (Mastrobuoni, 2019; Blanes-i Vidal and Mastrobuoni, 2018), we find declines in victimization and reported crime with no evidence of spillovers to neighboring municipalities. In line with this pattern, households report better police performance and greater trust in police, and administrative data show that arrests increase even as crime declines, including in municipalities with resource increases similar to those in control municipalities. Taken together, these findings suggest that beat policing improved public security through stronger deterrence and more effective prevention, response, and crime resolution under stable territorial assignment.

Our findings contribute to two related strands of the literature on police organization and public safety. First, they provide some of the first causal evidence on the effects of beat policing on crime and security perceptions. Beat policing has become one of the most widely adopted policing models globally (National Academies of Sciences, Engineering, and Medicine, 2018; Ministerio del Interior y Seguridad Pública, 2014), yet robust causal evidence on its effectiveness remains scarce. We show that this model can generate substantial reductions in crime victimization, reported crime, and insecurity perceptions.

These results also speak to recent work on other policing models. Hot-spot policing reallocates police resources toward high-crime locations and has been shown to reduce crime in targeted areas, though often with displacement to nearby places (Di Tella and Schargrodsky, 2004; Blattman et al., 2021a; Draca et al., 2011; Weisburd, 2021). Beat policing differs fun-

damentally. Rather than concentrating police presence in selected locations—where the same officers need not return repeatedly—it provides continuous, territory-wide coverage through permanent officer-to-area assignment. This distinction matters in our setting, where many quadrants were not crime hot spots and the reform generated municipality-wide declines in crime and insecurity perceptions, with no evidence of spillovers to neighboring municipalities. Community policing, another model that has received growing attention in the economics literature, emphasizes citizen participation in defining policing priorities and, in some cases, in implementation itself. While it has been shown to increase trust in police in developed countries, evidence from low- and middle-income settings points to more limited effects on crime and trust (Blair et al., 2021; Tobón, 2021). Beat policing also differs from that approach. By permanently assigning officers to stable territorial units, it strengthens local knowledge and brings police closer to communities without requiring direct citizen participation in strategy or operations. Our findings show that this model can reduce crime and insecurity perceptions while increasing trust in police institutions in a middle-income country.

A second contribution is to the broader literature showing that specific features of police organization matter for police effectiveness and public security outcomes. Research has shown that faster police response times improve crime clearance (Blanes-i-Vidal and Kirchmaier, 2018), that the effectiveness of police presence depends on how this is organized and deployed rather than simply on their intensity (Facchetti, 2025; Mastrobuoni, 2019; Blanes-i Vidal and Mastrobuoni, 2018), that shifts in police prioritization across crime types shape criminal behavior (Adda et al., 2014), and that better information systems and technology can improve police performance (Mastrobuoni, 2020). Earlier work also found that areas covered by smaller, more localized police departments outperform, in terms of security, those covered by larger centralized ones, hypothesizing that this might be related, among other factors, to closer citizen-police engagement (Ostrom et al., 1973, 1978). We add to this literature by studying a policing model that combines several of these margins within a single organizational reform. By assigning officers permanently to smaller territorial units rather than rotating them across larger ones, beat policing is designed to increase police visibility, strengthen knowledge of local conditions, and improve the capacity of officers to adapt their practices to local crime patterns. In line with these mechanisms, we show that the adoption of *Plan Cuadrante* increased perceived police presence in neighborhoods and raised police effectiveness, as arrests increased even while victimization and reported crime declined. These organizational improvements translated into meaningful gains in public safety and security perceptions.

A third contribution concerns the literature on police resources, which documents elasticities of crime with respect to officers and expenditure ranging from -0.1 to -2 , with larger effects in developed settings and for violent crimes (Chalfin and McCrary, 2017; Levitt, 1997; Di Tella and Schargrodsky, 2004). We complement this body of work by showing that the quantity of resources is not the only margin that matters. How those resources are deployed matters as well. *Plan Cuadrante* improved public safety outcomes even in municipalities that experienced no increase in police resources relative to controls, suggesting that reorganizing deployment can raise public safety and police effectiveness without expanding the resource base.

We also contribute to the literature on police practices, institutional trust, and citizen cooperation. A persistent challenge is that security perceptions often diverge from actual crime trends (Ajzenman et al., 2023a), limiting the scope for improving both simultaneously. Rising crime erodes institutional trust (Blanco, 2013), while police misconduct undermines coopera-

tion (Ang et al., 2025).¹ Our results show that beat policing can improve both dimensions—security and security perceptions—by bringing officers into sustained engagement with local communities through permanent territorial assignment. We document not only increases in police trust and evaluations of police performance, but also substantial reductions in private security investment, indicating that households revised their threat assessments in response to the program.

Finally, our findings speak to a broader question about how organizational design shapes public-service effectiveness. A growing literature shows that the spatial organization of service provision matters. In policing, we show that a model organizing officers around fixed territorial units—allowing them to build local knowledge, coordinate with local actors, and adapt practices accordingly—can substantially improve both crime control and institutional legitimacy. This aligns with evidence from health and education showing that bringing providers closer to the populations they serve can improve outcomes (Besley and Coate, 2003; Bardhan, 2002; Bloom et al., 2015; Basurto et al., 2020). In a rigorously identified setting, we show that this type of organizational proximity generates meaningful gains in police effectiveness and citizen security.

The rest of the paper is organized as follows. Section 2 presents background information about Chilean police and the program we study. Section 3 introduces the data and empirical strategy. Section 4 presents the main results and discusses potential drivers behind our findings. Finally, section 5 concludes.

Institutional context and the *Plan Cuadrante*: the new section on the institutional context and the *Plan Cuadrante* reads as follows:

Carabineros, Chile’s national police force, was founded in 1927. Through most of our study period, Carabineros operated under the Ministry of Defense, which oversaw both military and police functions until domestic security responsibilities were transferred to the Ministry of Interior in 2011.²

Carabineros’ mission, functions, and structure are established by Law 18.961 and the Regulation on Carabineros’ Organization (Reglamento de Carabineros) (Biblioteca del Congreso Nacional de Chile, 1990; Ministerio de Defensa Nacional, República de Chile, 1989). Under this framework, Carabineros is a professional, disciplined, and hierarchical police institution whose mission is to preserve public order and security with a central emphasis on preventive policing. The General Director heads the strategic level, responsible for defining doctrine and institutional strategy, while the Subdirector coordinates implementation. The Directorate of Order and Security administers police functions at the tactical level.

Territorially, Carabineros is organized into zones that oversee *prefecturas*. Zones and *prefecturas* are primarily management layers. Zones map roughly one-to-one to Chile’s regional divisions, with two exceptions: the Metropolitan Region of Santiago—the most populated in the country—and La Araucanía—which has recently experienced significant social conflict—are each divided into two zones. Each zone is subdivided into *prefecturas*—54 nationwide—which supervise 227 *comisarías*, the main operational units at the tactical level. These *comisarías*

¹Racial disparities in police behavior are also well documented; see, e.g., Gonçalves et al. (2025), who develop a method to correct for selection bias in estimates of racial disparities in the use of force.

²Currently, Carabineros reports to the Ministry of Public Security, created in 2024 to focus exclusively on domestic security.

oversee smaller units—*subcomisaría*s, *tenencias*, *retenes*, *puestos*, and *avanzadas*—that operate within their assigned jurisdictions.

In practice, the *comisaría* is the basic unit of routine frontline policing and, in most municipalities, serves as the territorial anchor of police work. Officers conduct patrol, surveillance, prevention, and first response at this level. In most municipalities, a single *comisaría* covers the entire municipal territory, though larger municipalities may have multiple ones. Santiago, for instance, has four. All municipalities that adopted *Plan Cuadrante* during our study period had a *comisaría*. With the exception of 9 municipalities, each had a single *comisaría* covering the full municipal territory. On average, a *comisaría* in our sample covered 1,608 square kilometers and served 57,642 inhabitants.

Plan Cuadrante was introduced within this institutional framework without altering Carabineros’ hierarchical structure or its strategic and tactical levels. The reform emerged in a context of rising victimization and insecurity, with the explicit goal of reversing these trends (Ministerio del Interior y Seguridad Pública, 2014). Its central innovation was a shift toward a beat policing model. In the criminology literature, beat policing refers to assigning officers to fixed territorial areas so they can accumulate local knowledge, strengthen ties with residents, and respond more effectively to local crime patterns (Cordner, 1995; Skogan and Hartnett, 1997). In practice, the catchment area of police forces in municipalities adopting *Plan Cuadrante* was partitioned into smaller, fixed geographic units (quadrants), to which officers were permanently assigned. Rather than building new infrastructure, the reform redefined the territorial responsibility of officers in already existing police stations (Winchester et al., 2007).³ This represented a clear break from the pre-reform system, in which officers rotated across broad operational territories, very often spanning entire municipalities.

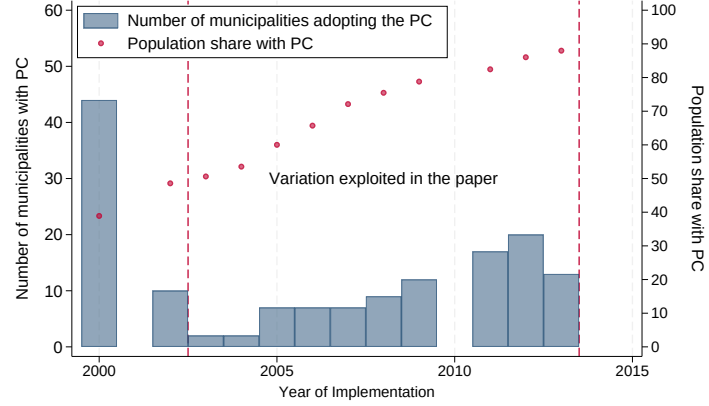
Carabineros began piloting *Plan Cuadrante* in 1998. The program expanded gradually through 2013, as shown in Panel (a) of Figure 1, reaching 150 urban municipalities and covering roughly 90% of Chile’s population.⁴ Online Appendix A provides further details about the rollout of the program. The rollout was designed to follow a municipality ranking based on a weighted index of victimization, police resource deficits, drug prevalence, unemployment, and demand for police services, with annual entry determined by available budget.⁵ In practice, however, Budget Office evaluations show that adoption diverged from this ranking for undocumented reasons, so the rule operated more as a loose guideline than as a binding assignment mechanism (Salas P. et al., 2014).

³In 2016, Carabineros launched the *Plan Cuadrante 2.0* program in those municipalities served by *Plan Cuadrante*. While this reform introduced additional reforms aimed at integrating the community into the provision of security and building capacity to gather and analyze crime data in police stations, this paper’s study period ends before these improvements were introduced.

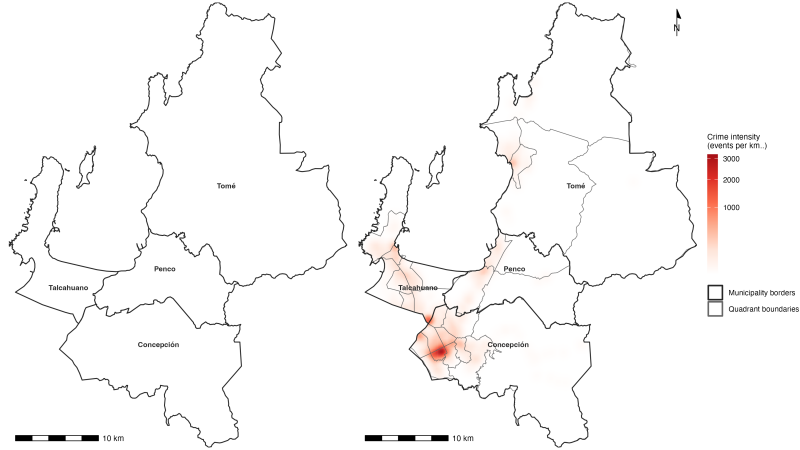
⁴Further details about the expansion of *Plan Cuadrante* are provided in Online Appendix A and Table A.1.

⁵The weights assigned to each component changed on a yearly basis (Winchester et al., 2007; Salas P. et al., 2014).

Figure 1: Implementation of *Plan Cuadrante*



a Adoption of *Plan Cuadrante*



b *Plan Cuadrante* in the Concepción area

Notes: Panel (a) shows the number of municipalities adopting the *Plan Cuadrante* each year (bars), while the circles above them illustrate the cumulative share of the Chilean population living in a municipality with the *Plan Cuadrante* already in place. Panel (b) depicts the metropolitan area of Concepción, the second-largest in Chile: the left map shows its four municipalities—the original police catchment areas—and the right map overlays the Carabineros quadrants together with a kernel-density estimate of all types of robberies and thefts recorded by the police (both reported crimes and arrests). It illustrates how *Plan Cuadrante* subdivided each catchment area (which corresponds to the territory of a municipality) into smaller quadrants, as well as the spatial concentration of crime within them.

Conceptually, police effectiveness can be viewed as a function of three inputs: total resources, territorial deployment, and policing strategy. Although in some municipalities police resources were increased significantly to operationalize the beat policing model (Winchester et al., 2007; Salas P. et al., 2014), the program primarily transformed the latter two elements. The creation of quadrants was accompanied by permanent geographic assignment of officers to specific areas. This change was operationally critical. By making officers responsible for their quadrant, it encouraged accumulation of local knowledge and stronger community ties, allowing them to adapt patrols and prevention activities to local crime patterns. Importantly, this was not driven by program-specific operational directive changes. Throughout the period we study, a single national set of operational guidelines applied to officers in both treated and control municipalities (Carabineros de Chile, 2018; Winchester et al., 2007). Thus, while the average effects we document on crime and crime perception likely reflect combined changes in the three elements we discuss—i.e., resources, deployment, and strategy—the deployment change itself was essential to operationalizing the reform. We discuss the drivers of the effects we document in Section 4.4.

Quadrant size and shape were determined by local police authorities based on factors such as road networks, geographic features, and available resources. In principle, resource allocation across quadrants followed demand for police services, measured in Equivalent Surveillance Units (UEVs); in practice, the station commander retained discretionary authority, as was also the case prior to *Plan Cuadrante* (Salas P. et al., 2014). Municipalities implementing *Plan Cuadrante* were subdivided into an average of seven quadrants, each typically staffed by approximately three officers per shift. On average, each quadrant covered 185 square kilometers and served 13,466 inhabitants. Panel (b) of Figure 1 illustrates this structure, showing the 29 quadrants created across the four municipalities of the Concepción metropolitan region—Chile’s second-largest metropolitan area and the largest in our sample. Quadrants represented approximately 14% of the original *comisaría* territory. Officers continued to operate from the same police station but were permanently assigned to a specific sub-area within its jurisdiction. While Carabineros’ hierarchical structure limits frontline officer decision-making authority, each quadrant was assigned a Quadrant Delegate (*Delegado de Cuadrante*), a non-commissioned officer, responsible for immediate operational decisions within the quadrant.

Plan Cuadrante may affect public safety through two complementary channels: deterrence and police effectiveness, understood as the capacity to respond to and resolve crimes. The changes induced by *Plan Cuadrante*—permanently assigning officers to well-defined neighborhoods and, in some municipalities, increasing available resources—can increase both police visibility and local knowledge. As officers repeatedly patrol the same area, they become familiar with its streets, residents, and routines, which can raise the perceived probability of identification and apprehension and reduce the expected returns to crime. At the same time, this deeper knowledge of local territory and communities can improve police response and case resolution once crimes occur. The reform therefore likely affects both dimensions. Consistent with this interpretation, in Section 4 we document declines in victimization and in reported crime—patterns consistent with deterrence—together with increases in arrests, consistent with improved police effectiveness.

Although beat policing shares features with other policing approaches, such as hot-spot policing and community-based policing, which have been widely studied, it has distinctive features that make it worth studying as an independent approach. Hot-spot policing concentrates resources in high-crime areas and does not focus on building local knowledge and community

ties. It is typically implemented through targeted interventions at specific locations rather than systematic territorial organization, and the same officers are not necessarily sent repeatedly to the same areas. Evidence on hot-spot policing is mixed, with some studies showing crime reductions in targeted areas but also potential displacement effects (Braga et al., 2019). *Plan Cuadrante*, by contrast, subdivided all urban municipalities into quadrants with permanent officer assignment. As Panel (b) of Figure 1 illustrates, some quadrants contained crime hot-spots, but many did not. The reform thus addressed entire urban territories rather than specific high-crime areas, potentially mitigating displacement effects associated with hot-spot policing.

Community-based policing emphasizes building relationships with residents and addressing underlying social issues, and typically involves active community participation in defining strategy and priorities—in some cases, even assuming surveillance and prevention responsibilities. While increasing trust in the police and strengthening community ties were among the declared goals of *Plan Cuadrante*, and may have resulted from its territorial reorganization, the community was not formally involved in the provision of security.⁶

In recent decades, police departments worldwide have transitioned toward beat policing reforms (National Academies of Sciences, Engineering, and Medicine, 2018; Ministerio del Interior y Seguridad Pública, 2014), yet the effectiveness of this model has not been rigorously assessed. Although the beat policing approach introduced through *Plan Cuadrante* shares features with programs previously studied in the economic literature, its distinctive characteristics—comprehensive territorial reorganization and systematic local officer specialization—make it a valuable case for studying how deployment and policing strategies not yet examined in the economics literature affect crime and crime perception.

Mechanisms: The revised section on mechanisms discusses the drivers of the effects we document, organized around the framework you suggested. Specifically, we investigate the three channels through which *Plan Cuadrante* may affect public security, namely changes in police resources, territorial deployment, and policing strategy. Please note also that figures including data on police resources have changed slightly, as more detailed information on police vehicles and officers became available after the previous manuscript was submitted. The new section reads as follows:

4.4 What Is Driving the Effects of *Plan Cuadrante*?

To reduce crime and improve security perceptions, police forces need to strengthen both prevention and police effectiveness. Both margins can be shaped by resources, territorial deployment, and policing strategy. This section examines which components of *Plan Cuadrante* are most likely to drive the effects documented in Section 4.

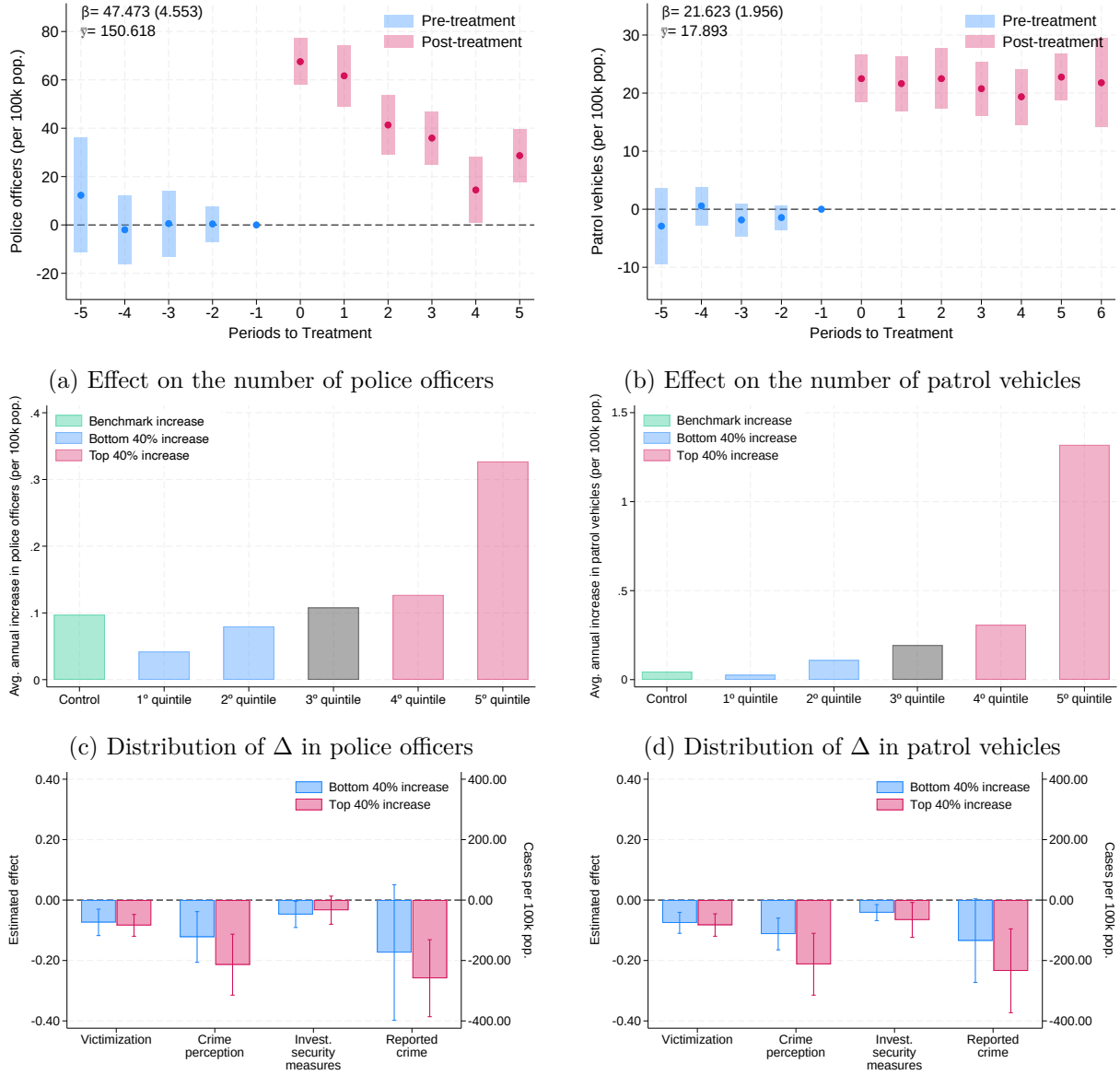
The central objective of *Plan Cuadrante* was to introduce a beat policing model by reorganizing deployment. Police operational territories that had typically matched municipal boundaries

⁶The *Plan Cuadrante*’s original design contemplated a formal community integration model (MICC) with a Community Affairs Office (*Oficina de Asuntos Comunitarios*) and a dedicated officer in each police station. However, this model was not implemented during our study period. Independent program evaluations from the Budget Office document that these Community Affairs Offices did not exist and had no dedicated personnel, budget, or structured activities in the period we study (Winchester et al., 2007; Salas P. et al., 2014). The community integration model was piloted in four police stations during 2012–2013, none of which are in our analytical sample. Financing was then discontinued in early 2014 and only restored and expanded after our study period ends.

were subdivided into quadrants, and officers were permanently assigned to these quadrants. During our study period, national operational guidelines remained the same in adopting and non-adopting municipalities. Still, this new deployment likely affected policing strategy at a more operational level, because it allowed officers to adapt patrol routes, schedules, and prevention practices using the local knowledge that the program sought to build.

Although changes in resources were not the core of the reform, implementation contemplated increases in municipalities that lacked the personnel and patrol capacity required to move to this model. During the period we study, Chile was expanding public spending on security, and police resources increased in most municipalities regardless of whether or when they adopted *Plan Cuadrante*. On average, however, municipalities adopting the program experienced larger increases in officers and patrol vehicles, as shown in panels (a) and (b) of Figure 4. Since increases in police resources have been shown to reduce crime in some contexts (Evans and Owens, 2007), this pattern raises a central question for interpretation, namely whether our estimates reflect resource expansion or the deployment and strategy changes introduced by *Plan Cuadrante*. We address this question through a set of exercises that exploit heterogeneity in resource increases across municipalities and over time.

Figure 4: Plan Cuadrante and Changes in Police Resources



(e) Effects by the intensity of Δ in police officers (f) Effects by the intensity of Δ in patrol vehicles

Notes: Panels (a) and (b) show the effect of *Plan Cuadrante* on police officers and patrol vehicles per 100,000 inhabitants. Panels (c) and (d) show the distribution of annual changes in officers and patrol vehicles across municipalities. Panels (e) and (f) report effects by the intensity of resource changes, comparing municipalities in the bottom 40% and top 40% of the distribution. The estimation follows Callaway and Sant'Anna (2021). The graphs display estimated coefficients and 95% confidence intervals. Standard errors are clustered at the municipality level using wild bootstrapping.

Panels (c) and (d) of Figure 4 show that the average increase in police resources masks substantial heterogeneity. Municipalities in the bottom 40% of the resource-change distribution (blue bars) experienced increases close to the average annual variation observed among never treated and not-yet-treated municipalities (green bar). By contrast, municipalities in the top 40% (red bars) experienced much larger increases, with officers rising by around 22% and patrol vehicles

by around 80%. If the effects of *Plan Cuadrante* were driven by resource increases alone, we would not expect effects in municipalities whose resource increases were similar to those of the control group. Yet panels (e) and (f) of Figure 4 show sizeable effects for most crime and crime-perception outcomes even in that low-increase group. While there may be a resource gradient for some crime-perception outcomes, the effects on twelve-month victimization and private investment in security among municipalities with control-like resource increases are close in magnitude to those in municipalities with larger increases. Reassuringly, municipalities in the top and bottom 40% of the resource-increase distribution do not display differential pre-treatment trends in outcomes or police resources, which suggests that this heterogeneity is not driven by differential pre-existing trends.⁷

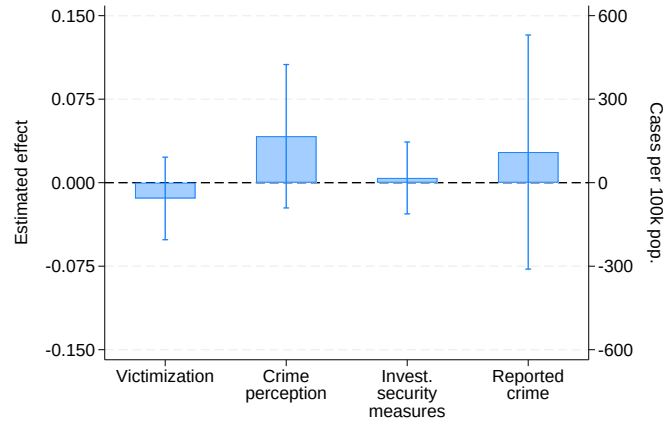
Although we cannot fully separate the beat policing component from the resource component of the reform—resources may have been important to enable effective implementation of the program, or may explain part of the effects in some municipalities—the fact that municipalities with resource increases similar to those experienced by control municipalities still display sizeable effects indicates that the effects of *Plan Cuadrante* are not driven entirely by resources.⁸⁹

⁷We test for the joint significance of the pre-treatment coefficients between the two groups, separately for the split based on the increase in police officers and the split based on the increase in patrol vehicles. The resulting p-values are, respectively, 0.823 and 0.048 for victimization, 0.059 and 0.379 for crime perception, 0.820 and 0.157 for investment in security measures, and 0.213 and 0.419 for reported crime, as well as 0.616 and 0.444 for police officers and 0.405 and 0.157 for patrol vehicles.

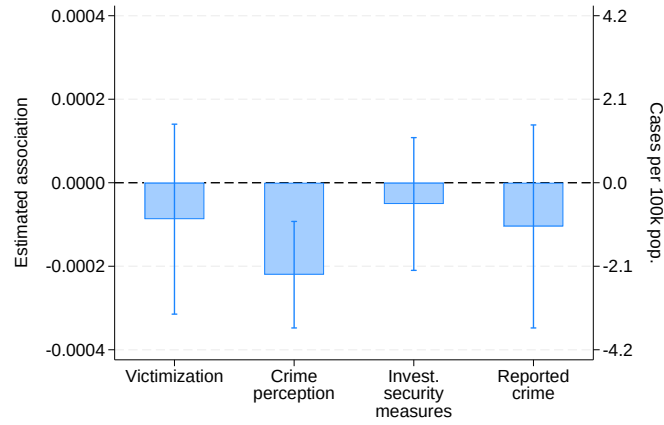
⁸We also re-estimate the results in Figure 2 but controlling for police resources. Online Appendix Table B.3 shows that the magnitude of the estimated effects remains similar for most variables, although the statistical significance is reduced in some estimations. This likely reflects a mechanical feature of the Callaway and Sant’Anna (2021) estimation procedure: control variables are not incorporated parametrically, but are instead used to construct the comparison group via outcome regression or propensity-score weighting, so the resulting estimation uncertainty propagates to the ATT standard errors. Note, in addition, that police resources are a bad control—i.e., at least part of their variation is a result of the adoption of *Plan Cuadrante*—so these results should be interpreted cautiously.

⁹Table B.4 in Online Appendix B examines which factors are associated with a larger increase in police resources upon the adoption of *Plan Cuadrante*.

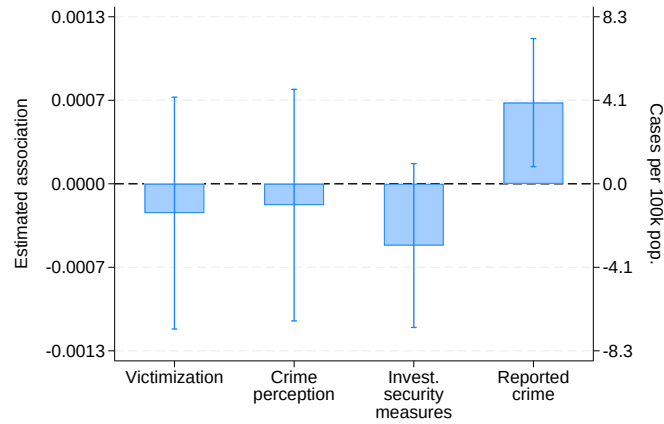
Figure 5: Police Resources and Public Safety



a Effects of *Seguridad Ciudadana*



b Association between police officers and effectiveness



c Association between patrol vehicles and effectiveness

Notes: Panel (a) reports the pooled effect of *Seguridad Ciudadana*, a program that expanded local security personnel to support patrolling and deterrence but did not replicate the deployment-and-strategy features of *Plan Cuadrante*. Panels (b) and (c) report within-municipality associations between police resources and police-effectiveness outcomes among the 41 municipalities of the Santiago Metropolitan Region, all of which had adopted *Plan Cuadrante* by 2000. The regressions include municipality and year fixed effects. Standard errors are clustered at the municipality level using the wild bootstrapping procedure.

To further investigate whether resource increases alone can explain our results, we assess the effects of *Seguridad Ciudadana*, a program that expanded local security personnel to support patrolling and deterrence, but did not replicate the core deployment-and-strategy features of *Plan Cuadrante*. Using a difference-in-differences specification following Callaway and Sant’Anna (2021), panel (a) of Figure 5 shows no significant effects of these resource changes on police effectiveness. Further details about this exercise and the corresponding event-study estimates are reported in Online Appendix B and Online Appendix Figure B.4.

Finally, we study whether variation in police resources in municipalities that had already adopted *Plan Cuadrante* improves public safety and security perceptions. Using ENUSC data and information on resources for 2006–2012 in the 41 municipalities of the Santiago Metropolitan Region—all of which had adopted *Plan Cuadrante* by 2000—we exploit variation in officers and patrol vehicles that is unrelated to program adoption. Panels (b) and (c) of Figure 5 show the association between police resources and our main outcomes, conditional on municipality and year fixed effects. With one exception—the association between police officers and crime perception—none of these associations is statistically significant at conventional confidence levels.¹⁰

Taken together, these results suggest that resource increases alone do not account for the effects of *Plan Cuadrante*. We next turn to evidence that speaks more directly to the beat-policing components of the reform.

Panel (a) of Figure 4 shows that, following the adoption of *Plan Cuadrante*, households are much more likely to report increased police presence in their neighborhoods. The share of households perceiving increased police presence rises by 35 percentage points (156%), the share reporting that police pass by their house daily rises by 22 percentage points (59%), the share reporting never seeing police patrols near their house falls by 10 percentage points (57%), and complaints about lack of police surveillance fall by around 5 percentage points (20%). This evidence is consistent with both larger resources and the new deployment model. Yet the latter estimate—the only police-presence outcome for which we can separately estimate effects in municipalities with small and large resource increases—is similar in magnitude across the two groups, which suggests that the increase in police presence is not explained only by larger resource increases (see Online Appendix Figure B.3).¹¹ More importantly, the broader pattern of results suggests that *Plan Cuadrante* did more than simply increase patrolling intensity. Existing work shows that police presence can reduce crime while officers are present, but that its overall effects may be attenuated when crime is displaced across places or time windows (Mastrobuoni, 2019; Blanes-i Vidal and Mastrobuoni, 2018). Our results point to something broader. As discussed in Section 4, we find actual declines in victimization and reported crime, together with no evidence of spillovers to neighboring municipalities. This pattern is more consistent with officers accumulating local knowledge and adapting prevention strategies to local conditions under stable territorial assignment than with a purely mechanical increase in patrol intensity.

Panel (b) of Figure 4 reinforces this interpretation. Households in early-adopting municipalities

¹⁰Online Appendix Figure B.5 reports the same exercise and also shows similar results for the broader set of municipalities that adopted *Plan Cuadrante* before 2006 and never-treated municipalities (red bars); see Online Appendix B for further details.

¹¹We cannot estimate heterogeneous effects for other police-presence outcomes, such as the reporting of police passing by or the share of households perceiving an increase in police presence, because those variables are available only in ENUSC rounds 2003, 2005, and 2006, while resource data start in 2006.

report better police performance across multiple dimensions. They are 8.1 percentage points (16%) more likely to rate police performance as good or very good, 7.6 percentage points (11%) more likely to describe police as efficient, 5 percentage points (6%) more likely to report that police are helpful, and 4.9 percentage points (7%) more likely to report that police are well informed. They are also 3.9 percentage points (10%) less likely to perceive police as corrupt, 7.9 percentage points (21%) less likely to believe police fail to control crime, and 7.6 percentage points (22%) less likely to report that police do not perform their duties effectively.¹²

The administrative evidence points in the same direction. Panel (c) of Figure 4 shows that arrests per 100,000 inhabitants increase by 48 (14%) in our main estimates, even as crime declines overall.¹³ This effect is also significant among municipalities with resource increases similar to those in the control group (see Online Appendix Figure B.3). Consistent with this pattern, panel (d) of Figure 4 shows that the arrest rate per reported crime also increases by 1.7 percentage points (10%), suggesting that a higher share of crimes reported to the police are being solved. Because these gains appear even where resources did not increase more than in control municipalities, they are difficult to reconcile with a pure resource story and are more naturally interpreted as improvements in police effectiveness under the deployment and strategy changes introduced by *Plan Cuadrante*. The corresponding event-study estimates for arrests, the arrest rate per reported crime, and trust in the police in Figure 4 confirm that these changes emerge only after adoption.

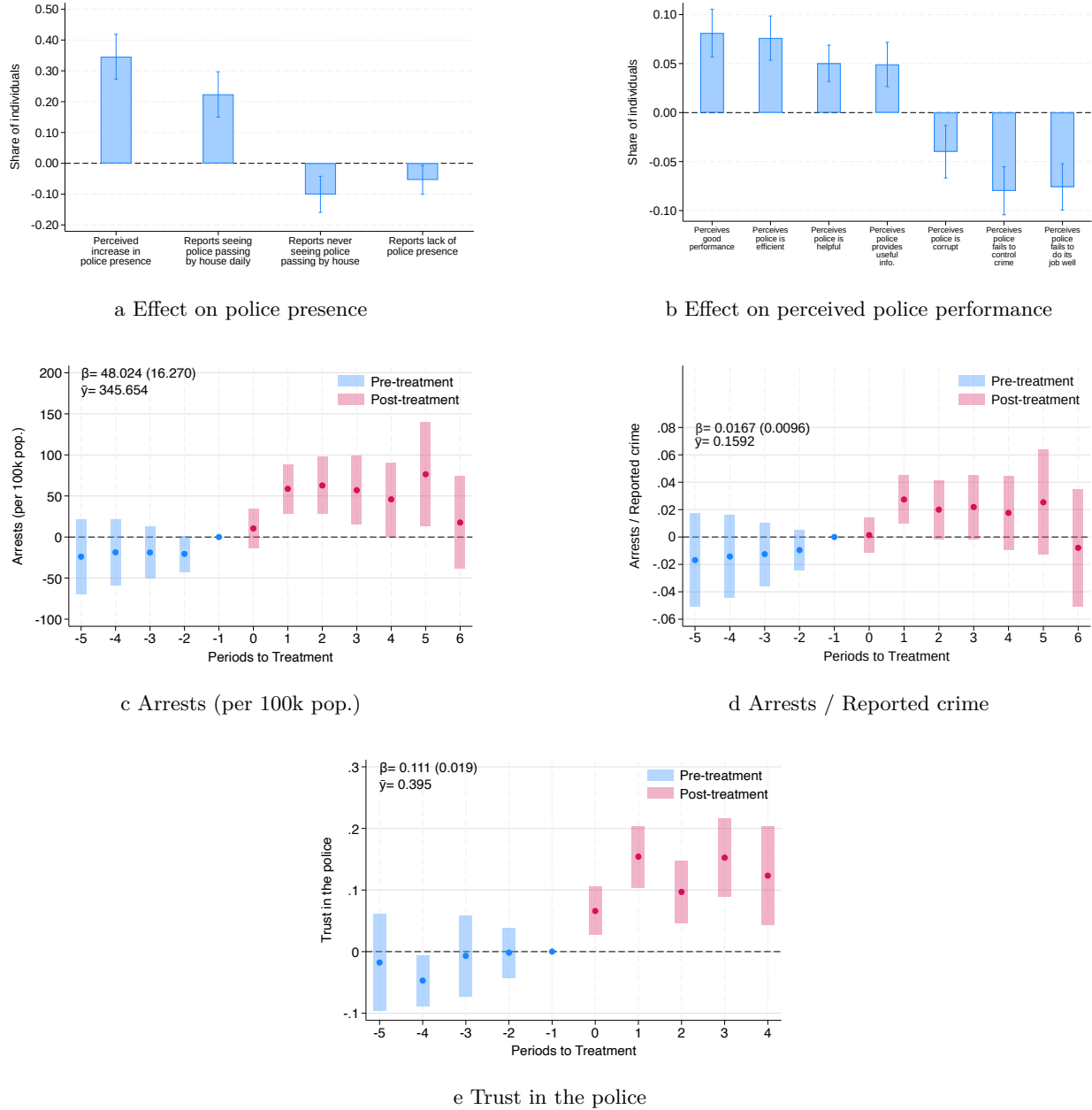
Finally, panel (e) of Figure 4 shows a sharp increase in trust in the police in our main specification (Section 4), consistent with one of the objectives of beat policing, namely bringing officers closer to the community and allowing them to accumulate local knowledge through repeated interaction. Online Appendix Figure B.3 shows that this effect remains sizeable both in municipalities that experienced resource increases similar to the control group and in those with larger increases.

Overall, although resources were likely important for implementation and may explain part of the effects in some municipalities, the results in this section indicate that resource changes alone are not enough to explain our findings. The evidence is more consistent with a combination of deployment changes and the strategy adjustments enabled by local knowledge accumulation under permanent assignment to quadrants.

¹²Information on police performance is only available in ENUSC round 2005. Using this round, we compare perceived police performance in municipalities that adopted *Plan Cuadrante* in 2005 or earlier and in municipalities adopting after 2005. Figure B.6 in Online Appendix B shows that the number of years of exposure to *Plan Cuadrante* is positively correlated with perceived police performance. While these exercises on perceived police performance should be interpreted with caution, we show in the same appendix that the timing of future adoption is not correlated with perceived police performance in 2005 for municipalities that had not yet adopted the program by 2005.

¹³As with reported crime, this event study is restricted to 5 pre-treatment and 7 post-treatment periods for symmetry with the ENUSC-based analysis; Online Appendix Figure C.8 reports the results for arrests using the full set of pre-treatment and post-treatment periods.

Figure 6: *Plan Cuadrante* and Changes in Police Performance



Notes: Panel (a) reports the effects on perceived police presence. Panel (b) reports perceived police performance in 2005, comparing municipalities where *Plan Cuadrante* was already implemented before 2006 with municipalities where it was implemented after 2005. Panels (c), (d), and (e) report dynamic event-study estimates for arrests per 100,000 inhabitants, the ratio of arrests to reported crime, and trust in the police. The estimation follows Callaway and Sant'Anna (2021). The graphs display estimated coefficients and 95% confidence intervals. Standard errors are clustered at the municipality level using the wild bootstrapping procedure.

Conclusion: the new conclusion reads as follows:

Beat policing—i.e., assigning the same officers to smaller, well-defined territorial units rather

than rotating them over larger ones—has become a widespread strategy to address crime and maintain public safety, yet rigorous causal evidence on its effectiveness remains scarce. Using administrative and survey data from Chile, we show that *Plan Cuadrante*, a reform that introduced beat policing in urban municipalities, substantially reduced victimization and crime perception, while increasing trust in the police. Although its implementation required increasing police resources in some municipalities, we show that resource increases alone cannot account for the observed improvements. The beat policing elements of the reform—permanent assignment to quadrants enabling officers to build local knowledge—seem to be key drivers of its effectiveness.

The reform not only reduced crime incidence but also insecurity perceptions. As a result, households in adopting municipalities reduced their investment in private security measures. Crime rates and perceived victimization risk do not always move together (Ajzenman et al., 2023b), yet *Plan Cuadrante* reduced both simultaneously. This is an important result as misalignments between actual and perceived risk can lead households to overinvest in private security and distort public expenditure priorities (Mastrorocco and Minale, 2018; Ajzenman et al., 2023b).

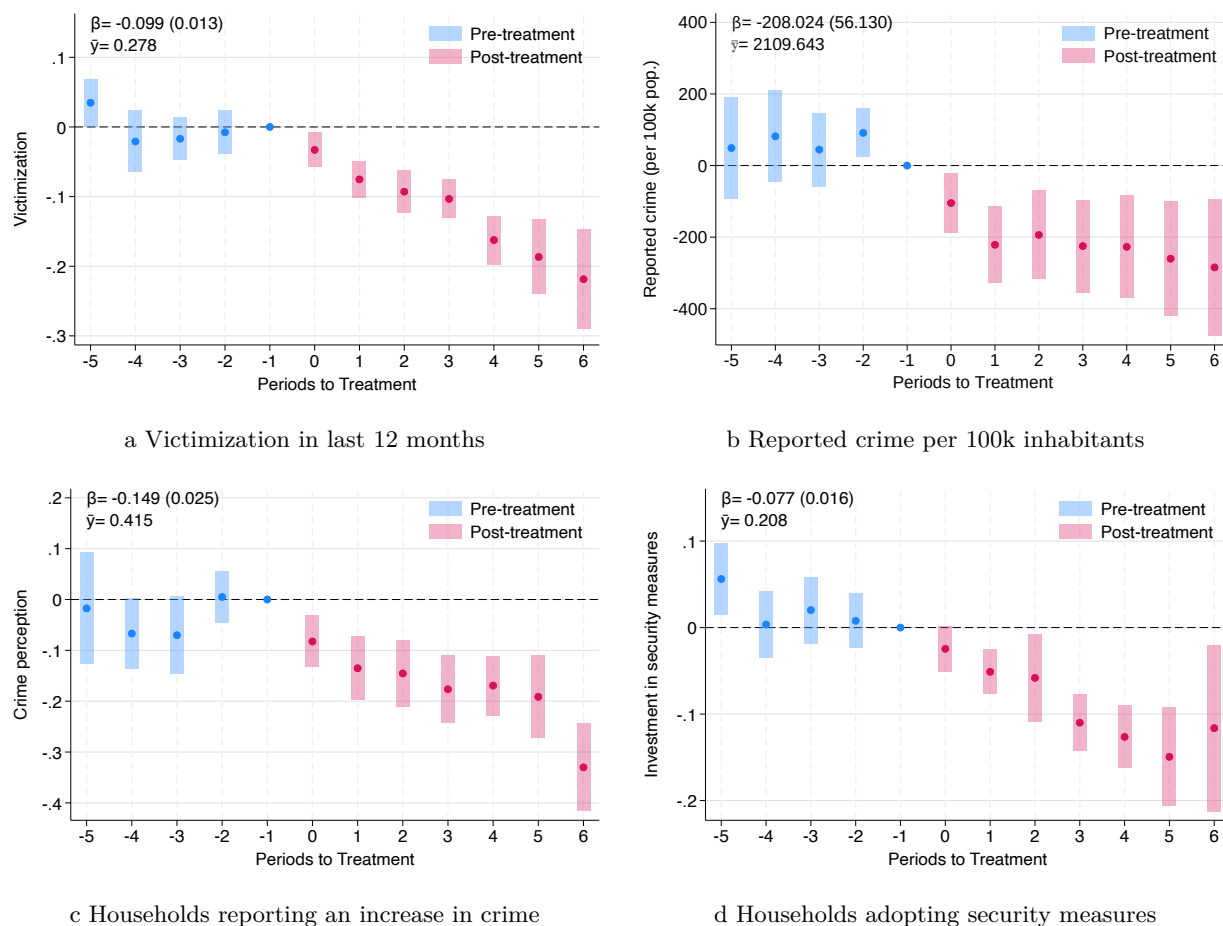
The reform also increased trust in the police—an outcome that has proven particularly difficult to achieve in low- and middle-income countries. Community policing, long regarded as the primary strategy for building police legitimacy, has shown limited effects on both crime and trust in such settings (Blair et al., 2021; Tobón, 2021). Beat-policing takes a different approach. Rather than relying on community co-production of security, it builds proximity and trust organically through sustained, stable territorial engagement. Our results suggest this organizational logic can be effective even outside developed-country contexts.

Taken together, these results contribute to a broader body of evidence that highlights that the spatial organization of police resources matters. By assigning officers permanently to smaller territorial units, beat policing increases visibility, deepens local knowledge, and improves the capacity of officers to prevent crime and respond to it effectively. This pattern echoes findings from health and education showing that aligning frontline service providers more closely with the communities they serve enhances outcomes (Besley and Coate, 2003; Bardhan, 2002; Bloom et al., 2015; Basurto et al., 2020).

While *Plan Cuadrante* was implemented under specific conditions, those conditions help clarify when beat policing is likely to succeed. Two institutional features appear particularly important. First, the absence of significant organized crime during the study period allowed officers to operate repeatedly in their assigned areas without meaningful risk of intimidation or capture, enabling local knowledge and community ties to take root. Second, the relatively high level of professional training in the Chilean police provided a foundation for effectively implementing the reform. Neither of these features is idiosyncratic. They describe an institutional environment that is present, or could be developed, in many settings. Together, they suggest that the gains from beat policing are most likely to materialize when supported by a capable police force operating in a security environment where persistent local engagement is feasible.

Other figures and tables referenced above:

Figure 2: Effects of *Plan Cuadrante* on Crime Outcomes and Security Perceptions



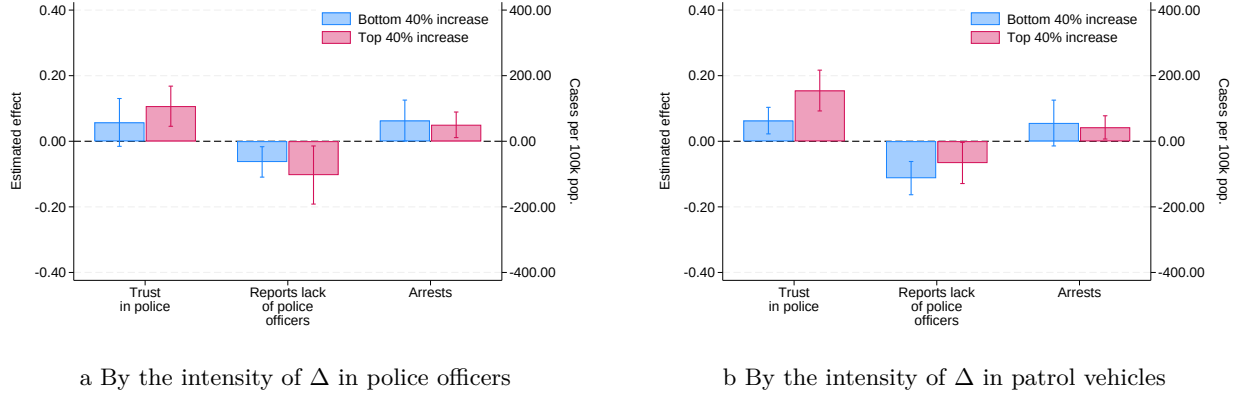
Notes: This figure shows the dynamic effects of the *Plan Cuadrante* on different measures of police effectiveness. The estimation is conducted using the procedure developed in Callaway and Sant'Anna (2021) for difference-in-differences with staggered treatment adoption. The dots in the figure represent the estimated coefficients and the bars behind them 95% confidence intervals. Standard errors are clustered at the municipality level using the wild bootstrapping procedure. The pooled effect and the mean outcome before the introduction of the treatment is also presented in each panel. Panel (a) reports the effect of *Plan Cuadrante* on the probability of household victimization in the last 12 months using the ENUSC survey. Panel (b) reports the effect on the number of crimes reported to the police in the municipality per 100,000 inhabitants using administrative data. Panel (c) reports the effect on the probability of perceiving an increase in crime using the ENUSC survey. Panel (d) reports the effect on the probability of adopting security measures in the last twelve months using the ENUSC survey. Panel (a) includes 93,041 observations; panel (b), 4,231; panel (c), 89,578; and panel (d), 93,041 observations.

Table A.1: Implementation of *Plan Cuadrante* by Year

Year	N	Municipalities
1998–2000	44	Metropolitan Region of Santiago (44 municipalities)
2002	10	Chiguayante, Concepción, Hualpén, Quilpué, San Antonio, San Pedro de la Paz, Talcahuano, Valparaíso, Villa Alemana, Viña del Mar
2003	2	Padre Las Casas, Temuco
2004	2	Antofagasta, Copiapó
2005	7	Alto Hospicio, Arica, Calama, Curicó, Iquique, Linares, Talca
2006	7	Coquimbo, La Serena, Los Andes, Ovalle, Rancagua, San Felipe, San Fernando
2007	7	Chillán, Chillán Viejo, Los Ángeles, Osorno, Puerto Montt, Punta Arenas, Valdivia
2008	9	Ancud, Castro, Constitución, Coronel, Lota, Peñaflor, Quillota, Rengo, Villarrica
2009	12	Angol, Aysén, Calera, Concón, Coyhaique, La Unión, Padre Hurtado, Parral, Penco, San Carlos, Tomé, Vallenar
2011	17	Arauco, Cañete, Cartagena, Cauquenes, Curanilahue, La Ligua, Limache, Molina, Nueva Imperial, Pucón, Puerto Varas, Río Bueno, San Clemente, San Javier, San Vicente, Santa Cruz, Victoria
2012	20	Cabrero, Calbuco, Casablanca, Chimbarongo, Collipulli, Curacaví, El Monte, Graneros, Illapel, Isla de Maipo, Lautaro, Machalí, Mostazal, Mulchén, Nacimiento, Natales, Panguipulli, Quellón, Quintero, Tocopilla
2013	13	Bulnes, Laja, Las Cabras, Lebu, Llaillay, Loncoche, Longaví, Los Álamos, Maule, Requínoa, Teno, Vicuña, Yumbel
Total	150	

Note: N is the number of municipalities that adopted *Plan Cuadrante* in each year. Data shared by Carabineros de Chile.

Figure B.3: Effects of *Plan Cuadrante* by Resource Increase: Trust, Complaints, and Arrests



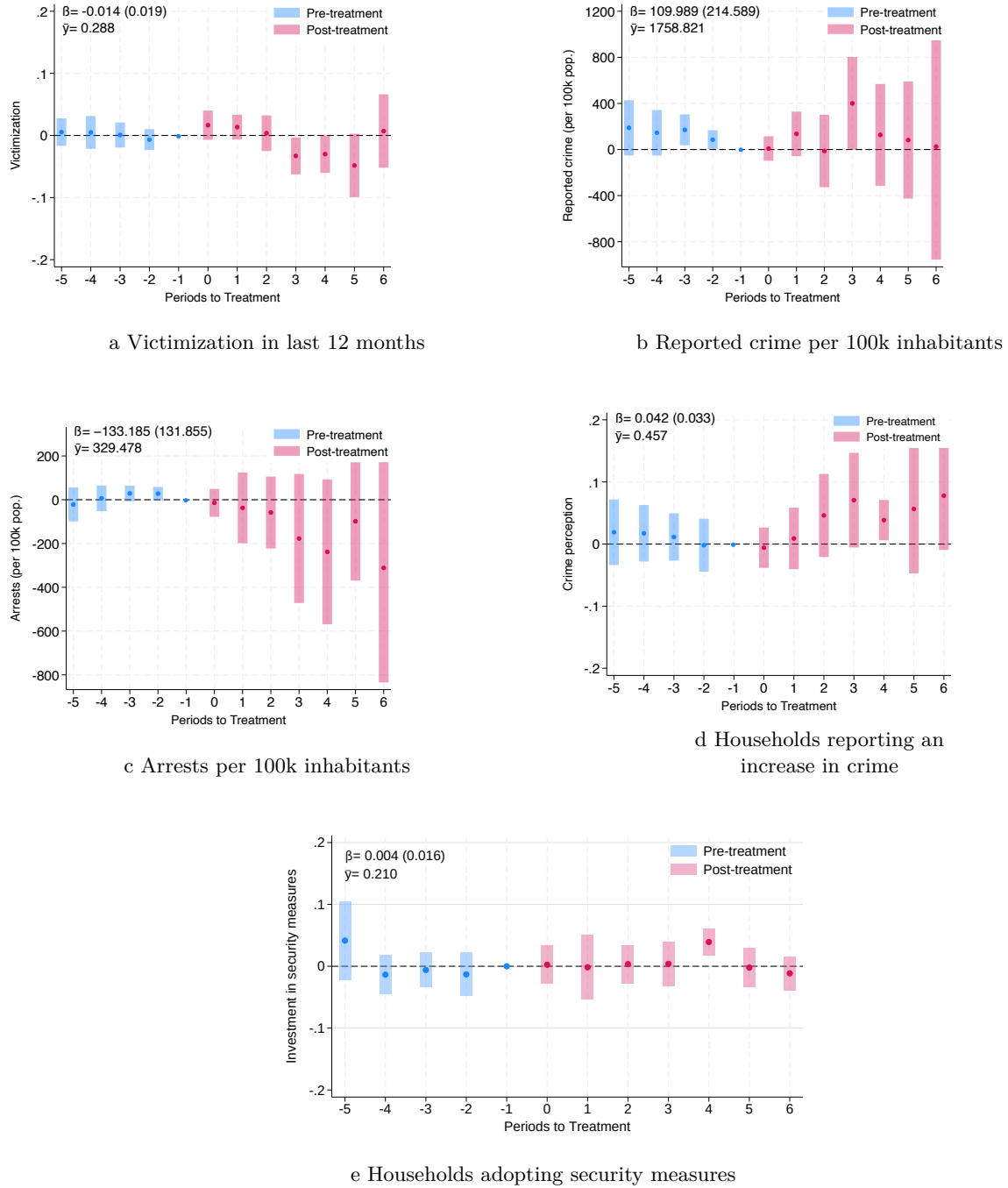
Notes: This figure complements Figure 4 by reporting, for the three outcomes not shown there, the effect of *Plan Cuadrante* estimated separately for municipalities in the bottom 40% (blue) and top 40% (red) of the resource-increase distribution. Panel (a) splits municipalities by the increase in police officers and panel (b) by the increase in patrol vehicles. In each panel, arrests (administrative data, in cases per 100,000 inhabitants) are read on the right axis; the other two outcomes, from the ENUSC survey (trust in the police, and reports about the lack of police officers), are read on the left axis. Effects are estimated using the difference-in-differences estimator for staggered treatment adoption developed in Callaway and Sant'Anna (2021); bars are point estimates and whiskers 95% confidence intervals, with standard errors clustered at the municipality level using the wild bootstrapping procedure.

Table B.3: Effects of *Plan Cuadrante* Controlling for Police Resources

	(1)	(2)	(3)	(4)	(5)
<i>Panel A. Victimization</i>					
ATT	-0.099*** (0.013)	-0.077*** (0.013)	-0.086** (0.043)	-0.094*** (0.028)	-0.087* (0.053)
N	93,041	53,973	53,553	72,017	53,553
<i>Panel B. Crime perception</i>					
ATT	-0.149*** (0.025)	-0.140*** (0.032)	-0.112** (0.057)	-0.082** (0.033)	-0.111 (0.106)
N	89,578	52,160	51,746	69,434	51,746
<i>Panel C. Investment in security measures</i>					
ATT	-0.077*** (0.016)	-0.066*** (0.019)	-0.109*** (0.041)	-0.081*** (0.019)	-0.168*** (0.064)
N	93,041	53,973	53,553	72,017	53,553
<i>Panel D. Reported crime (per 100k pop.)</i>					
ATT	-208.024*** (56.130)	-157.010*** (56.224)	29.230 (147.261)	-261.439 (170.954)	-245.555 (165.221)
N	4,231	2,963	1,911	2,248	1,911
Sample	Full	Resource yrs	Resource yrs	Resource yrs	Resource yrs
Police officers (per 100k pop.)			✓		✓
Patrol vehicles (per 100k pop.)				✓	✓

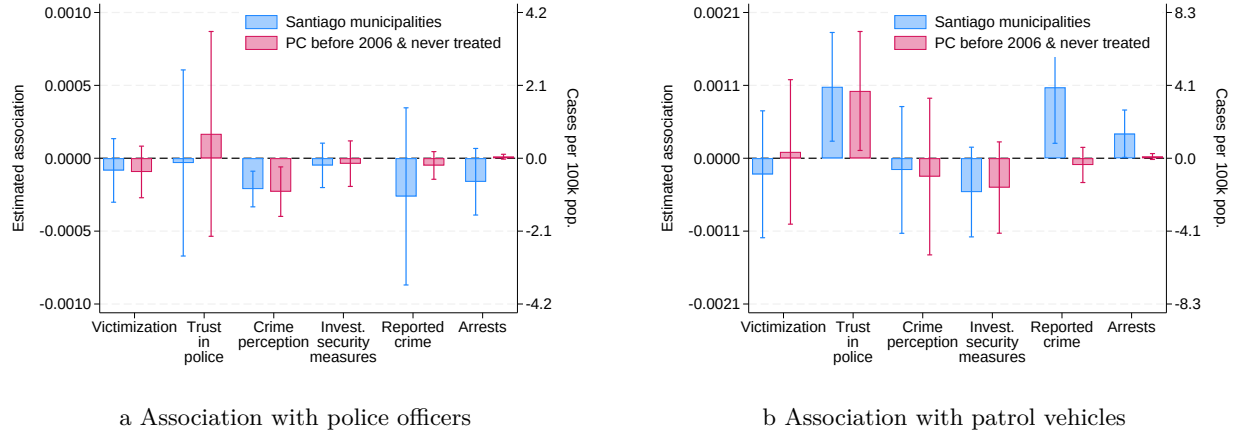
Note: ATT estimates of the effect of *Plan Cuadrante* on each outcome. Column (1) is the baseline (full estimation sample, no controls) and reproduces the headline estimate of the corresponding estimator. Column (2) restricts the estimation to the years in which police-resource data are available (from 2006 onward). No control variables are used in this estimation. Columns (3)–(5) add time-varying controls for police resources per 100,000 inhabitants: officers only; patrol vehicles only; and officers and patrol vehicles jointly (entered as separate regressors). Patrol vehicles are measured, per 100,000 inhabitants, as all patrol automobiles (radiopatrulla, auto patrullero, auto policial, automovil, station wagon, jeep and camioneta) plus motorcycles. Standard errors in parentheses; row N reports the estimation-sample size. Estimation follows Callaway and Sant’Anna (2021); standard errors clustered at the municipality level using wild bootstrapping procedures. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Figure B.4: Effects of *Seguridad Ciudadana* on Crime Outcomes and Security Perceptions



Notes: This figure shows the dynamic effects of the *Seguridad Ciudadana* program on different measures of crime and security perceptions. The estimation is conducted using the procedure developed in Callaway and Sant'Anna (2021) for difference-in-differences with staggered treatment adoption. The dots in the figure represent the estimated coefficients and the bars behind them 95% confidence intervals. Standard errors are clustered at the municipality level using wild bootstrapping. The pooled effect and the mean outcome before the introduction of the treatment for the outcome is also presented in each panel. Panel (a) reports the effect of *Seguridad Ciudadana* on the probability of household victimization in the last 12 months using the ENUSC survey. Panel (b) reports the effect on the number of crimes reported to the police in the municipality per 100,000 inhabitants using administrative data. Panel (c) reports the effect on the number of arrests in the municipality per 100,000 inhabitants using administrative data. Panel (d) reports the effect on the probability of perceiving an increase in crime using the ENUSC survey. Panel (e) reports the effect on the probability of adopting security measures in the last twelve months using the ENUSC survey. Panel (a) includes 210,102 observations; panel (b), 2,234; panel (c), 1,908; panel (d), 203,078; and panel (e), 74,316 observations.

Figure B.5: Association between Police Resources and Police Effectiveness (No *Plan Cuadrante* Variation)



Notes: The figure shows the estimated within-municipality association between police resources (per 100,000 inhabitants) and the main outcomes, in municipalities where *Plan Cuadrante* does not vary over the sample period. Estimates come from OLS regressions of each outcome on the resource measure with municipality and year fixed effects; standard errors are clustered at the municipality level. Blue bars use the 41 metropolitan municipalities of Santiago, all of which adopted *Plan Cuadrante* in 2000; red bars use the broader set of municipalities that adopted *Plan Cuadrante* before 2006 together with the never-treated municipalities surveyed in ENUSC (2006–2012). Reported crime and arrests (administrative data, in cases per 100,000 inhabitants) are read on the right axis; all other outcomes, from the ENUSC survey (victimization, trust in the police, crime perception, and investment in security measures), are read on the left axis. Panel (a) reports associations with the number of police officers; panel (b) with the number of patrol vehicles. Bars are point estimates with 95% confidence intervals.

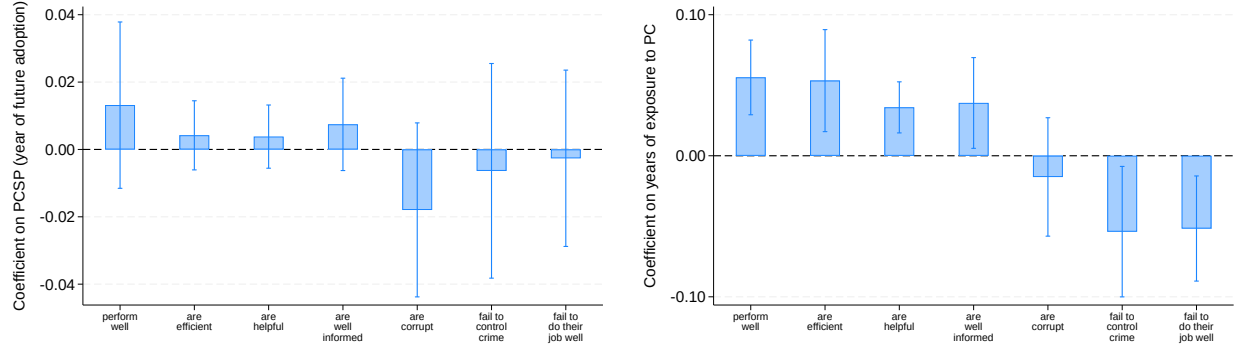
Table B.4: Predictors of the Increase in Police Resources under *Plan Cuadrante*

	Administrative sample		ENUSC sample	
	Police officers (1)	Patrol vehicles (2)	Police officers (3)	Patrol vehicles (4)
Dep. var: avg. annual increase in resources (share of pre-adoption level)				
Reported crime (per 100k pop.)	−0.203** (0.099)	−0.199** (0.097)	−0.319* (0.182)	0.272* (0.137)
Arrests (per 100k pop.)	0.099 (0.136)	−0.041 (0.129)	0.003 (0.186)	0.297 (0.223)
Victimization			−0.017* (0.010)	−0.025* (0.013)
Crime perception			0.005 (0.017)	0.008 (0.014)
Trust in the police			0.024 (0.018)	0.027 (0.019)
Investment in security measures			0.004 (0.010)	−0.003 (0.009)
Police officers (per 100k pop.)	−0.207 (0.155)	0.155 (0.129)	−0.888*** (0.154)	0.571*** (0.136)
Patrol vehicles (per 100k pop.)	−0.129 (0.129)	−0.139 (0.126)	0.537** (0.212)	−0.507*** (0.169)
Log population	−0.529*** (0.145)	−0.203* (0.112)	−0.193 (0.189)	−0.626** (0.269)
Poverty rate (2004)	0.170 (0.113)	0.308** (0.126)	0.027 (0.108)	−0.017 (0.108)
Observations	65	65	8,781	8,781
R-squared	0.342	0.282	0.614	0.545

Note: OLS regressions of the average annual increase in police officers (columns 1 and 3) and patrol vehicles (columns 2 and 4) per 100,000 inhabitants, from the year before adoption of *Plan Cuadrante* to 2012. As in panels (a) and (b) of Figure 4, the increase is expressed as a share of the resources available the year before adoption. Columns 1–2 use the administrative sample at the municipality level; columns 3–4 use ENUSC survey respondents interviewed before 2007 (survey outcomes at the respondent level, other regressors as pre-2006 municipality aggregates). All variables are standardized. Standard errors clustered by municipality in parentheses.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

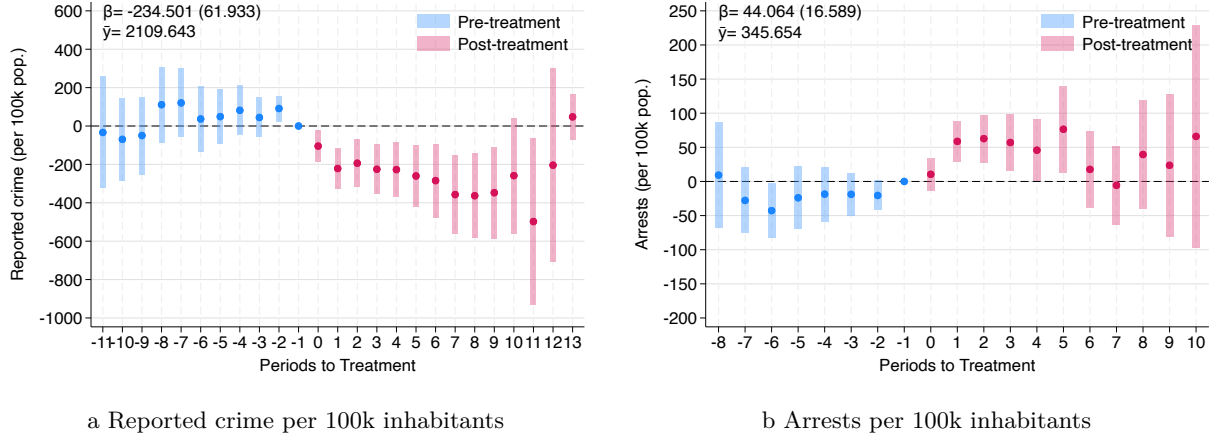
Figure B.6: Timing of Adoption of *Plan Cuadrante* and Perceived Police Performance



a Year of adoption and police performance before adoption b Years exposed to *Plan Cuadrante* (dose response)

Notes: The figure considers seven measures of police performance: the share of respondents who perceive police performance as good, and who perceive the police to be efficient, helpful, well informed, corrupt, failing to control crime, and failing to do their job well. Using the sample of municipalities that adopted *Plan Cuadrante* after 2005, panel (a) examines whether the future timing of adoption of *Plan Cuadrante* is correlated with perceived police performance in 2005, before the introduction of the program. Panel (b) reestimates the same comparison as Figure 4, replacing the binary adoption indicator with the municipality's number of years of exposure to *Plan Cuadrante* as of 2005. In both panels, each bar plots the coefficient from a separate regression of one performance measure. The graphs display the estimated coefficients and 95% confidence intervals.

Figure C.8: Effects of *Plan Cuadrante* on Reported Crime and Arrests Using the Full Set of Post-treatment and Pre-treatment Periods



Notes: This figure shows the dynamic effects of *Plan Cuadrante* on the number of crimes reported to the police in the municipality per 100,000 inhabitants (panel a) and on the number of arrests in the municipality per 100,000 inhabitants (panel b), using administrative data. The estimation is conducted using the procedure developed in Callaway and Sant'Anna (2021) for difference-in-differences with staggered treatment adoption. The dots in the figure represent the estimated coefficients and the bars behind them 95% confidence intervals. Standard errors are clustered at the municipality level using the wild bootstrapping procedure. The pooled effect and the mean outcome before the introduction of the treatment are also presented in each panel. Unlike Figure 2 in the main text, we do not cap the event study to the same 5 pre-treatment and 6 post-treatment periods used for symmetry with the ENUSC-based analysis. Instead, we exploit the full set of post-treatment and pre-treatment periods. Panel (a) includes 4,371 observations and panel (b) 3,372 observations.

Editor Comment 2

2. The SUTVA/spillovers point by R2 may be partly linked to the previous issue. Explaining the nature of externalities— which ones you capture, which ones you allow for, and which ones you assume away in your strategy—is key.

Response to Editor Comment 2:

Thank you for this comment. We agree with you and the referees that spatial spillovers could be a concern in our setting. Although the original manuscript already included a brief spillover check, we appreciate Reviewer 2’s suggestion to expand this analysis and provide much stronger reassurance that spillovers do not bias our estimates. We have therefore added a new subsection in the main text and a new appendix with detailed discussion of each exercise. Taken together, the new evidence shows that *Plan Cuadrante* did not displace crime to neighboring municipalities or divert police resources from control to treated areas. We discuss the four exercises below and provide the new manuscript sections.

The first exercise relies on administrative data on reported crime, arrests, and police resources. Since these records are available for all municipalities in the country, we can study what happened around all the municipalities that adopted *Plan Cuadrante* in the period we analyze. Building on our main specification but defining the outcome at the neighboring municipality level, we show that the adoption of *Plan Cuadrante* neither displaced crime from treated to nearby control municipalities nor diverted resources from control to treated areas, both of which would lead us to overestimate the program’s effects. These results hold both in the universe of municipalities observed in the administrative records and in the ENUSC subsample in which we study victimization, crime perception, and household investment in security measures.

Second, following the suggestions of Reviewer 2, we implemented variations of the methods developed in Crépon et al. (2013) and Blattman et al. (2021b). In the spirit of Crépon et al. (2013), we estimate the effect of a higher share of neighboring municipalities adopting *Plan Cuadrante* on crime, arrests, and resources in control municipalities, finding no significant effects on any of these outcomes. Next, following Blattman et al. (2021b), we conduct a randomization inference exercise in which treatment status and adoption dates are randomly assigned 1,000 times, constructing placebo distributions of treatment and spillover effects. The true treatment effect on crime falls in the bottom 1% of the placebo distribution, while the estimated spillover effects fall within the 40th–45th percentile of the placebo distribution, providing no evidence of meaningful spillovers on crime, arrests, or resources onto neighboring untreated municipalities.

The exercises described in the previous paragraph rely on administrative records from the police, which are available for all municipalities adopting *Plan Cuadrante* in the period we study and for all of their neighboring municipalities. This broad coverage allows us to estimate spillovers credibly using the full set of administrative outcomes. By contrast, outcomes measured in ENUSC are observed for only 46 adopting municipalities and 1 never-treated municipality. This makes spillover analyses more challenging, because coverage of neighboring municipalities is limited: only 10 of these municipalities share a border with another municipality surveyed in ENUSC, and only 6 share a border with either a never-treated municipality or a treated municipality that adopted the program in a different year. Given this limited coverage, the Crépon et al. (2013) and Blattman et al. (2021b) exercises are implemented only on the administrative outcomes, for which coverage of neighboring municipalities is sufficient. For the outcomes measured only in ENUSC, we instead estimate the spillover effects of early-adopting ENUSC municipalities on their late-adopting ENUSC

neighbors before the latter’s own adoption, and once again find no evidence of spatial spillovers.

Below, we provide the sections we added to the updated manuscript:

Spatial Spillovers Section: the new section on spatial spillovers reads as follows:

4.2 *Plan Cuadrante* and Spatial Spillovers

Beyond parallel trends, a causal interpretation of our estimates requires the absence of spillover effects from treated to control municipalities, since such spillovers could bias treatment effects upward. This concern is relevant in our setting. Previous work on deployment-based policing reforms, such as hot-spot policing, shows that crime may be displaced to nearby areas (Blattman et al., 2021a). Spillovers could also arise through the reallocation of police resources from control to treated municipalities.

In our setting, spillovers are less likely for several reasons. First, as discussed in Section 2, *Plan Cuadrante* introduced beat policing in Chile by reorganizing deployment across the full municipal territory rather than concentrating resources in specific hot spots. Second, Chilean municipalities are large—on average 101,000 inhabitants and 2,000 square kilometers (Gobierno de Chile, 2024)—and criminals typically do not travel such long distances to commit crimes, which makes displacement to other municipalities unlikely (Kirchmaier et al., 2023). Third, and more directly related to police resources, police regulations assign officers to *comisarías*—broader operational units that usually cover an entire municipality—for fixed periods, which limits rapid reallocation of officers across jurisdictions.¹⁴

We nonetheless assess spillovers empirically through three complementary exercises. Online Appendix C reports the full results. First, we estimate the effect of *Plan Cuadrante* on police-reported crime and police resources in untreated municipalities whose neighbors adopt the program. Online Appendix Figure C.12 summarizes these estimates and shows no evidence of spatial spillovers.

Second, following Crépon et al. (2013), we test whether higher exposure to *Plan Cuadrante* among neighboring municipalities affects outcomes in control municipalities. Online Appendix Figures C.13 and C.14—which define neighbor exposure as the share and the number of treated neighbors, respectively—show no meaningful effects across the outcomes we study, again suggesting the absence of spatial spillovers.

Third, following Blattman et al. (2021a), we implement a randomization-inference exercise in which treatment status and adoption dates are randomly reassigned 1,000 times. Online Appendix Figure C.15 shows that estimated spillover effects lie near the center of the placebo distribution—between the 40th and 45th percentiles—which provides further evidence against spatial spillovers. By contrast, the estimated effects for treated municipalities lie in the bottom first percentile of the placebo distribution, consistent with *Plan Cuadrante* significantly reducing crime in treated municipalities.

The exercises above evaluate spatial spillovers using administrative records for the extended municipality sample. We complement this evidence with two additional analyses focused on ENUSC municipalities. First, we re-estimate the administrative-outcome spillover analysis—

¹⁴See the Transfer Manual for Carabineros de Chile Personnel available at https://www.carabineros.cl/transparencia/og/pdf/OG_2707_13112019.pdf

for reported crime, arrests, police officers, and patrol vehicles—restricting the sample to neighbors of the 46 ENUSC municipalities (Online Appendix Figure C.16). Second, we test for spillovers directly in ENUSC outcomes. Because ENUSC coverage is limited—46 treated municipalities and 1 never-treated municipality, with only 10 sharing a border and 6 of those treated in different years—we estimate spillover effects from early-adopting ENUSC municipalities onto late-adopting ENUSC neighbors before the latter adopt the program (Online Appendix Figure C.17). Both exercises yield reassuringly null effects.

Taken together, these results indicate that spatial spillovers are unlikely to threaten identification in our setting.

Spatial Spillovers Appendix: the new Online Appendix C.9. reads as follows:

C.9 Testing for Spatial Spillovers

This Online Appendix details the specifications behind the spillover tests summarized in Section 4.2.

The baseline test estimates the effect of *Plan Cuadrante* on reported crime, arrests, police officers, and patrol vehicles in neighboring untreated municipalities,

$$Outcome_{mt} = \delta_1 \text{Plan Cuadrante near } municip_{.mt} + Municipality_m + Year_t + u_{mt} \quad (1)$$

where the sample is restricted to municipalities that never adopted the program, each is assigned a treatment date equal to the earliest adoption among its treated neighbors, and δ_1 is estimated dynamically with the staggered difference-in-differences estimator of Callaway and Sant’Anna (2021). Because these administrative outcomes are observed for every municipality, we estimate the specification on the full national sample of never treated municipalities (Online Appendix Figure C.12).

For the intensity-based test in the spirit of Crépon et al. (2013), we replace the binary neighbor-treatment indicator with a continuous measure of exposure and restrict the sample to never-treated municipalities,

$$Outcome_{mt} = \delta_1 \text{Share neighbors with Plan Cuadrante}_{mt} + Municipality_m + Year_t + u_{mt} \quad (2)$$

Because the treatment is continuous, we estimate it with the heterogeneity-robust estimator of Chaisemartin and D’Haultfoeuille (2020), measuring exposure both as the share of treated neighbors (Online Appendix Figure C.13) and as their number (Online Appendix Figure C.14).

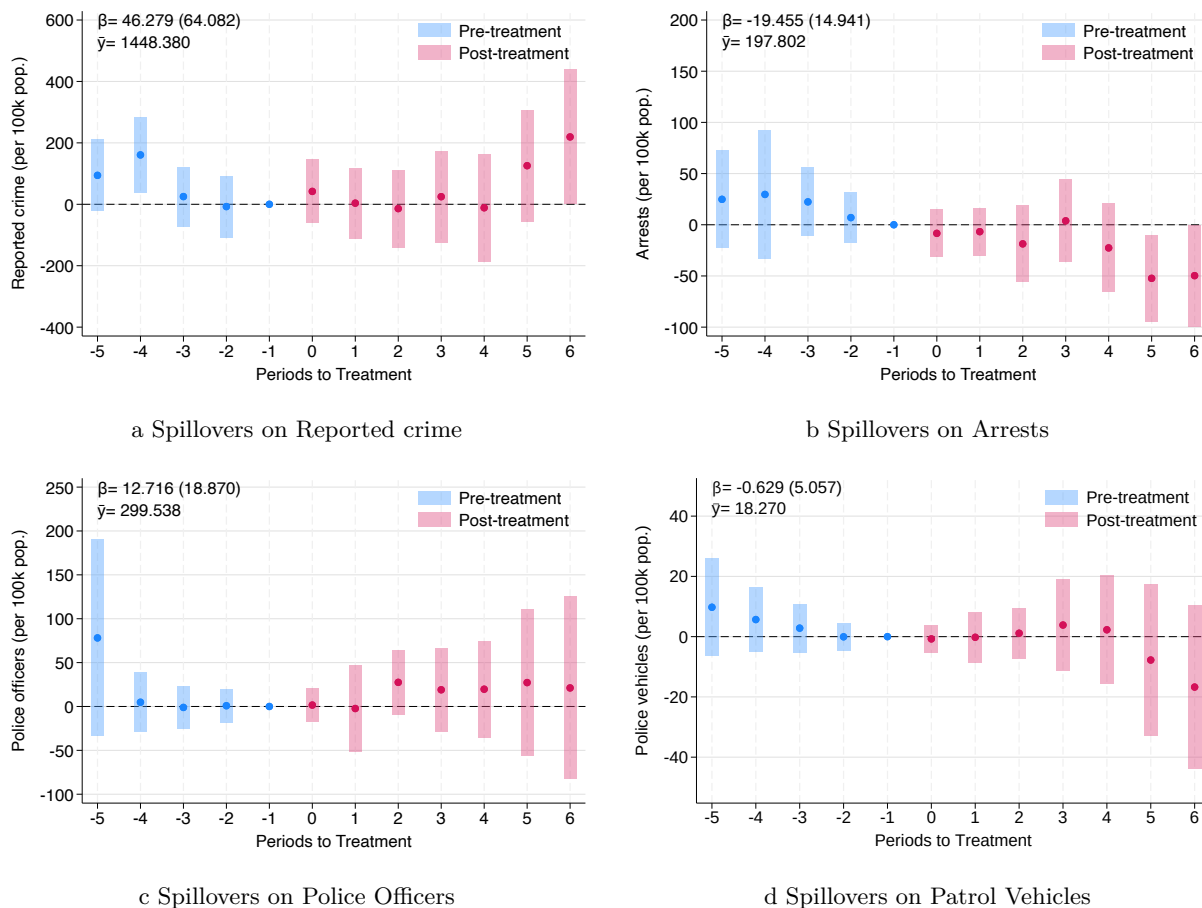
For the randomization-inference test, we reassign adoption dates and never-treated status across municipalities 1,000 times and re-estimate the direct and spillover effects under the sharp null, yielding the placebo distributions in Online Appendix Figure C.15. For this exercise we rely on the estimator of Chaisemartin and D’Haultfoeuille (2020) rather than that of Callaway and Sant’Anna (2021): re-estimating the latter over a thousand permutations would require prohibitive computational power and running time.

Finally, we re-estimate the baseline administrative specification on the municipalities neighboring the 46 ENUSC municipalities (Online Appendix Figure C.16); and, for the outcomes measured only in ENUSC, we estimate the spillover of early-adopting ENUSC municipalities on their late-adopting ENUSC neighbors before the latter’s own adoption (Online Appendix

Figure C.17). The six focal late-adopters, each shown with its earliest-adopting ENUSC neighbor and that neighbor's adoption year, are Molina (Curicó, 2005), Tocopilla (Iquique, 2005), San Carlos (Chillán, 2007), Padre Hurtado (Peñaflor, 2008), Calera (Quillota, 2008), and Limaiche (Quillota, 2008). Across all of these specifications, the estimated spillover effects are small and statistically indistinguishable from zero.

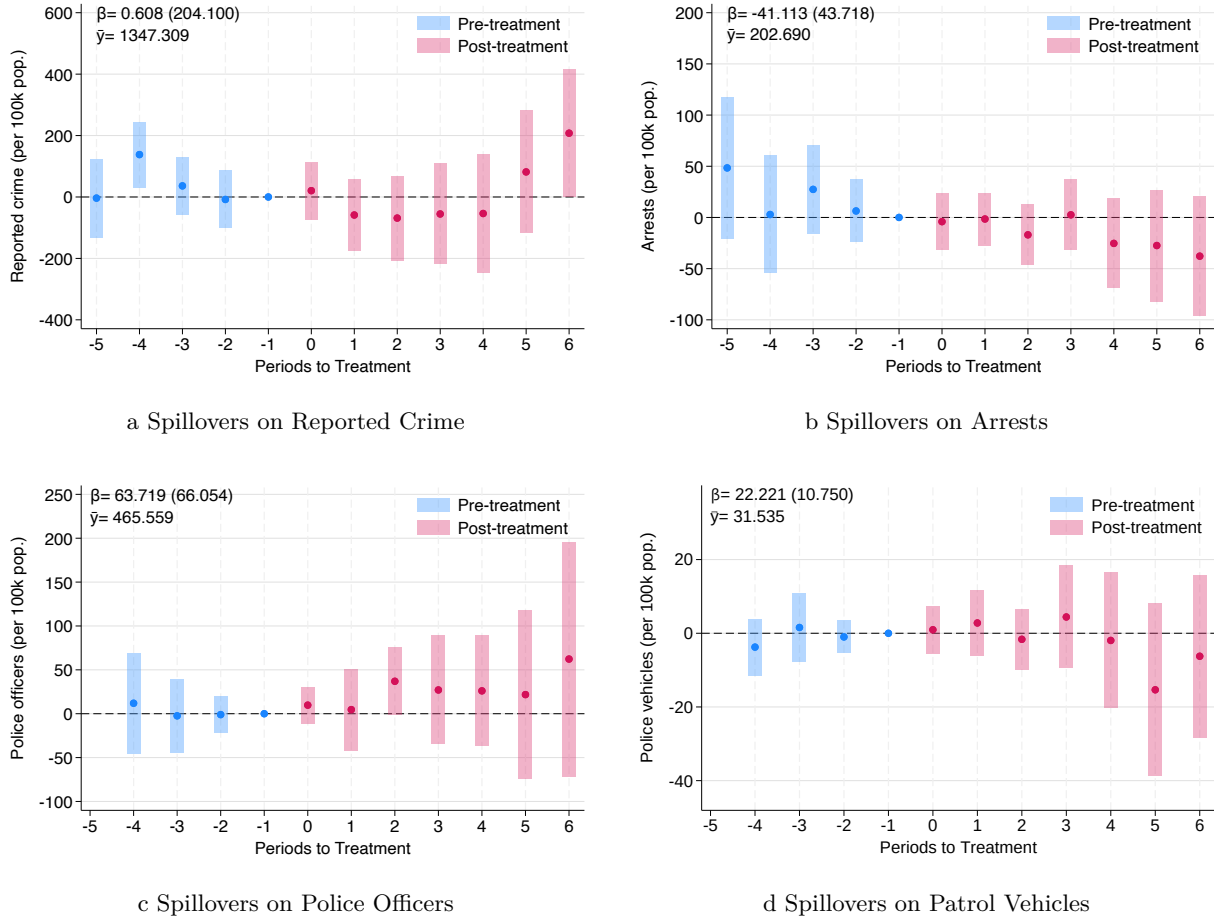
Other figures and tables referenced above:

Figure C.12: Geographic Spillovers of *Plan Cuadrante* on Neighboring Untreated Municipalities



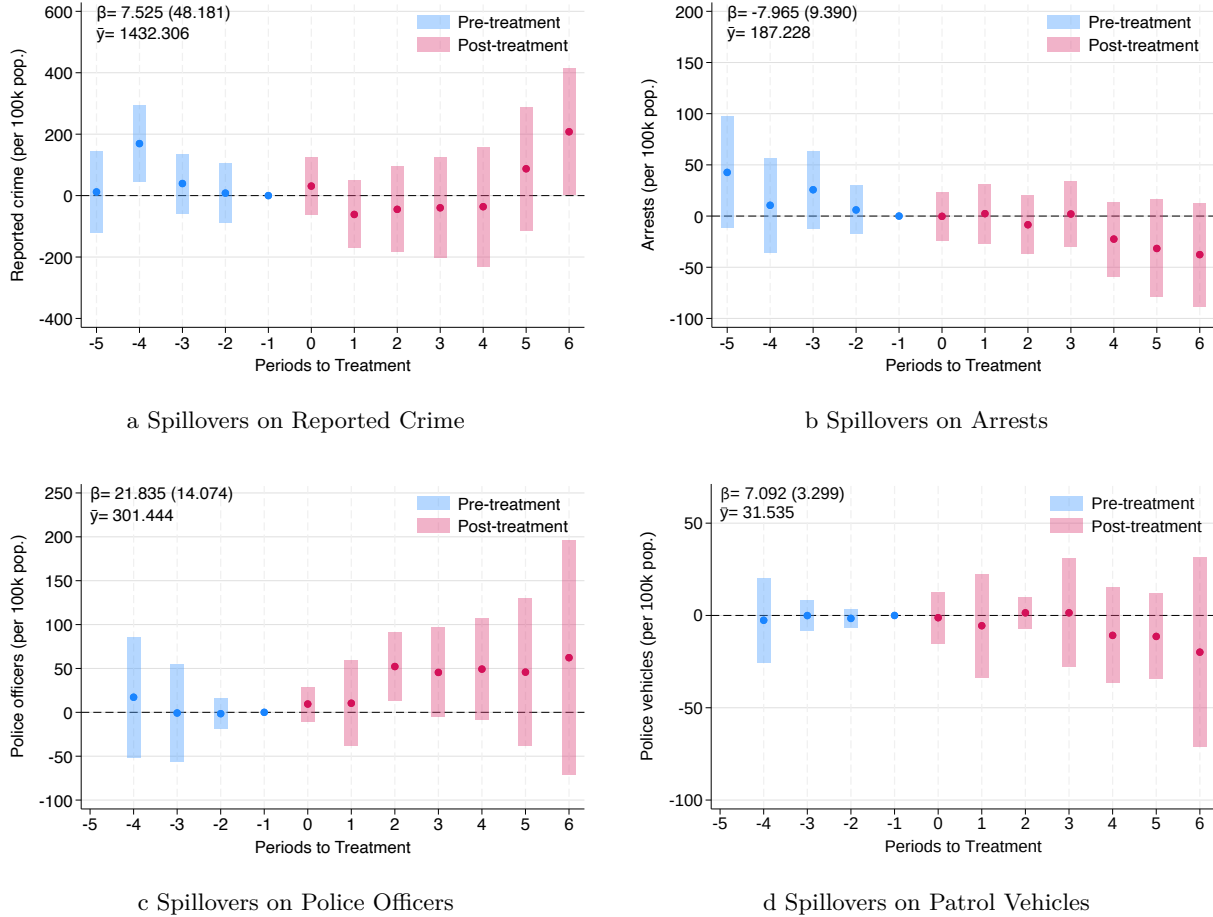
Notes: This figure reports the spillover effects of *Plan Cuadrante* on the number of crimes reported to the police in the municipality per 100,000 inhabitants (panel a), the number of arrests in the municipality per 100,000 inhabitants (panel b), the number of police officers per 100,000 inhabitants (panel c), and the number of patrol vehicles per 100,000 inhabitants (panel d). Panels (a)–(d) include 2,859, 2,166, 1,144, and 1,942 observations respectively. Specifically, the figure shows the dynamic effects of the *Plan Cuadrante* for neighbor municipalities which never introduced the program. The estimation is conducted using the procedure developed in Callaway and Sant’Anna (2021) for difference-in-differences with staggered treatment adoption, using as the treatment introduction date the earliest date of a neighboring municipality. The dots in the figure represent the estimated coefficients and the bars behind them 95% confidence intervals. Standard errors are clustered at the municipality level using the wild bootstrapping procedure. The pooled effect and the mean outcome before the introduction of the treatment for every outcome is also presented in each panel.

Figure C.13: Geographic Spillovers of *Plan Cuadrante* by Share of Treated Neighbors



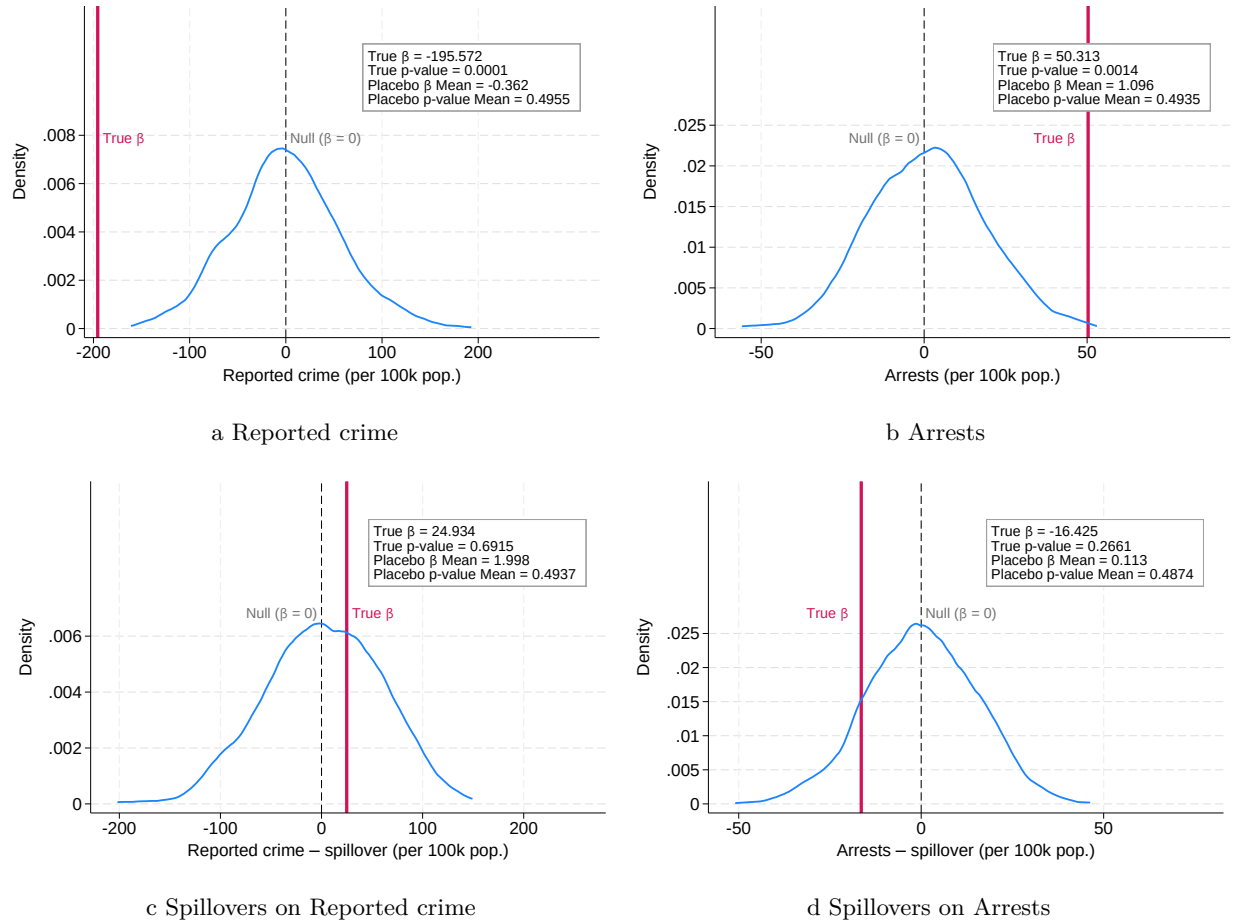
Notes: This figure reports the geographic spillover effects of *Plan Cuadrante* on four outcomes for never-treated municipalities: reported crime per 100,000 inhabitants (panel a), arrests per 100,000 inhabitants (panel b), the number of police officers per 100,000 inhabitants (panel c), and the number of patrol vehicles per 100,000 inhabitants (panel d). Unlike the dummy spillover specification, the treatment variable is now continuous and equals the share of neighboring municipalities that have implemented *Plan Cuadrante* by year t , following the spillover-intensity approach of Crépon et al. (2013). The sample is restricted to municipalities that never adopted *Plan Cuadrante*. Estimation is performed using the heterogeneity-robust estimator of Chaisemartin and D'Haultfœuille (2020), which accommodates a continuous, time-varying treatment intensity in a staggered design. The dots represent the estimated dynamic effects and the bars behind them denote 95% confidence intervals. Standard errors are clustered at the municipality level. The pooled effect and the mean of the outcome among municipalities with no treated neighbors are reported in each panel.

Figure C.14: Geographic Spillovers of *Plan Cuadrante* by Number of Treated Neighbors



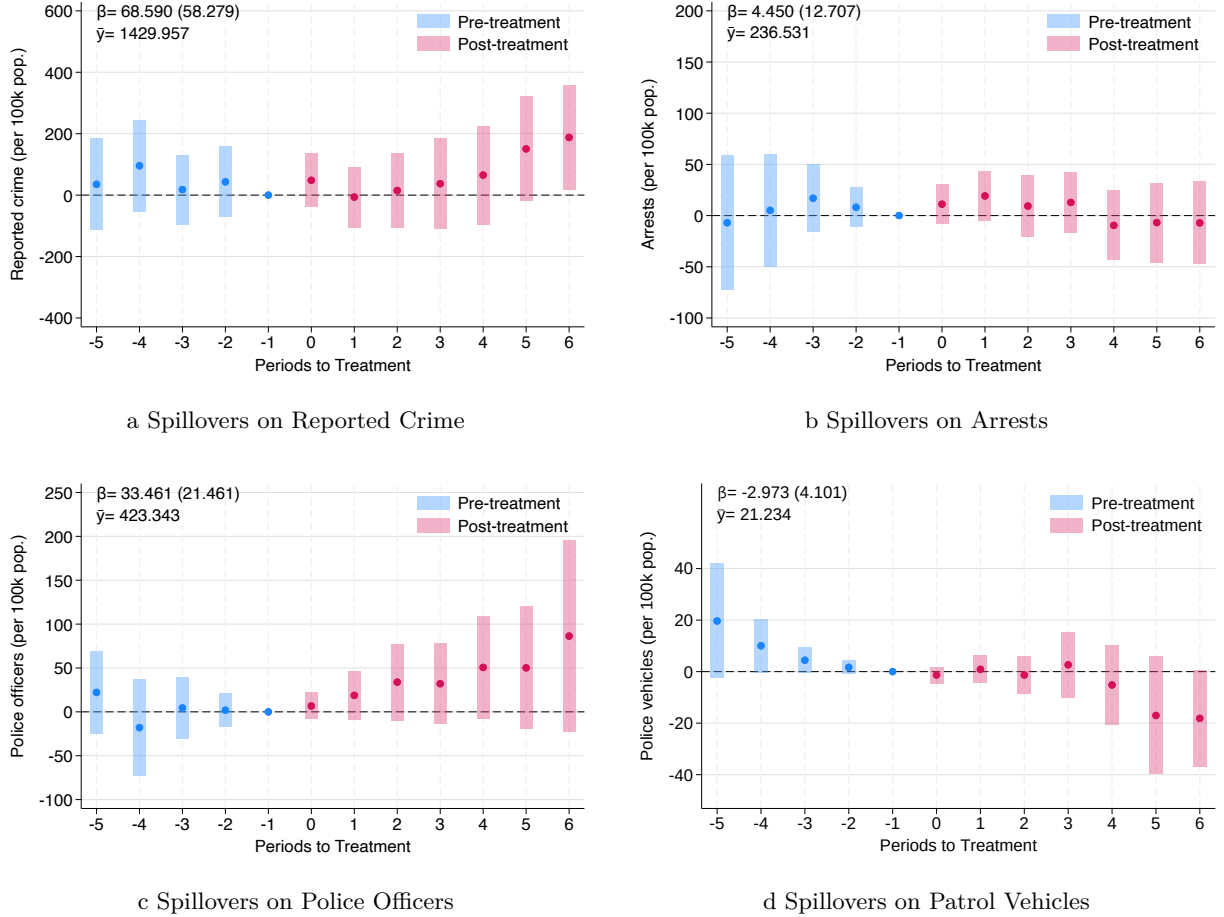
Notes: This figure reports the geographic spillover effects of *Plan Cuadrante* on four outcomes for never-treated municipalities: reported crime per 100,000 inhabitants (panel a), arrests per 100,000 inhabitants (panel b), the number of police officers per 100,000 inhabitants (panel c), and the number of patrol vehicles per 100,000 inhabitants (panel d). Unlike the dummy spillover specification, the treatment variable is now continuous and equals the number of neighboring municipalities that have implemented *Plan Cuadrante* by year t , following the spillover-intensity approach of Crépon et al. (2013). The sample is restricted to municipalities that never adopted *Plan Cuadrante*. Estimation is performed using the heterogeneity-robust estimator of Chaisemartin and D'Haultfœuille (2020), which accommodates a continuous, time-varying treatment intensity in a staggered design. The dots represent the estimated dynamic effects and the bars behind them denote 95% confidence intervals. Standard errors are clustered at the municipality level. The pooled effect and the mean of the outcome among municipalities with no treated neighbors are reported in each panel.

Figure C.15: Randomization Inference for the Direct and Spillover Effects of *Plan Cuadrante*



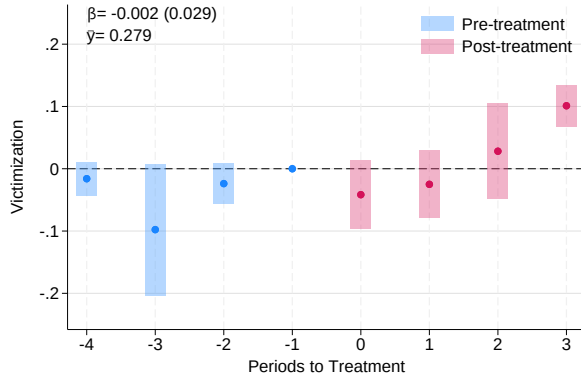
Notes: This figure presents randomization inference results under the sharp null of no treatment effect, using the heterogeneity-robust estimator of Chaisemartin and D'Haultfœuille (2020). Each curve is the kernel density of 1,000 placebo ATT estimates obtained by jointly permuting *Plan Cuadrante* adoption dates and never-treated status across municipalities (drawing from the empirical distribution of (date, status) pairs, so the marginal counts of treated and never-treated municipalities are preserved in every permutation). In each permutation the variable PCSP is shuffled across all 346 municipalities and a new time-varying treatment indicator is computed before re-estimating the dynamic effects. The solid colored vertical line marks the point estimate from the true rollout; the dashed black line marks the null expectation $\beta = 0$. The boxes report the true point estimate, its asymptotic two-sided p -value, and the mean of the placebo β and placebo p -value distributions.

Figure C.16: Geographic Spillovers of *Plan Cuadrante* for Neighbors of the ENUSC Municipalities (Administrative Outcomes)

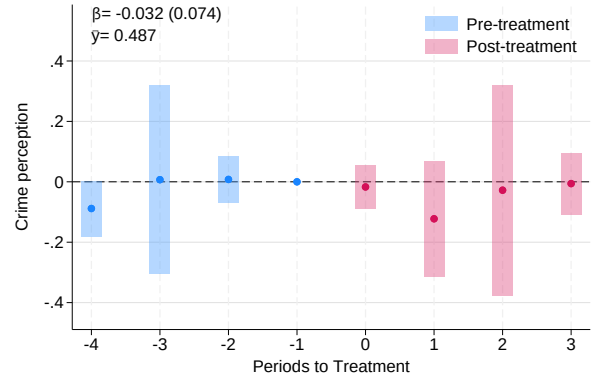


Notes: This figure reports the spillover effects of *Plan Cuadrante* on the number of crimes reported to the police in the municipality per 100,000 inhabitants (panel a), on the number of arrests in the municipality per 100,000 inhabitants (panel b), on the number of police officers (panel c), and on the number of patrol vehicles (panel d). Specifically, the figure shows the dynamic effects of *Plan Cuadrante* for neighboring never-treated municipalities to the 46 municipalities included in the ENUSC sample. The estimation is conducted using the procedure developed in Callaway and Sant'Anna (2021) for difference-in-differences with staggered treatment adoption, using as the treatment introduction date the earliest date of a neighboring municipality. The dots in the figure represent the estimated coefficients and the bars behind them 95% confidence intervals. Standard errors are clustered at the municipality level using the wild bootstrap procedure. The pooled effect and the mean outcome before the introduction of the treatment for each outcome are also presented in each panel.

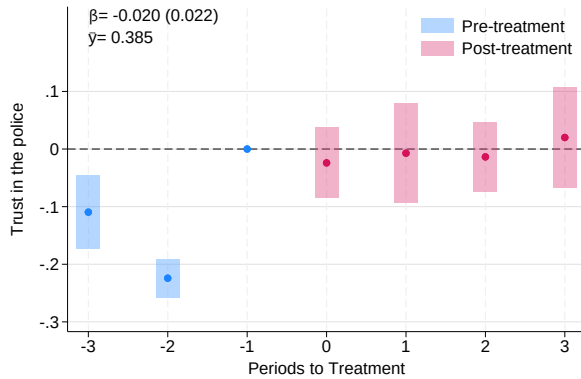
Figure C.17: Geographic Spillovers of *Plan Cuadrante* on ENUSC Survey Outcomes (Early- to Late-Adopting Municipalities)



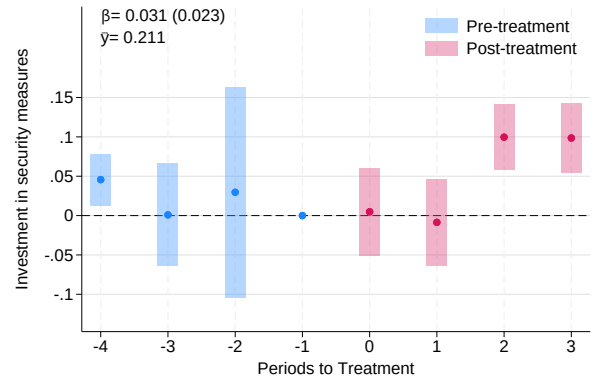
a Spillovers on Victimization in last 12 months



b Spillovers on Households reporting an increase in crime



c Spillovers on Trust in the police



d Spillovers on Households adopting security measures

Notes: This figure presents the geographic spillover effects of *Plan Cuadrante* from early-adopting ENUSC municipalities on their late-adopting ENUSC neighbours, using the ENUSC sample. Panel (a) reports the effect of *Plan Cuadrante* on the probability of household victimization in the last 12-months using the ENUSC survey. Panel (b) reports the effect on crime perception using the ENUSC survey. Panel (c) reports the effect on trust in the police using the ENUSC survey. Panel (d) reports the results on the probability of purchasing private security measures in the last twelve months using the ENUSC survey. The focal treated sample consists of six late-adopting ENUSC municipalities—Molina, Tocopilla, San Carlos, Padre Hurtado, Calera, and Limache—each assigned a spillover-onset date equal to the year its earliest-adopting ENUSC neighbour received *Plan Cuadrante*: Curicó (2005) for Molina, Iquique (2005) for Tocopilla, Chillán (2007) for San Carlos, Peñaflor (2008) for Padre Hurtado, and Quillota (2008) for both Calera and Limache. Estimation is performed using the difference-in-differences estimator for staggered treatment adoption of Callaway and Sant’Anna (2021). The dots represent the estimated dynamic effects and the bars behind them denote 95% confidence intervals. Standard errors are clustered at the municipality level. The pooled effect and the mean of the outcome among municipalities with no treated neighbors are reported in each panel.

Editor Comment 3

You will find many other detailed and insightful comments in the reviewers' reports. Please read them carefully, as many of the points raised could lead to significant improvements in the paper. However, your paper should remain your own, and I do not expect you to modify it in response to every single comment. I would like you to prepare a detailed, separate reply for each reviewer and revise the paper accordingly. Additionally, please prepare a separate note for me, outlining the changes made during the revision process.

Response to Editor Comment 3:

Thank you for this guidance. We found the reviewers' reports extremely useful and have carefully read and considered all of the comments they raised. We believe we have been able to address the vast majority of them, substantially strengthening the paper.

As requested, we have prepared a detailed and separate reply, explaining how we have responded to each comment and describing the corresponding changes in the manuscript. In addition, we submit a revised version of the manuscript incorporating all of the changes discussed in the response documents.

References

- Adda, J., B. McConnell, and I. Rasul (2014). Crime and the depenalization of cannabis possession: Evidence from a policing experiment. *Journal of Political Economy* 122(5), 1130–1202.
- Ajzenman, N., P. Dominguez, and R. Undurraga (2023a). Immigration, crime, and crime (mis) perceptions. *American Economic Journal: Applied Economics*.
- Ajzenman, N., P. Dominguez, and R. Undurraga (2023b, October). Immigration, crime, and crime (mis)perceptions. *American Economic Journal: Applied Economics* 15(4), 142–76.
- Ang, D., P. Bencsik, J. Bruhn, and E. Derenoncourt (2025, March). Community engagement with law enforcement after high-profile acts of police violence. *American Economic Review: Insights* 7(1), 124–42.
- Bardhan, P. (2002). Decentralization of governance and development. *Journal of Economic Perspectives* 16(4), 185–205.
- Basurto, M. P., P. Dupas, and J. Robinson (2020). Decentralization and efficiency of subsidy targeting: Evidence from chiefs in rural malawi. *Journal of Public Economics* 185, 104047.

- Besley, T. and S. Coate (2003). Centralized versus decentralized provision of local public goods: a political economy approach. *Journal of public economics* 87(12), 2611–2637.
- Biblioteca del Congreso Nacional de Chile (1990). Ley orgánica constitucional de carabineros de chile (ley 18.961). Promulgada el 27 de febrero de 1990; publicada el 7 de marzo de 1990; última versión consolidada consultada en la Biblioteca del Congreso Nacional.
- Blair, G., J. M. Weinstein, F. Christia, E. Arias, E. Badran, R. A. Blair, A. Cheema, A. Farooqui, T. Fetzer, G. Grossman, et al. (2021). Community policing does not build citizen trust in police or reduce crime in the global south. *Science* 374(6571), eabd3446.
- Blanco, L. R. (2013, None). The impact of crime on trust in institutions in mexico. *European Journal of Political Economy* 32(C), 38–55.
- Blanes-i-Vidal, J. and T. Kirchmaier (2018). The effect of police response time on crime clearance rates. *The Review of Economic Studies* 85(2), 855–891.
- Blanes-i Vidal, J. and G. Mastrobuoni (2018, Mar). Police patrols and crime. IZA Discussion Paper 11393, Institute of Labor Economics (IZA), Bonn.
- Blattman, C., D. P. Green, D. Ortega, and S. Tobón (2021a, 02). Place-Based Interventions at Scale: The Direct and Spillover Effects of Policing and City Services on Crime. *Journal of the European Economic Association* 19(4), 2022–2051.
- Blattman, C., D. P. Green, D. Ortega, and S. Tobón (2021b, 02). Place-Based Interventions at Scale: The Direct and Spillover Effects of Policing and City Services on Crime. *Journal of the European Economic Association* 19(4), 2022–2051.
- Bloom, N., R. Lemos, R. Sadun, and J. V. Reenen (2015). Does management matter in schools? *The Economic Journal* 125(584), 647–674.
- Braga, A. A., B. Turchan, A. V. Papachristos, and D. M. Hureau (2019). Hot spots policing of small geographic areas effects on crime. *Campbell Systematic Reviews* 15(3).
- Cafferata, F. and C. Scartascini (2021). Combating crime in latin america and the caribbean: What public policies do citizens want? Inter american development bank.

- Callaway, B. and P. H. Sant’Anna (2021). Difference-in-differences with multiple time periods. *Journal of Econometrics* 225(2), 200–230.
- Carabineros de Chile (2018). *Manual Operativo: Plan Cuadrante de Seguridad Preventiva 2.0*. Carabineros de Chile. Aprobado conforme a Orden General N° 1960 del 14 de septiembre de 2010, actualizado para el PCSP 2.0.
- Chaisemartin, C. and X. D’Haultfœuille (2020, September). Two-way fixed effects estimators with heterogeneous treatment effects. *American Economic Review* 110(9), 2964–96.
- Chalfin, A. and J. McCrary (2017, March). Criminal deterrence: A review of the literature. *Journal of Economic Literature* 55(1), 5–48.
- Cordner, G. W. (1995). Community policing: Elements and effects. *Police Forum* 5(3), 1–8.
- Crépon, B., E. Duflo, M. Gurgand, R. Rathelot, and P. Zamora (2013). Do labor market policies have displacement effects? Evidence from a clustered randomized experiment. *The Quarterly Journal of Economics* 128(2), 531–580.
- Di Tella, R. and E. Schargrodsky (2004, March). Do police reduce crime? estimates using the allocation of police forces after a terrorist attack. *American Economic Review* 94(1), 115–133.
- Draca, M., S. Machin, and R. Witt (2011, August). Panic on the streets of london: Police, crime, and the july 2005 terror attacks. *American Economic Review* 101(5), 2157–81.
- Evans, W. N. and E. G. Owens (2007). Cops and crime. *Journal of Public Economics* 91(1), 181–201.
- Facchetti, E. (2025). Police infrastructure, police performance, and crime: Evidence from austerity cuts. IFS Working Paper W24/16, Institute for Fiscal Studies.
- Gallup (2023). Gallup global law and order report 2023.
- Gobierno de Chile (2024). Subsecretaria de Desarrollo Regional y Administrativo. Sistema Nacional de información Municipal. <https://datos.sinim.gov.cl>. Consultado el 6 de agosto de 2024.
- Gonçalves, F., S. Mello, and E. Weisburst (2025, 01). Selection bias and racial disparities in police use of force. *SSRN Electronic Journal*.

- Kirchmaier, T., M. Langella, and A. Manning (2023). Commuting for crime. *The Economic Journal* 134(659), 1173–1198.
- Levitt, S. D. (1997). Using electoral cycles in police hiring to estimate the effect of police on crime. *The American Economic Review* 87(3), 270–290.
- Mastrobuoni, G. (2019). Police disruption and performance: Evidence from recurrent redeployments within a city. *Journal of Public Economics* 176, 18–31.
- Mastrobuoni, G. (2020, 03). Crime is terribly revealing: Information technology and police productivity. *The Review of Economic Studies* 87(6), 2727–2753.
- Mastrorocco, N. and L. Minale (2018). News media and crime perceptions: Evidence from a natural experiment. *Journal of Public Economics* 165, 230–255.
- Ministerio de Defensa Nacional, República de Chile (1989). Reglamento de organización de carabineros de Chile n° 1. Decreto N° 77, de 1 de marzo de 1989.
- Ministerio del Interior y Seguridad Pública (2014). Evaluación del Programa Plan Cuadrante de Seguridad Preventiva. Technical report.
- National Academies of Sciences, Engineering, and Medicine (2018). *Proactive Policing: Effects on Crime and Communities*. Washington, DC: The National Academies Press.
- Ostrom, E., R. B. Parks, and G. P. Whitaker (1973). Do we really want to consolidate urban police forces? A reappraisal of some old assertions. *Public Administration Review* 33(5), 423–432.
- Ostrom, E., R. B. Parks, and G. P. Whitaker (1978). *Patterns of Metropolitan Policing*. Cambridge, MA: Ballinger Publishing Company.
- Salas P., R., J. Fernández L., and J. Muñoz V. (2014, July). Evaluación de programas gubernamentales: Informe final corregido. programa plan cuadrante de seguridad preventiva. Evaluación de programas gubernamentales, División de Control de Gestión, Dirección de Presupuestos, Ministerio de Hacienda, Gobierno de Chile. Ministerio Responsable: Ministerio del Interior y Seguridad Pública. Período de evaluación: 2010–2013.

- Skogan, W. G. and S. M. Hartnett (1997). *Community Policing, Chicago Style*. New York: Oxford University Press.
- Tobón, S. (2021). Community policing in the developing world. *Science* 374(6571), 1046–1047.
- Weisburd, S. (2021, 05). Police Presence, Rapid Response Rates, and Crime Prevention. *The Review of Economics and Statistics* 103(2), 280–293.
- Winchester, L., G. Zambrano, and J. Le-Bert (2007, June). Informe final de evaluación: Programa plan cuadrante de seguridad preventiva. Evaluación de programas gubernamentales, Dirección de Presupuestos, Ministerio de Hacienda, Gobierno de Chile. Ministerio Responsable: Ministerio de Defensa. Servicio Responsable: Carabineros de Chile. Período de evaluación: 2004–2006.

Referee 1 Suggestions, Comments and Questions

Thank you very much for your thoughtful and constructive feedback, which has substantially improved the paper. In particular, your comments on the institutional framing of *Plan Cuadrante* were particularly useful: they led us to contact Carabineros de Chile directly and substantially expand Section 2 with new institutional detail. We also found the literature you pointed us to very helpful in sharpening our contribution. Your comments on mechanisms were equally valuable and led us to substantially reframe that discussion. More generally, we have carefully considered all your comments and hope that we adequately addressed your questions and suggestions in the revised manuscript and in the response document below.

Comment R1.1

Institutional setting and framing: I think the paper needs to explain the setting in more detail. For instance, how policing was organized before the reform and where officers were based? The paper states quadrants were subdivided into seven units with about 3 officers on average, but what were the corresponding figures prior to the reform? Were officers simply reallocated (e.g. centralized and moved from one place to another) or did the reform also entail an increase in the overall level of resources for front-line policing? This is key for understanding what the counterfactual would have been and therefore for interpreting the policy's effects (see point 2 below).

Response to R1.1:

We thank the reviewer for this comment and have substantially expanded Section 2. To address this point, we contacted Carabineros de Chile and held multiple meetings with its Research Department. These interactions gave us access to documents on the design and implementation of the program, as well as evaluation reports produced by the Chilean Budget Office (DIPRES).

Prior to *Plan Cuadrante*, most police stations were responsible for the entire territory of their municipality, without any formal subdivision. Given that municipalities averaged more than 2,000 square kilometers and 100,000 inhabitants, this represented a large operational area. Officers were not permanently assigned to specific sub-areas within the municipality but instead rotated across the full territory in response to incoming calls and operational needs, coordinated centrally by the provincial communications center (CENCO). Although local commanders retained discretion over deployment decisions, this discretion was typically exercised reactively rather than used to systematically assign officers to fixed sub-areas within the municipality, as confirmed in our discussions with Carabineros' research department.

During the period studied, *Plan Cuadrante* introduced two main changes. The first and most fundamental change was the permanent territorial assignment of officers to smaller, fixed catchment areas (quadrants). Rather than rotating across the full territory covered by the police station (typically the municipality), each officer or patrol unit was assigned to a specific quadrant as their primary area of responsibility, enabling the accumulation of local knowledge about the territory, its residents, and its crime patterns. This policing approach, known in the criminology literature as beat policing, is widely adopted around the world (National Academies of Sciences, Engineering, and Medicine, 2018; Ministerio del Interior y Seguridad Pública, 2014), yet its effects have not been

rigorously assessed.

The second change was an increase in police resources in some municipalities. Some of the municipalities joining *Plan Cuadrante* received additional officers and patrol vehicles with the goal of closing their pre-existing resource deficit, measured through a standardized methodology based on Equivalent Surveillance Units (UEs) that quantified the gap between the demand for police services and the available supply (Winchester et al., 2007; Salas P. et al., 2014). This increase was financed primarily through a national expansion of Carabineros during the 2000s and 2010s, driven by successive legislative commitments that increased officer recruitment, benefiting both treated and control municipalities. The relocation of existing officers from control to treated municipalities was rare, as strict regulatory constraints on officer transfers made it very difficult to relocate officers before completing their assigned posting period.¹⁵ The additional resources allocated specifically to *Plan Cuadrante* municipalities were intended to close their deficit to the level required for the program’s successful implementation, though budget constraints meant this goal was rarely fully achieved (Salas P. et al., 2014). Notably, while some treated municipalities received large resource increases, the bottom 40% of the distribution of resource increases among treated municipalities received increases very similar to those observed in the control group.

Whether there was an increase in the proportion of resources allocated to front-line policing is something we cannot directly observe. However, this was not a distinguishing feature of *Plan Cuadrante* and there was a single national set of operational guidelines applied to officers in both treated and control municipalities (Carabineros de Chile, 2018; Winchester et al., 2007). Any potential shift in the allocation of resources toward front-line policing would therefore reflect discretionary decisions by local station commanders, rather than an explicit tactical change embedded in the program.

The counterfactual is therefore a municipality of similar characteristics where officers continued to rotate across large catchment areas without permanent sub-area assignments and without the additional resources brought by *Plan Cuadrante* to some municipalities. To disentangle the role of the permanent territorial assignment from that of additional resources, we leverage the substantial variation in the size of the resource increase across treated municipalities and show that the effects of *Plan Cuadrante* are large and statistically significant even among municipalities that experienced no resource increases relative to control municipalities, suggesting that while resources might be key for the effective implementation of beat policing reforms or might explain part of the effects of *Plan Cuadrante* in some municipalities, the permanent assignment to smaller catchment areas plays an important role beyond the simple resource channel.¹⁶ In line with this result, we show that increases in police resources that do not involve the beat policing component in Chile had limited effects on the same outcomes.

Below we report the new Section 2 describing the institutional context and the *Plan Cuadrante*:

Carabineros, Chile’s national police force, was founded in 1927. Through most of our study period, Carabineros operated under the Ministry of Defense, which oversaw both military and police functions until domestic security responsibilities were transferred to the Ministry of

¹⁵See the Transfer Manual for Carabineros de Chile Personnel available at https://www.carabineros.cl/transparencia/og/pdf/OG_2707_13112019.pdf

¹⁶We requested access to data on the demand and supply of UEs to study the role of resources more directly, but this information was not shared. While our analysis does not yield a clean causal estimate of the resource component in isolation, the fact that the program had similarly large effects in municipalities whose resource increases were similar to those in the control group suggests that the beat policing component is key.

Interior in 2011.¹⁷

Carabineros' mission, functions, and structure are established by Law 18.961 and the Regulation on Carabineros' Organization (Reglamento de Carabineros) (Biblioteca del Congreso Nacional de Chile, 1990; Ministerio de Defensa Nacional, República de Chile, 1989). Under this framework, Carabineros is a professional, disciplined, and hierarchical police institution whose mission is to preserve public order and security with a central emphasis on preventive policing. The General Director heads the strategic level, responsible for defining doctrine and institutional strategy, while the Subdirector coordinates implementation. The Directorate of Order and Security administers police functions at the tactical level.

Territorially, Carabineros is organized into zones that oversee *prefecturas*. Zones and *prefecturas* are primarily management layers. Zones map roughly one-to-one to Chile's regional divisions, with two exceptions: the Metropolitan Region of Santiago—the most populated in the country—and La Araucanía—which has recently experienced significant social conflict—are each divided into two zones. Each zone is subdivided into *prefecturas*—54 nationwide—which supervise 227 *comisaría*s, the main operational units at the tactical level. These *comisaría*s oversee smaller units—*subcomisaría*s, *tenencias*, *retenes*, *puestos*, and *avanzadas*—that operate within their assigned jurisdictions.

In practice, the *comisaría* is the basic unit of routine frontline policing and, in most municipalities, serves as the territorial anchor of police work. Officers conduct patrol, surveillance, prevention, and first response at this level. In most municipalities, a single *comisaría* covers the entire municipal territory, though larger municipalities may have multiple ones. Santiago, for instance, has four. All municipalities that adopted *Plan Cuadrante* during our study period had a *comisaría*. With the exception of 9 municipalities, each had a single *comisaría* covering the full municipal territory. On average, a *comisaría* in our sample covered 1,608 square kilometers and served 57,642 inhabitants.

Plan Cuadrante was introduced within this institutional framework without altering Carabineros' hierarchical structure or its strategic and tactical levels. The reform emerged in a context of rising victimization and insecurity, with the explicit goal of reversing these trends (Ministerio del Interior y Seguridad Pública, 2014). Its central innovation was a shift toward a beat policing model. In the criminology literature, beat policing refers to assigning officers to fixed territorial areas so they can accumulate local knowledge, strengthen ties with residents, and respond more effectively to local crime patterns (Cordner, 1995; Skogan and Hartnett, 1997). In practice, the catchment area of police forces in municipalities adopting *Plan Cuadrante* was partitioned into smaller, fixed geographic units (quadrants), to which officers were permanently assigned. Rather than building new infrastructure, the reform redefined the territorial responsibility of officers in already existing police stations (Winchester et al., 2007).¹⁸ This represented a clear break from the pre-reform system, in which officers rotated across broad operational territories, very often spanning entire municipalities.

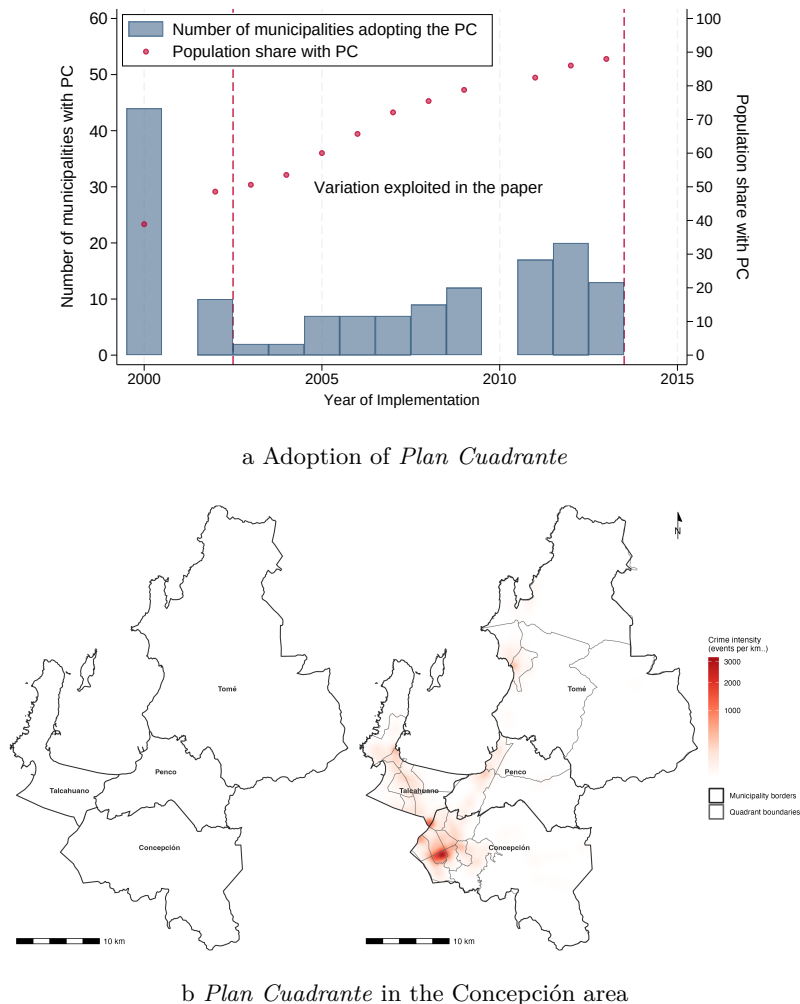
Carabineros began piloting *Plan Cuadrante* in 1998. The program expanded gradually through 2013, as shown in Panel (a) of Figure 1, reaching 150 urban municipalities and covering roughly

¹⁷Currently, Carabineros reports to the Ministry of Public Security, created in 2024 to focus exclusively on domestic security.

¹⁸In 2016, Carabineros launched the *Plan Cuadrante 2.0* program in those municipalities served by *Plan Cuadrante*. While this reform introduced additional reforms aimed at integrating the community into the provision of security and building capacity to gather and analyze crime data in police stations, this paper's study period ends before these improvements were introduced.

90% of Chile’s population.¹⁹ Online Appendix A provides further details about the rollout of the program. The rollout was designed to follow a municipality ranking based on a weighted index of victimization, police resource deficits, drug prevalence, unemployment, and demand for police services, with annual entry determined by available budget.²⁰ In practice, however, Budget Office evaluations show that adoption diverged from this ranking for undocumented reasons, so the rule operated more as a loose guideline than as a binding assignment mechanism (Salas P. et al., 2014).

Figure 1: Implementation of *Plan Cuadrante*



Notes: Panel (a) shows the number of municipalities adopting the *Plan Cuadrante* each year (bars), while the circles above them illustrate the cumulative share of the Chilean population living in a municipality with the *Plan Cuadrante* already in place. Panel (b) depicts the metropolitan area of Concepción, the second-largest in Chile: the left map shows its four municipalities—the original police catchment areas—and the right map overlays the Carabineros quadrants together with a kernel-density estimate of all types of robberies and thefts recorded by the police (both reported crimes and arrests). It illustrates how *Plan Cuadrante* subdivided each catchment area (which corresponds to the territory of a municipality) into smaller quadrants, as well as the spatial concentration of crime within them.

¹⁹Further details about the expansion of *Plan Cuadrante* are provided in Online Appendix A and Table A.1.

²⁰The weights assigned to each component changed on a yearly basis (Winchester et al., 2007; Salas P. et al., 2014).

Conceptually, police effectiveness can be viewed as a function of three inputs: total resources, territorial deployment, and policing strategy. Although in some municipalities police resources were increased significantly to operationalize the beat policing model (Winchester et al., 2007; Salas P. et al., 2014), the program primarily transformed the latter two elements. The creation of quadrants was accompanied by permanent geographic assignment of officers to specific areas. This change was operationally critical. By making officers responsible for their quadrant, it encouraged accumulation of local knowledge and stronger community ties, allowing them to adapt patrols and prevention activities to local crime patterns. Importantly, this was not driven by program-specific operational directive changes. Throughout the period we study, a single national set of operational guidelines applied to officers in both treated and control municipalities (Carabineros de Chile, 2018; Winchester et al., 2007). Thus, while the average effects we document on crime and crime perception likely reflect combined changes in the three elements we discuss—i.e., resources, deployment, and strategy—the deployment change itself was essential to operationalizing the reform. We discuss the drivers of the effects we document in Section 4.4.

Quadrant size and shape were determined by local police authorities based on factors such as road networks, geographic features, and available resources. In principle, resource allocation across quadrants followed demand for police services, measured in Equivalent Surveillance Units (UEVs); in practice, the station commander retained discretionary authority, as was also the case prior to *Plan Cuadrante* (Salas P. et al., 2014). Municipalities implementing *Plan Cuadrante* were subdivided into an average of seven quadrants, each typically staffed by approximately three officers per shift. On average, each quadrant covered 185 square kilometers and served 13,466 inhabitants. Panel (b) of Figure 1 illustrates this structure, showing the 29 quadrants created across the four municipalities of the Concepción metropolitan region—Chile’s second-largest metropolitan area and the largest in our sample. Quadrants represented approximately 14% of the original *comisaría* territory. Officers continued to operate from the same police station but were permanently assigned to a specific sub-area within its jurisdiction. While Carabineros’ hierarchical structure limits frontline officer decision-making authority, each quadrant was assigned a Quadrant Delegate (*Delegado de Cuadrante*), a non-commissioned officer, responsible for immediate operational decisions within the quadrant.

Plan Cuadrante may affect public safety through two complementary channels: deterrence and police effectiveness, understood as the capacity to respond to and resolve crimes. The changes induced by *Plan Cuadrante*—permanently assigning officers to well-defined neighborhoods and, in some municipalities, increasing available resources—can increase both police visibility and local knowledge. As officers repeatedly patrol the same area, they become familiar with its streets, residents, and routines, which can raise the perceived probability of identification and apprehension and reduce the expected returns to crime. At the same time, this deeper knowledge of local territory and communities can improve police response and case resolution once crimes occur. The reform therefore likely affects both dimensions. Consistent with this interpretation, in Section 4 we document declines in victimization and in reported crime—patterns consistent with deterrence—together with increases in arrests, consistent with improved police effectiveness.

Although beat policing shares features with other policing approaches, such as hot-spot policing and community-based policing, which have been widely studied, it has distinctive features that make it worth studying as an independent approach. Hot-spot policing concentrates resources in high-crime areas and does not focus on building local knowledge and community

ties. It is typically implemented through targeted interventions at specific locations rather than systematic territorial organization, and the same officers are not necessarily sent repeatedly to the same areas. Evidence on hot-spot policing is mixed, with some studies showing crime reductions in targeted areas but also potential displacement effects (Braga et al., 2019). *Plan Cuadrante*, by contrast, subdivided all urban municipalities into quadrants with permanent officer assignment. As Panel (b) of Figure 1 illustrates, some quadrants contained crime hot-spots, but many did not. The reform thus addressed entire urban territories rather than specific high-crime areas, potentially mitigating displacement effects associated with hot-spot policing.

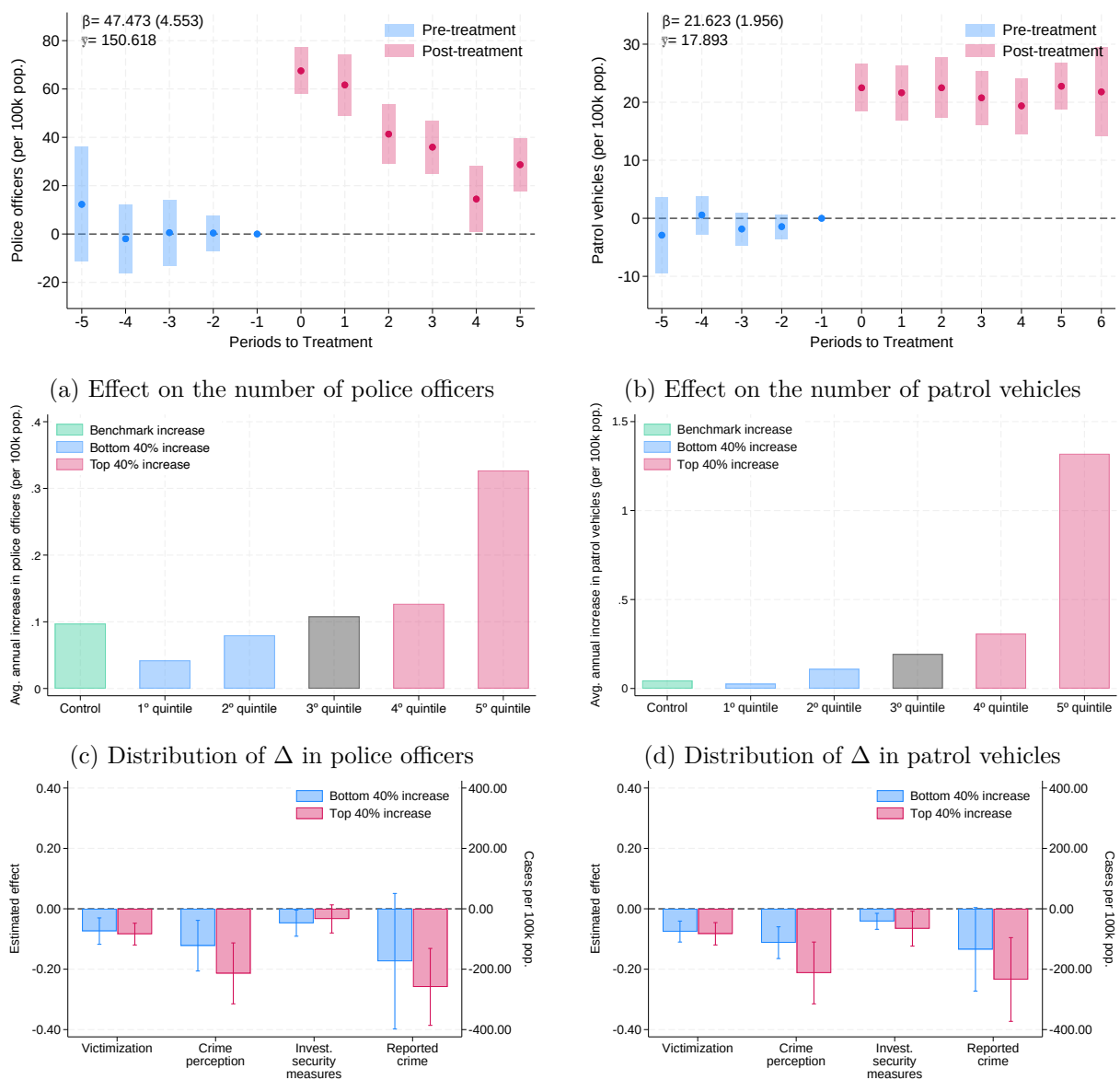
Community-based policing emphasizes building relationships with residents and addressing underlying social issues, and typically involves active community participation in defining strategy and priorities—in some cases, even assuming surveillance and prevention responsibilities. While increasing trust in the police and strengthening community ties were among the declared goals of *Plan Cuadrante*, and may have resulted from its territorial reorganization, the community was not formally involved in the provision of security.²¹

In recent decades, police departments worldwide have transitioned toward beat policing reforms (National Academies of Sciences, Engineering, and Medicine, 2018; Ministerio del Interior y Seguridad Pública, 2014), yet the effectiveness of this model has not been rigorously assessed. Although the beat policing approach introduced through *Plan Cuadrante* shares features with programs previously studied in the economic literature, its distinctive characteristics—comprehensive territorial reorganization and systematic local officer specialization—make it a valuable case for studying how deployment and policing strategies not yet examined in the economics literature affect crime and crime perception.

²¹The *Plan Cuadrante*’s original design contemplated a formal community integration model (MICC) with a Community Affairs Office (*Oficina de Asuntos Comunitarios*) and a dedicated officer in each police station. However, this model was not implemented during our study period. Independent program evaluations from the Budget Office document that these Community Affairs Offices did not exist and had no dedicated personnel, budget, or structured activities in the period we study (Winchester et al., 2007; Salas P. et al., 2014). The community integration model was piloted in four police stations during 2012–2013, none of which are in our analytical sample. Financing was then discontinued in early 2014 and only restored and expanded after our study period ends.

Other figures and tables referenced above:

Figure 5: Plan Cuadrante and Changes in Police Resources



(e) Effects by the intensity of Δ in police officers (f) Effects by the intensity of Δ in patrol vehicles

Notes: Panels (a) and (b) show the effect of *Plan Cuadrante* on police officers and patrol vehicles per 100,000 inhabitants. Panels (c) and (d) show the distribution of annual changes in officers and patrol vehicles across municipalities. Panels (e) and (f) report effects by the intensity of resource changes, comparing municipalities in the bottom 40% and top 40% of the distribution. The estimation follows Callaway and Sant'Anna (2021). The graphs display estimated coefficients and 95% confidence intervals. Standard errors are clustered at the municipality level using wild bootstrapping.

Table A.1: Implementation of *Plan Cuadrante* by Year

Year	N	Municipalities
1998–2000	44	Metropolitan Region of Santiago (44 municipalities)
2002	10	Chiguayante, Concepción, Hualpén, Quilpué, San Antonio, San Pedro de la Paz, Talcahuano, Valparaíso, Villa Alemana, Viña del Mar
2003	2	Padre Las Casas, Temuco
2004	2	Antofagasta, Copiapó
2005	7	Alto Hospicio, Arica, Calama, Curicó, Iquique, Linares, Talca
2006	7	Coquimbo, La Serena, Los Andes, Ovalle, Rancagua, San Felipe, San Fernando
2007	7	Chillán, Chillán Viejo, Los Ángeles, Osorno, Puerto Montt, Punta Arenas, Valdivia
2008	9	Ancud, Castro, Constitución, Coronel, Lota, Peñaflor, Quillota, Rengo, Villarrica
2009	12	Angol, Aysén, Calera, Concón, Coyhaique, La Unión, Padre Hurtado, Parral, Penco, San Carlos, Tomé, Vallenar
2011	17	Arauco, Cañete, Cartagena, Cauquenes, Curanilahue, La Ligua, Limache, Molina, Nueva Imperial, Pucón, Puerto Varas, Río Bueno, San Clemente, San Javier, San Vicente, Santa Cruz, Victoria
2012	20	Cabrero, Calbuco, Casablanca, Chimbarongo, Collipulli, Curacaví, El Monte, Graneros, Illapel, Isla de Maipo, Lautaro, Machalí, Mostazal, Mulchén, Nacimiento, Natales, Panguipulli, Quellón, Quintero, Tocopilla
2013	13	Bulnes, Laja, Las Cabras, Lebu, Llaillay, Loncoche, Longaví, Los Álamos, Maule, Requínoa, Teno, Vicuña, Yumbel
Total	150	

Note: N is the number of municipalities that adopted *Plan Cuadrante* in each year. Data shared by Carabineros de Chile.

Comment R1.2

Relatedly, I find the label “geographic specialization” potentially misleading, as it may suggest policing efforts concentrated in specific city areas, which is not how the paper currently reads. In the economics of crime literature, similar interventions are more often described as “hot-spot policing,” in contrast with approaches that aim to maintain constant response times. Can the authors directly compare this reform to these? Alternatively, if the reform went beyond territorial reallocation to include changes in personnel assignments (e.g., shifting away from office to street duties) or shifts in decision-making from central to local levels, this should be explained clearly, since it would ultimately shape the paper’s contribution (see also point 3 below).

Response to R1.2:

We thank the reviewer for this comment. We agree that “geographic specialization” may be misleading, and we have changed the terminology used to describe the program. The intervention we study is known in the criminological literature and police practice as beat policing, the permanent assignment of officers to smaller, fixed catchment areas rather than rotating across larger ones. Beat policing has been widely implemented across police forces worldwide and connects directly to Ostrom’s seminal hypothesis about why smaller police departments outperform larger ones (Ostrom et al., 1973). Despite its relevance, it has not been rigorously evaluated in the economics or criminology literatures. We used “geographic specialization”, reasoning that it would be more intuitive for an economics audience and could connect better with the literature on local public service delivery, but we recognize its limitations. In the revised manuscript, we use the term “beat policing” and connect it to the literature on territorial/local specialization in public service delivery (Besley and Coate, 2003), which we believe makes the concept more intuitive for an economics audience. If the editor or reviewer is not convinced by the use of “beat policing” either, we would welcome guidance on preferred terminology. Alternative formulations we have considered include “territorial reorganization” and “local police assignment”.

Regarding the comparison with hot spot policing, *Plan Cuadrante* is not hot spot policing. The latter approach concentrates resources in high-crime areas but does not necessarily involve permanent beat assignments, nor does it focus on building local knowledge and community ties. It is typically implemented through targeted interventions at specific locations rather than systematic territorial organization, and the same officers are not necessarily sent to the same locations repeatedly. Evidence on its effectiveness is mixed, with some studies finding crime reductions in targeted areas but others documenting potential displacement to neighboring zones (Braga et al., 2019). *Plan Cuadrante*, by contrast, subdivided entire urban territories into quadrants with *permanent* officer assignment, ensuring stable and knowledgeable coverage across the full territory rather than concentrating resources where crime is highest. As a result, while some quadrants contained crime hot-spots, many did not, and this comprehensive territorial approach potentially mitigates the displacement effects associated with hot-spot policing.

We discuss these differences explicitly in the updated Section 2 of the manuscript:

Although beat policing shares features with other policing approaches, such as hot-spot policing and community-based policing, which have been widely studied, it has distinctive features that make it worth studying as an independent approach. Hot-spot policing concentrates resources in high-crime areas and does not focus on building local knowledge and community

ties. It is typically implemented through targeted interventions at specific locations rather than systematic territorial organization, and the same officers are not necessarily sent repeatedly to the same areas. Evidence on hot-spot policing is mixed, with some studies showing crime reductions in targeted areas but also potential displacement effects (Braga et al., 2019). *Plan Cuadrante*, by contrast, subdivided all urban municipalities into quadrants with permanent officer assignment. As Panel (b) of Figure 1 illustrates, some quadrants contained crime hot-spots, but many did not. The reform thus addressed entire urban territories rather than specific high-crime areas, potentially mitigating displacement effects associated with hot-spot policing.

Regarding response time reforms, program documentation does not include any explicit objective of reducing response times (Winchester et al., 2007; Salas P. et al., 2014). The reform did not build new police stations; rather, it redefined the territorial responsibilities of officers within already existing police stations (Winchester et al., 2007). Throughout our study period, a single national set of operational guidelines applied to officers in both treated and control municipalities, with no formal directive to change police tactics to improve response times (Carabineros de Chile, 2018; Winchester et al., 2007). Nonetheless, response times could have improved as a byproduct of the increase in patrol vehicles or police officers that accompanied the program in some municipalities. While we lack information on response times to directly test their role, the fact that the program had similarly large effects on victimization in municipalities where resource increases were similar to those in the control group suggests that response times are unlikely to be the main mechanism driving the effects. The mechanism we propose operates through a different channel: by fixing the same officers to the same territory rather than rotating them across a larger territory, the reform generates accumulated local knowledge that increases police effectiveness and deterrence.

Regarding whether the reform included decentralization of decision-making, the program formally created the role of Quadrant Delegate, a non-commissioned officer with primary operational responsibility for their assigned quadrant, which represented a certain degree of decentralization of day-to-day operational decisions from the station commander to the frontline officer. This autonomy was however constrained by the hierarchical command structure within police stations, which remained unchanged.

While Plan Cuadrante does not entail any explicit change in police tactics during the period studied (Salas P. et al., 2014), and officers operate under the same operational guidelines without any formal directive to shift personnel from office to street duties, in some municipalities the program was accompanied by an increase in police resources, driven primarily by new hires rather than reallocation, which plausibly increased the number of officers on the street. Consistent with this, we show that Plan Cuadrante increased perceived police presence in neighborhoods. However, we also show that the program had large effects on perceived police presence in municipalities where the increase in the number of officers was very similar to that in the control group, suggesting that the reduction in the catchment area plays an important role in police visibility beyond the resource channel.

Finally, Plan Cuadrante should not be confused with community policing. While the program’s original design envisaged a formal community integration component, independent program evaluations document that this component existed only on paper throughout our study period (Winchester et al., 2007; Salas P. et al., 2014). The community integration model (MICC) was piloted in only four police stations in 2012–2013 (in police stations not included in the analytical sample) before its financing agreement was discontinued in early 2014, and its large-scale implementation only materialized with Plan Cuadrante 2.0 after the end of our study period. Any improvement in police-community relations during the period we study therefore did not result from an ex-

PLICIT tactical change or the formal integration of the community into the prevention strategy, but rather from the more continuous and stable contact that naturally arises when the same officers are permanently assigned to the same territory.

To address these questions, we have edited substantially the introduction and section 2, where we provide details about Plan Cuadrante and compare it to other policing approaches such as hot-spot policing or community policing.

Below we report the new **Section 2** describing the institutional context and the *Plan Cuadrante*:

Carabineros, Chile’s national police force, was founded in 1927. Through most of our study period, Carabineros operated under the Ministry of Defense, which oversaw both military and police functions until domestic security responsibilities were transferred to the Ministry of Interior in 2011.²²

Carabineros’ mission, functions, and structure are established by Law 18.961 and the Regulation on Carabineros’ Organization (Reglamento de Carabineros) (Biblioteca del Congreso Nacional de Chile, 1990; Ministerio de Defensa Nacional, República de Chile, 1989). Under this framework, Carabineros is a professional, disciplined, and hierarchical police institution whose mission is to preserve public order and security with a central emphasis on preventive policing. The General Director heads the strategic level, responsible for defining doctrine and institutional strategy, while the Subdirector coordinates implementation. The Directorate of Order and Security administers police functions at the tactical level.

Territorially, Carabineros is organized into zones that oversee *prefecturas*. Zones and *prefecturas* are primarily management layers. Zones map roughly one-to-one to Chile’s regional divisions, with two exceptions: the Metropolitan Region of Santiago—the most populated in the country—and La Araucanía—which has recently experienced significant social conflict—are each divided into two zones. Each zone is subdivided into *prefecturas*—54 nationwide—which supervise 227 *comisaría*s, the main operational units at the tactical level. These *comisaría*s oversee smaller units—*subcomisaría*s, *tenencias*, *retenes*, *puestos*, and *avanzadas*—that operate within their assigned jurisdictions.

In practice, the *comisaría* is the basic unit of routine frontline policing and, in most municipalities, serves as the territorial anchor of police work. Officers conduct patrol, surveillance, prevention, and first response at this level. In most municipalities, a single *comisaría* covers the entire municipal territory, though larger municipalities may have multiple ones. Santiago, for instance, has four. All municipalities that adopted *Plan Cuadrante* during our study period had a *comisaría*. With the exception of 9 municipalities, each had a single *comisaría* covering the full municipal territory. On average, a *comisaría* in our sample covered 1,608 square kilometers and served 57,642 inhabitants.

Plan Cuadrante was introduced within this institutional framework without altering Carabineros’ hierarchical structure or its strategic and tactical levels. The reform emerged in a context of rising victimization and insecurity, with the explicit goal of reversing these trends (Ministerio del Interior y Seguridad Pública, 2014). Its central innovation was a shift toward a beat policing model. In the criminology literature, beat policing refers to assigning officers to fixed territorial areas so they can accumulate local knowledge, strengthen ties with residents,

²²Currently, Carabineros reports to the Ministry of Public Security, created in 2024 to focus exclusively on domestic security.

and respond more effectively to local crime patterns (Cordner, 1995; Skogan and Hartnett, 1997). In practice, the catchment area of police forces in municipalities adopting *Plan Cuadrante* was partitioned into smaller, fixed geographic units (quadrants), to which officers were permanently assigned. Rather than building new infrastructure, the reform redefined the territorial responsibility of officers in already existing police stations (Winchester et al., 2007).²³ This represented a clear break from the pre-reform system, in which officers rotated across broad operational territories, very often spanning entire municipalities.

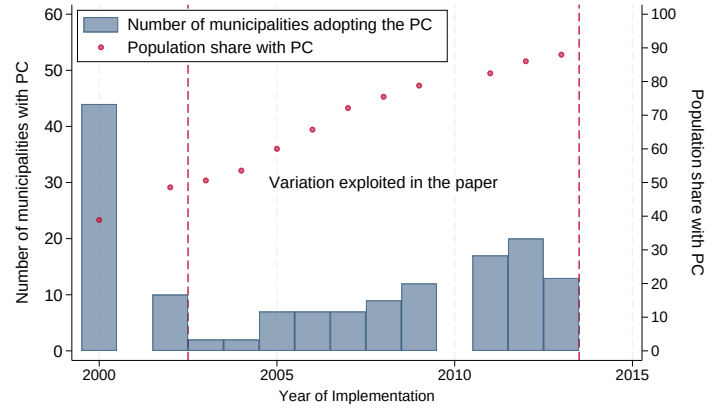
Carabineros began piloting *Plan Cuadrante* in 1998. The program expanded gradually through 2013, as shown in Panel (a) of Figure 1, reaching 150 urban municipalities and covering roughly 90% of Chile’s population.²⁴ Online Appendix A provides further details about the rollout of the program. The rollout was designed to follow a municipality ranking based on a weighted index of victimization, police resource deficits, drug prevalence, unemployment, and demand for police services, with annual entry determined by available budget.²⁵ In practice, however, Budget Office evaluations show that adoption diverged from this ranking for undocumented reasons, so the rule operated more as a loose guideline than as a binding assignment mechanism (Salas P. et al., 2014).

²³In 2016, Carabineros launched the *Plan Cuadrante 2.0* program in those municipalities served by *Plan Cuadrante*. While this reform introduced additional reforms aimed at integrating the community into the provision of security and building capacity to gather and analyze crime data in police stations, this paper’s study period ends before these improvements were introduced.

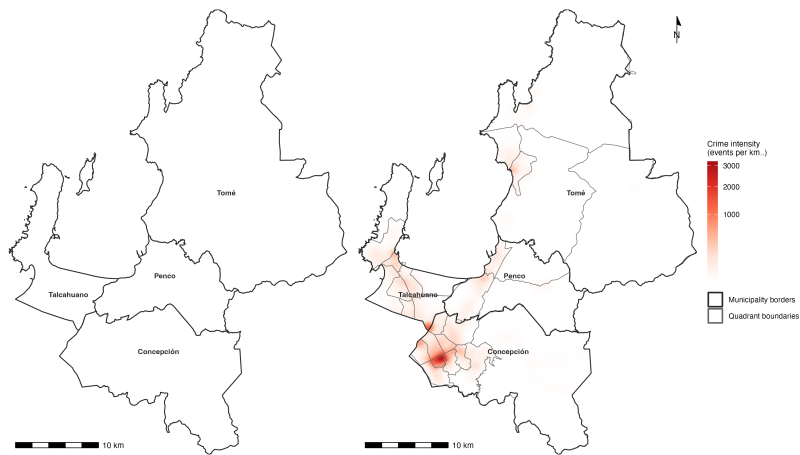
²⁴Further details about the expansion of *Plan Cuadrante* are provided in Online Appendix A and Table A.1.

²⁵The weights assigned to each component changed on a yearly basis (Winchester et al., 2007; Salas P. et al., 2014).

Figure 1: Implementation of *Plan Cuadrante*



a Adoption of *Plan Cuadrante*



b *Plan Cuadrante* in the Concepción area

Notes: Panel (a) shows the number of municipalities adopting the *Plan Cuadrante* each year (bars), while the circles above them illustrate the cumulative share of the Chilean population living in a municipality with the *Plan Cuadrante* already in place. Panel (b) depicts the metropolitan area of Concepción, the second-largest in Chile: the left map shows its four municipalities—the original police catchment areas—and the right map overlays the Carabineros quadrants together with a kernel-density estimate of all types of robberies and thefts recorded by the police (both reported crimes and arrests). It illustrates how *Plan Cuadrante* subdivided each catchment area (which corresponds to the territory of a municipality) into smaller quadrants, as well as the spatial concentration of crime within them.

Conceptually, police effectiveness can be viewed as a function of three inputs: total resources, territorial deployment, and policing strategy. Although in some municipalities police resources were increased significantly to operationalize the beat policing model (Winchester et al., 2007; Salas P. et al., 2014), the program primarily transformed the latter two elements. The creation of quadrants was accompanied by permanent geographic assignment of officers to specific areas. This change was operationally critical. By making officers responsible for their quadrant, it encouraged accumulation of local knowledge and stronger community ties, allowing them to adapt patrols and prevention activities to local crime patterns. Importantly, this was not driven by program-specific operational directive changes. Throughout the period we study, a single national set of operational guidelines applied to officers in both treated and control municipalities (Carabineros de Chile, 2018; Winchester et al., 2007). Thus, while the average effects we document on crime and crime perception likely reflect combined changes in the three elements we discuss—i.e., resources, deployment, and strategy—the deployment change itself was essential to operationalizing the reform. We discuss the drivers of the effects we document in Section 4.4.

Quadrant size and shape were determined by local police authorities based on factors such as road networks, geographic features, and available resources. In principle, resource allocation across quadrants followed demand for police services, measured in Equivalent Surveillance Units (UEVs); in practice, the station commander retained discretionary authority, as was also the case prior to *Plan Cuadrante* (Salas P. et al., 2014). Municipalities implementing *Plan Cuadrante* were subdivided into an average of seven quadrants, each typically staffed by approximately three officers per shift. On average, each quadrant covered 185 square kilometers and served 13,466 inhabitants. Panel (b) of Figure 1 illustrates this structure, showing the 29 quadrants created across the four municipalities of the Concepción metropolitan region—Chile’s second-largest metropolitan area and the largest in our sample. Quadrants represented approximately 14% of the original *comisaría* territory. Officers continued to operate from the same police station but were permanently assigned to a specific sub-area within its jurisdiction. While Carabineros’ hierarchical structure limits frontline officer decision-making authority, each quadrant was assigned a Quadrant Delegate (*Delegado de Cuadrante*), a non-commissioned officer, responsible for immediate operational decisions within the quadrant.

Plan Cuadrante may affect public safety through two complementary channels: deterrence and police effectiveness, understood as the capacity to respond to and resolve crimes. The changes induced by *Plan Cuadrante*—permanently assigning officers to well-defined neighborhoods and, in some municipalities, increasing available resources—can increase both police visibility and local knowledge. As officers repeatedly patrol the same area, they become familiar with its streets, residents, and routines, which can raise the perceived probability of identification and apprehension and reduce the expected returns to crime. At the same time, this deeper knowledge of local territory and communities can improve police response and case resolution once crimes occur. The reform therefore likely affects both dimensions. Consistent with this interpretation, in Section 4 we document declines in victimization and in reported crime—patterns consistent with deterrence—together with increases in arrests, consistent with improved police effectiveness.

Although beat policing shares features with other policing approaches, such as hot-spot policing and community-based policing, which have been widely studied, it has distinctive features that make it worth studying as an independent approach. Hot-spot policing concentrates resources in high-crime areas and does not focus on building local knowledge and community

ties. It is typically implemented through targeted interventions at specific locations rather than systematic territorial organization, and the same officers are not necessarily sent repeatedly to the same areas. Evidence on hot-spot policing is mixed, with some studies showing crime reductions in targeted areas but also potential displacement effects (Braga et al., 2019). *Plan Cuadrante*, by contrast, subdivided all urban municipalities into quadrants with permanent officer assignment. As Panel (b) of Figure 1 illustrates, some quadrants contained crime hot-spots, but many did not. The reform thus addressed entire urban territories rather than specific high-crime areas, potentially mitigating displacement effects associated with hot-spot policing.

Community-based policing emphasizes building relationships with residents and addressing underlying social issues, and typically involves active community participation in defining strategy and priorities—in some cases, even assuming surveillance and prevention responsibilities. While increasing trust in the police and strengthening community ties were among the declared goals of *Plan Cuadrante*, and may have resulted from its territorial reorganization, the community was not formally involved in the provision of security.²⁶

In recent decades, police departments worldwide have transitioned toward beat policing reforms (National Academies of Sciences, Engineering, and Medicine, 2018; Ministerio del Interior y Seguridad Pública, 2014), yet the effectiveness of this model has not been rigorously assessed. Although the beat policing approach introduced through *Plan Cuadrante* shares features with programs previously studied in the economic literature, its distinctive characteristics—comprehensive territorial reorganization and systematic local officer specialization—make it a valuable case for studying how deployment and policing strategies not yet examined in the economics literature affect crime and crime perception.

And below the updated **Introduction**:

Crime imposes significant social costs, straining the criminal justice system and increasing both public and private security expenditures. Given these stakes, crime consistently ranks among the public’s top concerns and features prominently in political debate (Cafferata and Scartascini, 2021; Gallup, 2023). Police forces play a central role in controlling crime and maintaining public safety. While there is extensive evidence on the effects of police resources on crime (see, for instance, Chalfin and McCrary, 2017; Levitt, 1997; Di Tella and Scharrotsky, 2004), we know much less about how the deployment of those resources, and the strategy changes enabled by different territorial assignments, shape police effectiveness. Studies of public service provision show that decentralization and local specialization can improve performance by aligning operations with local needs (see, for instance, Besley and Coate, 2003). This suggests that policing models that foster local knowledge and allow officers to adapt practices to local conditions may be more effective. Beat policing reflects these insights. Rather than rotating officers across large areas, it permanently assigns each officer to a smaller territorial unit, enabling the accumulation of local knowledge and adaptation to local crime patterns and community dynamics (Cordner, 1995; Skogan and Hartnett, 1997). This model

²⁶The *Plan Cuadrante*’s original design contemplated a formal community integration model (MICC) with a Community Affairs Office (*Oficina de Asuntos Comunitarios*) and a dedicated officer in each police station. However, this model was not implemented during our study period. Independent program evaluations from the Budget Office document that these Community Affairs Offices did not exist and had no dedicated personnel, budget, or structured activities in the period we study (Winchester et al., 2007; Salas P. et al., 2014). The community integration model was piloted in four police stations during 2012–2013, none of which are in our analytical sample. Financing was then discontinued in early 2014 and only restored and expanded after our study period ends.

has become increasingly common across police departments worldwide (National Academies of Sciences, Engineering, and Medicine, 2018; Ministerio del Interior y Seguridad Pública, 2014). Yet despite its widespread adoption, rigorous causal evidence on its effects on public safety and security perceptions remains scarce.

This paper combines rich administrative and survey data from Chile to provide causal evidence that beat policing can significantly improve public security and police effectiveness along multiple dimensions. We exploit quasi-random variation from the staggered adoption of a large reform, *Plan Cuadrante*, which introduced a beat-policing model in urban areas by dividing police operational areas—typically municipalities—into smaller zones—quadrants—and permanently assigning officers to them. This permanent assignment was designed to help officers accumulate local knowledge of territorial features, crime patterns, and communities, and to adapt their practices accordingly. Using a difference-in-differences strategy, we show that adopting municipalities experienced significant declines in victimization, reported crime, and perceived insecurity.

Plan Cuadrante was implemented in municipalities with at least 70% urban population and rolled out between 1998 and 2013 across the 150 municipalities that met this condition. Its central objective was to reduce victimization and insecurity perceptions. Rather than changing police hierarchy or operational doctrine, the reform reorganized frontline deployment by subdividing police operational areas—typically municipalities—into quadrants and permanently assigning officers to each of them. On average, municipalities were divided into seven quadrants, with size and boundaries defined by local police authorities based on road networks, geographic features, and available resources.

To examine the effects of *Plan Cuadrante*, we use household-level data from the National Urban Security Survey (ENUSC), which reports crime victimization, security perceptions, trust in police, and private security investment for roughly 25,000 urban households across 101 municipalities each year. We complement these data with administrative records on arrests and crimes reported to the police. To leverage staggered adoption, we employ a difference-in-differences strategy comparing early and late adopters using the procedure developed in Callaway and Sant’Anna (2021). Analyses based on survey data focus on 46 urban municipalities that adopted the program between 2005 and 2012 and one never-treated municipality; together, they represent over one-quarter of Chile’s population. Analyses based on administrative data use a broader sample of 96 municipalities that adopted between 2003 and 2013 and 196 never-treated municipalities, representing over half of Chile’s population.

We find that households in adopting municipalities become 10 percentage points (36%) less likely to experience crime in the previous 12 months, with effects appearing within one year and growing over time. Consistent with this finding, we document a 10% decline in crimes reported to the police. These estimates may understate the reform’s true impact because not all crimes are reported and, by bringing police closer to communities, the program could have increased reporting rates. Nevertheless, the alignment between survey-based and administrative estimates provides reassuring evidence of the reform’s effectiveness.

Beyond reducing crime, the reform also reduced insecurity perceptions and private investment in security. This is noteworthy because actual crime and crime perceptions can diverge, potentially leading to suboptimal security expenditures. For example, Ajzenman et al. (2023) show that rising immigration in Chile between 2000 and 2016 did not increase crime but did increase insecurity perceptions.

A potential identification concern is that crime reductions in adopting municipalities could reflect displacement to neighboring areas rather than genuine effectiveness gains. Hot-spot policing, which concentrates police resources in high-crime locations, has been shown to generate this type of spatial displacement (Blattman et al., 2021). In our setting, spillovers are less likely because municipalities are large geographic areas and beat policing reorganized deployment across the full municipal territory rather than concentrating resources in selected locations. Nevertheless, we directly assess spatial spillovers through three complementary exercises. We find no evidence that *Plan Cuadrante* displaced crime to neighboring municipalities, diverted police resources from control to treated areas, or generated spillovers in victimization and insecurity outcomes in the ENUSC sample. These results support the interpretation that improvements in treated municipalities reflect genuine effectiveness gains rather than crime displacement.

To improve public safety and security perceptions, police work must strengthen prevention and crime resolution. Both dimensions can be affected by police resources, territorial deployment, and policing strategy. In *Plan Cuadrante*, the core beat-policing components were the reorganization of deployment and the operational strategy adjustments enabled by that reorganization. At the same time, some municipalities experienced sizeable increases in police resources, which raises the question of whether resource expansion alone can explain the effects we estimate.

During our study period, public expenditure on security was rising, and most municipalities experienced increases in police resources. Police resources were not the central component of *Plan Cuadrante*, but adopting the model required a minimum operational capacity, which in some municipalities implied increases in officers and patrol vehicles above those observed elsewhere. Thus, part of the effects we document could reflect resource increases. However, changes in police resources alone do not explain our findings. Municipalities that adopted *Plan Cuadrante* but experienced resource increases similar to those in control municipalities still show large and statistically significant improvements in public safety outcomes. Moreover, for most outcomes, these effects are close in magnitude to those in municipalities where resources increased more substantially. In addition, variation in police resources unrelated to *Plan Cuadrante* adoption does not seem to generate comparable gains. Taken together, these patterns indicate that while resource increases helped enable implementation and may explain part of the effects in some municipalities, the beat-policing components of the reform were central drivers of the overall gains.

Against this background, the core institutional change introduced by *Plan Cuadrante* was on the deployment margin. The reform permanently assigned officers to smaller, fixed quadrants while leaving national operational guidelines unchanged across treated and control municipalities throughout our study period (Carabineros de Chile, 2018; Winchester et al., 2007). The Chilean police is a highly hierarchical institution with centrally defined doctrine and limited officer discretion, so adoption did not involve a top-down redesign of policing strategy. Instead, any strategic adjustments likely emerged from the new deployment itself. By remaining in the same territory, officers accumulated local knowledge of the area and its community and could adapt day-to-day practices—patrol routes and prevention activities—to local conditions. In this sense, *Plan Cuadrante* changed police work primarily through deployment, with strategy evolving endogenously within a common institutional framework.

The evidence supports this interpretation. Following the adoption of *Plan Cuadrante*, households report higher police presence in their neighborhoods, a pattern that is consistent with

both resource expansion and deployment changes. However, our broader results suggest effects beyond patrolling intensity alone. While earlier work finds that police presence can reduce crime while officers are present but may be attenuated by displacement across places or time windows (Mastrobuoni, 2019; Blanes-i Vidal and Mastrobuoni, 2018), we find declines in victimization and reported crime with no evidence of spillovers to neighboring municipalities. In line with this pattern, households report better police performance and greater trust in police, and administrative data show that arrests increase even as crime declines, including in municipalities with resource increases similar to those in control municipalities. Taken together, these findings suggest that beat policing improved public security through stronger deterrence and more effective prevention, response, and crime resolution under stable territorial assignment.

Our findings contribute to two related strands of the literature on police organization and public safety. First, they provide some of the first causal evidence on the effects of beat policing on crime and security perceptions. Beat policing has become one of the most widely adopted policing models globally (National Academies of Sciences, Engineering, and Medicine, 2018; Ministerio del Interior y Seguridad Pública, 2014), yet robust causal evidence on its effectiveness remains scarce. We show that this model can generate substantial reductions in crime victimization, reported crime, and insecurity perceptions.

These results also speak to recent work on other policing models. Hot-spot policing reallocates police resources toward high-crime locations and has been shown to reduce crime in targeted areas, though often with displacement to nearby places (Di Tella and Schargrodsky, 2004; Blattman et al., 2021; Draca et al., 2011; Weisburd, 2021). Beat policing differs fundamentally. Rather than concentrating police presence in selected locations—where the same officers need not return repeatedly—it provides continuous, territory-wide coverage through permanent officer-to-area assignment. This distinction matters in our setting, where many quadrants were not crime hot spots and the reform generated municipality-wide declines in crime and insecurity perceptions, with no evidence of spillovers to neighboring municipalities. Community policing, another model that has received growing attention in the economics literature, emphasizes citizen participation in defining policing priorities and, in some cases, in implementation itself. While it has been shown to increase trust in police in developed countries, evidence from low- and middle-income settings points to more limited effects on crime and trust (Blair et al., 2021; Tobón, 2021). Beat policing also differs from that approach. By permanently assigning officers to stable territorial units, it strengthens local knowledge and brings police closer to communities without requiring direct citizen participation in strategy or operations. Our findings show that this model can reduce crime and insecurity perceptions while increasing trust in police institutions in a middle-income country.

A second contribution is to the broader literature showing that specific features of police organization matter for police effectiveness and public security outcomes. Research has shown that faster police response times improve crime clearance (Blanes-i-Vidal and Kirchmaier, 2018), that the effectiveness of police presence depends on how this is organized and deployed rather than simply on their intensity (Facchetti, 2025; Mastrobuoni, 2019; Blanes-i Vidal and Mastrobuoni, 2018), that shifts in police prioritization across crime types shape criminal behavior (Adda et al., 2014), and that better information systems and technology can improve police performance (Mastrobuoni, 2020). Earlier work also found that areas covered by smaller, more localized police departments outperform, in terms of security, those covered by larger centralized ones, hypothesizing that this might be related, among other factors, to closer citizen-police engagement (Ostrom et al., 1973, 1978). We add to this literature by studying a policing model

that combines several of these margins within a single organizational reform. By assigning officers permanently to smaller territorial units rather than rotating them across larger ones, beat policing is designed to increase police visibility, strengthen knowledge of local conditions, and improve the capacity of officers to adapt their practices to local crime patterns. In line with these mechanisms, we show that the adoption of *Plan Cuadrante* increased perceived police presence in neighborhoods and raised police effectiveness, as arrests increased even while victimization and reported crime declined. These organizational improvements translated into meaningful gains in public safety and security perceptions.

A third contribution concerns the literature on police resources, which documents elasticities of crime with respect to officers and expenditure ranging from -0.1 to -2 , with larger effects in developed settings and for violent crimes (Chalfin and McCrary, 2017; Levitt, 1997; Di Tella and Schargrodsky, 2004). We complement this body of work by showing that the quantity of resources is not the only margin that matters. How those resources are deployed matters as well. *Plan Cuadrante* improved public safety outcomes even in municipalities that experienced no increase in police resources relative to controls, suggesting that reorganizing deployment can raise public safety and police effectiveness without expanding the resource base.

We also contribute to the literature on police practices, institutional trust, and citizen cooperation. A persistent challenge is that security perceptions often diverge from actual crime trends (Ajzenman et al., 2023), limiting the scope for improving both simultaneously. Rising crime erodes institutional trust (Blanco, 2013), while police misconduct undermines cooperation (Ang et al., 2025).²⁷ Our results show that beat policing can improve both dimensions—security and security perceptions—by bringing officers into sustained engagement with local communities through permanent territorial assignment. We document not only increases in police trust and evaluations of police performance, but also substantial reductions in private security investment, indicating that households revised their threat assessments in response to the program.

Finally, our findings speak to a broader question about how organizational design shapes public-service effectiveness. A growing literature shows that the spatial organization of service provision matters. In policing, we show that a model organizing officers around fixed territorial units—allowing them to build local knowledge, coordinate with local actors, and adapt practices accordingly—can substantially improve both crime control and institutional legitimacy. This aligns with evidence from health and education showing that bringing providers closer to the populations they serve can improve outcomes (Besley and Coate, 2003; Bardhan, 2002; Bloom et al., 2015; Basurto et al., 2020). In a rigorously identified setting, we show that this type of organizational proximity generates meaningful gains in police effectiveness and citizen security.

The rest of the paper is organized as follows. Section 2 presents background information about Chilean police and the program we study. Section 3 introduces the data and empirical strategy. Section 4 presents the main results and discusses potential drivers behind our findings. Finally, section 5 concludes.

²⁷Racial disparities in police behavior are also well documented; see, e.g., Gonçalves et al. (2025), who develop a method to correct for selection bias in estimates of racial disparities in the use of force.

Other table referenced above:

Table A.1: Implementation of *Plan Cuadrante* by Year

Year	N	Municipalities
1998–2000	44	Metropolitan Region of Santiago (44 municipalities)
2002	10	Chiguayante, Concepción, Hualpén, Quilpué, San Antonio, San Pedro de la Paz, Talcahuano, Valparaíso, Villa Alemana, Viña del Mar
2003	2	Padre Las Casas, Temuco
2004	2	Antofagasta, Copiapó
2005	7	Alto Hospicio, Arica, Calama, Curicó, Iquique, Linares, Talca
2006	7	Coquimbo, La Serena, Los Andes, Ovalle, Rancagua, San Felipe, San Fernando
2007	7	Chillán, Chillán Viejo, Los Ángeles, Osorno, Puerto Montt, Punta Arenas, Valdivia
2008	9	Ancud, Castro, Constitución, Coronel, Lota, Peñaflor, Quillota, Rengo, Villarrica
2009	12	Angol, Aysén, Calera, Concón, Coyhaique, La Unión, Padre Hurtado, Parral, Penco, San Carlos, Tomé, Vallenar
2011	17	Arauco, Cañete, Cartagena, Cauquenes, Curanilahue, La Ligua, Limache, Molina, Nueva Imperial, Pucón, Puerto Varas, Río Bueno, San Clemente, San Javier, San Vicente, Santa Cruz, Victoria
2012	20	Cabrero, Calbuco, Casablanca, Chimbarongo, Collipulli, Curacaví, El Monte, Graneros, Illapel, Isla de Maipo, Lautaro, Machalí, Mostazal, Mulchén, Nacimiento, Natales, Panguipulli, Quellón, Quintero, Tocopilla
2013	13	Bulnes, Laja, Las Cabras, Lebu, Llaillay, Loncoche, Longaví, Los Álamos, Maule, Requínoa, Teno, Vicuña, Yumbel
Total	150	

Note: *N* is the number of municipalities that adopted *Plan Cuadrante* in each year. Data shared by Carabineros de Chile.

Comment R1.3

Policy impacts and increases in police resources: At present, the paper provides too little detail, leaving the reader unsure about how to interpret results of a reform that changed multiple aspects of policing.

1. The policy seems to have changed for some cities the allocation of resources conditional on total levels, and for some others the aggregate levels too. The authors attempt to address this in Figure 5, however a few points are still unclear. In Panels C and D, the first bar seems to suggest that even the control group experienced increases in the number of officers. Could the authors clarify why this is the case, and how large these increases were relative to baseline levels? In addition, Panel E and F show the effects on crime perceptions and reported crime for municipalities with small versus large increases in police forces. The point estimates differ, and the overlapping confidence intervals may simply reflect low statistical power (given a sample size of $N=200/300$). Thus, the conclusion that “resource increases alone were not sufficient to improve police effectiveness” seems premature. Perhaps more importantly, this would contrast with much of the existing literature on police strength and crime, and it would be helpful if the authors clarified how they interpret their findings in light of this.

Response to R1.3:

Thanks for the opportunity to clarify this point. Increasingly concerned about growing crime and crime perception, successive Chilean governments substantially increased police resources during the 2000s and 2010s.²⁸ As a result, both treatment and control municipalities experienced substantial increases in resources during the study period, which explains why the control group also shows increases in the number of officers in Figure 4. For example, during this period, the average yearly increase in the number of officers in a never treated or not yet treated municipality was 9.7%, and that of patrol vehicles was 6%.

A key premise for the successful territorial reorganization introduced by *Plan Cuadrante* was that municipalities had sufficient resources to staff the new quadrants. With this goal in mind, some adopting municipalities received significant increases in officers and patrol vehicles relative to the control group, though budget constraints meant that the target of closing pre-existing deficits—measured in Equivalent Surveillance Units (UEs)—was rarely fully achieved, and the size of the deficit does not appear to have been the main criterion determining the increase a municipality received (Winchester et al., 2007; Salas P. et al., 2014).²⁹ The additional officers were predominantly newly hired rather than reallocated from control municipalities, as transfers across Carabineros units are centrally regulated and subject to minimum permanence periods of up to ten years (Salas P. et al., 2014). As shown in Figure 4, the resource increase was nonetheless highly uneven: while municipalities in the bottom 40% of the resource-increase distribution upon joining *Plan Cuadrante* received increases in resources very similar to those observed in never-treated or not-yet-treated municipalities, municipalities in the top 40% experienced substantially larger increases, with officers rising by around 22% per year and patrol vehicles by around 80% per year. Reassuringly, municipalities receiving larger versus smaller resource increases—whether the split is based on the increase in police officers or in patrol vehicles—do not display differential pre-treatment trends in

²⁸Between 2004 and 2006, the Lagos government incorporated 2,970 new Carabineros. The Bachelet government (2006–2010) added a further 6,000 officers, and the Piñera government (2010–2014) committed to 10,000 additional officers, a target achieved through a combination of new hires and the reallocation of existing officers from administrative to operational duties.

²⁹We requested access to information on the demand, supply, and deficit of UEs from Carabineros, but this information was not shared.

either the outcomes of interest or in police resources themselves, which alleviates concerns that this split might itself be confounded by pre-existing resource or crime trends across municipalities.³⁰

Regarding panels (c) and (d) of Figure 4, we agree that the overlapping confidence intervals may partly reflect limited statistical power, and we have been careful not to overstate our conclusions on this point in the revised manuscript. We do not argue that resources play no role: they likely enable the successful implementation of the program, and for some outcomes such as crime perception the effects may indeed be larger in municipalities with greater resource increases. In the absence of a natural experiment, we cannot estimate the causal effect of resources in isolation in the context of *Plan Cuadrante*.³¹ Our argument is different: the beat policing component of *Plan Cuadrante*—the division of the municipal territory into quadrants with each officer in the municipality permanently assigned to one quadrant rather than rotating across the full municipality—plays a key role in explaining the results, over and above the resource channel. This is supported by the fact that municipalities with no resource increases relative to the control group (municipalities in the bottom 40% of the resource-increase distribution) still display sizeable and statistically significant effects for most outcomes. For some outcomes such as victimization or household investments in prevention measures, the magnitude is very similar to the effect of the program in municipalities that receive large resource increases. While for crime perception the effect is smaller, it is also significant at conventional confidence levels. On the other hand, the effect on reported crime has the expected sign and is sizeable, although statistically insignificant, which might be attributable to lower statistical power in this analysis, since unlike the survey-based outcomes it is conducted at the municipality rather than at the household level. In line with the relevance of the beat policing component, we also show in the updated manuscript that increases in police resources unrelated to *Plan Cuadrante* have limited effects on the same outcomes as discussed below.

We also want to clarify that we do not view our results as being in tension with the existing literature on police strength and crime. As Chalfin (2025) notes in his review of the literature, the effectiveness of additional police resources depends crucially on how they are organized and deployed, suggesting that organizational reforms of the kind studied here can have important effects on public safety independently of changes in resource levels.

In the revised manuscript we have substantially expanded Section 2 to provide further details on the components of *Plan Cuadrante*:

Carabineros, Chile’s national police force, was founded in 1927. Through most of our study period, Carabineros operated under the Ministry of Defense, which oversaw both military and police functions until domestic security responsibilities were transferred to the Ministry of Interior in 2011.³²

Carabineros’ mission, functions, and structure are established by Law 18.961 and the Regulation on Carabineros’ Organization (Reglamento de Carabineros) (Biblioteca del Congreso Nacional de Chile, 1990; Ministerio de Defensa Nacional, República de Chile, 1989). Under

³⁰The p-values for the joint test of equal pre-trends between the two groups are, for the officers-based split and the patrol-vehicles-based split respectively: 0.823 and 0.048 for victimization, 0.059 and 0.379 for crime perception, 0.820 and 0.157 for investment in security measures, 0.213 and 0.419 for reported crime, 0.616 and 0.444 for police officers, and 0.405 and 0.157 for patrol vehicles.

³¹We requested information on the supply and demand of UVEs, which might provide a proxy for the planned increase in resources at the time of program adoption, but this information was not shared.

³²Currently, Carabineros reports to the Ministry of Public Security, created in 2024 to focus exclusively on domestic security.

this framework, Carabineros is a professional, disciplined, and hierarchical police institution whose mission is to preserve public order and security with a central emphasis on preventive policing. The General Director heads the strategic level, responsible for defining doctrine and institutional strategy, while the Subdirector coordinates implementation. The Directorate of Order and Security administers police functions at the tactical level.

Territorially, Carabineros is organized into zones that oversee *prefecturas*. Zones and *prefecturas* are primarily management layers. Zones map roughly one-to-one to Chile's regional divisions, with two exceptions: the Metropolitan Region of Santiago—the most populated in the country—and La Araucanía—which has recently experienced significant social conflict—are each divided into two zones. Each zone is subdivided into *prefecturas*—54 nationwide—which supervise 227 *comisarías*, the main operational units at the tactical level. These *comisarías* oversee smaller units—*subcomisarías*, *tenencias*, *retenes*, *puestos*, and *avanzadas*—that operate within their assigned jurisdictions.

In practice, the *comisaría* is the basic unit of routine frontline policing and, in most municipalities, serves as the territorial anchor of police work. Officers conduct patrol, surveillance, prevention, and first response at this level. In most municipalities, a single *comisaría* covers the entire municipal territory, though larger municipalities may have multiple ones. Santiago, for instance, has four. All municipalities that adopted *Plan Cuadrante* during our study period had a *comisaría*. With the exception of 9 municipalities, each had a single *comisaría* covering the full municipal territory. On average, a *comisaría* in our sample covered 1,608 square kilometers and served 57,642 inhabitants.

Plan Cuadrante was introduced within this institutional framework without altering Carabineros' hierarchical structure or its strategic and tactical levels. The reform emerged in a context of rising victimization and insecurity, with the explicit goal of reversing these trends (Ministerio del Interior y Seguridad Pública, 2014). Its central innovation was a shift toward a beat policing model. In the criminology literature, beat policing refers to assigning officers to fixed territorial areas so they can accumulate local knowledge, strengthen ties with residents, and respond more effectively to local crime patterns (Cordner, 1995; Skogan and Hartnett, 1997). In practice, the catchment area of police forces in municipalities adopting *Plan Cuadrante* was partitioned into smaller, fixed geographic units (quadrants), to which officers were permanently assigned. Rather than building new infrastructure, the reform redefined the territorial responsibility of officers in already existing police stations (Winchester et al., 2007).³³ This represented a clear break from the pre-reform system, in which officers rotated across broad operational territories, very often spanning entire municipalities.

Carabineros began piloting *Plan Cuadrante* in 1998. The program expanded gradually through 2013, as shown in Panel (a) of Figure 1, reaching 150 urban municipalities and covering roughly 90% of Chile's population.³⁴ Online Appendix A provides further details about the rollout of the program. The rollout was designed to follow a municipality ranking based on a weighted index of victimization, police resource deficits, drug prevalence, unemployment, and demand for police services, with annual entry determined by available budget.³⁵ In practice, however,

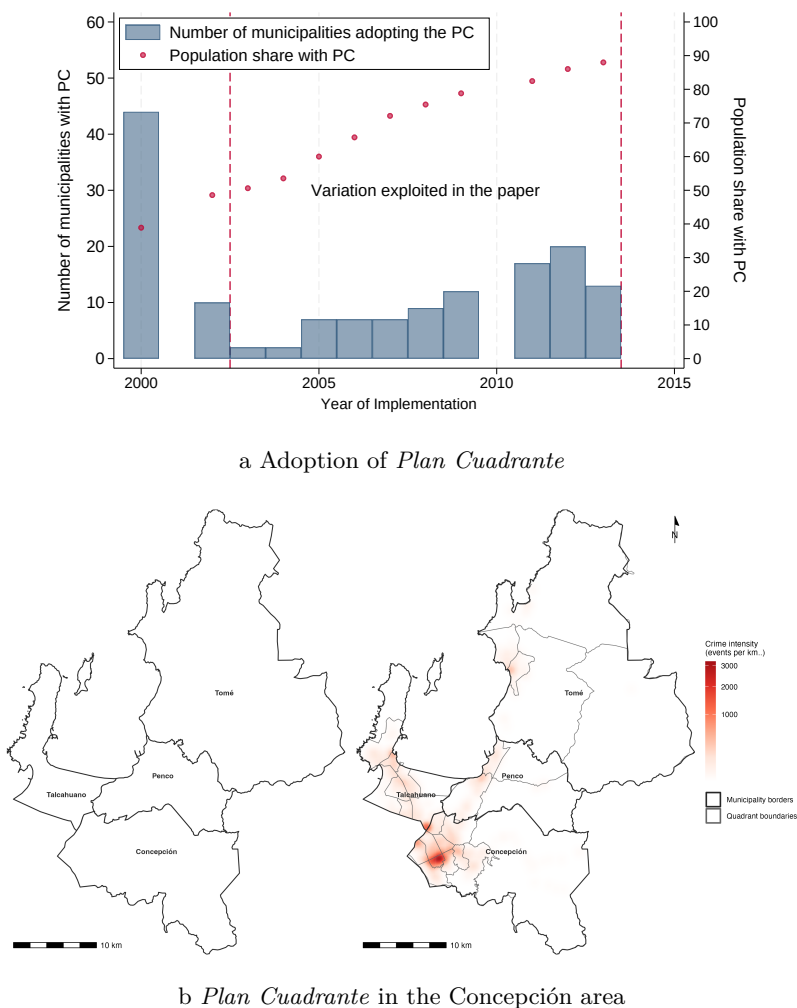
³³In 2016, Carabineros launched the *Plan Cuadrante 2.0* program in those municipalities served by *Plan Cuadrante*. While this reform introduced additional reforms aimed at integrating the community into the provision of security and building capacity to gather and analyze crime data in police stations, this paper's study period ends before these improvements were introduced.

³⁴Further details about the expansion of *Plan Cuadrante* are provided in Online Appendix A and Table A.1.

³⁵The weights assigned to each component changed on a yearly basis (Winchester et al., 2007; Salas P. et al., 2014).

Budget Office evaluations show that adoption diverged from this ranking for undocumented reasons, so the rule operated more as a loose guideline than as a binding assignment mechanism (Salas P. et al., 2014).

Figure 1: Implementation of *Plan Cuadrante*



Notes: Panel (a) shows the number of municipalities adopting the *Plan Cuadrante* each year (bars), while the circles above them illustrate the cumulative share of the Chilean population living in a municipality with the *Plan Cuadrante* already in place. Panel (b) depicts the metropolitan area of Concepción, the second-largest in Chile: the left map shows its four municipalities—the original police catchment areas—and the right map overlays the Carabineros quadrants together with a kernel-density estimate of all types of robberies and thefts recorded by the police (both reported crimes and arrests). It illustrates how *Plan Cuadrante* subdivided each catchment area (which corresponds to the territory of a municipality) into smaller quadrants, as well as the spatial concentration of crime within them.

Conceptually, police effectiveness can be viewed as a function of three inputs: total resources, territorial deployment, and policing strategy. Although in some municipalities police resources were increased significantly to operationalize the beat policing model (Winchester et al., 2007; Salas P. et al., 2014), the program primarily transformed the latter two elements. The creation of quadrants was accompanied by permanent geographic assignment of officers to specific areas. This change was operationally critical. By making officers responsible for their quadrant, it encouraged accumulation of local knowledge and stronger community ties, allowing them to adapt patrols and prevention activities to local crime patterns. Importantly, this was not driven by program-specific operational directive changes. Throughout the period we study, a single national set of operational guidelines applied to officers in both treated and control municipalities (Carabineros de Chile, 2018; Winchester et al., 2007). Thus, while the average effects we document on crime and crime perception likely reflect combined changes in the three elements we discuss—i.e., resources, deployment, and strategy—the deployment change itself was essential to operationalizing the reform. We discuss the drivers of the effects we document in Section 4.4.

Quadrant size and shape were determined by local police authorities based on factors such as road networks, geographic features, and available resources. In principle, resource allocation across quadrants followed demand for police services, measured in Equivalent Surveillance Units (UEVs); in practice, the station commander retained discretionary authority, as was also the case prior to *Plan Cuadrante* (Salas P. et al., 2014). Municipalities implementing *Plan Cuadrante* were subdivided into an average of seven quadrants, each typically staffed by approximately three officers per shift. On average, each quadrant covered 185 square kilometers and served 13,466 inhabitants. Panel (b) of Figure 1 illustrates this structure, showing the 29 quadrants created across the four municipalities of the Concepción metropolitan region—Chile’s second-largest metropolitan area and the largest in our sample. Quadrants represented approximately 14% of the original *comisaría* territory. Officers continued to operate from the same police station but were permanently assigned to a specific sub-area within its jurisdiction. While Carabineros’ hierarchical structure limits frontline officer decision-making authority, each quadrant was assigned a Quadrant Delegate (*Delegado de Cuadrante*), a non-commissioned officer, responsible for immediate operational decisions within the quadrant.

Plan Cuadrante may affect public safety through two complementary channels: deterrence and police effectiveness, understood as the capacity to respond to and resolve crimes. The changes induced by *Plan Cuadrante*—permanently assigning officers to well-defined neighborhoods and, in some municipalities, increasing available resources—can increase both police visibility and local knowledge. As officers repeatedly patrol the same area, they become familiar with its streets, residents, and routines, which can raise the perceived probability of identification and apprehension and reduce the expected returns to crime. At the same time, this deeper knowledge of local territory and communities can improve police response and case resolution once crimes occur. The reform therefore likely affects both dimensions. Consistent with this interpretation, in Section 4 we document declines in victimization and in reported crime—patterns consistent with deterrence—together with increases in arrests, consistent with improved police effectiveness.

Although beat policing shares features with other policing approaches, such as hot-spot policing and community-based policing, which have been widely studied, it has distinctive features that make it worth studying as an independent approach. Hot-spot policing concentrates resources in high-crime areas and does not focus on building local knowledge and community

ties. It is typically implemented through targeted interventions at specific locations rather than systematic territorial organization, and the same officers are not necessarily sent repeatedly to the same areas. Evidence on hot-spot policing is mixed, with some studies showing crime reductions in targeted areas but also potential displacement effects (Braga et al., 2019). *Plan Cuadrante*, by contrast, subdivided all urban municipalities into quadrants with permanent officer assignment. As Panel (b) of Figure 1 illustrates, some quadrants contained crime hot-spots, but many did not. The reform thus addressed entire urban territories rather than specific high-crime areas, potentially mitigating displacement effects associated with hot-spot policing.

Community-based policing emphasizes building relationships with residents and addressing underlying social issues, and typically involves active community participation in defining strategy and priorities—in some cases, even assuming surveillance and prevention responsibilities. While increasing trust in the police and strengthening community ties were among the declared goals of *Plan Cuadrante*, and may have resulted from its territorial reorganization, the community was not formally involved in the provision of security.³⁶

In recent decades, police departments worldwide have transitioned toward beat policing reforms (National Academies of Sciences, Engineering, and Medicine, 2018; Ministerio del Interior y Seguridad Pública, 2014), yet the effectiveness of this model has not been rigorously assessed. Although the beat policing approach introduced through *Plan Cuadrante* shares features with programs previously studied in the economic literature, its distinctive characteristics—comprehensive territorial reorganization and systematic local officer specialization—make it a valuable case for studying how deployment and policing strategies not yet examined in the economics literature affect crime and crime perception.

We have also expanded the analysis of the role of the resource increase component in the Mechanism section and in the appendix:

To reduce crime and improve security perceptions, police forces need to strengthen both prevention and police effectiveness. Both margins can be shaped by resources, territorial deployment, and policing strategy. This section examines which components of *Plan Cuadrante* are most likely to drive the effects documented in Section 4.

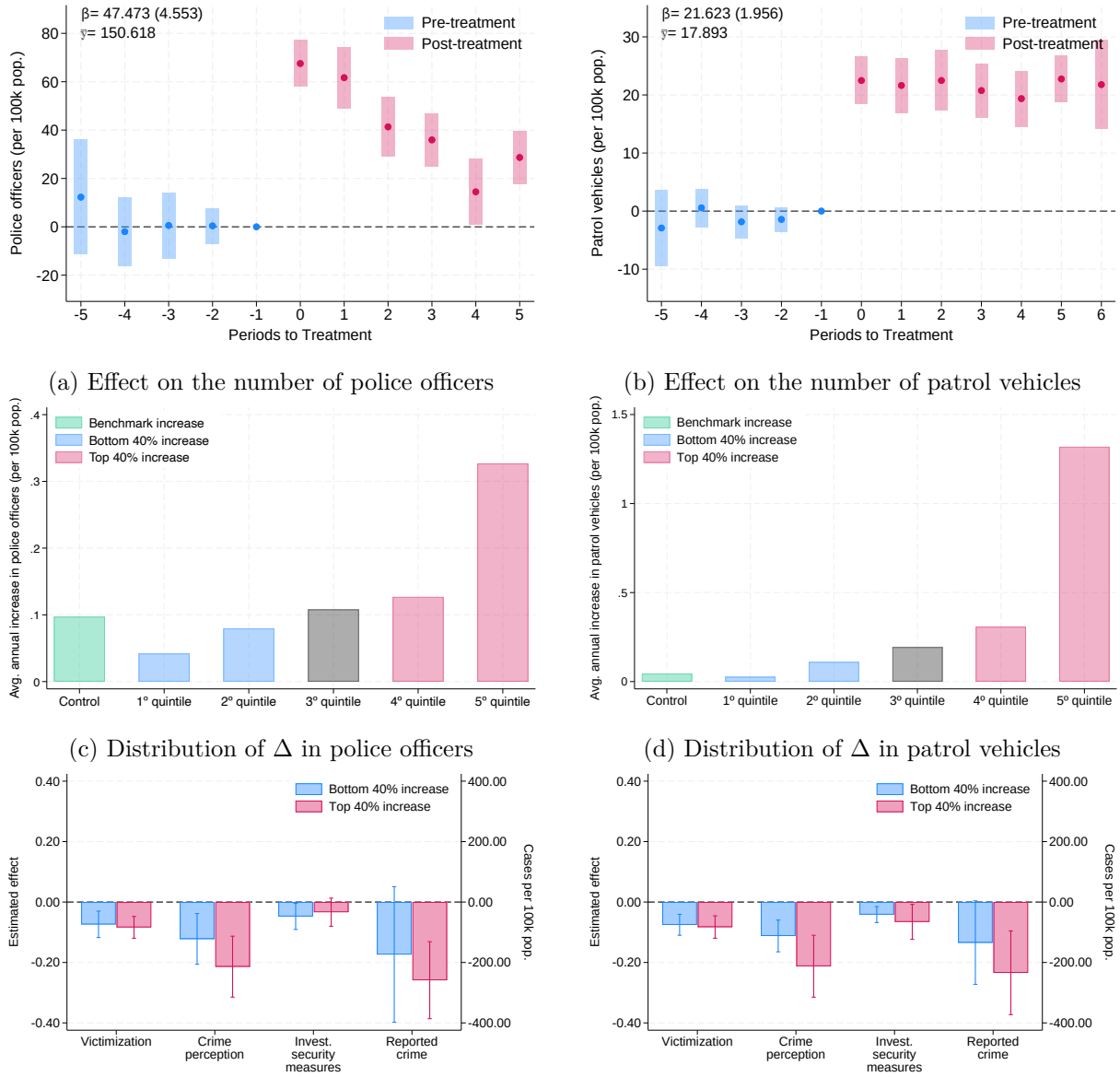
The central objective of *Plan Cuadrante* was to introduce a beat policing model by reorganizing deployment. Police operational territories that had typically matched municipal boundaries were subdivided into quadrants, and officers were permanently assigned to these quadrants. During our study period, national operational guidelines remained the same in adopting and non-adopting municipalities. Still, this new deployment likely affected policing strategy at a more operational level, because it allowed officers to adapt patrol routes, schedules, and prevention practices using the local knowledge that the program sought to build.

Although changes in resources were not the core of the reform, implementation contemplated increases in municipalities that lacked the personnel and patrol capacity required to move to this model. During the period we study, Chile was expanding public spending on security, and

³⁶The *Plan Cuadrante*’s original design contemplated a formal community integration model (MICC) with a Community Affairs Office (*Oficina de Asuntos Comunitarios*) and a dedicated officer in each police station. However, this model was not implemented during our study period. Independent program evaluations from the Budget Office document that these Community Affairs Offices did not exist and had no dedicated personnel, budget, or structured activities in the period we study (Winchester et al., 2007; Salas P. et al., 2014). The community integration model was piloted in four police stations during 2012–2013, none of which are in our analytical sample. Financing was then discontinued in early 2014 and only restored and expanded after our study period ends.

police resources increased in most municipalities regardless of whether or when they adopted *Plan Cuadrante*. On average, however, municipalities adopting the program experienced larger increases in officers and patrol vehicles, as shown in panels (a) and (b) of Figure 4. Since increases in police resources have been shown to reduce crime in some contexts (Evans and Owens, 2007), this pattern raises a central question for interpretation, namely whether our estimates reflect resource expansion or the deployment and strategy changes introduced by *Plan Cuadrante*. We address this question through a set of exercises that exploit heterogeneity in resource increases across municipalities and over time.

Figure 4: Plan Cuadrante and Changes in Police Resources



(e) Effects by the intensity of Δ in police officers (f) Effects by the intensity of Δ in patrol vehicles

Notes: Panels (a) and (b) show the effect of *Plan Cuadrante* on police officers and patrol vehicles per 100,000 inhabitants. Panels (c) and (d) show the distribution of annual changes in officers and patrol vehicles across municipalities. Panels (e) and (f) report effects by the intensity of resource changes, comparing municipalities in the bottom 40% and top 40% of the distribution. The estimation follows Callaway and Sant'Anna (2021). The graphs display estimated coefficients and 95% confidence intervals. Standard errors are clustered at the municipality level using wild bootstrapping.

Panels (c) and (d) of Figure 4 show that the average increase in police resources masks substantial heterogeneity. Municipalities in the bottom 40% of the resource-change distribution (blue bars) experienced increases close to the average annual variation observed among never treated and not-yet-treated municipalities (green bar). By contrast, municipalities in the top 40% (red bars) experienced much larger increases, with officers rising by around 22% and patrol vehicles

by around 80%. If the effects of *Plan Cuadrante* were driven by resource increases alone, we would not expect effects in municipalities whose resource increases were similar to those of the control group. Yet panels (e) and (f) of Figure 4 show sizeable effects for most crime and crime-perception outcomes even in that low-increase group. While there may be a resource gradient for some crime-perception outcomes, the effects on twelve-month victimization and private investment in security among municipalities with control-like resource increases are close in magnitude to those in municipalities with larger increases. Reassuringly, municipalities in the top and bottom 40% of the resource-increase distribution do not display differential pre-treatment trends in outcomes or police resources, which suggests that this heterogeneity is not driven by differential pre-existing trends.³⁷

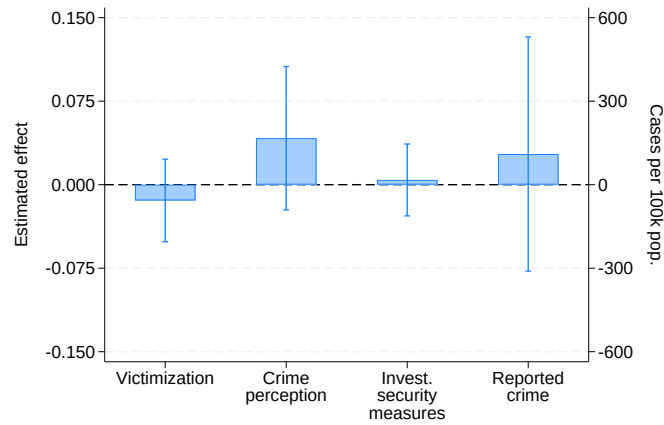
Although we cannot fully separate the beat policing component from the resource component of the reform—resources may have been important to enable effective implementation of the program, or may explain part of the effects in some municipalities—the fact that municipalities with resource increases similar to those experienced by control municipalities still display sizeable effects indicates that the effects of *Plan Cuadrante* are not driven entirely by resources.³⁸³⁹

³⁷We test for the joint significance of the pre-treatment coefficients between the two groups, separately for the split based on the increase in police officers and the split based on the increase in patrol vehicles. The resulting p-values are, respectively, 0.823 and 0.048 for victimization, 0.059 and 0.379 for crime perception, 0.820 and 0.157 for investment in security measures, and 0.213 and 0.419 for reported crime, as well as 0.616 and 0.444 for police officers and 0.405 and 0.157 for patrol vehicles.

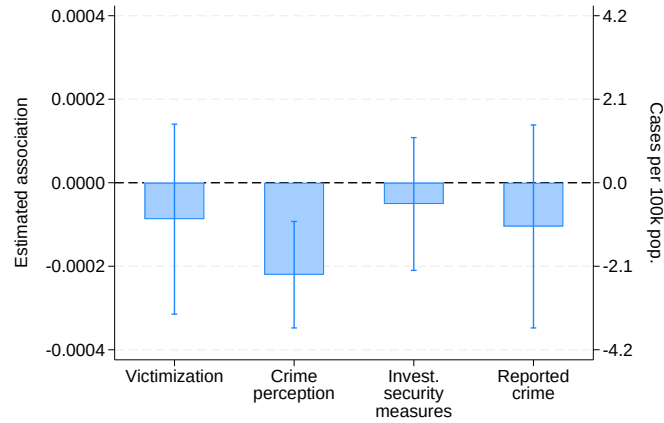
³⁸We also re-estimate the results in Figure 2 but controlling for police resources. Online Appendix Table B.3 shows that the magnitude of the estimated effects remains similar for most variables, although the statistical significance is reduced in some estimations. This likely reflects a mechanical feature of the Callaway and Sant’Anna (2021) estimation procedure: control variables are not incorporated parametrically, but are instead used to construct the comparison group via outcome regression or propensity-score weighting, so the resulting estimation uncertainty propagates to the ATT standard errors. Note, in addition, that police resources are a bad control—i.e., at least part of their variation is a result of the adoption of *Plan Cuadrante*—so these results should be interpreted cautiously.

³⁹Table B.4 in Online Appendix B examines which factors are associated with a larger increase in police resources upon the adoption of *Plan Cuadrante*.

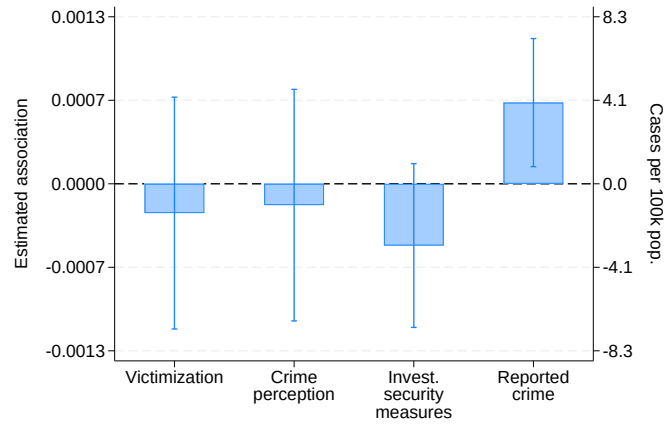
Figure 5: Police Resources and Public Safety



a Effects of *Seguridad Ciudadana*



b Association between police officers and effectiveness



c Association between patrol vehicles and effectiveness

Notes: Panel (a) reports the pooled effect of *Seguridad Ciudadana*, a program that expanded local security personnel to support patrolling and deterrence but did not replicate the deployment-and-strategy features of *Plan Cuadrante*. Panels (b) and (c) report within-municipality associations between police resources and police-effectiveness outcomes among the 41 municipalities of the Santiago Metropolitan Region, all of which had adopted *Plan Cuadrante* by 2000. The regressions include municipality and year fixed effects. Standard errors are clustered at the municipality level using the wild bootstrapping procedure.

To further investigate whether resource increases alone can explain our results, we assess the effects of *Seguridad Ciudadana*, a program that expanded local security personnel to support patrolling and deterrence, but did not replicate the core deployment-and-strategy features of *Plan Cuadrante*. Using a difference-in-differences specification following Callaway and Sant’Anna (2021), panel (a) of Figure 5 shows no significant effects of these resource changes on police effectiveness. Further details about this exercise and the corresponding event-study estimates are reported in Online Appendix B and Online Appendix Figure B.4.

Finally, we study whether variation in police resources in municipalities that had already adopted *Plan Cuadrante* improves public safety and security perceptions. Using ENUSC data and information on resources for 2006–2012 in the 41 municipalities of the Santiago Metropolitan Region—all of which had adopted *Plan Cuadrante* by 2000—we exploit variation in officers and patrol vehicles that is unrelated to program adoption. Panels (b) and (c) of Figure 5 show the association between police resources and our main outcomes, conditional on municipality and year fixed effects. With one exception—the association between police officers and crime perception—none of these associations is statistically significant at conventional confidence levels.⁴⁰

Taken together, these results suggest that resource increases alone do not account for the effects of *Plan Cuadrante*. We next turn to evidence that speaks more directly to the beat-policing components of the reform.

Panel (a) of Figure 4 shows that, following the adoption of *Plan Cuadrante*, households are much more likely to report increased police presence in their neighborhoods. The share of households perceiving increased police presence rises by 35 percentage points (156%), the share reporting that police pass by their house daily rises by 22 percentage points (59%), the share reporting never seeing police patrols near their house falls by 10 percentage points (57%), and complaints about lack of police surveillance fall by around 5 percentage points (20%). This evidence is consistent with both larger resources and the new deployment model. Yet the latter estimate—the only police-presence outcome for which we can separately estimate effects in municipalities with small and large resource increases—is similar in magnitude across the two groups, which suggests that the increase in police presence is not explained only by larger resource increases (see Online Appendix Figure B.3).⁴¹ More importantly, the broader pattern of results suggests that *Plan Cuadrante* did more than simply increase patrolling intensity. Existing work shows that police presence can reduce crime while officers are present, but that its overall effects may be attenuated when crime is displaced across places or time windows (Mastrobuoni, 2019; Blanes-i Vidal and Mastrobuoni, 2018). Our results point to something broader. As discussed in Section 4, we find actual declines in victimization and reported crime, together with no evidence of spillovers to neighboring municipalities. This pattern is more consistent with officers accumulating local knowledge and adapting prevention strategies to local conditions under stable territorial assignment than with a purely mechanical increase in patrol intensity.

Panel (b) of Figure 4 reinforces this interpretation. Households in early-adopting municipalities

⁴⁰Online Appendix Figure B.5 reports the same exercise and also shows similar results for the broader set of municipalities that adopted *Plan Cuadrante* before 2006 and never-treated municipalities (red bars); see Online Appendix B for further details.

⁴¹We cannot estimate heterogeneous effects for other police-presence outcomes, such as the reporting of police passing by or the share of households perceiving an increase in police presence, because those variables are available only in ENUSC rounds 2003, 2005, and 2006, while resource data start in 2006.

report better police performance across multiple dimensions. They are 8.1 percentage points (16%) more likely to rate police performance as good or very good, 7.6 percentage points (11%) more likely to describe police as efficient, 5 percentage points (6%) more likely to report that police are helpful, and 4.9 percentage points (7%) more likely to report that police are well informed. They are also 3.9 percentage points (10%) less likely to perceive police as corrupt, 7.9 percentage points (21%) less likely to believe police fail to control crime, and 7.6 percentage points (22%) less likely to report that police do not perform their duties effectively.⁴²

The administrative evidence points in the same direction. Panel (c) of Figure 4 shows that arrests per 100,000 inhabitants increase by 48 (14%) in our main estimates, even as crime declines overall.⁴³ This effect is also significant among municipalities with resource increases similar to those in the control group (see Online Appendix Figure B.3). Consistent with this pattern, panel (d) of Figure 4 shows that the arrest rate per reported crime also increases by 1.7 percentage points (10%), suggesting that a higher share of crimes reported to the police are being solved. Because these gains appear even where resources did not increase more than in control municipalities, they are difficult to reconcile with a pure resource story and are more naturally interpreted as improvements in police effectiveness under the deployment and strategy changes introduced by *Plan Cuadrante*. The corresponding event-study estimates for arrests, the arrest rate per reported crime, and trust in the police in Figure 4 confirm that these changes emerge only after adoption.

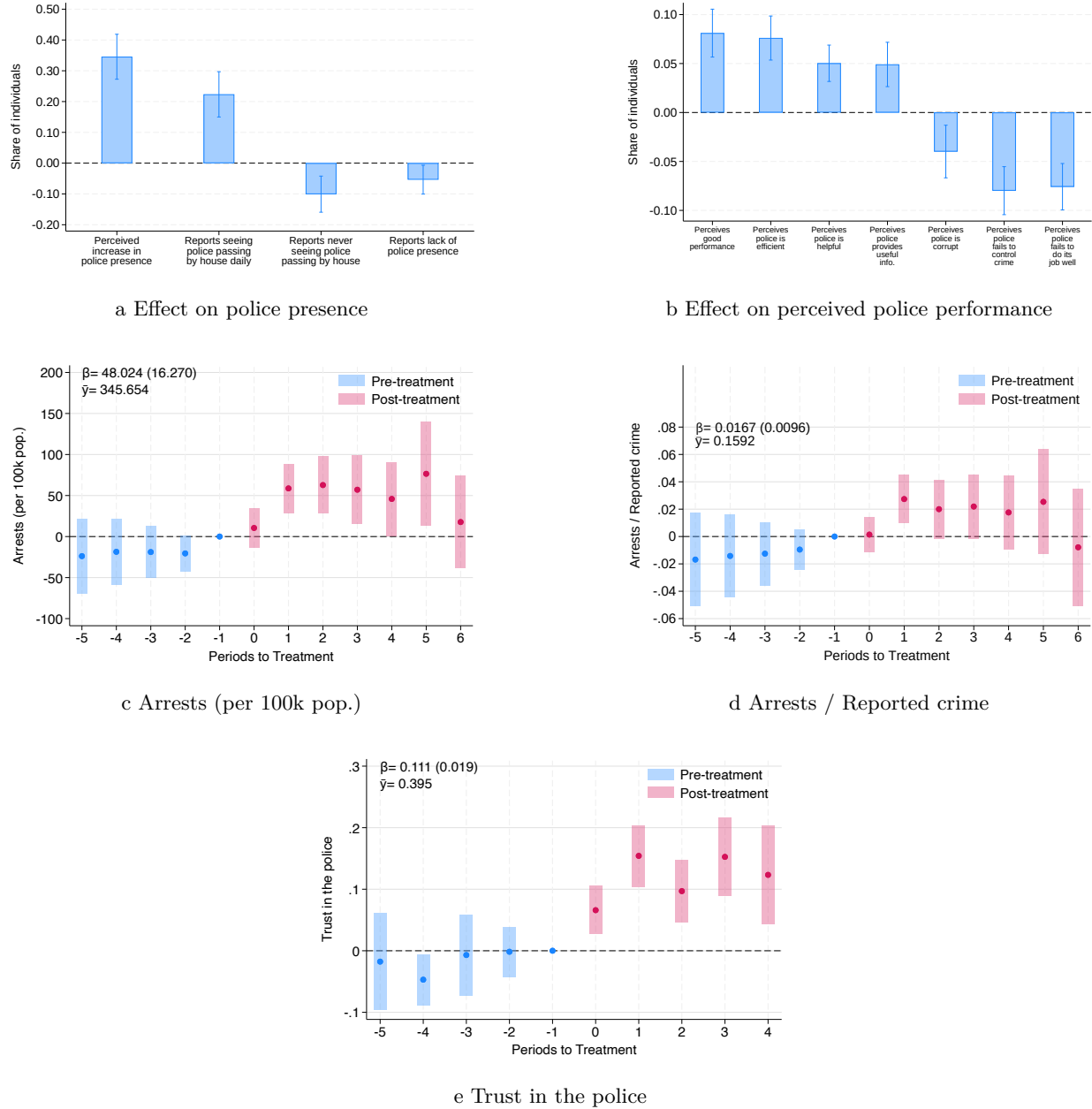
Finally, panel (e) of Figure 4 shows a sharp increase in trust in the police in our main specification (Section 4), consistent with one of the objectives of beat policing, namely bringing officers closer to the community and allowing them to accumulate local knowledge through repeated interaction. Online Appendix Figure B.3 shows that this effect remains sizeable both in municipalities that experienced resource increases similar to the control group and in those with larger increases.

Overall, although resources were likely important for implementation and may explain part of the effects in some municipalities, the results in this section indicate that resource changes alone are not enough to explain our findings. The evidence is more consistent with a combination of deployment changes and the strategy adjustments enabled by local knowledge accumulation under permanent assignment to quadrants.

⁴²Information on police performance is only available in ENUSC round 2005. Using this round, we compare perceived police performance in municipalities that adopted *Plan Cuadrante* in 2005 or earlier and in municipalities adopting after 2005. Figure B.6 in Online Appendix B shows that the number of years of exposure to *Plan Cuadrante* is positively correlated with perceived police performance. While these exercises on perceived police performance should be interpreted with caution, we show in the same appendix that the timing of future adoption is not correlated with perceived police performance in 2005 for municipalities that had not yet adopted the program by 2005.

⁴³As with reported crime, this event study is restricted to 5 pre-treatment and 7 post-treatment periods for symmetry with the ENUSC-based analysis; Online Appendix Figure C.8 reports the results for arrests using the full set of pre-treatment and post-treatment periods.

Figure 6: *Plan Cuadrante* and Changes in Police Performance



Notes: Panel (a) reports the effects on perceived police presence. Panel (b) reports perceived police performance in 2005, comparing municipalities where *Plan Cuadrante* was already implemented before 2006 with municipalities where it was implemented after 2005. Panels (c), (d), and (e) report dynamic event-study estimates for arrests per 100,000 inhabitants, the ratio of arrests to reported crime, and trust in the police. The estimation follows Callaway and Sant'Anna (2021). The graphs display estimated coefficients and 95% confidence intervals. Standard errors are clustered at the municipality level using the wild bootstrapping procedure.

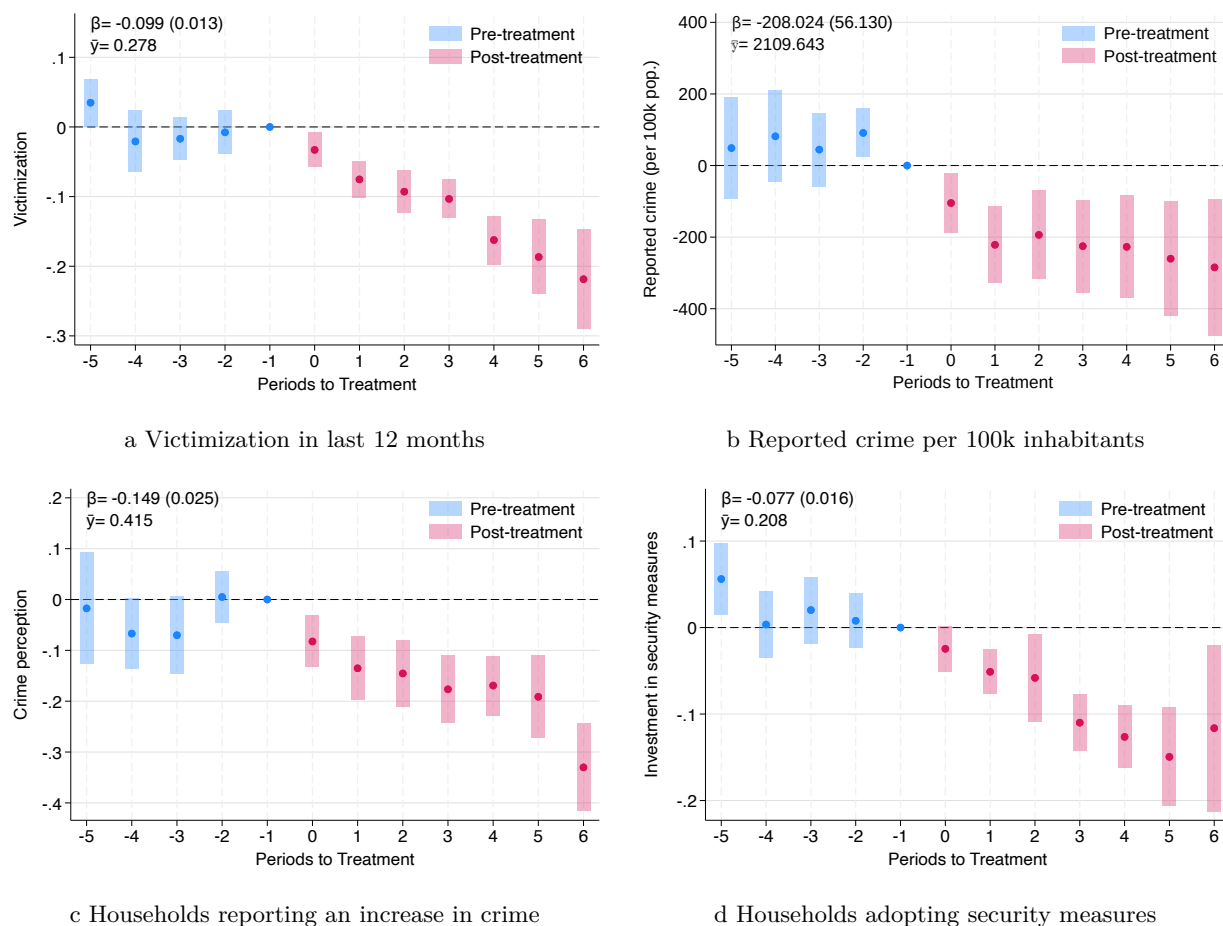
These additional exercises are reported in the “Effects of *Plan Cuadrante* and Changes in Police Resources” and “Police Resources and Police Effectiveness” subsections of Online Appendix B.

Finally, we have also rewritten the relevant parts of the paper to ensure a more precise interpretation

of the results, making clear throughout that we do not claim resources play no role, but rather that the beat policing component played a pivotal role in driving the observed effects. This precise framing now appears, among other places, in the Introduction (“these findings show that while police resources might be important in enabling the implementation of the beat policing model, or might explain part of the effects in some municipalities, the beat policing component of the reform...was a key driver of its effectiveness”), in Section 4.4 (“the fact that municipalities with negligible resource increases relative to the control group still display sizeable effects rules out the hypothesis that the effects of *Plan Cuadrante* are driven entirely by resources”), and in the Conclusion.

Other figures and tables referenced above:

Figure 2: Effects of *Plan Cuadrante* on Crime Outcomes and Security Perceptions



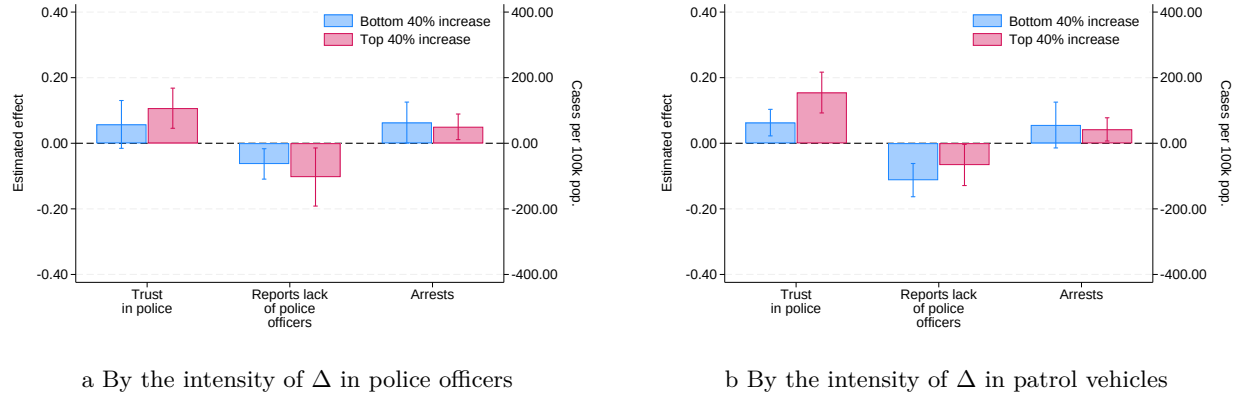
Notes: This figure shows the dynamic effects of the *Plan Cuadrante* on different measures of police effectiveness. The estimation is conducted using the procedure developed in Callaway and Sant'Anna (2021) for difference-in-differences with staggered treatment adoption. The dots in the figure represent the estimated coefficients and the bars behind them 95% confidence intervals. Standard errors are clustered at the municipality level using the wild bootstrapping procedure. The pooled effect and the mean outcome before the introduction of the treatment is also presented in each panel. Panel (a) reports the effect of *Plan Cuadrante* on the probability of household victimization in the last 12 months using the ENUSC survey. Panel (b) reports the effect on the number of crimes reported to the police in the municipality per 100,000 inhabitants using administrative data. Panel (c) reports the effect on the probability of perceiving an increase in crime using the ENUSC survey. Panel (d) reports the effect on the probability of adopting security measures in the last twelve months using the ENUSC survey. Panel (a) includes 93,041 observations; panel (b), 4,231; panel (c), 89,578; and panel (d), 93,041 observations.

Table A.1: Implementation of *Plan Cuadrante* by Year

Year	N	Municipalities
1998–2000	44	Metropolitan Region of Santiago (44 municipalities)
2002	10	Chiguayante, Concepción, Hualpén, Quilpué, San Antonio, San Pedro de la Paz, Talcahuano, Valparaíso, Villa Alemana, Viña del Mar
2003	2	Padre Las Casas, Temuco
2004	2	Antofagasta, Copiapó
2005	7	Alto Hospicio, Arica, Calama, Curicó, Iquique, Linares, Talca
2006	7	Coquimbo, La Serena, Los Andes, Ovalle, Rancagua, San Felipe, San Fernando
2007	7	Chillán, Chillán Viejo, Los Ángeles, Osorno, Puerto Montt, Punta Arenas, Valdivia
2008	9	Ancud, Castro, Constitución, Coronel, Lota, Peñaflor, Quillota, Rengo, Villarrica
2009	12	Angol, Aysén, Calera, Concón, Coyhaique, La Unión, Padre Hurtado, Parral, Penco, San Carlos, Tomé, Vallenar
2011	17	Arauco, Cañete, Cartagena, Cauquenes, Curanilahue, La Ligua, Limache, Molina, Nueva Imperial, Pucón, Puerto Varas, Río Bueno, San Clemente, San Javier, San Vicente, Santa Cruz, Victoria
2012	20	Cabrero, Calbuco, Casablanca, Chimbarongo, Collipulli, Curacaví, El Monte, Graneros, Illapel, Isla de Maipo, Lautaro, Machalí, Mostazal, Mulchén, Nacimiento, Natales, Panguipulli, Quellón, Quintero, Tocopilla
2013	13	Bulnes, Laja, Las Cabras, Lebu, Llaillay, Loncoche, Longaví, Los Álamos, Maule, Requínoa, Teno, Vicuña, Yumbel
Total	150	

Note: N is the number of municipalities that adopted *Plan Cuadrante* in each year. Data shared by Carabineros de Chile.

Figure B.3: Effects of *Plan Cuadrante* by Resource Increase: Trust, Complaints, and Arrests



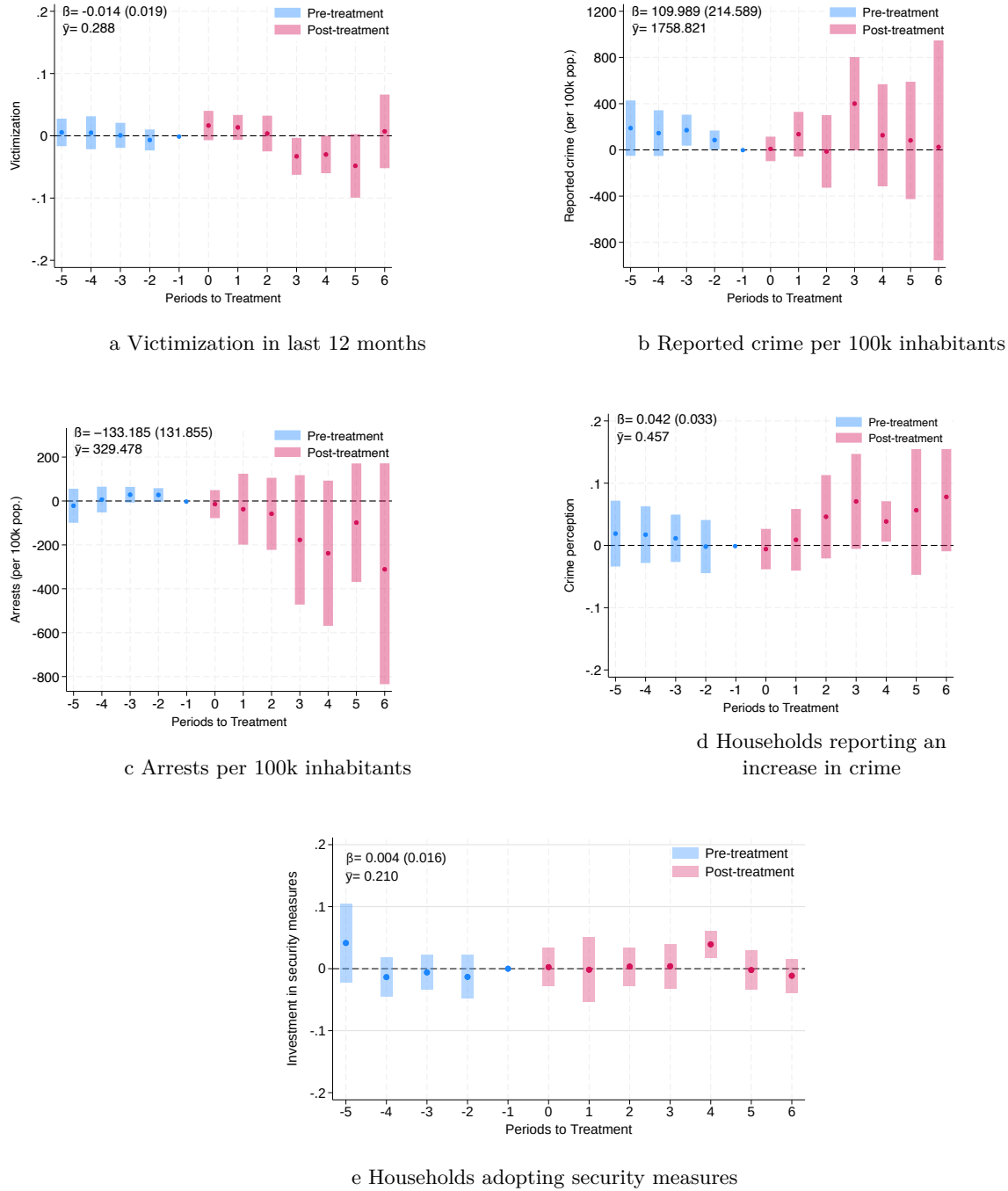
Notes: This figure complements Figure 4 by reporting, for the three outcomes not shown there, the effect of *Plan Cuadrante* estimated separately for municipalities in the bottom 40% (blue) and top 40% (red) of the resource-increase distribution. Panel (a) splits municipalities by the increase in police officers and panel (b) by the increase in patrol vehicles. In each panel, arrests (administrative data, in cases per 100,000 inhabitants) are read on the right axis; the other two outcomes, from the ENUSC survey (trust in the police, and reports about the lack of police officers), are read on the left axis. Effects are estimated using the difference-in-differences estimator for staggered treatment adoption developed in Callaway and Sant'Anna (2021); bars are point estimates and whiskers 95% confidence intervals, with standard errors clustered at the municipality level using the wild bootstrapping procedure.

Table B.3: Effects of *Plan Cuadrante* Controlling for Police Resources

	(1)	(2)	(3)	(4)	(5)
<i>Panel A. Victimization</i>					
ATT	-0.099*** (0.013)	-0.077*** (0.013)	-0.086** (0.043)	-0.094*** (0.028)	-0.087* (0.053)
N	93,041	53,973	53,553	72,017	53,553
<i>Panel B. Crime perception</i>					
ATT	-0.149*** (0.025)	-0.140*** (0.032)	-0.112** (0.057)	-0.082** (0.033)	-0.111 (0.106)
N	89,578	52,160	51,746	69,434	51,746
<i>Panel C. Investment in security measures</i>					
ATT	-0.077*** (0.016)	-0.066*** (0.019)	-0.109*** (0.041)	-0.081*** (0.019)	-0.168*** (0.064)
N	93,041	53,973	53,553	72,017	53,553
<i>Panel D. Reported crime (per 100k pop.)</i>					
ATT	-208.024*** (56.130)	-157.010*** (56.224)	29.230 (147.261)	-261.439 (170.954)	-245.555 (165.221)
N	4,231	2,963	1,911	2,248	1,911
Sample	Full	Resource yrs	Resource yrs	Resource yrs	Resource yrs
Police officers (per 100k pop.)			✓		✓
Patrol vehicles (per 100k pop.)				✓	✓

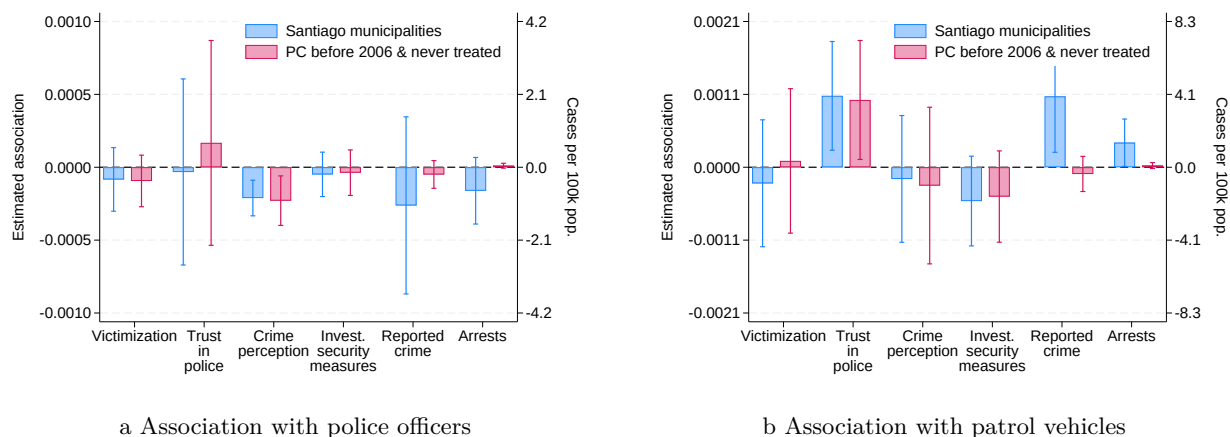
Note: ATT estimates of the effect of *Plan Cuadrante* on each outcome. Column (1) is the baseline (full estimation sample, no controls) and reproduces the headline estimate of the corresponding estimator. Column (2) restricts the estimation to the years in which police-resource data are available (from 2006 onward). No control variables are used in this estimation. Columns (3)–(5) add time-varying controls for police resources per 100,000 inhabitants: officers only; patrol vehicles only; and officers and patrol vehicles jointly (entered as separate regressors). Patrol vehicles are measured, per 100,000 inhabitants, as all patrol automobiles (radiopatrulla, auto patrullero, auto policial, automovil, station wagon, jeep and camioneta) plus motorcycles. Standard errors in parentheses; row N reports the estimation-sample size. Estimation follows Callaway and Sant’Anna (2021); standard errors clustered at the municipality level using wild bootstrapping procedures. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Figure B.4: Effects of *Seguridad Ciudadana* on Crime Outcomes and Security Perceptions



Notes: This figure shows the dynamic effects of the *Seguridad Ciudadana* program on different measures of crime and security perceptions. The estimation is conducted using the procedure developed in Callaway and Sant'Anna (2021) for difference-in-differences with staggered treatment adoption. The dots in the figure represent the estimated coefficients and the bars behind them 95% confidence intervals. Standard errors are clustered at the municipality level using wild bootstrapping. The pooled effect and the mean outcome before the introduction of the treatment for the outcome is also presented in each panel. Panel (a) reports the effect of *Seguridad Ciudadana* on the probability of household victimization in the last 12 months using the ENUSC survey. Panel (b) reports the effect on the number of crimes reported to the police in the municipality per 100,000 inhabitants using administrative data. Panel (c) reports the effect on the number of arrests in the municipality per 100,000 inhabitants using administrative data. Panel (d) reports the effect on the probability of perceiving an increase in crime using the ENUSC survey. Panel (e) reports the effect on the probability of adopting security measures in the last twelve months using the ENUSC survey. Panel (a) includes 210,102 observations; panel (b), 2,234; panel (c), 1,908; panel (d), 203,078; and panel (e), 74,316 observations.

Figure B.5: Association between Police Resources and Police Effectiveness (No *Plan Cuadrante* Variation)



Notes: The figure shows the estimated within-municipality association between police resources (per 100,000 inhabitants) and the main outcomes, in municipalities where *Plan Cuadrante* does not vary over the sample period. Estimates come from OLS regressions of each outcome on the resource measure with municipality and year fixed effects; standard errors are clustered at the municipality level. Blue bars use the 41 metropolitan municipalities of Santiago, all of which adopted *Plan Cuadrante* in 2000; red bars use the broader set of municipalities that adopted *Plan Cuadrante* before 2006 together with the never-treated municipalities surveyed in ENUSC (2006–2012). Reported crime and arrests (administrative data, in cases per 100,000 inhabitants) are read on the right axis; all other outcomes, from the ENUSC survey (victimization, trust in the police, crime perception, and investment in security measures), are read on the left axis. Panel (a) reports associations with the number of police officers; panel (b) with the number of patrol vehicles. Bars are point estimates with 95% confidence intervals.

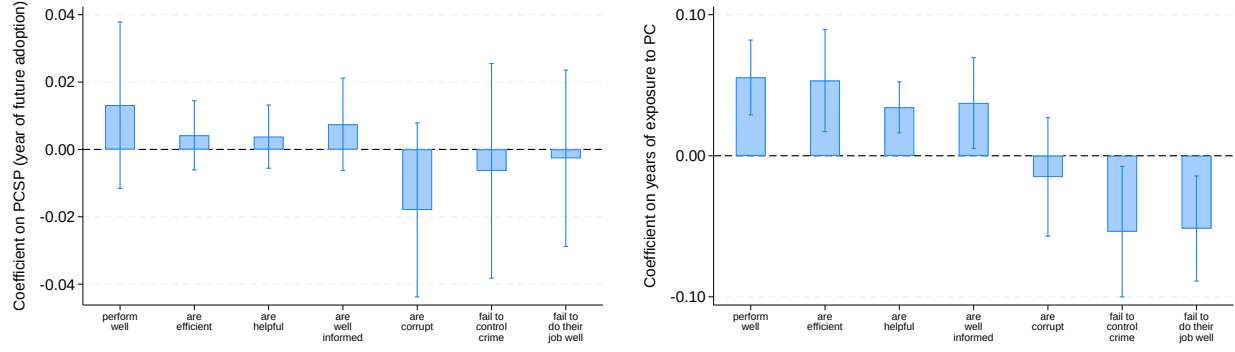
Table B.4: Predictors of the Increase in Police Resources under *Plan Cuadrante*

	Administrative sample		ENUSC sample	
	Police officers (1)	Patrol vehicles (2)	Police officers (3)	Patrol vehicles (4)
Dep. var: avg. annual increase in resources (share of pre-adoption level)				
Reported crime (per 100k pop.)	−0.203** (0.099)	−0.199** (0.097)	−0.319* (0.182)	0.272* (0.137)
Arrests (per 100k pop.)	0.099 (0.136)	−0.041 (0.129)	0.003 (0.186)	0.297 (0.223)
Victimization			−0.017* (0.010)	−0.025* (0.013)
Crime perception			0.005 (0.017)	0.008 (0.014)
Trust in the police			0.024 (0.018)	0.027 (0.019)
Investment in security measures			0.004 (0.010)	−0.003 (0.009)
Police officers (per 100k pop.)	−0.207 (0.155)	0.155 (0.129)	−0.888*** (0.154)	0.571*** (0.136)
Patrol vehicles (per 100k pop.)	−0.129 (0.129)	−0.139 (0.126)	0.537** (0.212)	−0.507*** (0.169)
Log population	−0.529*** (0.145)	−0.203* (0.112)	−0.193 (0.189)	−0.626** (0.269)
Poverty rate (2004)	0.170 (0.113)	0.308** (0.126)	0.027 (0.108)	−0.017 (0.108)
Observations	65	65	8,781	8,781
R-squared	0.342	0.282	0.614	0.545

Note: OLS regressions of the average annual increase in police officers (columns 1 and 3) and patrol vehicles (columns 2 and 4) per 100,000 inhabitants, from the year before adoption of *Plan Cuadrante* to 2012. As in panels (a) and (b) of Figure 4, the increase is expressed as a share of the resources available the year before adoption. Columns 1–2 use the administrative sample at the municipality level; columns 3–4 use ENUSC survey respondents interviewed before 2007 (survey outcomes at the respondent level, other regressors as pre-2006 municipality aggregates). All variables are standardized. Standard errors clustered by municipality in parentheses.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Figure B.6: Timing of Adoption of *Plan Cuadrante* and Perceived Police Performance

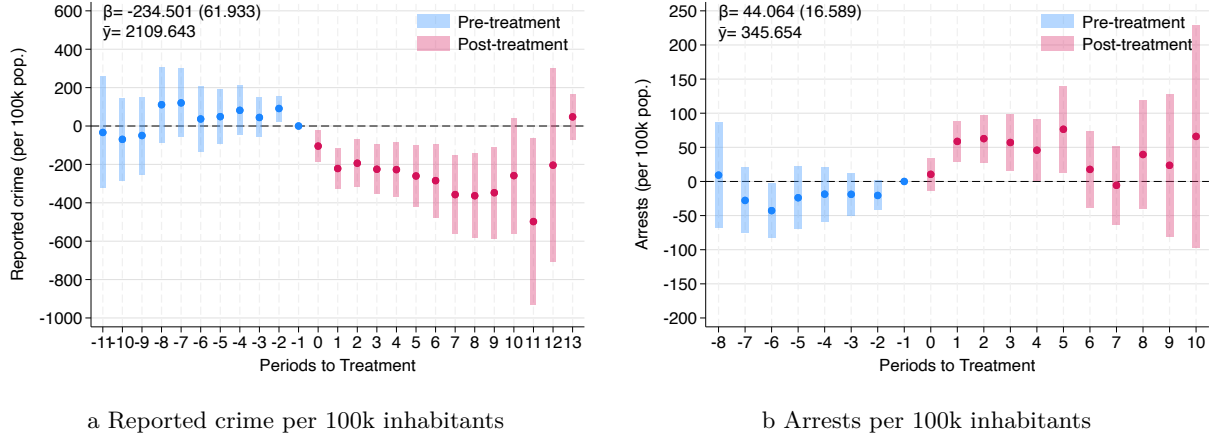


a Year of adoption and police performance before adoption

b Years exposed to *Plan Cuadrante* (dose response)

Notes: The figure considers seven measures of police performance: the share of respondents who perceive police performance as good, and who perceive the police to be efficient, helpful, well informed, corrupt, failing to control crime, and failing to do their job well. Using the sample of municipalities that adopted *Plan Cuadrante* after 2005, panel (a) examines whether the future timing of adoption of *Plan Cuadrante* is correlated with perceived police performance in 2005, before the introduction of the program. Panel (b) reestimates the same comparison as Figure 4, replacing the binary adoption indicator with the municipality's number of years of exposure to *Plan Cuadrante* as of 2005. In both panels, each bar plots the coefficient from a separate regression of one performance measure. The graphs display the estimated coefficients and 95% confidence intervals.

Figure C.8: Effects of *Plan Cuadrante* on Reported Crime and Arrests Using the Full Set of Post-treatment and Pre-treatment Periods



Notes: This figure shows the dynamic effects of *Plan Cuadrante* on the number of crimes reported to the police in the municipality per 100,000 inhabitants (panel a) and on the number of arrests in the municipality per 100,000 inhabitants (panel b), using administrative data. The estimation is conducted using the procedure developed in Callaway and Sant'Anna (2021) for difference-in-differences with staggered treatment adoption. The dots in the figure represent the estimated coefficients and the bars behind them 95% confidence intervals. Standard errors are clustered at the municipality level using the wild bootstrapping procedure. The pooled effect and the mean outcome before the introduction of the treatment are also presented in each panel. Unlike Figure 2 in the main text, we do not cap the event study to the same 5 pre-treatment and 6 post-treatment periods used for symmetry with the ENUSC-based analysis. Instead, we exploit the full set of post-treatment and pre-treatment periods. Panel (a) includes 4,371 observations and panel (b) 3,372 observations.

Comment R1.4

2. I think Table C2 is a first step in the right direction. However, Equation (5) should include both controls for N of officers and cars, so that it can approximate a police production function as closely as possible. Ideally, one would like to run a specification that includes the policy indicator together with N of officers and cars, but I understand that the policy dummy drops out in a regression at the municipality-year level. A viable option would be to interact treatment dummy with an indicator for municipalities that experienced larger increases in officers or vehicles. Alternatively, a mediation analysis could help quantify the extent to which resource increases account for the total observed effects of the policy. Finally, I am not fully convinced by the statement “lack of statistical association between police resources and the main outcomes” since several coefficients in Table C2 appear significant. This also suggests that resources may play a more substantive role than currently acknowledged.

Response to R1.4:

Thanks for the suggestions. Let us start by clarifying our main claim, which we have stated more precisely in the revised manuscript. We do not argue that the police resources brought by *Plan Cuadrante* play no role. They likely enable the successful implementation of the program, and for some outcomes the effects may indeed be larger in municipalities that received greater resource increases. In the absence of a natural experiment that would allow us to isolate the causal effect of resources from the organizational reform, we cannot fully disentangle the two channels in municipalities that received large resource increases. Our argument is different: the beat policing component of *Plan Cuadrante*—the division of the municipal territory into quadrants with the permanent assignment of officers to those territories—plays a key role in explaining the results, over and above the resource channel. This is supported by the evidence in Figure 4: while some treated municipalities received substantial increases in officers and patrol vehicles, others experienced changes very similar to those in the control group, and the program had large and statistically significant effects in this latter group across most outcomes. Reassuringly, municipalities receiving larger versus smaller resource increases—whether the split is based on the increase in police officers or in patrol vehicles—do not display differential pre-treatment trends in either the outcomes of interest or in police resources themselves, alleviating concerns that this split might itself be confounded by pre-existing differences across municipalities.⁴⁴ We also note that our results are not in tension with the existing literature on police strength and crime. As Chalfin (2025) notes, the effectiveness of additional police resources depends crucially on how they are organized and deployed, suggesting that organizational reforms like *Plan Cuadrante* can substantially improve public safety independently of changes in resource levels.

To provide further evidence, we split the sample of treated municipalities into those that experienced larger versus smaller resource increases with the adoption of *Plan Cuadrante* and estimate the ATT separately for each subgroup. This subgroup analysis is in the spirit of the interaction suggested by the reviewer, although a direct interaction between the treatment dummy and a resource-increase indicator is not feasible within the Callaway and Sant’Anna framework.⁴⁵ Additionally, we estimate the effect of *Seguridad Ciudadana*, a program that increased police resources without any organizational component, and find no significant effects on public safety outcomes. Similarly,

⁴⁴The p-values for the joint test of equal pre-trends between the two groups are, for the officers-based split and the patrol-vehicles-based split respectively: 0.823 and 0.048 for victimization, 0.059 and 0.379 for crime perception, 0.820 and 0.157 for investment in security measures, 0.213 and 0.419 for reported crime, 0.616 and 0.444 for police officers, and 0.405 and 0.157 for patrol vehicles.

⁴⁵Using OLS in this framework is not convenient due to the problem of negative weights (Goodman-Bacon, 2021).

we do not find a significant association between police resources and most public safety outcomes in the post-*Plan Cuadrante* period.

Following the reviewer’s recommendation, we estimate in the table below the main regression of the effect of *Plan Cuadrante* controlling for both the number of officers and the number of patrol vehicles, separately and jointly, to approximate a police production function as closely as possible. Three caveats are worth noting. First, police resources are affected by *Plan Cuadrante* and are therefore potentially bad controls, but including them allows us to approximate the effect of the territorial reorganization component while holding resource levels constant. Second, the output of the estimation does not yield point estimates for the control variables. The only coefficient yield in each estimation is the ATT of *Plan Cuadrante*. Third, information on resources is not available for the full sample, so adding these controls reduces the sample size. The results are reported in Table B.3 and show that, overall, the point estimates remain broadly stable when we add both patrol vehicles and police officers as controls. The statistical significance of some coefficients, however, is reduced. We believe this reflects a mechanical feature of the Callaway and Sant’Anna estimation procedure rather than weaker underlying effects. In this framework, control variables are not incorporated parametrically as in standard OLS but are instead used to construct the comparison group through outcome regression or propensity score weighting, introducing an additional estimation step that inflates mechanically the standard errors of the ATT estimates. Unlike in OLS, where adding controls typically reduces standard errors by absorbing residual variance, in the Callaway and Sant’Anna framework the uncertainty from estimating the outcome model or the propensity score propagates to the final estimates. The stability of the point estimates across specifications with and without controls for most outcomes supports this interpretation.

Taken together, while the results cannot rule out that police resources enable the successful implementation of the program, drive part of the effects of *Plan Cuadrante* in some municipalities, they consistently highlight that the beat policing component of *Plan Cuadrante*, namely the division of the territory into quadrants with the permanent assignment of officers to those territories, plays a crucial role in explaining the success of the program.

Finally, regarding the recommendation to use mediation analysis, the procedure proposed by Imai et al. (2010) is unfortunately not directly applicable in the context of staggered difference-in-differences estimators, which limits our ability to formally quantify the share of the total effect attributable to the resource channel.

We have added a footnote referencing the suggested exercise in Section 4.4 of the updated manuscript:

We also re-estimate the results in Figure 2 but controlling for police resources. Online Appendix Table B.3 shows that the magnitude of the estimated effects remains similar for most variables, although the statistical significance is reduced in some estimations. This likely reflects a mechanical feature of the Callaway and Sant’Anna (2021) estimation procedure: control variables are not incorporated parametrically, but are instead used to construct the comparison group via outcome regression or propensity-score weighting, so the resulting estimation uncertainty propagates to the ATT standard errors. Note, however, that police resources are a bad control—i.e., at least part of their variation is a result of the adoption of *Plan Cuadrante*—so these results should be interpreted cautiously.

Also, we have added the following paragraphs in Appendix B of the updated manuscript with the suggested exercise:

From the “Effects of Plan Cuadrante and Changes in Police Resources” subsection of Appendix

B:

Finally, we re-estimate the main results of the study reported in Figure 2, controlling for the number of police officers and the number of patrol vehicles in the municipality.

The results of this analysis are reported in Table B.3. The magnitude of the estimated effects remains similar for most variables, although the statistical significance is reduced in some estimations. This likely reflects a mechanical feature of the Callaway and Sant’Anna (2021) estimation procedure: control variables are not incorporated parametrically, but are instead used to construct the comparison group via outcome regression or propensity-score weighting, so the resulting estimation uncertainty propagates to the ATT standard errors. Note, however, that these resource variables are bad controls, as they are affected by the adoption of *Plan Cuadrante*, so these results should be interpreted with caution.

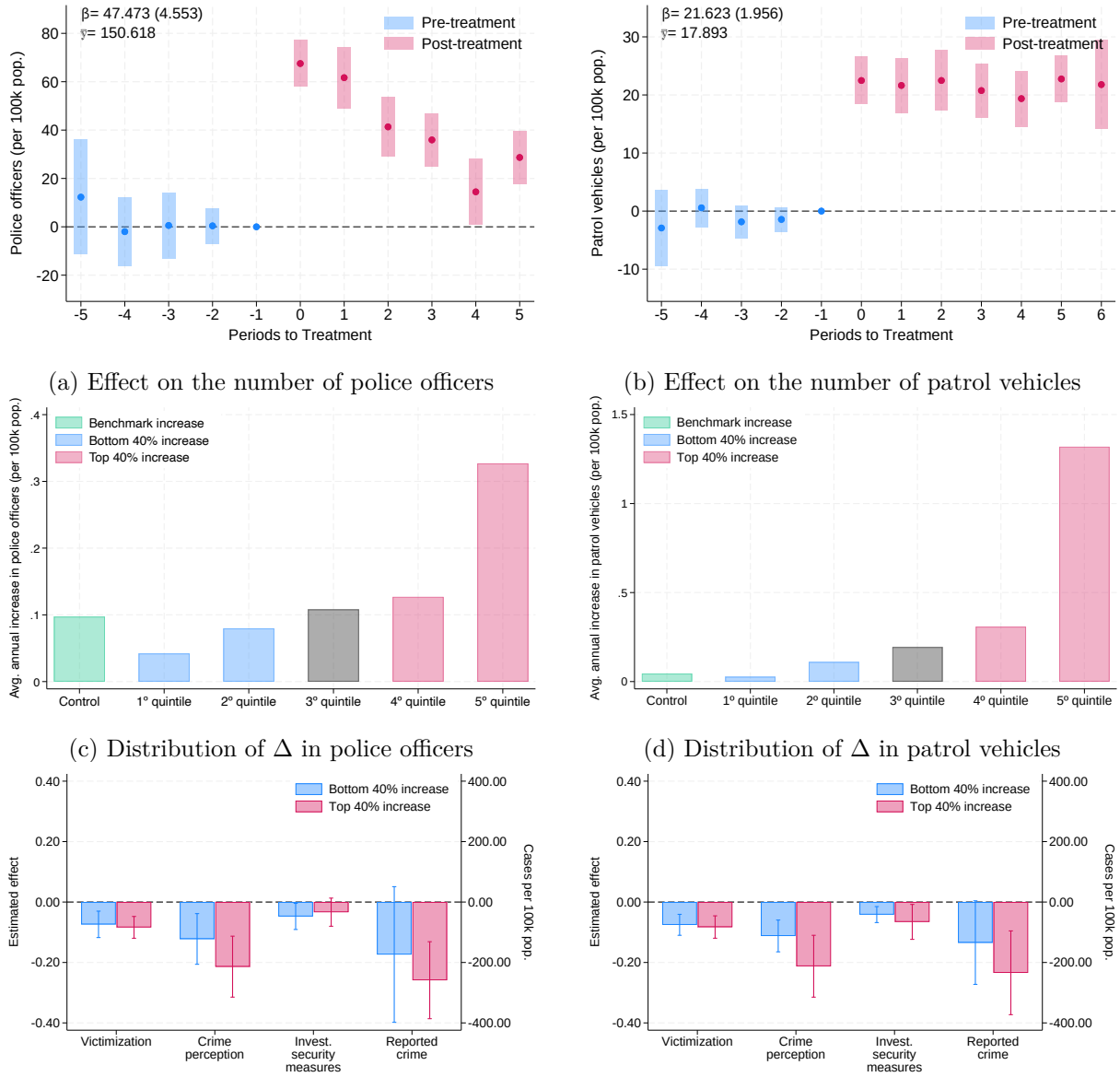
Table B.3: Effects of *Plan Cuadrante* Controlling for Police Resources

	(1)	(2)	(3)	(4)	(5)
<i>Panel A. Victimization</i>					
ATT	-0.099*** (0.013)	-0.077*** (0.013)	-0.086** (0.043)	-0.094*** (0.028)	-0.087* (0.053)
N	93,041	53,973	53,553	72,017	53,553
<i>Panel B. Crime perception</i>					
ATT	-0.149*** (0.025)	-0.140*** (0.032)	-0.112** (0.057)	-0.082** (0.033)	-0.111 (0.106)
N	89,578	52,160	51,746	69,434	51,746
<i>Panel C. Investment in security measures</i>					
ATT	-0.077*** (0.016)	-0.066*** (0.019)	-0.109*** (0.041)	-0.081*** (0.019)	-0.168*** (0.064)
N	93,041	53,973	53,553	72,017	53,553
<i>Panel D. Reported crime (per 100k pop.)</i>					
ATT	-208.024*** (56.130)	-157.010*** (56.224)	29.230 (147.261)	-261.439 (170.954)	-245.555 (165.221)
N	4,231	2,963	1,911	2,248	1,911
Sample	Full	Resource yrs	Resource yrs	Resource yrs	Resource yrs
Police officers (per 100k pop.)			✓		✓
Patrol vehicles (per 100k pop.)				✓	✓

Note: ATT estimates of the effect of *Plan Cuadrante* on each outcome. Column (1) is the baseline (full estimation sample, no controls) and reproduces the headline estimate of the corresponding estimator. Column (2) restricts the estimation to the years in which police-resource data are available (from 2006 onward). No control variables are used in this estimation. Columns (3)–(5) add time-varying controls for police resources per 100,000 inhabitants: officers only; patrol vehicles only; and officers and patrol vehicles jointly (entered as separate regressors). Patrol vehicles are measured, per 100,000 inhabitants, as all patrol automobiles (radiopatrulla, auto patrullero, auto policial, automovil, station wagon, jeep and camioneta) plus motorcycles. Standard errors in parentheses; row N reports the estimation-sample size. Estimation follows Callaway and Sant’Anna (2021); standard errors clustered at the municipality level using wild bootstrapping procedures. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Other figure referenced above:

Figure 5: Plan Cuadrante and Changes in Police Resources



(e) Effects by the intensity of Δ in police officers (f) Effects by the intensity of Δ in patrol vehicles

Notes: Panels (a) and (b) show the effect of *Plan Cuadrante* on police officers and patrol vehicles per 100,000 inhabitants. Panels (c) and (d) show the distribution of annual changes in officers and patrol vehicles across municipalities. Panels (e) and (f) report effects by the intensity of resource changes, comparing municipalities in the bottom 40% and top 40% of the distribution. The estimation follows Callaway and Sant'Anna (2021). The graphs display estimated coefficients and 95% confidence intervals. Standard errors are clustered at the municipality level using wild bootstrapping.

Comment R1.5

3. The paper should also discuss subsequent adjustments to the policy in 2007, and the potential disruption created by the 2010 earthquake. Both could have affected treatment quality/intensity (or even the different components?) as well as local perceptions, and it is important to explain how these events are handled in the empirical section. For example, is it possible to exclude the affected areas, or to test whether the earthquake generated changes in perceptions that confound the results? At the same time, could the 2007 adjustment and 2010 earthquake be exploited as sources of heterogeneity to shed light on mechanisms if they were different in nature?

Response to R1.5:

In 2007, an evaluation conducted by DIPRES (Chile's Directorate of Budget, an independent government body) recommended the formalization of new specialized roles within each police station, including a Head of Operations Office, a Community Affairs Officer, and officers responsible for drug prevention, domestic violence, and judicial orders (Winchester et al., 2007). However, these changes were more formal than operational, and were not effectively implemented at scale until the introduction of *Plan Cuadrante 2.0*, outside our study period. The 2014 DIPRES evaluation documents, for example, that while the Community Affairs Officer role was formally created in some municipalities, dedicated personnel and budget were not assigned, and the community integration model (MICC) was only piloted in 4 to 6 police stations in 2012–2013 (none of these stations were in municipalities in our analytical sample) before its financing agreement was discontinued in early 2014 (Salas P. et al., 2014; Carabineros de Chile, 2018). Similarly, the Operations Office lacked the digitalization, personnel, and dedicated resources needed to fulfill its analytical functions until the launch of *Plan Cuadrante 2.0* (Salas P. et al., 2014; Carabineros de Chile, 2018). Our results therefore capture the effects of the original *Plan Cuadrante*, characterized primarily by the reduction of police catchment areas and the permanent territorial assignment of officers, without contamination from the more substantive operational reforms introduced later.

Regarding the disruption that the 2010 earthquake in Concepcion could have on the implementation of *Plan Cuadrante*: The 2010 earthquake halted the expansion of the program. No new municipalities joined the program in 2010. Moreover, it could potentially confound our estimates on outcomes if it differentially affected treated and control municipalities, either by disrupting police operations and thus reducing treatment intensity, or by independently affecting crime perceptions and security outcomes in ways unrelated to *Plan Cuadrante*. To address this concern, in the revised manuscript we re-estimate our main results excluding municipalities more exposed to the earthquake, defined as those experiencing a Modified Mercalli Intensity greater than 7. This corresponds to 21 municipalities in the ENUSC sample and 158 municipalities in the administrative data sample. The results, reported in Figure C.11, are nearly identical to the main estimates, suggesting that the 2010 earthquake does not confound our estimates.

We have added the following footnote in Section 4.1 of the updated manuscript:

These results are robust to excluding municipalities most affected by the 2010 Maule earthquake, defined as those with a Modified Mercalli Intensity greater than 7; see Figure C.11 in Online Appendix C for details.

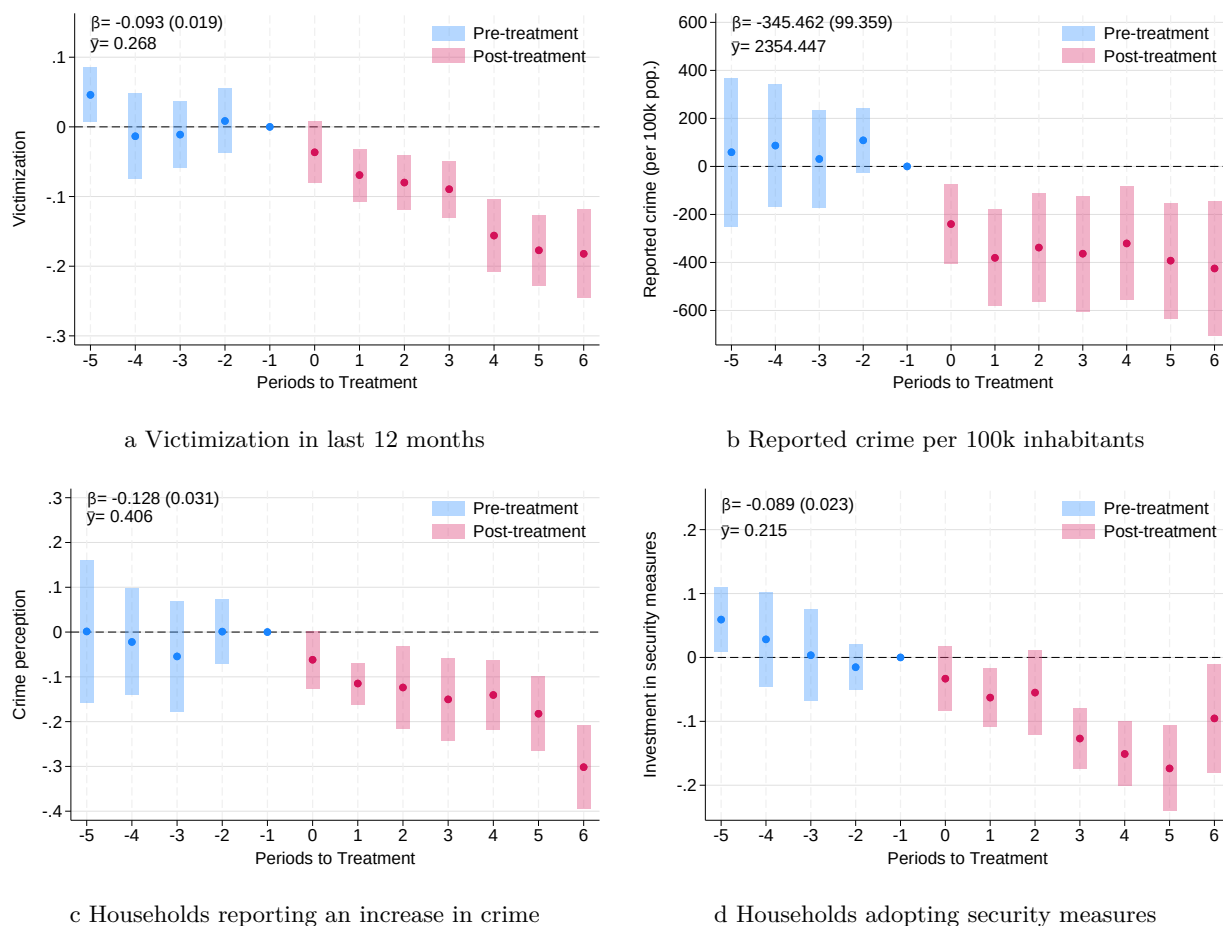
We also add the following subsection, “Robustness to the 2010 Earthquake,” to Online Appendix C reporting this exercise in detail:

The 2010 Maule earthquake could confound our estimates if it differentially affected treated and control municipalities, either by disrupting police operations and thus reducing treatment intensity, or by independently affecting crime perceptions and security outcomes in ways unrelated to *Plan Cuadrante*. To address this concern, we re-estimate our main results excluding municipalities more exposed to the earthquake, defined as those experiencing a Modified Mercalli Intensity greater than 7. This corresponds to 21 municipalities in the ENUSC sample and 158 municipalities in the administrative data sample. The results, reported in Figure C.11, are nearly identical to the main estimates, confirming that the 2010 earthquake does not drive our findings.

Regarding the suggestion to exploit the earthquake as a source of heterogeneity, we agree this is an interesting analysis that might help shed light on the mechanisms. To assess the heterogeneous effects of *Plan Cuadrante* by exposure to the earthquake, we estimate the effects of the program separately for municipalities more exposed to the earthquake (Modified Mercalli Intensity greater than 7) and municipalities with a Modified Mercalli Intensity of 7 or lower. The results are reported in Figure S1. They show relatively similar effects across the two groups for both victimization and crime perception. The effects on investment in security measures and reported crime appear somehow larger for municipalities less exposed to the earthquake, although they remain statistically significant for the more affected municipalities as well.

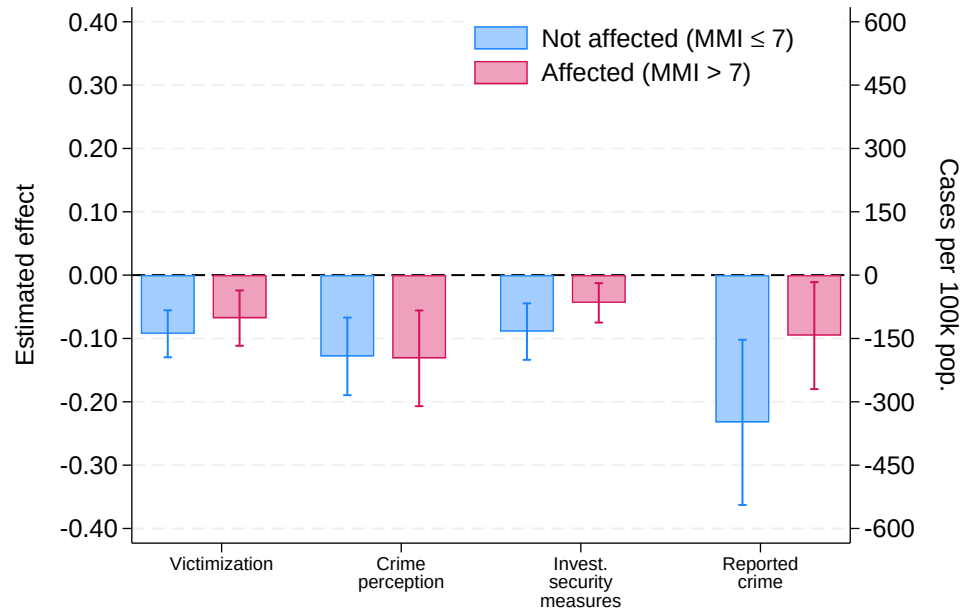
Other figure referenced above:

Figure C.11: Effects of *Plan Cuadrante* excluding Municipalities Affected by the 2010 Earthquake



Notes: This figure replicates the results of Figure 2 after excluding municipalities affected by the 2010 Maule earthquake, defined as those with a Modified Mercalli Intensity (MMI) score strictly greater than 7. The estimation is conducted using the procedure developed in Callaway and Sant'Anna (2021) for difference-in-differences with staggered treatment adoption. The dots in the figure represent the estimated coefficients and the bars behind them 95% confidence intervals. Standard errors are clustered at the municipality level using the wild bootstrapping procedure. The pooled effect and the mean outcome before the introduction of the treatment is also presented in each panel. Panel (a) reports the effect of *Plan Cuadrante* on the probability of household victimization in the last 12 months using the ENUSC survey. Panel (b) reports the effect on the number of crimes reported to the police in the municipality per 100,000 inhabitants using administrative data. Panel (c) reports the effect on the probability of perceiving an increase in crime using the ENUSC survey. Panel (d) reports the effect on the probability of adopting security measures in the last twelve months using the ENUSC survey.

Figure S1: Effects of *Plan Cuadrante* by Exposure to the 2010 Earthquake



Notes: This figure reports the heterogeneous effects of *Plan Cuadrante* by exposure to the 2010 Maule earthquake, estimated separately for municipalities more exposed to the earthquake—defined as those with a Modified Mercalli Intensity (MMI) greater than 7, “affected” (red bars)—and municipalities with an MMI of 7 or lower (“not affected”, blue bars), using all adopters. For each exposure group, we estimate the pooled post-treatment effect of the program with the difference-in-differences estimator for staggered adoption of Callaway and Sant’Anna (2021), comparing municipalities of the same exposure level to each other so that the earthquake’s direct effect on outcomes is differenced out within each group. Victimization is the probability of household victimization in the last 12 months; crime perception is the probability of perceiving an increase in crime; and investment in security measures is the probability of adopting security measures in the last twelve months—all from the ENUSC survey and read on the left axis. Reported crime is the number of crimes reported to the police per 100,000 inhabitants, from administrative data, and read on the right axis. Bars represent the average post-treatment effect and the whiskers the 95% confidence intervals; standard errors are clustered at the municipality level using the wild bootstrapping procedure.

Comment R1.6

4. In addition, to assess the impacts of the reform the authors look at the effects on the perceptions of police performance by comparing perceptions between early and late adopters (Figure 4). To interpret this exercise, it would first be necessary that the authors provide evidence that adoption timing was not driven by differences in other pre-policy changes in local outcomes. Even if not all survey questions are available in every year, a regression of outcomes on the number of years of exposure to the policy could provide an alternative way to address this.

Response to R1.6:

This is an excellent point, thank you very much. The reviewer noted that, to interpret this exercise, it would be necessary to show that adoption timing was not driven by differences in other pre-policy changes. Following reviewer's suggestion, we regress perceived police performance outcomes on the year of adoption of *Plan Cuadrante*, restricting the sample to municipalities that had not yet adopted the program as of 2005, the year for which information on perceived police performance is available. The results, reported in panel (a) of Figure B.6, show that perceived police performance before *Plan Cuadrante* did not predict the timing of adoption.

Following your suggestion, as an additional exercise, we also reestimate the comparison reported in panel (a) of Figure 4, measuring years of exposure to *Plan Cuadrante* in 2005 rather than simply whether the municipality had adopted the program. The results, reported in panel (b) of Figure B.6, show the expected correlations, consistent with an increasing effect of *Plan Cuadrante* on perceived police performance.

We have included the following footnote in Section 4.4 of the manuscript to reference these results:

Information on police performance is only available in ENUSC round 2005. Using this round, we compare perceived police performance in municipalities that adopted *Plan Cuadrante* in 2005 or earlier and in municipalities adopting after 2005. Figure B.6 in Online Appendix B shows that the number of years of exposure to *Plan Cuadrante* is positively correlated with perceived police performance. While these exercises on perceived police performance should be interpreted with caution, we show in the same appendix that the timing of future adoption is not correlated with perceived police performance in 2005 for municipalities that had not yet adopted the program by 2005.

And we added the following subsection, "Adoption Timing and Perceived Police Performance," to Online Appendix B reporting this exercise in detail:

Information on perceived police performance is only available in the 2005 round of ENUSC, so the comparison in Figure 4 relies on a single cross-section, comparing municipalities that had already adopted *Plan Cuadrante* by 2005 (early adopters) with those that had not yet done so (late adopters). We consider seven measures of perceived police performance: the share of respondents who perceive police performance as good, and who perceive the police to be efficient, helpful, well informed, corrupt, failing to control crime, and failing to do their job well.

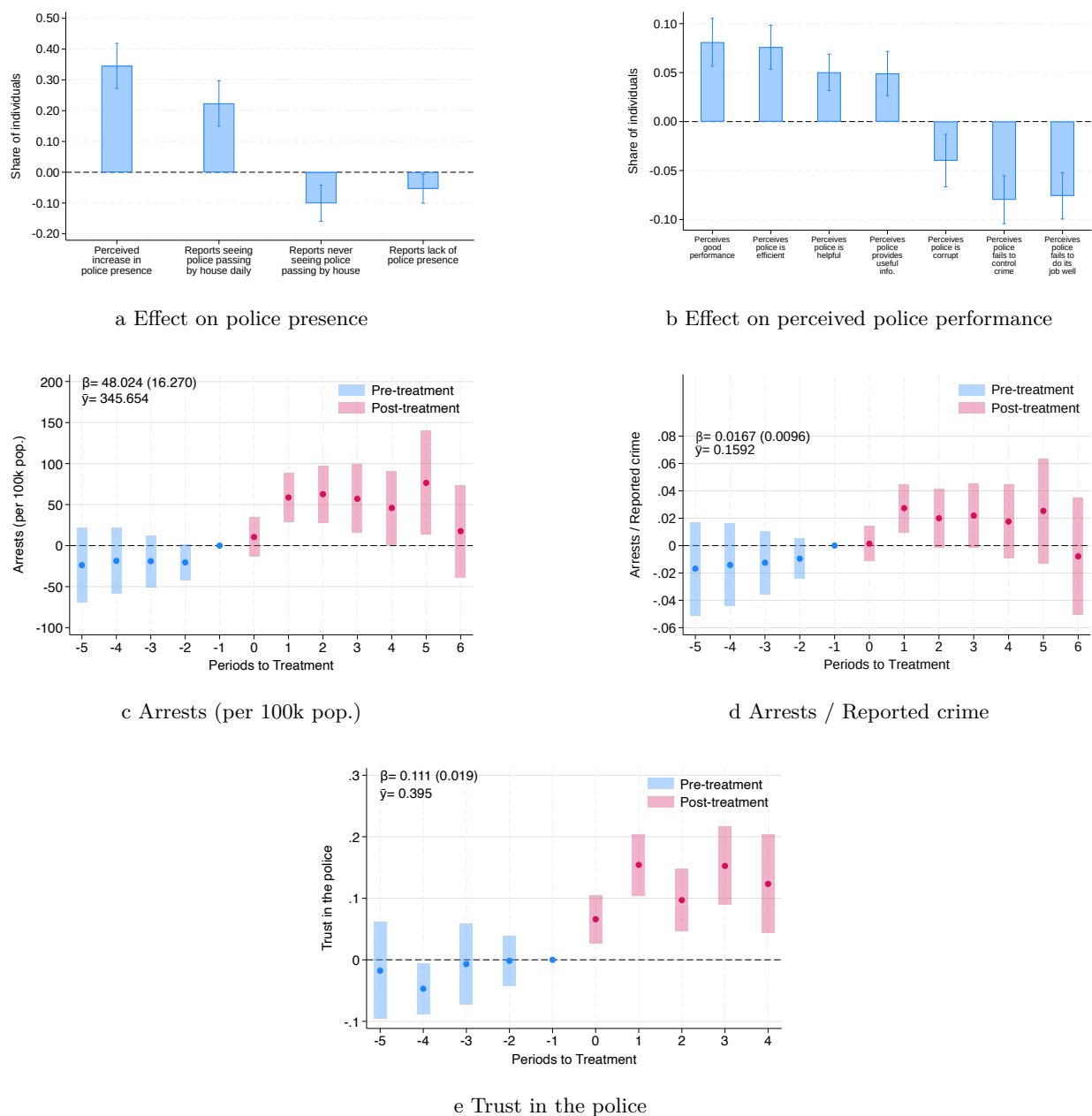
We start by assessing whether adoption timing is correlated with police performance. Using the sample of municipalities that introduced *Plan Cuadrante* after 2005, Panel (a) of Figure B.6 examines the correlation between adoption timing and perceived police performance in

2005. The estimated coefficients are close to zero and statistically insignificant at conventional confidence levels, indicating that how soon a not-yet-treated municipality would go on to adopt the program is unrelated to its perceived police performance before treatment.

In Panel (b) of Figure B.6 we re-estimate the analysis conducted in Figure 4 but measuring years of exposure to *Plan Cuadrante* in 2005 rather than simply whether they have adopted it or not. The results are qualitatively similar to those estimates reported in Figure 4, showing strong correlations between years of exposure to *Plan Cuadrante* and perceived police performance.

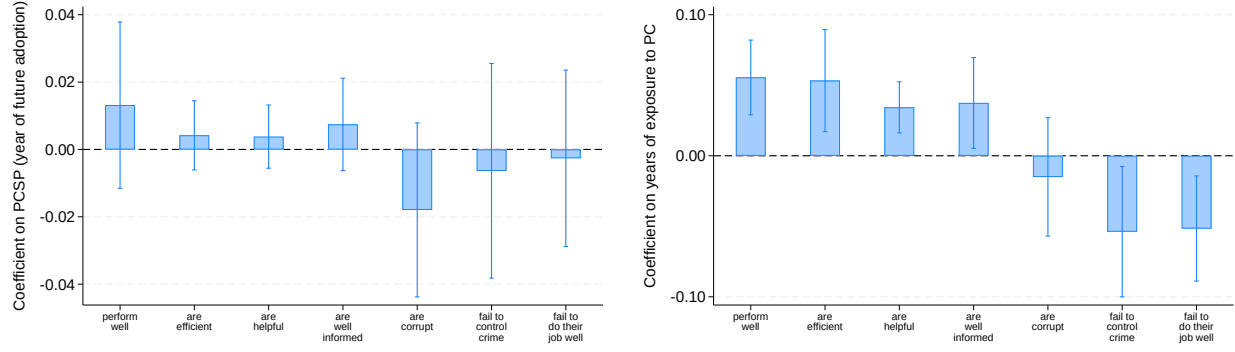
Other figures referenced above:

Figure 4: *Plan Cuadrante* and Changes in Police Performance



Notes: Panel (a) reports the effects on perceived police presence. Panel (b) reports perceived police performance in 2005, comparing municipalities where *Plan Cuadrante* was already implemented before 2006 with municipalities where it was implemented after 2005. Panels (c), (d), and (e) report dynamic event-study estimates for arrests per 100,000 inhabitants, the ratio of arrests to reported crime, and trust in the police. The estimation follows Callaway and Sant'Anna (2021). The graphs display estimated coefficients and 95% confidence intervals. Standard errors are clustered at the municipality level using the wild bootstrapping procedure.

Figure B.6: Timing of Adoption of *Plan Cuadrante* and Perceived Police Performance



a Year of adoption and police performance before adoption b Years exposed to *Plan Cuadrante* (dose response)

Notes: The figure considers seven measures of police performance: the share of respondents who perceive police performance as good, and who perceive the police to be efficient, helpful, well informed, corrupt, failing to control crime, and failing to do their job well. Using the sample of municipalities that adopted *Plan Cuadrante* after 2005, panel (a) examines whether the future timing of adoption of *Plan Cuadrante* is correlated with perceived police performance in 2005, before the introduction of the program. Panel (b) reestimates the same comparison as Figure 4, replacing the binary adoption indicator with the municipality's number of years of exposure to *Plan Cuadrante* as of 2005. In both panels, each bar plots the coefficient from a separate regression of one performance measure. The graphs display the estimated coefficients and 95% confidence intervals.

Comment R1.7

Contribution: Once the ingredients of the policy are clarified, the paper's contribution should be expanded, because at present it does not fully engage with the most recent literature. Beyond the large body of work on police size and crime, a few recent papers have examined how the allocation of police resources across space and time affects police effectiveness (among others, Adda et al. (2014), Blanes i Vidal and Mastrobuoni (2018), Mastrobuoni (2019, 2020)). Furthermore, it should also connect more explicitly with the growing body of work on the impact of policing on community-related outcomes such as trust, perceptions, and willingness to cooperate (e.g. Blanco et al. (2013), Ang et al. (2025), Facchetti (2025), Gonçalves et al. (2025)).

I appreciate the effort to relate to work that inform how to organize state presence in space. However, to make this a convincing contribution, the authors would need to discuss explicitly whether centralizing policing over a larger jurisdiction improves the allocation of resources compared to spreading them, and I am not sure if they can currently do it. A cost-benefit analysis, or even a marginal value of public funds (MVPF) estimate, could help.

Response to R1.7:

Thank you for these suggestions. We have read all the cited studies carefully and agree that they are highly relevant and should have been included in our discussion. They are now integrated into the motivation of the paper.

Adda et al. (2014), Blanes-i Vidal and Mastrobuoni (2018), Mastrobuoni (2019), Facchetti (2025), and Mastrobuoni (2020) were particularly helpful in sharpening our understanding of how the allocation of police resources and efforts across space, time and crimes affects policing outcomes and public safety, the mechanisms driving these effects, and of the role that technology can play in improving police effectiveness. We now engage more explicitly with this literature when discussing the contributions of our results.

Blanco (2013) and Ang et al. (2025) helped us better motivate the importance of community trust in the police, its drivers, and how it shapes policing outcomes, and Gonçalves et al. (2025) documented police misconduct. We now connect our findings more explicitly to this growing literature, highlighting how the intervention we examine can increase trust in the police and the beneficial secondary effects of it.

Below are the paragraphs in the updated introduction where we have integrated the relevant literature and highlighted the contributions of our results:

Our findings contribute to two related strands of the literature on police organization and public safety. First, they provide some of the first causal evidence on the effects of beat policing on crime and security perceptions. Beat policing has become one of the most widely adopted policing models globally (National Academies of Sciences, Engineering, and Medicine, 2018; Ministerio del Interior y Seguridad Pública, 2014), yet robust causal evidence on its effectiveness remains scarce. We show that this model can generate substantial reductions in crime victimization, reported crime, and insecurity perceptions.

These results also speak to recent work on other policing models. Hot-spot policing reallocates police resources toward high-crime locations and has been shown to reduce crime in targeted areas, though often with displacement to nearby places (Di Tella and Schargrodsky,

2004; Blattman et al., 2021; Draca et al., 2011; Weisburd, 2021). Beat policing differs fundamentally. Rather than concentrating police presence in selected locations—where the same officers need not return repeatedly—it provides continuous, territory-wide coverage through permanent officer-to-area assignment. This distinction matters in our setting, where many quadrants were not crime hot spots and the reform generated municipality-wide declines in crime and insecurity perceptions, with no evidence of spillovers to neighboring municipalities. Community policing, another model that has received growing attention in the economics literature, emphasizes citizen participation in defining policing priorities and, in some cases, in implementation itself. While it has been shown to increase trust in police in developed countries, evidence from low- and middle-income settings points to more limited effects on crime and trust (Blair et al., 2021; Tobón, 2021). Beat policing also differs from that approach. By permanently assigning officers to stable territorial units, it strengthens local knowledge and brings police closer to communities without requiring direct citizen participation in strategy or operations. Our findings show that this model can reduce crime and insecurity perceptions while increasing trust in police institutions in a middle-income country.

A second contribution is to the broader literature showing that specific features of police organization matter for police effectiveness and public security outcomes. Research has shown that faster police response times improve crime clearance (Blanes-i-Vidal and Kirchmaier, 2018), that the effectiveness of police presence depends on how this is organized and deployed rather than simply on their intensity (Facchetti, 2025; Mastrobuoni, 2019; Blanes-i Vidal and Mastrobuoni, 2018), that shifts in police prioritization across crime types shape criminal behavior (Adda et al., 2014), and that better information systems and technology can improve police performance (Mastrobuoni, 2020). Earlier work also found that areas covered by smaller, more localized police departments outperform, in terms of security, those covered by larger centralized ones, hypothesizing that this might be related, among other factors, to closer citizen-police engagement (Ostrom et al., 1973, 1978). We add to this literature by studying a policing model that combines several of these margins within a single organizational reform. By assigning officers permanently to smaller territorial units rather than rotating them across larger ones, beat policing is designed to increase police visibility, strengthen knowledge of local conditions, and improve the capacity of officers to adapt their practices to local crime patterns. In line with these mechanisms, we show that the adoption of *Plan Cuadrante* increased perceived police presence in neighborhoods and raised police effectiveness, as arrests increased even while victimization and reported crime declined. These organizational improvements translated into meaningful gains in public safety and security perceptions.

A third contribution concerns the literature on police resources, which documents elasticities of crime with respect to officers and expenditure ranging from -0.1 to -2 , with larger effects in developed settings and for violent crimes (Chalfin and McCrary, 2017; Levitt, 1997; Di Tella and Schargrodsky, 2004). We complement this body of work by showing that the quantity of resources is not the only margin that matters. How those resources are deployed matters as well. *Plan Cuadrante* improved public safety outcomes even in municipalities that experienced no increase in police resources relative to controls, suggesting that reorganizing deployment can raise public safety and police effectiveness without expanding the resource base.

We also contribute to the literature on police practices, institutional trust, and citizen cooperation. A persistent challenge is that security perceptions often diverge from actual crime trends (Ajzenman et al., 2023), limiting the scope for improving both simultaneously. Rising crime erodes institutional trust (Blanco, 2013), while police misconduct undermines coopera-

tion (Ang et al., 2025).⁴⁶ Our results show that beat policing can improve both dimensions—security and security perceptions—by bringing officers into sustained engagement with local communities through permanent territorial assignment. We document not only increases in police trust and evaluations of police performance, but also substantial reductions in private security investment, indicating that households revised their threat assessments in response to the program.

Finally, our findings speak to a broader question about how organizational design shapes public-service effectiveness. A growing literature shows that the spatial organization of service provision matters. In policing, we show that a model organizing officers around fixed territorial units—allowing them to build local knowledge, coordinate with local actors, and adapt practices accordingly—can substantially improve both crime control and institutional legitimacy. This aligns with evidence from health and education showing that bringing providers closer to the populations they serve can improve outcomes (Besley and Coate, 2003; Bardhan, 2002; Bloom et al., 2015; Basurto et al., 2020). In a rigorously identified setting, we show that this type of organizational proximity generates meaningful gains in police effectiveness and citizen security.

Regarding the connection of our results with the literature on state service provision organization, we agree that addressing whether centralizing policing over a larger jurisdiction—or, alternatively, spreading officers across smaller, fixed territorial units—improves the allocation of police resources is central to the contribution of the paper. Our findings speak directly to this question. The organizational logic of *Plan Cuadrante* is precisely one of territorial or local specialization: rather than maintaining a centralized dispatch model in which officers patrol in rotation large, undifferentiated areas, the reform assigns officers permanently to small, fixed quadrants, enabling them to build local knowledge and adapt their practices to local crime patterns. Municipalities receiving large increases in police officers and patrol vehicles experienced improvements in victimization similar to those in municipalities receiving resource increases similar to control, indicating that the scale of resource infusion alone does not drive effectiveness. This result supports the view that the organizational model—with permanent assignment to small territorial areas—is a key driver of effectiveness. While our design does not allow a direct budget-constrained comparison between a centralized rotating model and a beat policing model, these findings suggest that spreading officers across smaller fixed units, as *Plan Cuadrante* does, generates efficiency gains in resource allocation.

Regarding the request for a cost-benefit analysis or an MVPF estimate, we agree that this would be valuable in principle, but we do not think we can credibly implement it with the data currently available. On the cost side, the budget documents we were able to access (including Budget Office reports) do not cleanly separate baseline police expenditure that would have occurred in the absence of *Plan Cuadrante* from the truly incremental expenditure required to implement and operate the reform. On the benefit side, we also lack a credible context-specific willingness-to-pay measure that would allow us to monetize the welfare gains from the observed reductions in victimization and insecurity. These limitations rule out a credible cost-benefit or MVPF calculation.

That said, an important feature of our setting is that *Plan Cuadrante* is fundamentally an organizational reform of police deployment. Many municipalities that adopted the program experienced resource increases similar to those observed in control municipalities, suggesting that a substantial part of those resources would likely have been deployed even without the reform. In this sense, the

⁴⁶Racial disparities in police behavior are also well documented; see, e.g., Gonçalves et al. (2025), who develop a method to correct for selection bias in estimates of racial disparities in the use of force.

gains associated with the beat-policing component are largely linked to how existing police capacity is organized and deployed across territory, rather than to large additional spending above the cost of maintaining a police force. We therefore interpret our findings as evidence that organizational redesign can generate sizeable improvements in public safety even when incremental budgetary costs are limited.

Comment R1.8

Mechanisms: I encourage the authors to expand the discussion of mechanisms, that at present is not very well developed. First, the authors should be explicit on what they mean with “police effectiveness”. Typically, effectiveness is proxied by arrests or clearance rates. This would also help separate two channels: deterrence and police effectiveness. Deterrence would change the incentives to commit crime, and thus would be reflected in reductions in crime, whereas the latter would be reflected in the ability of the police to deal with crime, which would show up in arrests conditional on crime. Furthermore, to shed light on this, it would be useful to consider whether the reform affected response times, patrolling patterns, or any other proxies that capture the ability of the police to solve crime, such as improved technology. An alternative channel is that the reform increased the salience of policing and thus raised citizen awareness. For instance, was the policy advertised? Of course, these mechanisms are not mutually exclusive, and I recognize that disentangling them empirically may be difficult, but a more systematic discussion would be very valuable.

Response to R1.8:

Thanks for the specific and clear guidance to improve the analysis of mechanisms.

We revised the manuscript so that the mechanism discussion now distinguishes more clearly between deterrence and police effectiveness as two conceptually distinct but empirically intertwined channels through which beat policing may improve public safety and security perceptions. In the revised framing, police effectiveness is no longer used as a label for the main outcomes; instead, it refers to the capacity of police to prevent and resolve crime. In practice, with our data we cannot cleanly separate how much of the total effect comes from each channel, because they can reinforce each other: stronger preventive action and higher police presence can deter crime, and improved capacity to solve crimes can also deter crime by increasing the expected probability of apprehension; at the same time, lower crime incidence may allow police to concentrate effort on crime resolution. While we were not granted access to patrolling patterns or response times, we provide evidence consistent with both channels operating. Specifically, *Plan Cuadrante* increases perceived police presence (Panel (a) of Figure 4), reduces household victimization and reported crime (Panels (a) and (b) of Figure 2), increases arrests (Panel (c) of Figure 4), and increases arrests per reported crime (Panel (d) of Figure 4). The latter result should nonetheless be interpreted with caution, as the composition of reported crimes may also change with the program, affecting the ratio independently of any change in police effectiveness. Accordingly, in Section 4.4 the focus is not to isolate deterrence from effectiveness, but to disentangle which components of *Plan Cuadrante*—resources, territorial deployment, and policing strategy—are most likely to explain the effects we document.

Regarding the possibility that the reform was accompanied by an advertisement campaign that increased the salience of policing and raised citizen awareness, we believe this is unlikely for two reasons. First, *Plan Cuadrante* was an internal operational reform of Carabineros with no formal public communication component in its design. While the program received some media attention when it was broadly rolled out across Santiago in 2000, we find no evidence of targeted information

campaigns at the municipality level when individual municipalities subsequently joined the program, and the staggered rollout over more than a decade makes a sustained media salience mechanism implausible. Second, and more importantly, if the improvement in crime were driven by media coverage or information campaigns, one would expect the effects to be concentrated shortly after adoption and to fade over time. Instead, the dynamic estimates show that the effects for most outcomes grow and persist over time.

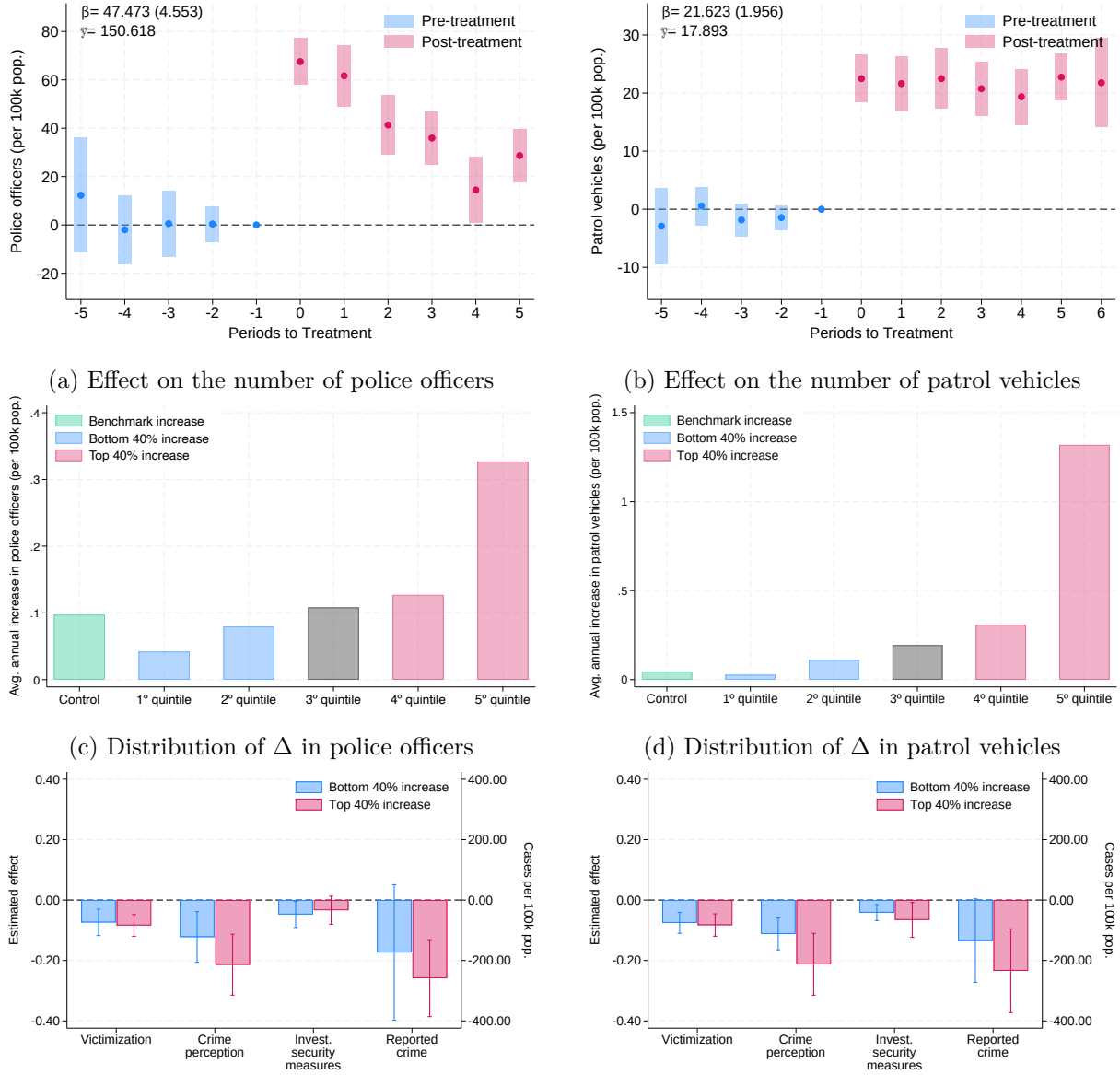
Below we provide the updated mechanism section:

To reduce crime and improve security perceptions, police forces need to strengthen both prevention and police effectiveness. Both margins can be shaped by resources, territorial deployment, and policing strategy. This section examines which components of *Plan Cuadrante* are most likely to drive the effects documented in Section 4.

The central objective of *Plan Cuadrante* was to introduce a beat policing model by reorganizing deployment. Police operational territories that had typically matched municipal boundaries were subdivided into quadrants, and officers were permanently assigned to these quadrants. During our study period, national operational guidelines remained the same in adopting and non-adopting municipalities. Still, this new deployment likely affected policing strategy at a more operational level, because it allowed officers to adapt patrol routes, schedules, and prevention practices using the local knowledge that the program sought to build.

Although changes in resources were not the core of the reform, implementation contemplated increases in municipalities that lacked the personnel and patrol capacity required to move to this model. During the period we study, Chile was expanding public spending on security, and police resources increased in most municipalities regardless of whether or when they adopted *Plan Cuadrante*. On average, however, municipalities adopting the program experienced larger increases in officers and patrol vehicles, as shown in panels (a) and (b) of Figure 4. Since increases in police resources have been shown to reduce crime in some contexts (Evans and Owens, 2007), this pattern raises a central question for interpretation, namely whether our estimates reflect resource expansion or the deployment and strategy changes introduced by *Plan Cuadrante*. We address this question through a set of exercises that exploit heterogeneity in resource increases across municipalities and over time.

Figure 4: Plan Cuadrante and Changes in Police Resources



(e) Effects by the intensity of Δ in police officers (f) Effects by the intensity of Δ in patrol vehicles

Notes: Panels (a) and (b) show the effect of *Plan Cuadrante* on police officers and patrol vehicles per 100,000 inhabitants. Panels (c) and (d) show the distribution of annual changes in officers and patrol vehicles across municipalities. Panels (e) and (f) report effects by the intensity of resource changes, comparing municipalities in the bottom 40% and top 40% of the distribution. The estimation follows Callaway and Sant'Anna (2021). The graphs display estimated coefficients and 95% confidence intervals. Standard errors are clustered at the municipality level using wild bootstrapping.

Panels (c) and (d) of Figure 4 show that the average increase in police resources masks substantial heterogeneity. Municipalities in the bottom 40% of the resource-change distribution (blue bars) experienced increases close to the average annual variation observed among never treated and not-yet-treated municipalities (green bar). By contrast, municipalities in the top 40% (red bars) experienced much larger increases, with officers rising by around 22% and patrol vehicles

by around 80%. If the effects of *Plan Cuadrante* were driven by resource increases alone, we would not expect effects in municipalities whose resource increases were similar to those of the control group. Yet panels (e) and (f) of Figure 4 show sizeable effects for most crime and crime-perception outcomes even in that low-increase group. While there may be a resource gradient for some crime-perception outcomes, the effects on twelve-month victimization and private investment in security among municipalities with control-like resource increases are close in magnitude to those in municipalities with larger increases. Reassuringly, municipalities in the top and bottom 40% of the resource-increase distribution do not display differential pre-treatment trends in outcomes or police resources, which suggests that this heterogeneity is not driven by differential pre-existing trends.⁴⁷

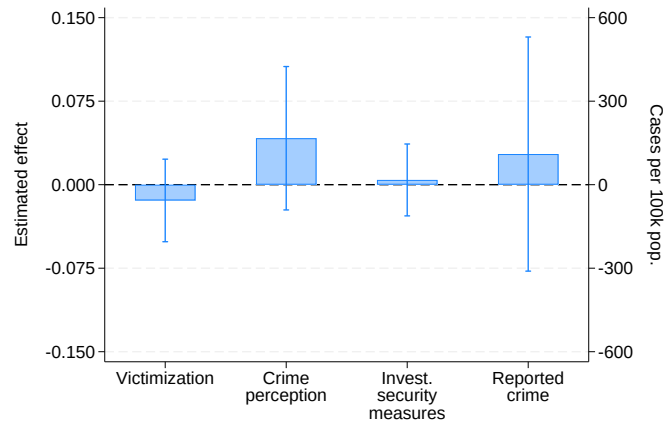
Although we cannot fully separate the beat policing component from the resource component of the reform—resources may have been important to enable effective implementation of the program, or may explain part of the effects in some municipalities—the fact that municipalities with resource increases similar to those experienced by control municipalities still display sizeable effects indicates that the effects of *Plan Cuadrante* are not driven entirely by resources.^{48,49}

⁴⁷We test for the joint significance of the pre-treatment coefficients between the two groups, separately for the split based on the increase in police officers and the split based on the increase in patrol vehicles. The resulting p-values are, respectively, 0.823 and 0.048 for victimization, 0.059 and 0.379 for crime perception, 0.820 and 0.157 for investment in security measures, and 0.213 and 0.419 for reported crime, as well as 0.616 and 0.444 for police officers and 0.405 and 0.157 for patrol vehicles.

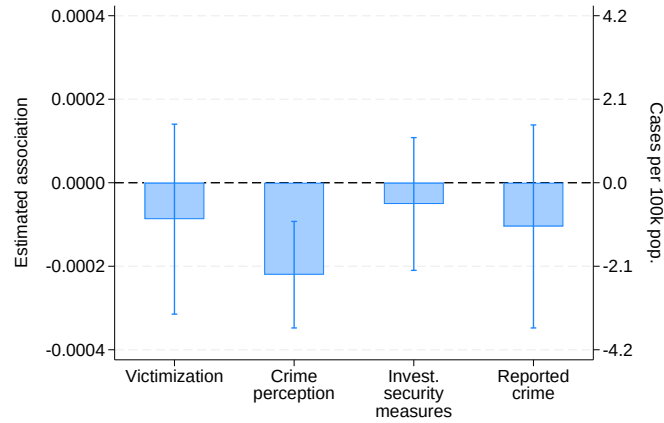
⁴⁸We also re-estimate the results in Figure 2 but controlling for police resources. Online Appendix Table B.3 shows that the magnitude of the estimated effects remains similar for most variables, although the statistical significance is reduced in some estimations. This likely reflects a mechanical feature of the Callaway and Sant’Anna (2021) estimation procedure: control variables are not incorporated parametrically, but are instead used to construct the comparison group via outcome regression or propensity-score weighting, so the resulting estimation uncertainty propagates to the ATT standard errors. Note, in addition, that police resources are a bad control—i.e., at least part of their variation is a result of the adoption of *Plan Cuadrante*—so these results should be interpreted cautiously.

⁴⁹Table B.4 in Online Appendix B examines which factors are associated with a larger increase in police resources upon the adoption of *Plan Cuadrante*.

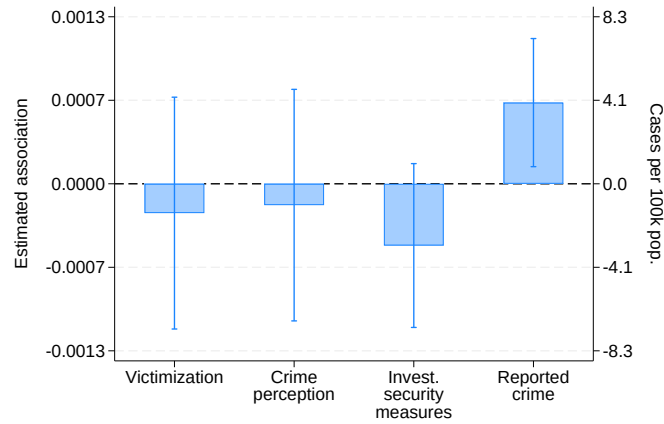
Figure 5: Police Resources and Public Safety



a Effects of *Seguridad Ciudadana*



b Association between police officers and effectiveness



c Association between patrol vehicles and effectiveness

Notes: Panel (a) reports the pooled effect of *Seguridad Ciudadana*, a program that expanded local security personnel to support patrolling and deterrence but did not replicate the deployment-and-strategy features of *Plan Cuadrante*. Panels (b) and (c) report within-municipality associations between police resources and police-effectiveness outcomes among the 41 municipalities of the Santiago Metropolitan Region, all of which had adopted *Plan Cuadrante* by 2000. The regressions include municipality and year fixed effects. Standard errors are clustered at the municipality level using the wild bootstrapping procedure.

To further investigate whether resource increases alone can explain our results, we assess the effects of *Seguridad Ciudadana*, a program that expanded local security personnel to support patrolling and deterrence, but did not replicate the core deployment-and-strategy features of *Plan Cuadrante*. Using a difference-in-differences specification following Callaway and Sant’Anna (2021), panel (a) of Figure 5 shows no significant effects of these resource changes on police effectiveness. Further details about this exercise and the corresponding event-study estimates are reported in Online Appendix B and Online Appendix Figure B.4.

Finally, we study whether variation in police resources in municipalities that had already adopted *Plan Cuadrante* improves public safety and security perceptions. Using ENUSC data and information on resources for 2006–2012 in the 41 municipalities of the Santiago Metropolitan Region—all of which had adopted *Plan Cuadrante* by 2000—we exploit variation in officers and patrol vehicles that is unrelated to program adoption. Panels (b) and (c) of Figure 5 show the association between police resources and our main outcomes, conditional on municipality and year fixed effects. With one exception—the association between police officers and crime perception—none of these associations is statistically significant at conventional confidence levels.⁵⁰

Taken together, these results suggest that resource increases alone do not account for the effects of *Plan Cuadrante*. We next turn to evidence that speaks more directly to the beat-policing components of the reform.

Panel (a) of Figure 4 shows that, following the adoption of *Plan Cuadrante*, households are much more likely to report increased police presence in their neighborhoods. The share of households perceiving increased police presence rises by 35 percentage points (156%), the share reporting that police pass by their house daily rises by 22 percentage points (59%), the share reporting never seeing police patrols near their house falls by 10 percentage points (57%), and complaints about lack of police surveillance fall by around 5 percentage points (20%). This evidence is consistent with both larger resources and the new deployment model. Yet the latter estimate—the only police-presence outcome for which we can separately estimate effects in municipalities with small and large resource increases—is similar in magnitude across the two groups, which suggests that the increase in police presence is not explained only by larger resource increases (see Online Appendix Figure B.3).⁵¹ More importantly, the broader pattern of results suggests that *Plan Cuadrante* did more than simply increase patrolling intensity. Existing work shows that police presence can reduce crime while officers are present, but that its overall effects may be attenuated when crime is displaced across places or time windows (Mastrobuoni, 2019; Blanes-i Vidal and Mastrobuoni, 2018). Our results point to something broader. As discussed in Section 4, we find actual declines in victimization and reported crime, together with no evidence of spillovers to neighboring municipalities. This pattern is more consistent with officers accumulating local knowledge and adapting prevention strategies to local conditions under stable territorial assignment than with a purely mechanical increase in patrol intensity.

Panel (b) of Figure 4 reinforces this interpretation. Households in early-adopting municipalities

⁵⁰Online Appendix Figure B.5 reports the same exercise and also shows similar results for the broader set of municipalities that adopted *Plan Cuadrante* before 2006 and never-treated municipalities (red bars); see Online Appendix B for further details.

⁵¹We cannot estimate heterogeneous effects for other police-presence outcomes, such as the reporting of police passing by or the share of households perceiving an increase in police presence, because those variables are available only in ENUSC rounds 2003, 2005, and 2006, while resource data start in 2006.

report better police performance across multiple dimensions. They are 8.1 percentage points (16%) more likely to rate police performance as good or very good, 7.6 percentage points (11%) more likely to describe police as efficient, 5 percentage points (6%) more likely to report that police are helpful, and 4.9 percentage points (7%) more likely to report that police are well informed. They are also 3.9 percentage points (10%) less likely to perceive police as corrupt, 7.9 percentage points (21%) less likely to believe police fail to control crime, and 7.6 percentage points (22%) less likely to report that police do not perform their duties effectively.⁵²

The administrative evidence points in the same direction. Panel (c) of Figure 4 shows that arrests per 100,000 inhabitants increase by 48 (14%) in our main estimates, even as crime declines overall.⁵³ This effect is also significant among municipalities with resource increases similar to those in the control group (see Online Appendix Figure B.3). Consistent with this pattern, panel (d) of Figure 4 shows that the arrest rate per reported crime also increases by 1.7 percentage points (10%), suggesting that a higher share of crimes reported to the police are being solved. Because these gains appear even where resources did not increase more than in control municipalities, they are difficult to reconcile with a pure resource story and are more naturally interpreted as improvements in police effectiveness under the deployment and strategy changes introduced by *Plan Cuadrante*. The corresponding event-study estimates for arrests, the arrest rate per reported crime, and trust in the police in Figure 4 confirm that these changes emerge only after adoption.

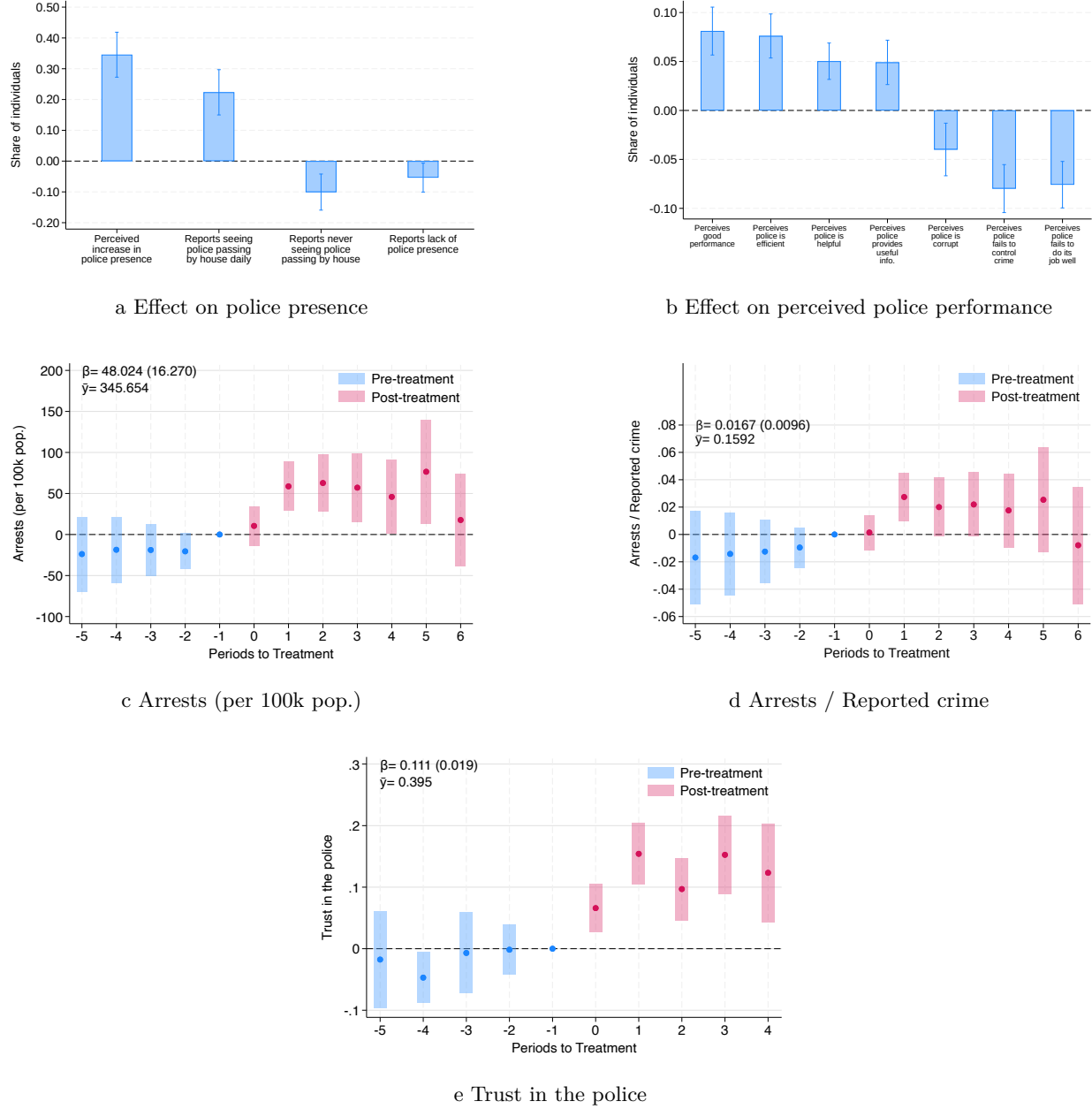
Finally, panel (e) of Figure 4 shows a sharp increase in trust in the police in our main specification (Section 4), consistent with one of the objectives of beat policing, namely bringing officers closer to the community and allowing them to accumulate local knowledge through repeated interaction. Online Appendix Figure B.3 shows that this effect remains sizeable both in municipalities that experienced resource increases similar to the control group and in those with larger increases.

Overall, although resources were likely important for implementation and may explain part of the effects in some municipalities, the results in this section indicate that resource changes alone are not enough to explain our findings. The evidence is more consistent with a combination of deployment changes and the strategy adjustments enabled by local knowledge accumulation under permanent assignment to quadrants.

⁵²Information on police performance is only available in ENUSC round 2005. Using this round, we compare perceived police performance in municipalities that adopted *Plan Cuadrante* in 2005 or earlier and in municipalities adopting after 2005. Figure B.6 in Online Appendix B shows that the number of years of exposure to *Plan Cuadrante* is positively correlated with perceived police performance. While these exercises on perceived police performance should be interpreted with caution, we show in the same appendix that the timing of future adoption is not correlated with perceived police performance in 2005 for municipalities that had not yet adopted the program by 2005.

⁵³As with reported crime, this event study is restricted to 5 pre-treatment and 7 post-treatment periods for symmetry with the ENUSC-based analysis; Online Appendix Figure C.8 reports the results for arrests using the full set of pre-treatment and post-treatment periods.

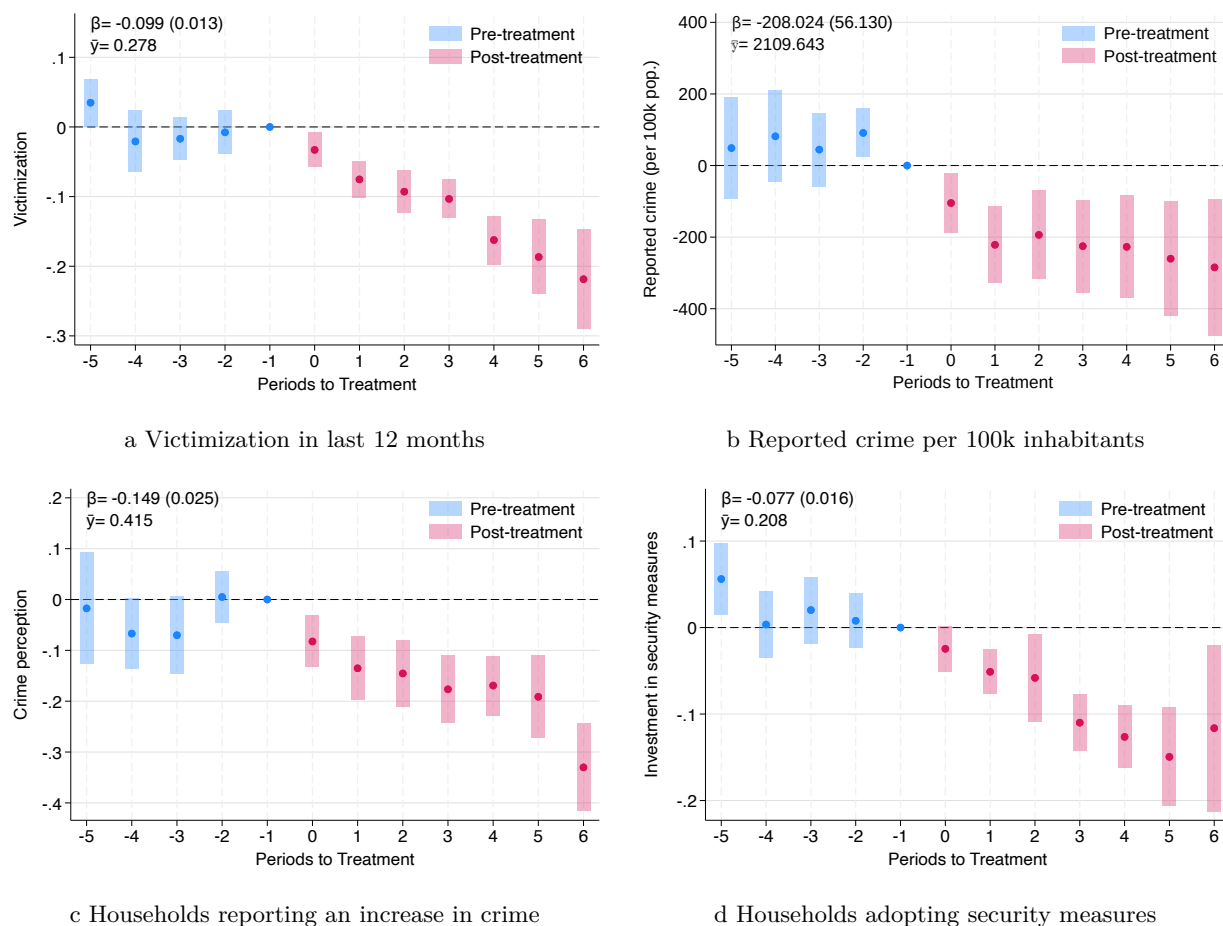
Figure 6: *Plan Cuadrante* and Changes in Police Performance



Notes: Panel (a) reports the effects on perceived police presence. Panel (b) reports perceived police performance in 2005, comparing municipalities where *Plan Cuadrante* was already implemented before 2006 with municipalities where it was implemented after 2005. Panels (c), (d), and (e) report dynamic event-study estimates for arrests per 100,000 inhabitants, the ratio of arrests to reported crime, and trust in the police. The estimation follows Callaway and Sant'Anna (2021). The graphs display estimated coefficients and 95% confidence intervals. Standard errors are clustered at the municipality level using the wild bootstrapping procedure.

Other figures and tables referenced above:

Figure 2: Effects of *Plan Cuadrante* on Crime Outcomes and Security Perceptions



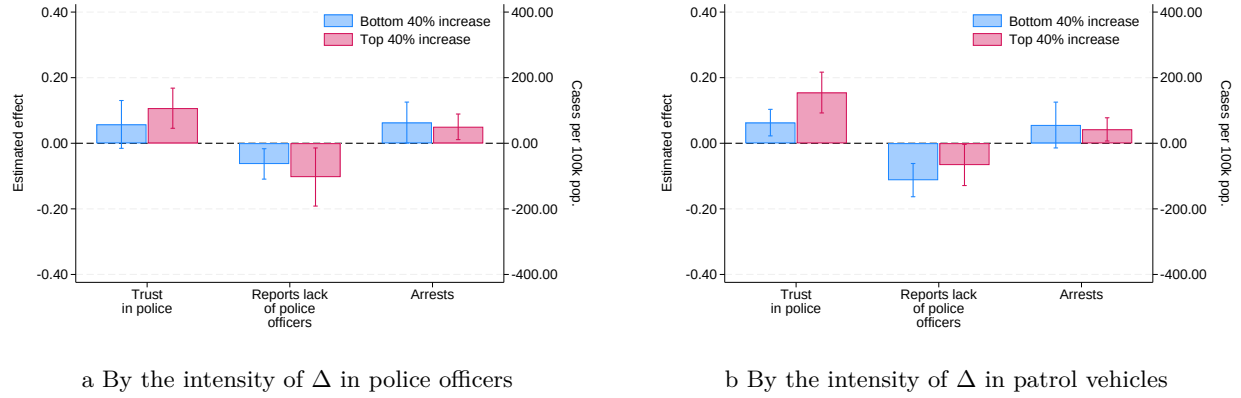
Notes: This figure shows the dynamic effects of the *Plan Cuadrante* on different measures of police effectiveness. The estimation is conducted using the procedure developed in Callaway and Sant'Anna (2021) for difference-in-differences with staggered treatment adoption. The dots in the figure represent the estimated coefficients and the bars behind them 95% confidence intervals. Standard errors are clustered at the municipality level using the wild bootstrapping procedure. The pooled effect and the mean outcome before the introduction of the treatment is also presented in each panel. Panel (a) reports the effect of *Plan Cuadrante* on the probability of household victimization in the last 12 months using the ENUSC survey. Panel (b) reports the effect on the number of crimes reported to the police in the municipality per 100,000 inhabitants using administrative data. Panel (c) reports the effect on the probability of perceiving an increase in crime using the ENUSC survey. Panel (d) reports the effect on the probability of adopting security measures in the last twelve months using the ENUSC survey. Panel (a) includes 93,041 observations; panel (b), 4,231; panel (c), 89,578; and panel (d), 93,041 observations.

Table B.3: Effects of *Plan Cuadrante* Controlling for Police Resources

	(1)	(2)	(3)	(4)	(5)
<i>Panel A. Victimization</i>					
ATT	-0.099*** (0.013)	-0.077*** (0.013)	-0.086** (0.043)	-0.094*** (0.028)	-0.087* (0.053)
N	93,041	53,973	53,553	72,017	53,553
<i>Panel B. Crime perception</i>					
ATT	-0.149*** (0.025)	-0.140*** (0.032)	-0.112** (0.057)	-0.082** (0.033)	-0.111 (0.106)
N	89,578	52,160	51,746	69,434	51,746
<i>Panel C. Investment in security measures</i>					
ATT	-0.077*** (0.016)	-0.066*** (0.019)	-0.109*** (0.041)	-0.081*** (0.019)	-0.168*** (0.064)
N	93,041	53,973	53,553	72,017	53,553
<i>Panel D. Reported crime (per 100k pop.)</i>					
ATT	-208.024*** (56.130)	-157.010*** (56.224)	29.230 (147.261)	-261.439 (170.954)	-245.555 (165.221)
N	4,231	2,963	1,911	2,248	1,911
Sample	Full	Resource yrs	Resource yrs	Resource yrs	Resource yrs
Police officers (per 100k pop.)			✓		✓
Patrol vehicles (per 100k pop.)				✓	✓

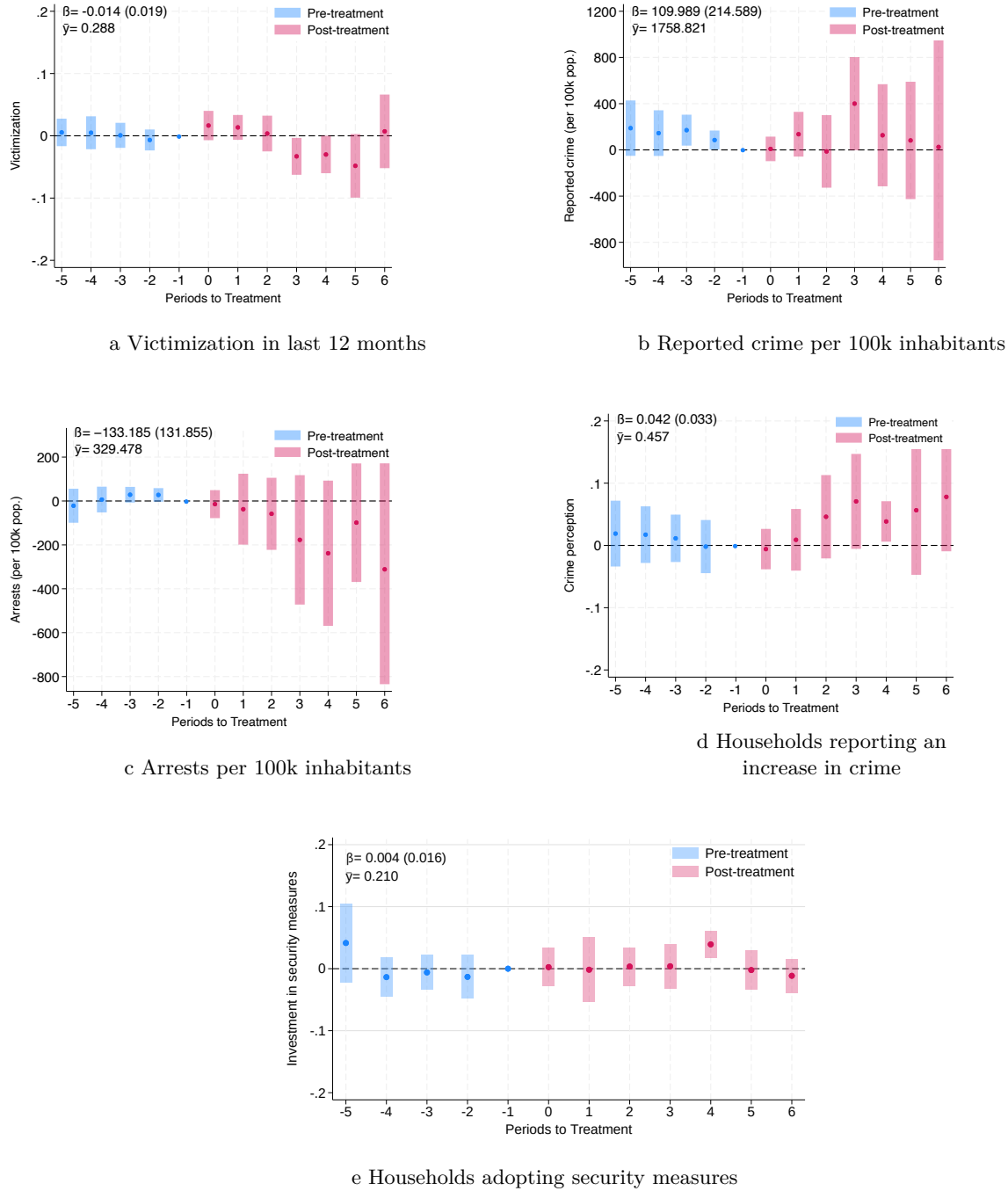
Note: ATT estimates of the effect of *Plan Cuadrante* on each outcome. Column (1) is the baseline (full estimation sample, no controls) and reproduces the headline estimate of the corresponding estimator. Column (2) restricts the estimation to the years in which police-resource data are available (from 2006 onward). No control variables are used in this estimation. Columns (3)–(5) add time-varying controls for police resources per 100,000 inhabitants: officers only; patrol vehicles only; and officers and patrol vehicles jointly (entered as separate regressors). Patrol vehicles are measured, per 100,000 inhabitants, as all patrol automobiles (radiopatrulla, auto patrullero, auto policial, automovil, station wagon, jeep and camioneta) plus motorcycles. Standard errors in parentheses; row N reports the estimation-sample size. Estimation follows Callaway and Sant’Anna (2021); standard errors clustered at the municipality level using wild bootstrapping procedures. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Figure B.3: Effects of *Plan Cuadrante* by Resource Increase: Trust, Complaints, and Arrests



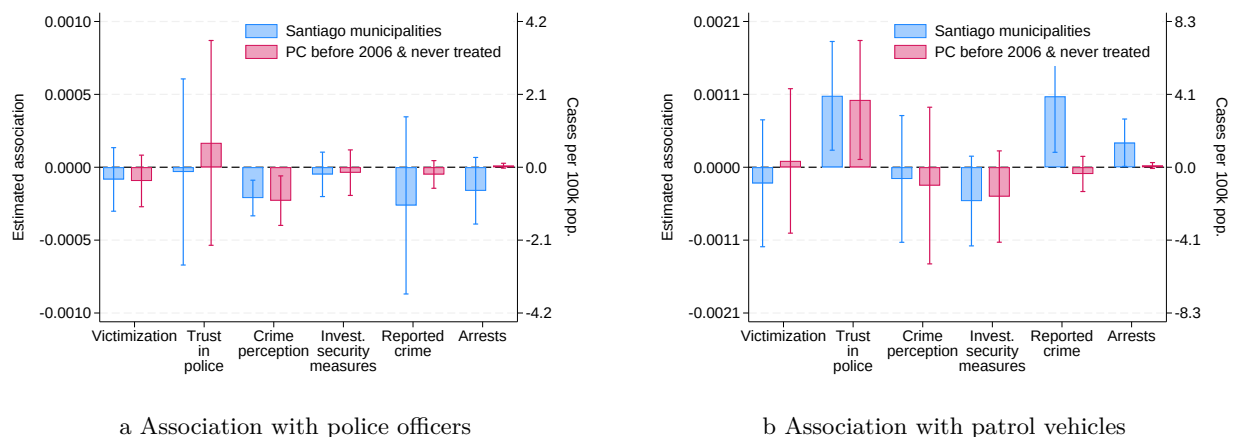
Notes: This figure complements Figure 4 by reporting, for the three outcomes not shown there, the effect of *Plan Cuadrante* estimated separately for municipalities in the bottom 40% (blue) and top 40% (red) of the resource-increase distribution. Panel (a) splits municipalities by the increase in police officers and panel (b) by the increase in patrol vehicles. In each panel, arrests (administrative data, in cases per 100,000 inhabitants) are read on the right axis; the other two outcomes, from the ENUSC survey (trust in the police, and reports about the lack of police officers), are read on the left axis. Effects are estimated using the difference-in-differences estimator for staggered treatment adoption developed in Callaway and Sant'Anna (2021); bars are point estimates and whiskers 95% confidence intervals, with standard errors clustered at the municipality level using the wild bootstrapping procedure.

Figure B.4: Effects of *Seguridad Ciudadana* on Crime Outcomes and Security Perceptions



Notes: This figure shows the dynamic effects of the *Seguridad Ciudadana* program on different measures of crime and security perceptions. The estimation is conducted using the procedure developed in Callaway and Sant'Anna (2021) for difference-in-differences with staggered treatment adoption. The dots in the figure represent the estimated coefficients and the bars behind them 95% confidence intervals. Standard errors are clustered at the municipality level using wild bootstrapping. The pooled effect and the mean outcome before the introduction of the treatment for the outcome is also presented in each panel. Panel (a) reports the effect of *Seguridad Ciudadana* on the probability of household victimization in the last 12 months using the ENUSC survey. Panel (b) reports the effect on the number of crimes reported to the police in the municipality per 100,000 inhabitants using administrative data. Panel (c) reports the effect on the number of arrests in the municipality per 100,000 inhabitants using administrative data. Panel (d) reports the effect on the probability of perceiving an increase in crime using the ENUSC survey. Panel (e) reports the effect on the probability of adopting security measures in the last twelve months using the ENUSC survey. Panel (a) includes 210,102 observations; panel (b), 2,234; panel (c), 1,908; panel (d), 203,078; and panel (e), 74,316 observations.

Figure B.5: Association between Police Resources and Police Effectiveness (No *Plan Cuadrante* Variation)



Notes: The figure shows the estimated within-municipality association between police resources (per 100,000 inhabitants) and the main outcomes, in municipalities where *Plan Cuadrante* does not vary over the sample period. Estimates come from OLS regressions of each outcome on the resource measure with municipality and year fixed effects; standard errors are clustered at the municipality level. Blue bars use the 41 metropolitan municipalities of Santiago, all of which adopted *Plan Cuadrante* in 2000; red bars use the broader set of municipalities that adopted *Plan Cuadrante* before 2006 together with the never-treated municipalities surveyed in ENUSC (2006–2012). Reported crime and arrests (administrative data, in cases per 100,000 inhabitants) are read on the right axis; all other outcomes, from the ENUSC survey (victimization, trust in the police, crime perception, and investment in security measures), are read on the left axis. Panel (a) reports associations with the number of police officers; panel (b) with the number of patrol vehicles. Bars are point estimates with 95% confidence intervals.

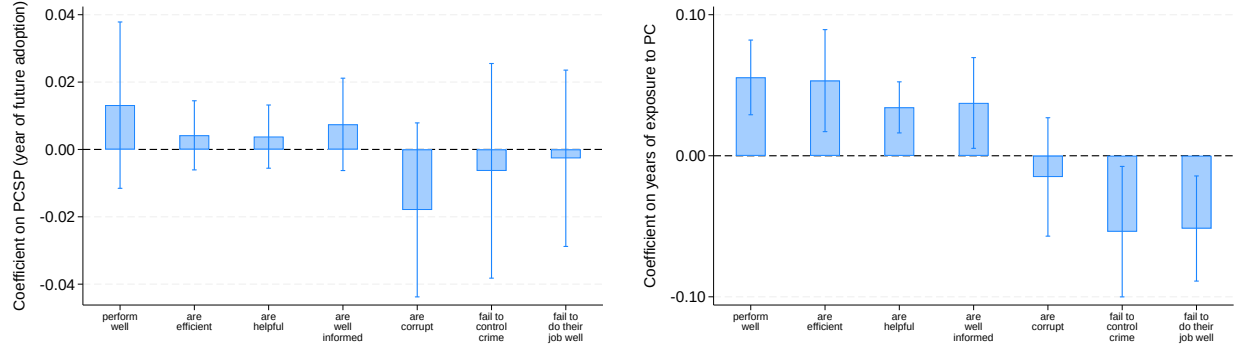
Table B.4: Predictors of the Increase in Police Resources under *Plan Cuadrante*

	Administrative sample		ENUSC sample	
	Police officers (1)	Patrol vehicles (2)	Police officers (3)	Patrol vehicles (4)
Dep. var: avg. annual increase in resources (share of pre-adoption level)				
Reported crime (per 100k pop.)	−0.203** (0.099)	−0.199** (0.097)	−0.319* (0.182)	0.272* (0.137)
Arrests (per 100k pop.)	0.099 (0.136)	−0.041 (0.129)	0.003 (0.186)	0.297 (0.223)
Victimization			−0.017* (0.010)	−0.025* (0.013)
Crime perception			0.005 (0.017)	0.008 (0.014)
Trust in the police			0.024 (0.018)	0.027 (0.019)
Investment in security measures			0.004 (0.010)	−0.003 (0.009)
Police officers (per 100k pop.)	−0.207 (0.155)	0.155 (0.129)	−0.888*** (0.154)	0.571*** (0.136)
Patrol vehicles (per 100k pop.)	−0.129 (0.129)	−0.139 (0.126)	0.537** (0.212)	−0.507*** (0.169)
Log population	−0.529*** (0.145)	−0.203* (0.112)	−0.193 (0.189)	−0.626** (0.269)
Poverty rate (2004)	0.170 (0.113)	0.308** (0.126)	0.027 (0.108)	−0.017 (0.108)
Observations	65	65	8,781	8,781
R-squared	0.342	0.282	0.614	0.545

Note: OLS regressions of the average annual increase in police officers (columns 1 and 3) and patrol vehicles (columns 2 and 4) per 100,000 inhabitants, from the year before adoption of *Plan Cuadrante* to 2012. As in panels (a) and (b) of Figure 4, the increase is expressed as a share of the resources available the year before adoption. Columns 1–2 use the administrative sample at the municipality level; columns 3–4 use ENUSC survey respondents interviewed before 2007 (survey outcomes at the respondent level, other regressors as pre-2006 municipality aggregates). All variables are standardized. Standard errors clustered by municipality in parentheses.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

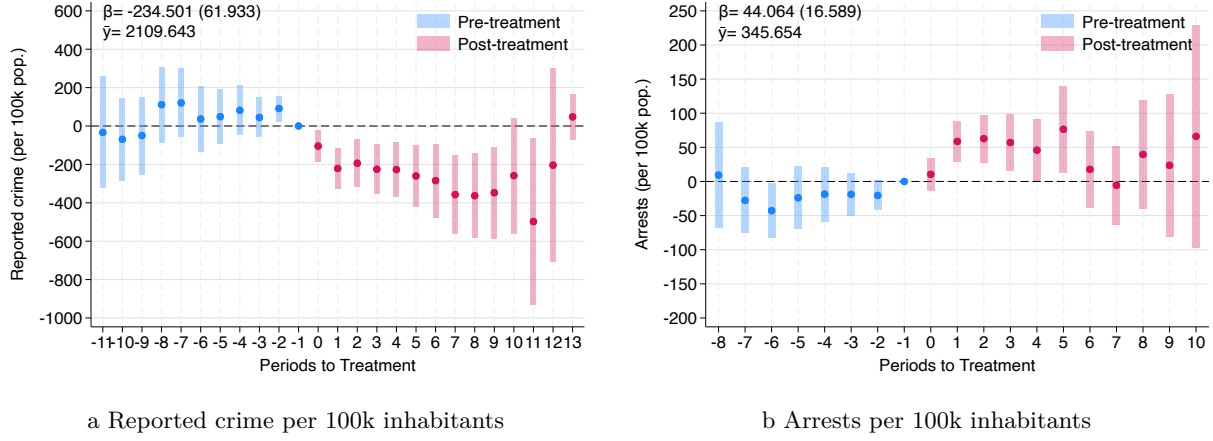
Figure B.6: Timing of Adoption of *Plan Cuadrante* and Perceived Police Performance



a Year of adoption and police performance before adoption b Years exposed to *Plan Cuadrante* (dose response)

Notes: The figure considers seven measures of police performance: the share of respondents who perceive police performance as good, and who perceive the police to be efficient, helpful, well informed, corrupt, failing to control crime, and failing to do their job well. Using the sample of municipalities that adopted *Plan Cuadrante* after 2005, panel (a) examines whether the future timing of adoption of *Plan Cuadrante* is correlated with perceived police performance in 2005, before the introduction of the program. Panel (b) reestimates the same comparison as Figure 4, replacing the binary adoption indicator with the municipality's number of years of exposure to *Plan Cuadrante* as of 2005. In both panels, each bar plots the coefficient from a separate regression of one performance measure. The graphs display the estimated coefficients and 95% confidence intervals.

Figure C.8: Effects of *Plan Cuadrante* on Reported Crime and Arrests Using the Full Set of Post-treatment and Pre-treatment Periods



Notes: This figure shows the dynamic effects of *Plan Cuadrante* on the number of crimes reported to the police in the municipality per 100,000 inhabitants (panel a) and on the number of arrests in the municipality per 100,000 inhabitants (panel b), using administrative data. The estimation is conducted using the procedure developed in Callaway and Sant'Anna (2021) for difference-in-differences with staggered treatment adoption. The dots in the figure represent the estimated coefficients and the bars behind them 95% confidence intervals. Standard errors are clustered at the municipality level using the wild bootstrapping procedure. The pooled effect and the mean outcome before the introduction of the treatment are also presented in each panel. Unlike Figure 2 in the main text, we do not cap the event study to the same 5 pre-treatment and 6 post-treatment periods used for symmetry with the ENUSC-based analysis. Instead, we exploit the full set of post-treatment and pre-treatment periods. Panel (a) includes 4,371 observations and panel (b) 3,372 observations.

Comment R1.9

Minor points:

1. Sample selection: I understand the final sample selection is driven by data availability, but sometimes I was confused by when the policy started and what years are covered by each data source. For example, survey data appear to be available from 2003, and administrative crime records also start in 2003 and run until 2018. Why, then, does the analysis stop in 2012? If this is due to survey availability, that should be made explicit, while the administrative data could still be shown descriptively for the later years. In addition, what distinguishes the municipalities included in the analysis from those excluded? How do early adopters compare to late adopters? For both the survey data and the police records, it would be useful to provide a comparison of characteristics between included and excluded municipalities to give readers confidence that the results are not being driven by sample selection.

Response to R1.9:

Thank you very much for the opportunity to clarify this point. This is related to the availability of data by year and municipality across our two main data sources, and to how the Callaway and Sant’Anna (2021) estimator is constructed. When the control group is formed from not-yet-treated units, the estimator only uses data up to the year in which the last treated unit in the data adopts the treatment.

Most outcomes in the paper rely on information from the ENUSC survey, which covers 101 urban municipalities. All of these municipalities were eventually treated except one. Because ENUSC was first conducted in 2003 and not administered in 2004, our main analytical sample includes the 46 urban municipalities that adopted *Plan Cuadrante* between 2005 and 2012—the year in which the last municipality surveyed in ENUSC adopted the program—plus the one never-treated municipality. Even if the ENUSC survey were available after 2012, using these data in the estimation would mean relying only on this single never-treated municipality as the control group for all subsequent periods, which would make the estimates unreliable. We therefore limit the study period in the main analysis to 2003–2012.

Regarding the outcomes constructed from administrative records: these are provided by the Undersecretary for Crime Prevention and are available yearly for all Chilean municipalities—reported crimes from 2002 to 2016, and arrests from 2005 to 2016. This comprehensive coverage results in a much larger number of never-treated units, allowing for a longer estimation period that includes all the data period and municipalities adopting the program up to 2013, when the last Chilean municipality joined *Plan Cuadrante*. In total, we use an analytical sample of 292 municipalities for reported crime and 281 for arrests. For simplicity and symmetry with the ENUSC-based analysis, our main specification nonetheless restricts the event study to 5 pre-treatment and 7 post-treatment periods. We show in Figure C.8 in the updated manuscript that the results are very similar when the full set of pre- and post-adoption periods is used instead.

Below we reproduce the relevant paragraphs from the Data subsection of the updated manuscript:

We combine administrative and survey data from *Carabineros*, the Undersecretary for Crime Prevention, and the National Institute of Statistics (INE).

Carabineros provided information on *Plan Cuadrante* implementation, which we use to define treatment status across municipalities over time. The National Institute of Statistics granted

access to the National Survey on Security and Crime (ENUSC), a repeated cross-section first conducted in 2003 and administered annually between October and December since 2005. The survey covers more than 25,000 households across 101 urban municipalities, collecting information on crime victimization, perceptions, trust in police, and household prevention measures over the previous 12 months.⁵⁴

Our empirical approach excludes always-treated municipalities (Callaway and Sant’Anna, 2021). Given ENUSC’s first implementation in 2003 and absence in 2004, our main sample includes households from 46 urban municipalities that adopted *Plan Cuadrante* between 2005 and 2012 plus one never-treated municipality.⁵⁵ The treated municipalities in this analytical sample account for approximately 27% of Chile’s population.

The Undersecretary for Crime Prevention provided data on the number of reported crimes (2002–2016) and arrests (2005–2016) for all municipalities. This comprehensive coverage allows us to expand the analytical sample: for reported crime, we include all municipalities adopting *Plan Cuadrante* after 2002 together with never-treated municipalities, for a total of 292 municipalities; for arrests, the corresponding sample includes all post-2005 adopters and never-treated municipalities, for a total of 281 municipalities. The treated municipalities in these analytical samples account for over half of Chile’s population.

The study period differs across the two data sources, as each covers a different time span and a different set of municipalities. For the ENUSC-based analysis, it is restricted to 2003–2012. The Callaway and Sant’Anna (2021) estimator uses not-yet-treated and never-treated municipalities as the comparison group. Since only one municipality in the survey sample is never treated, extending the analysis beyond 2012—the last year in which a municipality surveyed in ENUSC adopts the program—would leave only that single never-treated unit as the comparison group for subsequent periods, making the estimates unreliable. The administrative data, by contrast, cover a large number of never-treated municipalities that can serve as controls throughout the full data period, allowing us to exploit the complete administrative coverage—2002–2016 for reported crime and 2005–2016 for arrests—and to include all municipalities adopting the program from 2003 to 2013, when the last municipality joins *Plan Cuadrante*.

Together, the treated municipalities in our two analytical samples represent substantial shares of the Chilean population—approximately 27% in the ENUSC-based analysis and over half in the administrative data analysis—and are geographically distributed across the country rather than clustered in a single region. Tables A.2 and A.3 in Online Appendix A and Table B.1 and Figure B.1 in Online Appendix B provide a detailed characterization of the treated municipalities in each analytical sample relative to the average Chilean municipality.

Note the following footnote to Section 4 in the main manuscript:

Online Appendix Figure C.8 reports the results for reported crime and arrests using the full set of post-treatment and pre-treatment periods, which are very similar to those presented here.

and also the following subsection to Online Appendix C, “Effects of *Plan Cuadrante* Using the Full Set of Post-treatment and Pre-treatment Periods”:

⁵⁴The survey is administered to a randomly selected household member who must: a) live in the household, b) be 15 years or older, c) not be a domestic worker, and d) not have any disability preventing survey comprehension.

⁵⁵This sample excludes municipalities adopting *Plan Cuadrante* in 2013 as they are not covered by ENUSC. Also, there is only one never treated municipality covered by ENUSC.

For symmetry with the ENUSC-based analysis, the event studies for reported crime and arrests reported in the main text are restricted to 5 pre-treatment and 7 post-treatment periods. Because the administrative data are available for the full set of Chilean municipalities and therefore include a much larger number of never-treated municipalities than the ENUSC sample, we can estimate these effects over a longer window without exhausting the pool of valid comparison units. Figure C.8 reports the effects of *Plan Cuadrante* on reported crime and arrests using the full set of post-treatment and pre-treatment periods rather than capping the event study to the 5 pre-treatment and 7 post-treatment periods used in the main specification. The results are very similar to those presented in Section 4.1.

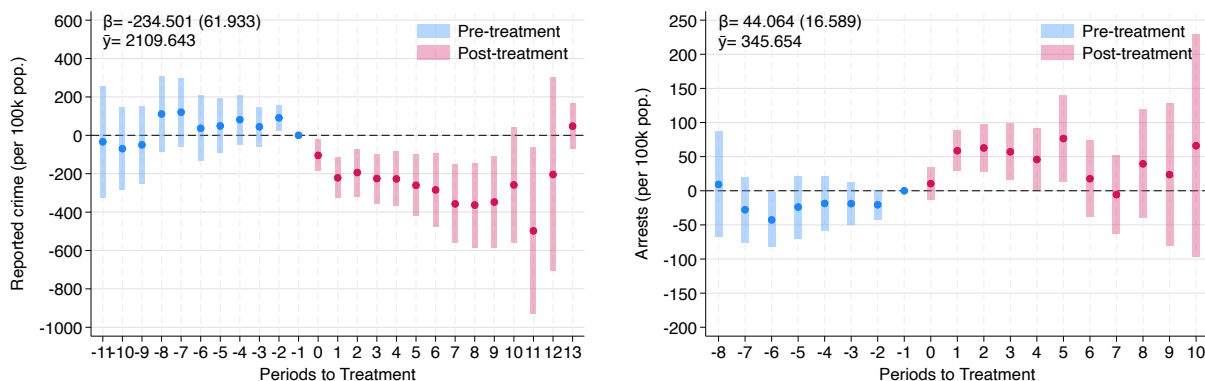
Finally, we have added the following subsection, “Characterization of the Analytical Samples,” to Online Appendix B, comparing the characteristics of the municipalities included in each analytical sample with those excluded from it or always treated:

The treated municipalities in our analytical samples are geographically distributed across the country. As shown in Figure B.1, the 46 treated municipalities in the ENUSC-based sample span Chile from north to south, covering a wide range of geographic and socioeconomic contexts, and together account for approximately 27% of Chile’s population. The 96 treated municipalities in the administrative data sample provide even broader national coverage, together accounting for over half of Chile’s population.

Table B.1 characterizes the treated municipalities in each analytical sample relative to all Chilean municipalities (Column 1), all *Plan Cuadrante* municipalities (Column 2), the always-treated municipalities excluded from our estimation samples (Column 3), and never-treated municipalities (Column 6). Columns (4) and (5) report characteristics for the treated municipalities in the ENUSC and administrative data analytical samples respectively. On several dimensions our samples broadly resemble the overall Chilean population: the poverty rate of the 46 treated municipalities in the ENUSC sample (21.7%) and the 96 treated municipalities in the administrative data sample (24.0%) are both close to the national average of 23.7%, and the crime rates of the treated ENUSC municipalities closely track the average across all *Plan Cuadrante* municipalities (2,202 vs. 2,132 per 100,000 inhabitants). They do differ on other dimensions—particularly crime rates relative to all Chilean municipalities and population density—but both differences have natural explanations. Higher crime rates are expected because *Plan Cuadrante* exclusively targeted urban municipalities, where crime is more prevalent. The lower-than-average population density in our samples (215 and 135 inhabitants per km², vs. a national mean of 870) reflects the fact that the most densely populated *Plan Cuadrante* municipalities—those of the Santiago metropolitan area—were the earliest adopters of the program and are therefore excluded from our analytical samples as always-treated.

Other figures and tables referenced above:

Figure C.8: Effects of *Plan Cuadrante* on Reported Crime and Arrests Using the Full Set of Post-treatment and Pre-treatment Periods



a Reported crime per 100k inhabitants

b Arrests per 100k inhabitants

Notes: This figure shows the dynamic effects of *Plan Cuadrante* on the number of crimes reported to the police in the municipality per 100,000 inhabitants (panel a) and on the number of arrests in the municipality per 100,000 inhabitants (panel b), using administrative data. The estimation is conducted using the procedure developed in Callaway and Sant'Anna (2021) for difference-in-differences with staggered treatment adoption. The dots in the figure represent the estimated coefficients and the bars behind them 95% confidence intervals. Standard errors are clustered at the municipality level using the wild bootstrapping procedure. The pooled effect and the mean outcome before the introduction of the treatment are also presented in each panel. Unlike Figure 2 in the main text, we do not cap the event study to the same 5 pre-treatment and 6 post-treatment periods used for symmetry with the ENUSC-based analysis. Instead, we exploit the full set of post-treatment and pre-treatment periods. Panel (a) includes 4,371 observations and panel (b) 3,372 observations.

Table A.2: Summary Statistics – ENUSC sample

	Mean	SD	Min	Max	N
<i>Household</i>					
Socioeconomic status	3.58	0.73	1	5	99,298
Victimization	0.26	0.44	0	1	99,298
Crime perception (neighborhood)	0.38	0.49	0	1	95,598
Crime perception (municipality)	0.66	0.48	0	1	96,794
Crime perception (country)	0.79	0.41	0	1	98,525
Trust in the police	0.44	0.50	0	1	51,445
Investment in security measures	0.21	0.40	0	1	99,298
<i>Plan Cuadrante</i> (year)	2007	1.85	2005	2012	97,847
<i>Household member interviewed</i>					
Female	0.55	0.50	0	1	99,298
Age	43.87	17.95	15	99	99,298
Primary school attendance	0.98	0.14	0	1	99,298
Secondary school attendance	0.72	0.45	0	1	99,298
University attendance	0.20	0.40	0	1	99,298

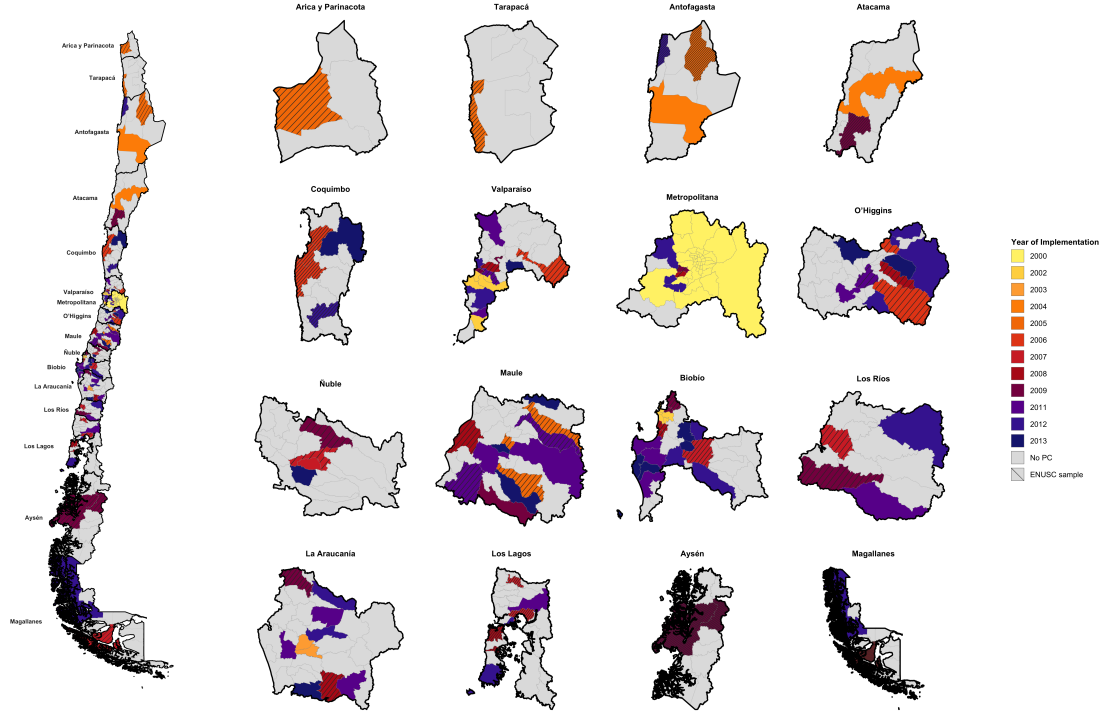
Note: This table presents summary statistics from the ENUSC sample, covering the 47 municipalities used in the survey-based analysis (46 that adopted *Plan Cuadrante* between 2005 and 2012 plus one never-treated municipality). Socioeconomic status is a 1 to 5 measure produced by ENUSC surveys and based on self-reported socioeconomic strata, being 1 the highest ranked and 5 the lowest. Crime perception is equal to 1 if the interviewed person reports that crime has increased in the last 12 months. The *Plan Cuadrante* (year) row is computed over treated municipalities only.

Table A.3: Summary Statistics – Administrative municipality level sample

	Mean	SD	Min	Max	N
<i>Panel A. Reported-crime sample (292 municipalities)</i>					
Reported crime (per 100k pop.)	1,732.39	1,036.68	0.00	11,434.40	4,371
Arrests (per 100k pop.)	332.07	271.75	0.00	1,712.30	3,504
Poverty rate (%)	25.74	9.72	5.65	58.33	3,720
Pop. density (pop./km ²)	63.51	140.47	0.09	1,109.58	4,065
<i>Plan Cuadrante</i> (year)	2010	2.82	2003	2013	1,440
<i>Panel B. Arrests sample (281 municipalities)</i>					
Reported crime (per 100k pop.)	1,684.74	1,014.67	0.00	11,434.40	4,209
Arrests (per 100k pop.)	313.21	251.23	0.00	1,712.30	3,372
Poverty rate (%)	26.01	9.72	5.65	58.33	3,570
Pop. density (pop./km ²)	57.90	127.30	0.09	1,109.58	3,900
<i>Plan Cuadrante</i> (year)	2010	2.26	2006	2013	1,275

Note: The table reports municipality-level summary statistics for the two administrative estimation samples used in the paper. Panel A covers the 292 municipalities in the reported-crime sample (municipalities adopting *Plan Cuadrante* from 2003 onward plus never-treated); Panel B the 281 municipalities in the arrests sample (adopters from 2006 onward plus never-treated — municipalities adopting in 2005 are excluded because arrest data begin that year, leaving no pre-treatment period). Reported crime and arrests are measured per 100,000 inhabitants over 2002–2016 and 2005–2016, respectively. Minimum reported crime and arrests of zero correspond to tiny, remote never-treated comunas (e.g., Juan Fernández, Cabo de Hornos). The *Plan Cuadrante* (year) row is computed over treated municipalities only. Data from the Undersecretary for Crime Prevention.

Figure B.1: Geographic Distribution of *Plan Cuadrante*



Notes: This figure shows the spatial rollout of *Plan Cuadrante* across Chilean municipalities. The national map (left) is displayed alongside the 16 regional maps (right) for closer inspection. The color of each treated municipality indicates its year of program adoption, ranging from 2000 (lightest) to 2013 (darkest), following the legend. Municipalities shown in light grey were never treated under PCSP. Diagonal hatching identifies the 46 treated municipalities in the analytical sample used to estimate the effects on ENUSC outcomes.

Table B.1: Baseline characteristics by subsample

	(1) All municipalities	PC municipalities				(6) Never- treated
		(2) All PC municipalities	(3) Always- treated	(4) Treated ENUSC	(5) Treated admin. data	
Reported crime (per 100k pop.)	1,650.23	2,132.29	2,437.25	2,202.06	1,976.59	1,282.47
Arrests (per 100k pop.)	267.62	442.08	606.01	459.05	358.11	134.11
Poverty rate (%)	23.73	20.50	14.88	21.73	23.96	26.82
Pop. density (pop./km ²)	870.28	1,904.14	4,593.32	215.52	134.50	27.02
Population	57,857.22	117,975.43	187,558.74	113,079.74	81,532.84	11,612.44
Municipalities	346	150	58	46	96	196

Note: Reported crime, poverty rate, population density, and population are measured as of 2003; arrests are measured as of 2005, the first year for which arrest data are available. Column (1) includes all municipalities in Chile. Column (2) includes all PC municipalities. Column (3) restricts to municipalities that adopted *Plan Cuadrante* before 2005, the always-treated in our sample, which are therefore excluded from our estimation samples (as depicted in Figure 1, panel (a)). Column (4) restricts to treated municipalities in the analytical sample using the ENUSC data. Column (5) restricts to treated municipalities included in the analytical sample using the administrative data. Column (6) includes never-treated municipalities.

Comment R1.10

2. Online Appendix C has the wrong reference on page 4.

Response to R1.10:

Thanks for spotting the mistake. We replaced with the correct reference in the updated manuscript.

References

- Adda, J., B. McConnell, and I. Rasul (2014). Crime and the depenalization of cannabis possession: Evidence from a policing experiment. *Journal of Political Economy* 122(5), 1130–1202.
- Ajzenman, N., P. Dominguez, and R. Undurraga (2023). Immigration, crime, and crime (mis) perceptions. *American Economic Journal: Applied Economics*.
- Ang, D., P. Bencsik, J. Bruhn, and E. Derenoncourt (2025, March). Community engagement with law enforcement after high-profile acts of police violence. *American Economic Review: Insights* 7(1), 124–42.
- Bardhan, P. (2002). Decentralization of governance and development. *Journal of Economic Perspectives* 16(4), 185–205.
- Basurto, M. P., P. Dupas, and J. Robinson (2020). Decentralization and efficiency of subsidy targeting: Evidence from chiefs in rural malawi. *Journal of Public Economics* 185, 104047.
- Besley, T. and S. Coate (2003). Centralized versus decentralized provision of local public goods: a political economy approach. *Journal of public economics* 87(12), 2611–2637.
- Biblioteca del Congreso Nacional de Chile (1990). Ley orgánica constitucional de carabineros de chile (ley 18.961). Promulgada el 27 de febrero de 1990; publicada el 7 de marzo de 1990; última versión consolidada consultada en la Biblioteca del Congreso Nacional.
- Blair, G., J. M. Weinstein, F. Christia, E. Arias, E. Badran, R. A. Blair, A. Cheema, A. Farooqui,

- T. Fetzer, G. Grossman, et al. (2021). Community policing does not build citizen trust in police or reduce crime in the global south. *Science* 374(6571), eabd3446.
- Blanco, L. R. (2013, None). The impact of crime on trust in institutions in mexico. *European Journal of Political Economy* 32(C), 38–55.
- Blanes-i-Vidal, J. and T. Kirchmaier (2018). The effect of police response time on crime clearance rates. *The Review of Economic Studies* 85(2), 855–891.
- Blanes-i Vidal, J. and G. Mastrobuoni (2018, Mar). Police patrols and crime. IZA Discussion Paper 11393, Institute of Labor Economics (IZA), Bonn.
- Blattman, C., D. P. Green, D. Ortega, and S. Tobón (2021, 02). Place-Based Interventions at Scale: The Direct and Spillover Effects of Policing and City Services on Crime. *Journal of the European Economic Association* 19(4), 2022–2051.
- Bloom, N., R. Lemos, R. Sadun, and J. V. Reenen (2015). Does management matter in schools? *The Economic Journal* 125(584), 647–674.
- Braga, A. A., B. Turchan, A. V. Papachristos, and D. M. Hureau (2019). Hot spots policing of small geographic areas effects on crime. *Campbell Systematic Reviews* 15(3).
- Cafferata, F. and C. Scartascini (2021). Combating crime in latin america and the caribbean: What public policies do citizens want? Inter american development bank.
- Callaway, B. and P. H. Sant’Anna (2021). Difference-in-differences with multiple time periods. *Journal of Econometrics* 225(2), 200–230.
- Carabineros de Chile (2018). *Manual Operativo: Plan Cuadrante de Seguridad Preventiva 2.0*. Carabineros de Chile. Aprobado conforme a Orden General N° 1960 del 14 de septiembre de 2010, actualizado para el PCSP 2.0.
- Chalfin, A. (2025). Investments in policing and community safety. *Annual Review of Criminology* 8, 403–429.
- Chalfin, A. and J. McCrary (2017, March). Criminal deterrence: A review of the literature. *Journal of Economic Literature* 55(1), 5–48.

- Cordner, G. W. (1995). Community policing: Elements and effects. *Police Forum* 5(3), 1–8.
- Di Tella, R. and E. Schargrodsky (2004, March). Do police reduce crime? estimates using the allocation of police forces after a terrorist attack. *American Economic Review* 94(1), 115–133.
- Draca, M., S. Machin, and R. Witt (2011, August). Panic on the streets of london: Police, crime, and the july 2005 terror attacks. *American Economic Review* 101(5), 2157–81.
- Evans, W. N. and E. G. Owens (2007). Cops and crime. *Journal of Public Economics* 91(1), 181–201.
- Facchetti, E. (2025). Police infrastructure, police performance, and crime: Evidence from austerity cuts. IFS Working Paper W24/16, Institute for Fiscal Studies.
- Gallup (2023). Gallup global law and order report 2023.
- Gonçalves, F., S. Mello, and E. Weisburst (2025, 01). Selection bias and racial disparities in police use of force. *SSRN Electronic Journal*.
- Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of Econometrics* 225(2), 254–277.
- Imai, K., L. Keele, and T. Yamamoto (2010). Identification, inference and sensitivity analysis for causal mediation effects. *Statistical Science* 25(1), 51–71.
- Levitt, S. D. (1997). Using electoral cycles in police hiring to estimate the effect of police on crime. *The American Economic Review* 87(3), 270–290.
- Mastrobuoni, G. (2019). Police disruption and performance: Evidence from recurrent redeployments within a city. *Journal of Public Economics* 176, 18–31.
- Mastrobuoni, G. (2020, 03). Crime is terribly revealing: Information technology and police productivity. *The Review of Economic Studies* 87(6), 2727–2753.
- Ministerio de Defensa Nacional, República de Chile (1989). Reglamento de organización de carabineros de chile n° 1. Decreto N° 77, de 1 de marzo de 1989.

- Ministerio del Interior y Seguridad Pública (2014). Evaluación del Programa Plan Cuadrante de Seguridad Preventiva. Technical report.
- National Academies of Sciences, Engineering, and Medicine (2018). *Proactive Policing: Effects on Crime and Communities*. Washington, DC: The National Academies Press.
- Ostrom, E., R. B. Parks, and G. P. Whitaker (1973). Do we really want to consolidate urban police forces? A reappraisal of some old assertions. *Public Administration Review* 33(5), 423–432.
- Ostrom, E., R. B. Parks, and G. P. Whitaker (1978). *Patterns of Metropolitan Policing*. Cambridge, MA: Ballinger Publishing Company.
- Salas P., R., J. Fernández L., and J. Muñoz V. (2014, July). Evaluación de programas gubernamentales: Informe final corregido. programa plan cuadrante de seguridad preventiva. Evaluación de programas gubernamentales, División de Control de Gestión, Dirección de Presupuestos, Ministerio de Hacienda, Gobierno de Chile. Ministerio Responsable: Ministerio del Interior y Seguridad Pública. Período de evaluación: 2010–2013.
- Skogan, W. G. and S. M. Hartnett (1997). *Community Policing, Chicago Style*. New York: Oxford University Press.
- Tobón, S. (2021). Community policing in the developing world. *Science* 374(6571), 1046–1047.
- Weisburd, S. (2021, 05). Police Presence, Rapid Response Rates, and Crime Prevention. *The Review of Economics and Statistics* 103(2), 280–293.
- Winchester, L., G. Zambrano, and J. Le-Bert (2007, June). Informe final de evaluación: Programa plan cuadrante de seguridad preventiva. Evaluación de programas gubernamentales, Dirección de Presupuestos, Ministerio de Hacienda, Gobierno de Chile. Ministerio Responsable: Ministerio de Defensa. Servicio Responsable: Carabineros de Chile. Período de evaluación: 2004–2006.

Referee 2 Suggestions, Comments and Questions

Thank you very much for your careful and critical feedback, which has substantially strengthened the paper. In particular, your concern about spatial spillovers was particularly useful: following your suggestions, we developed a new set of empirical exercises—summarized in a new Section 4.2 and detailed in Online Appendix C—that directly test for the displacement of crime and police resources to neighboring municipalities. Your comment on the institutional context of *Plan Cuadrante* was equally valuable and led us to substantially expand Section 2. We have carefully considered all your comments and hope that we adequately addressed them in the revised manuscript and in the response document below.

Comment R2.1

Framing of the paper: Section 2 lacks sufficient details on the design and implementation of “Plan Cuadrante”. In particular, I would like to see more information on the resources involved (e.g., number of officers, cars, etc.) and the rules or criteria for their allocation across quadrants and municipalities. This would not only strengthen the reader’s understanding of the intervention but also help in evaluating the plausibility of the proposed mechanisms.

Response to R2.1:

We thank the reviewer for this comment and agree that additional details on the design and implementation of *Plan Cuadrante* would strengthen the paper. In the revised manuscript, we have substantially expanded Section 2 and the mechanisms section to address these points.

During the 2000s and 2010s, the Chilean government substantially increased the overall number of Carabineros and other police resources such as patrol vehicles, and this increase benefited both treated and control municipalities. Specifically, our administrative data on police resources show that police resources per 100,000 inhabitants increased steadily over the study period even in municipalities that never adopted the program and in not-yet-treated municipalities. Between 2005 and 2012, the number of patrol vehicles per 100,000 inhabitants—patrol cars, radio-patrol cars, and motorcycles—grew by approximately 6% per year in control municipalities, and the number of police officers per 100,000 inhabitants grew by roughly 9.7% per year.

On top of this general trend, *Plan Cuadrante* aimed to provide treated municipalities with sufficient additional resources to successfully implement the program by closing their pre-existing police resource deficit, measured in Equivalent Surveillance Units (UVes). Panels (c) and (d) of Figure 4 show the pooled effect of *Plan Cuadrante* on resources over the study period: substantial increases in both police officers and patrol vehicles relative to the control group, with a strong effect among police officers that reduces over time and a more sustained effect among patrol vehicles. Panels (a) and (b) of Figure 4 show, however, that this average increase masks substantial heterogeneity across treated municipalities: while municipalities in the bottom 40% of the resource-change distribution experienced increases very similar to those observed in control municipalities, municipalities in the top 40% experienced substantially larger increases after joining *Plan Cuadrante*, with officers rising yearly by around 22% and patrol vehicles by around 80%.

But what drives this variation in resource increases across municipalities adopting *Plan Cuadrante*? In principle, it should reflect each municipality’s pre-existing resource deficit. However, program documents make clear that budgetary constraints meant that all municipalities joined *Plan Cuadrante* without fully closing this gap (Salas P. et al., 2014; Winchester et al., 2007), and program evaluations do not clearly establish that the size of this remaining deficit is what determined the resource increase each municipality received. We requested information on the demand, supply, and deficit of UVEs underlying this allocation from Carabineros, but this information was not provided.

Importantly, we find that municipalities in the bottom and in the top 40% of the resource-increase distribution upon adopting *Plan Cuadrante*—whether the split is based on the increase in police officers or in patrol vehicles—do not display differential pre-treatment trends in either the outcomes of interest or in police resources themselves.⁵⁶ In the “Factors Associated with the Increase in Police Resources” subsection of Online Appendix B we examine what pre-treatment levels of municipality characteristics are associated with a larger increase in police resources at the time of adopting *Plan Cuadrante*. In general, the pre-treatment crime, victimization, and other survey outcomes we examine do not appear to predict a larger resource increase upon adoption. Instead, the resource increase appears more closely related to structural municipality characteristics: less populous municipalities and, in the administrative sample, municipalities with higher poverty rates tend to experience larger increases in resources.

Regarding the allocation of resources within municipalities across quadrants, the program established that resources should be distributed proportionally to the demand of each quadrant. However, program evaluations explicitly acknowledge that this principle operated as a guideline rather than a binding rule, with the allocation ultimately remaining at the discretion of the Sub-commissioner of Services in each police station (Winchester et al., 2007; Salas P. et al., 2014). This is essentially the same discretionary mechanism operating in control municipalities, where resource allocation within the police station was similarly left to the judgment of the Sub-commissioner of Services. This is relevant for interpreting our results: what the program reliably changed was not the internal distribution of resources within the municipality, but the permanent territorial assignment of officers to smaller, fixed areas. This is the component we argue is central to the effects we find.

While we lack exogenous variation to perfectly disentangle the contribution of additional resources from that of the beat policing component, we leverage the substantial heterogeneity in resource increases across municipalities described above to test whether resources alone could be driving the effect. The effects of *Plan Cuadrante* in municipalities whose resource increases were similar to those in the control group are nearly identical to those in municipalities with larger increases, in terms of both victimization and household investment in crime prevention measures. While there may be a resource gradient in the effects of *Plan Cuadrante* for crime perception, the effects of the program for municipalities with resource increases similar to those of control municipalities remain sizable and relevant for this outcome as well. These results do not imply that resources play no role in crime reduction, or even in the effects of *Plan Cuadrante* in some municipalities—for instance, by enabling the successful implementation of the program. However, they clearly highlight that the beat policing component is central to the effects we document. Additional evidence on the role of police resources outside *Plan Cuadrante*, presented in Online Appendix B of the updated manuscript, confirms this interpretation.

⁵⁶The p-values for the joint test of equal pre-trends between the two groups are, for the officers-based split and the patrol-vehicles-based split respectively: 0.823 and 0.048 for victimization, 0.059 and 0.379 for crime perception, 0.820 and 0.157 for investment in security measures, 0.213 and 0.419 for reported crime, 0.616 and 0.444 for police officers, and 0.405 and 0.157 for patrol vehicles.

Below we report the new Section 2 describing the institutional context and the *Plan Cuadrante* reform:

Carabineros, Chile's national police force, was founded in 1927. Through most of our study period, Carabineros operated under the Ministry of Defense, which oversaw both military and police functions until domestic security responsibilities were transferred to the Ministry of Interior in 2011.⁵⁷

Carabineros' mission, functions, and structure are established by Law 18.961 and the Regulation on Carabineros' Organization (Reglamento de Carabineros) (Biblioteca del Congreso Nacional de Chile, 1990; Ministerio de Defensa Nacional, República de Chile, 1989). Under this framework, Carabineros is a professional, disciplined, and hierarchical police institution whose mission is to preserve public order and security with a central emphasis on preventive policing. The General Director heads the strategic level, responsible for defining doctrine and institutional strategy, while the Subdirector coordinates implementation. The Directorate of Order and Security administers police functions at the tactical level.

Territorially, Carabineros is organized into zones that oversee *prefecturas*. Zones and *prefecturas* are primarily management layers. Zones map roughly one-to-one to Chile's regional divisions, with two exceptions: the Metropolitan Region of Santiago—the most populated in the country—and La Araucanía—which has recently experienced significant social conflict—are each divided into two zones. Each zone is subdivided into *prefecturas*—54 nationwide—which supervise 227 *comisaría*s, the main operational units at the tactical level. These *comisaría*s oversee smaller units—*subcomisaría*s, *tenencias*, *retenes*, *puestos*, and *avanzadas*—that operate within their assigned jurisdictions.

In practice, the *comisaría* is the basic unit of routine frontline policing and, in most municipalities, serves as the territorial anchor of police work. Officers conduct patrol, surveillance, prevention, and first response at this level. In most municipalities, a single *comisaría* covers the entire municipal territory, though larger municipalities may have multiple ones. Santiago, for instance, has four. All municipalities that adopted *Plan Cuadrante* during our study period had a *comisaría*. With the exception of 9 municipalities, each had a single *comisaría* covering the full municipal territory. On average, a *comisaría* in our sample covered 1,608 square kilometers and served 57,642 inhabitants.

Plan Cuadrante was introduced within this institutional framework without altering Carabineros' hierarchical structure or its strategic and tactical levels. The reform emerged in a context of rising victimization and insecurity, with the explicit goal of reversing these trends (Ministerio del Interior y Seguridad Pública, 2014). Its central innovation was a shift toward a beat policing model. In the criminology literature, beat policing refers to assigning officers to fixed territorial areas so they can accumulate local knowledge, strengthen ties with residents, and respond more effectively to local crime patterns (Cordner, 1995; Skogan and Hartnett, 1997). In practice, the catchment area of police forces in municipalities adopting *Plan Cuadrante* was partitioned into smaller, fixed geographic units (quadrants), to which officers were permanently assigned. Rather than building new infrastructure, the reform redefined the territorial responsibility of officers in already existing police stations (Winchester et al., 2007).⁵⁸

⁵⁷Currently, Carabineros reports to the Ministry of Public Security, created in 2024 to focus exclusively on domestic security.

⁵⁸In 2016, Carabineros launched the *Plan Cuadrante 2.0* program in those municipalities served by *Plan Cuadrante*.

This represented a clear break from the pre-reform system, in which officers rotated across broad operational territories, very often spanning entire municipalities.

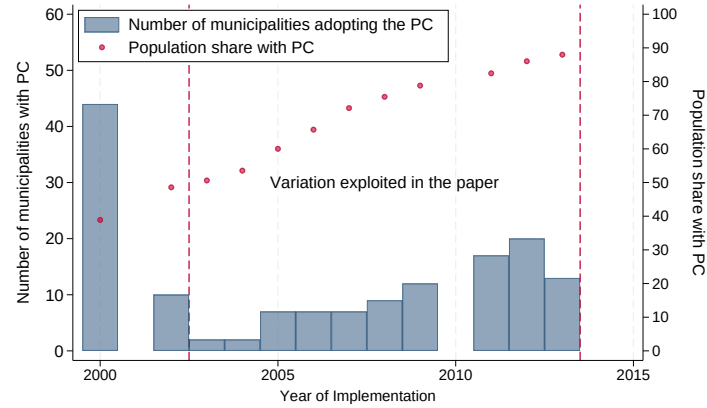
Carabineros began piloting *Plan Cuadrante* in 1998. The program expanded gradually through 2013, as shown in Panel (a) of Figure 1, reaching 150 urban municipalities and covering roughly 90% of Chile’s population.⁵⁹ Online Appendix A provides further details about the rollout of the program. The rollout was designed to follow a municipality ranking based on a weighted index of victimization, police resource deficits, drug prevalence, unemployment, and demand for police services, with annual entry determined by available budget.⁶⁰ In practice, however, Budget Office evaluations show that adoption diverged from this ranking for undocumented reasons, so the rule operated more as a loose guideline than as a binding assignment mechanism (Salas P. et al., 2014).

While this reform introduced additional reforms aimed at integrating the community into the provision of security and building capacity to gather and analyze crime data in police stations, this paper’s study period ends before these improvements were introduced.

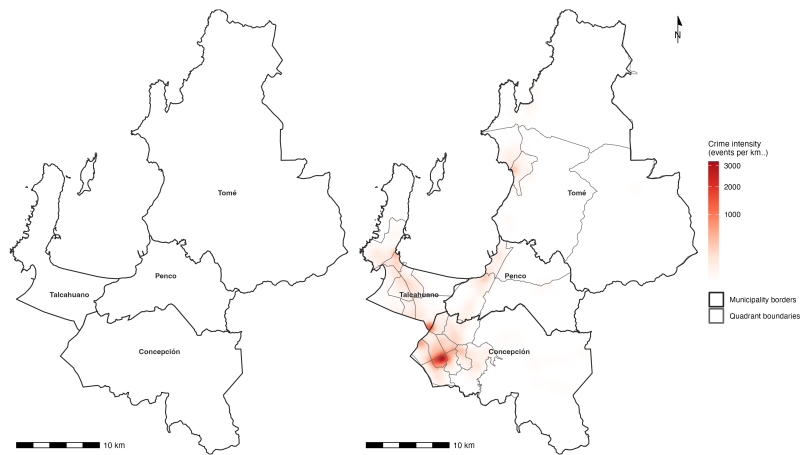
⁵⁹Further details about the expansion of *Plan Cuadrante* are provided in Online Appendix A and Table A.1.

⁶⁰The weights assigned to each component changed on a yearly basis (Winchester et al., 2007; Salas P. et al., 2014).

Figure 1: Implementation of *Plan Cuadrante*



a Adoption of *Plan Cuadrante*



b *Plan Cuadrante* in the Concepción area

Notes: Panel (a) shows the number of municipalities adopting the *Plan Cuadrante* each year (bars), while the circles above them illustrate the cumulative share of the Chilean population living in a municipality with the *Plan Cuadrante* already in place. Panel (b) depicts the metropolitan area of Concepción, the second-largest in Chile: the left map shows its four municipalities—the original police catchment areas—and the right map overlays the Carabineros quadrants together with a kernel-density estimate of all types of robberies and thefts recorded by the police (both reported crimes and arrests). It illustrates how *Plan Cuadrante* subdivided each catchment area (which corresponds to the territory of a municipality) into smaller quadrants, as well as the spatial concentration of crime within them.

Conceptually, police effectiveness can be viewed as a function of three inputs: total resources, territorial deployment, and policing strategy. Although in some municipalities police resources were increased significantly to operationalize the beat policing model (Winchester et al., 2007; Salas P. et al., 2014), the program primarily transformed the latter two elements. The creation of quadrants was accompanied by permanent geographic assignment of officers to specific areas. This change was operationally critical. By making officers responsible for their quadrant, it encouraged accumulation of local knowledge and stronger community ties, allowing them to adapt patrols and prevention activities to local crime patterns. Importantly, this was not driven by program-specific operational directive changes. Throughout the period we study, a single national set of operational guidelines applied to officers in both treated and control municipalities (Carabineros de Chile, 2018; Winchester et al., 2007). Thus, while the average effects we document on crime and crime perception likely reflect combined changes in the three elements we discuss—i.e., resources, deployment, and strategy—the deployment change itself was essential to operationalizing the reform. We discuss the drivers of the effects we document in Section 4.4.

Quadrant size and shape were determined by local police authorities based on factors such as road networks, geographic features, and available resources. In principle, resource allocation across quadrants followed demand for police services, measured in Equivalent Surveillance Units (UEVs); in practice, the station commander retained discretionary authority, as was also the case prior to *Plan Cuadrante* (Salas P. et al., 2014). Municipalities implementing *Plan Cuadrante* were subdivided into an average of seven quadrants, each typically staffed by approximately three officers per shift. On average, each quadrant covered 185 square kilometers and served 13,466 inhabitants. Panel (b) of Figure 1 illustrates this structure, showing the 29 quadrants created across the four municipalities of the Concepción metropolitan region—Chile’s second-largest metropolitan area and the largest in our sample. Quadrants represented approximately 14% of the original *comisaría* territory. Officers continued to operate from the same police station but were permanently assigned to a specific sub-area within its jurisdiction. While Carabineros’ hierarchical structure limits frontline officer decision-making authority, each quadrant was assigned a Quadrant Delegate (*Delegado de Cuadrante*), a non-commissioned officer, responsible for immediate operational decisions within the quadrant.

Plan Cuadrante may affect public safety through two complementary channels: deterrence and police effectiveness, understood as the capacity to respond to and resolve crimes. The changes induced by *Plan Cuadrante*—permanently assigning officers to well-defined neighborhoods and, in some municipalities, increasing available resources—can increase both police visibility and local knowledge. As officers repeatedly patrol the same area, they become familiar with its streets, residents, and routines, which can raise the perceived probability of identification and apprehension and reduce the expected returns to crime. At the same time, this deeper knowledge of local territory and communities can improve police response and case resolution once crimes occur. The reform therefore likely affects both dimensions. Consistent with this interpretation, in Section 4 we document declines in victimization and in reported crime—patterns consistent with deterrence—together with increases in arrests, consistent with improved police effectiveness.

Although beat policing shares features with other policing approaches, such as hot-spot policing and community-based policing, which have been widely studied, it has distinctive features that make it worth studying as an independent approach. Hot-spot policing concentrates resources in high-crime areas and does not focus on building local knowledge and community

ties. It is typically implemented through targeted interventions at specific locations rather than systematic territorial organization, and the same officers are not necessarily sent repeatedly to the same areas. Evidence on hot-spot policing is mixed, with some studies showing crime reductions in targeted areas but also potential displacement effects (Braga et al., 2019). *Plan Cuadrante*, by contrast, subdivided all urban municipalities into quadrants with permanent officer assignment. As Panel (b) of Figure 1 illustrates, some quadrants contained crime hot-spots, but many did not. The reform thus addressed entire urban territories rather than specific high-crime areas, potentially mitigating displacement effects associated with hot-spot policing.

Community-based policing emphasizes building relationships with residents and addressing underlying social issues, and typically involves active community participation in defining strategy and priorities—in some cases, even assuming surveillance and prevention responsibilities. While increasing trust in the police and strengthening community ties were among the declared goals of *Plan Cuadrante*, and may have resulted from its territorial reorganization, the community was not formally involved in the provision of security.⁶¹

In recent decades, police departments worldwide have transitioned toward beat policing reforms (National Academies of Sciences, Engineering, and Medicine, 2018; Ministerio del Interior y Seguridad Pública, 2014), yet the effectiveness of this model has not been rigorously assessed. Although the beat policing approach introduced through *Plan Cuadrante* shares features with programs previously studied in the economic literature, its distinctive characteristics—comprehensive territorial reorganization and systematic local officer specialization—make it a valuable case for studying how deployment and policing strategies not yet examined in the economics literature affect crime and crime perception.

We also include below Section 4.4, which summarizes the results of the analyses that aim to disentangle the role of resources from that of the beat policing component:

To reduce crime and improve security perceptions, police forces need to strengthen both prevention and police effectiveness. Both margins can be shaped by resources, territorial deployment, and policing strategy. This section examines which components of *Plan Cuadrante* are most likely to drive the effects documented in Section 4.

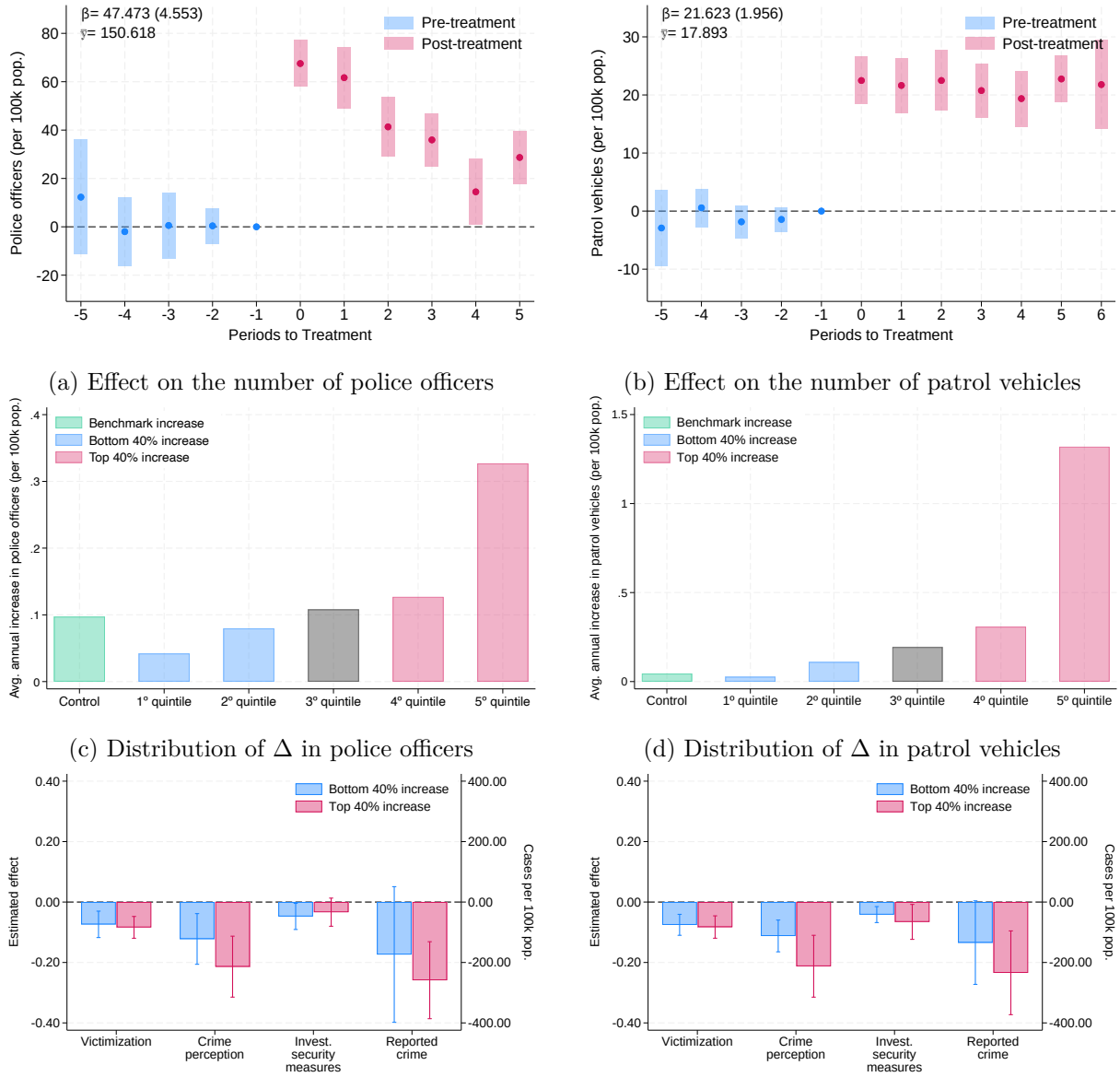
The central objective of *Plan Cuadrante* was to introduce a beat policing model by reorganizing deployment. Police operational territories that had typically matched municipal boundaries were subdivided into quadrants, and officers were permanently assigned to these quadrants. During our study period, national operational guidelines remained the same in adopting and non-adopting municipalities. Still, this new deployment likely affected policing strategy at a more operational level, because it allowed officers to adapt patrol routes, schedules, and prevention practices using the local knowledge that the program sought to build.

Although changes in resources were not the core of the reform, implementation contemplated increases in municipalities that lacked the personnel and patrol capacity required to move to

⁶¹The *Plan Cuadrante*’s original design contemplated a formal community integration model (MICC) with a Community Affairs Office (*Oficina de Asuntos Comunitarios*) and a dedicated officer in each police station. However, this model was not implemented during our study period. Independent program evaluations from the Budget Office document that these Community Affairs Offices did not exist and had no dedicated personnel, budget, or structured activities in the period we study (Winchester et al., 2007; Salas P. et al., 2014). The community integration model was piloted in four police stations during 2012–2013, none of which are in our analytical sample. Financing was then discontinued in early 2014 and only restored and expanded after our study period ends.

this model. During the period we study, Chile was expanding public spending on security, and police resources increased in most municipalities regardless of whether or when they adopted *Plan Cuadrante*. On average, however, municipalities adopting the program experienced larger increases in officers and patrol vehicles, as shown in panels (a) and (b) of Figure 4. Since increases in police resources have been shown to reduce crime in some contexts (Evans and Owens, 2007), this pattern raises a central question for interpretation, namely whether our estimates reflect resource expansion or the deployment and strategy changes introduced by *Plan Cuadrante*. We address this question through a set of exercises that exploit heterogeneity in resource increases across municipalities and over time.

Figure 4: Plan Cuadrante and Changes in Police Resources



(e) Effects by the intensity of Δ in police officers (f) Effects by the intensity of Δ in patrol vehicles

Notes: Panels (a) and (b) show the effect of *Plan Cuadrante* on police officers and patrol vehicles per 100,000 inhabitants. Panels (c) and (d) show the distribution of annual changes in officers and patrol vehicles across municipalities. Panels (e) and (f) report effects by the intensity of resource changes, comparing municipalities in the bottom 40% and top 40% of the distribution. The estimation follows Callaway and Sant'Anna (2021). The graphs display estimated coefficients and 95% confidence intervals. Standard errors are clustered at the municipality level using wild bootstrapping.

Panels (c) and (d) of Figure 4 show that the average increase in police resources masks substantial heterogeneity. Municipalities in the bottom 40% of the resource-change distribution (blue bars) experienced increases close to the average annual variation observed among never treated and not-yet-treated municipalities (green bar). By contrast, municipalities in the top 40% (red bars) experienced much larger increases, with officers rising by around 22% and patrol vehicles

by around 80%. If the effects of *Plan Cuadrante* were driven by resource increases alone, we would not expect effects in municipalities whose resource increases were similar to those of the control group. Yet panels (e) and (f) of Figure 4 show sizeable effects for most crime and crime-perception outcomes even in that low-increase group. While there may be a resource gradient for some crime-perception outcomes, the effects on twelve-month victimization and private investment in security among municipalities with control-like resource increases are close in magnitude to those in municipalities with larger increases. Reassuringly, municipalities in the top and bottom 40% of the resource-increase distribution do not display differential pre-treatment trends in outcomes or police resources, which suggests that this heterogeneity is not driven by differential pre-existing trends.⁶²

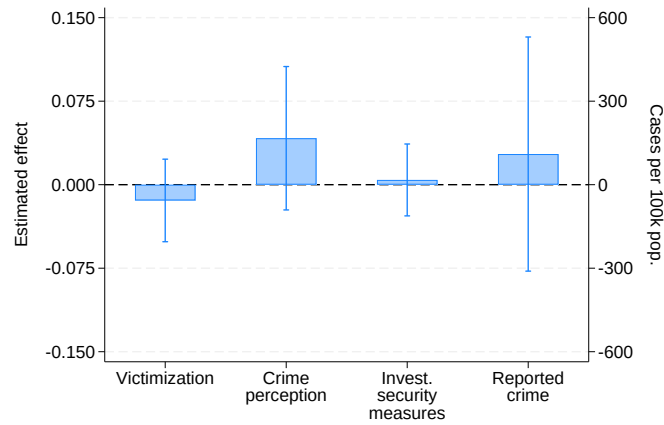
Although we cannot fully separate the beat policing component from the resource component of the reform—resources may have been important to enable effective implementation of the program, or may explain part of the effects in some municipalities—the fact that municipalities with resource increases similar to those experienced by control municipalities still display sizeable effects indicates that the effects of *Plan Cuadrante* are not driven entirely by resources.⁶³⁶⁴

⁶²We test for the joint significance of the pre-treatment coefficients between the two groups, separately for the split based on the increase in police officers and the split based on the increase in patrol vehicles. The resulting p-values are, respectively, 0.823 and 0.048 for victimization, 0.059 and 0.379 for crime perception, 0.820 and 0.157 for investment in security measures, and 0.213 and 0.419 for reported crime, as well as 0.616 and 0.444 for police officers and 0.405 and 0.157 for patrol vehicles.

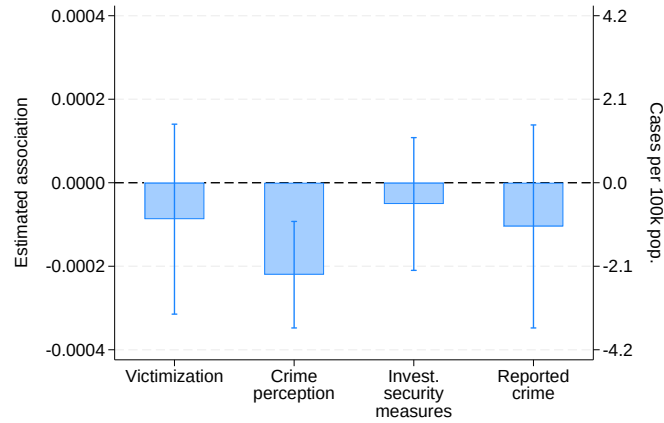
⁶³We also re-estimate the results in Figure 2 but controlling for police resources. Online Appendix Table B.3 shows that the magnitude of the estimated effects remains similar for most variables, although the statistical significance is reduced in some estimations. This likely reflects a mechanical feature of the Callaway and Sant’Anna (2021) estimation procedure: control variables are not incorporated parametrically, but are instead used to construct the comparison group via outcome regression or propensity-score weighting, so the resulting estimation uncertainty propagates to the ATT standard errors. Note, in addition, that police resources are a bad control—i.e., at least part of their variation is a result of the adoption of *Plan Cuadrante*—so these results should be interpreted cautiously.

⁶⁴Table B.4 in Online Appendix B examines which factors are associated with a larger increase in police resources upon the adoption of *Plan Cuadrante*.

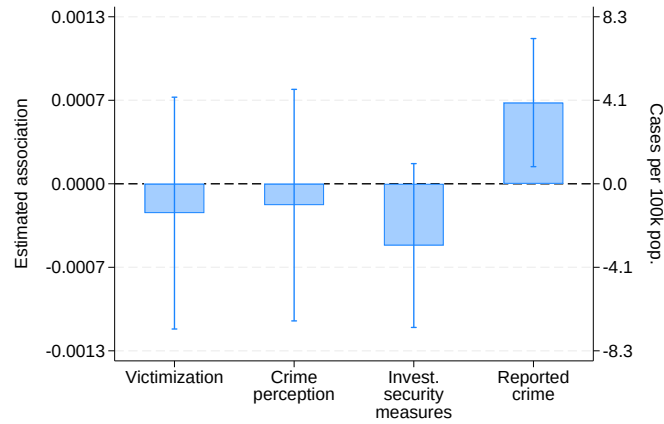
Figure 5: Police Resources and Public Safety



a Effects of *Seguridad Ciudadana*



b Association between police officers and effectiveness



c Association between patrol vehicles and effectiveness

Notes: Panel (a) reports the pooled effect of *Seguridad Ciudadana*, a program that expanded local security personnel to support patrolling and deterrence but did not replicate the deployment-and-strategy features of *Plan Cuadrante*. Panels (b) and (c) report within-municipality associations between police resources and police-effectiveness outcomes among the 41 municipalities of the Santiago Metropolitan Region, all of which had adopted *Plan Cuadrante* by 2000. The regressions include municipality and year fixed effects. Standard errors are clustered at the municipality level using the wild bootstrapping procedure.

To further investigate whether resource increases alone can explain our results, we assess the effects of *Seguridad Ciudadana*, a program that expanded local security personnel to support patrolling and deterrence, but did not replicate the core deployment-and-strategy features of *Plan Cuadrante*. Using a difference-in-differences specification following Callaway and Sant’Anna (2021), panel (a) of Figure 5 shows no significant effects of these resource changes on police effectiveness. Further details about this exercise and the corresponding event-study estimates are reported in Online Appendix B and Online Appendix Figure B.4.

Finally, we study whether variation in police resources in municipalities that had already adopted *Plan Cuadrante* improves public safety and security perceptions. Using ENUSC data and information on resources for 2006–2012 in the 41 municipalities of the Santiago Metropolitan Region—all of which had adopted *Plan Cuadrante* by 2000—we exploit variation in officers and patrol vehicles that is unrelated to program adoption. Panels (b) and (c) of Figure 5 show the association between police resources and our main outcomes, conditional on municipality and year fixed effects. With one exception—the association between police officers and crime perception—none of these associations is statistically significant at conventional confidence levels.⁶⁵

Taken together, these results suggest that resource increases alone do not account for the effects of *Plan Cuadrante*. We next turn to evidence that speaks more directly to the beat-policing components of the reform.

Panel (a) of Figure 4 shows that, following the adoption of *Plan Cuadrante*, households are much more likely to report increased police presence in their neighborhoods. The share of households perceiving increased police presence rises by 35 percentage points (156%), the share reporting that police pass by their house daily rises by 22 percentage points (59%), the share reporting never seeing police patrols near their house falls by 10 percentage points (57%), and complaints about lack of police surveillance fall by around 5 percentage points (20%). This evidence is consistent with both larger resources and the new deployment model. Yet the latter estimate—the only police-presence outcome for which we can separately estimate effects in municipalities with small and large resource increases—is similar in magnitude across the two groups, which suggests that the increase in police presence is not explained only by larger resource increases (see Online Appendix Figure B.3).⁶⁶ More importantly, the broader pattern of results suggests that *Plan Cuadrante* did more than simply increase patrolling intensity. Existing work shows that police presence can reduce crime while officers are present, but that its overall effects may be attenuated when crime is displaced across places or time windows (Mastrobuoni, 2019; Blanes-i Vidal and Mastrobuoni, 2018). Our results point to something broader. As discussed in Section 4, we find actual declines in victimization and reported crime, together with no evidence of spillovers to neighboring municipalities. This pattern is more consistent with officers accumulating local knowledge and adapting prevention strategies to local conditions under stable territorial assignment than with a purely mechanical increase in patrol intensity.

Panel (b) of Figure 4 reinforces this interpretation. Households in early-adopting municipalities

⁶⁵Online Appendix Figure B.5 reports the same exercise and also shows similar results for the broader set of municipalities that adopted *Plan Cuadrante* before 2006 and never-treated municipalities (red bars); see Online Appendix B for further details.

⁶⁶We cannot estimate heterogeneous effects for other police-presence outcomes, such as the reporting of police passing by or the share of households perceiving an increase in police presence, because those variables are available only in ENUSC rounds 2003, 2005, and 2006, while resource data start in 2006.

report better police performance across multiple dimensions. They are 8.1 percentage points (16%) more likely to rate police performance as good or very good, 7.6 percentage points (11%) more likely to describe police as efficient, 5 percentage points (6%) more likely to report that police are helpful, and 4.9 percentage points (7%) more likely to report that police are well informed. They are also 3.9 percentage points (10%) less likely to perceive police as corrupt, 7.9 percentage points (21%) less likely to believe police fail to control crime, and 7.6 percentage points (22%) less likely to report that police do not perform their duties effectively.⁶⁷

The administrative evidence points in the same direction. Panel (c) of Figure 4 shows that arrests per 100,000 inhabitants increase by 48 (14%) in our main estimates, even as crime declines overall.⁶⁸ This effect is also significant among municipalities with resource increases similar to those in the control group (see Online Appendix Figure B.3). Consistent with this pattern, panel (d) of Figure 4 shows that the arrest rate per reported crime also increases by 1.7 percentage points (10%), suggesting that a higher share of crimes reported to the police are being solved. Because these gains appear even where resources did not increase more than in control municipalities, they are difficult to reconcile with a pure resource story and are more naturally interpreted as improvements in police effectiveness under the deployment and strategy changes introduced by *Plan Cuadrante*. The corresponding event-study estimates for arrests, the arrest rate per reported crime, and trust in the police in Figure 4 confirm that these changes emerge only after adoption.

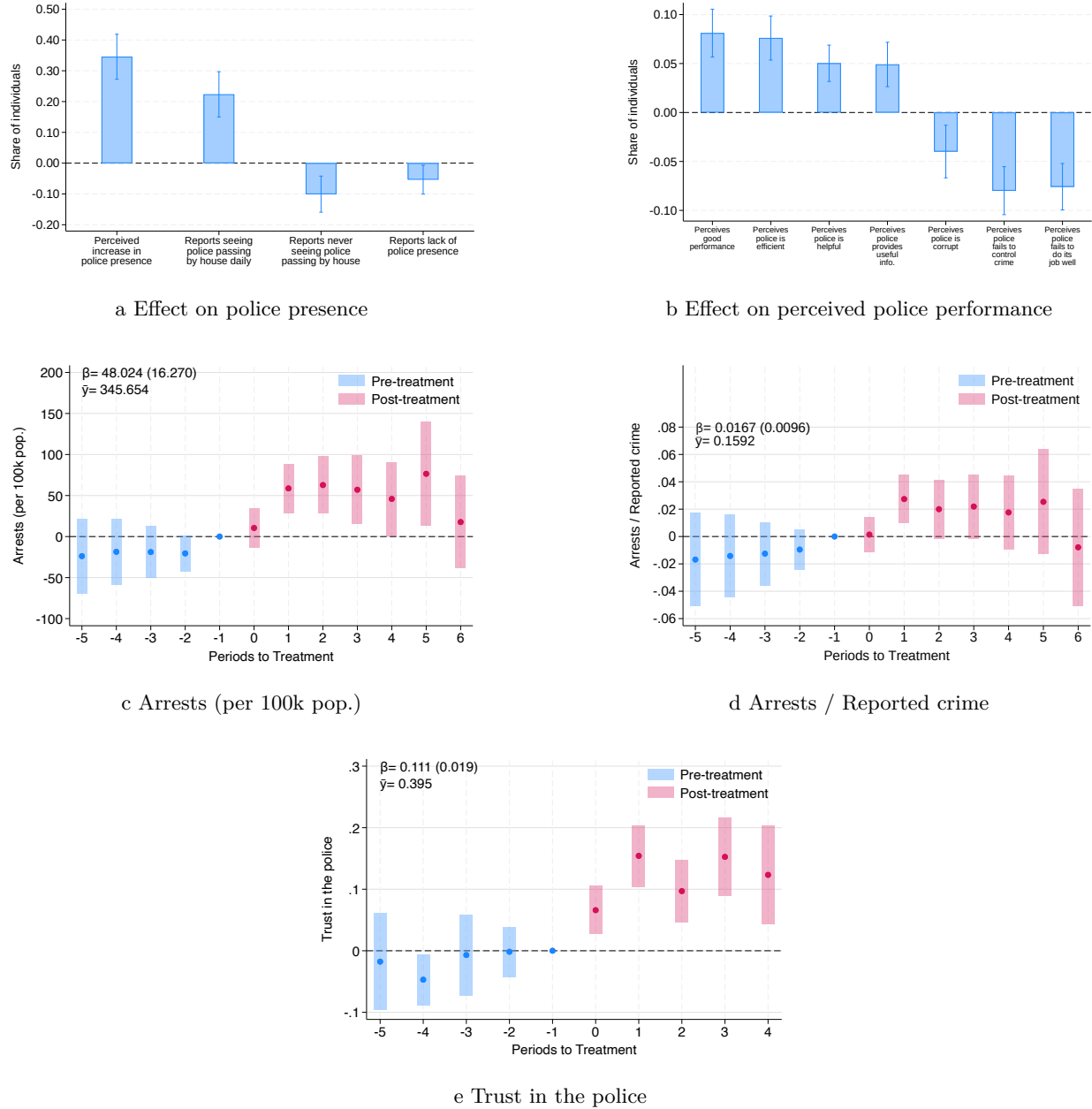
Finally, panel (e) of Figure 4 shows a sharp increase in trust in the police in our main specification (Section 4), consistent with one of the objectives of beat policing, namely bringing officers closer to the community and allowing them to accumulate local knowledge through repeated interaction. Online Appendix Figure B.3 shows that this effect remains sizeable both in municipalities that experienced resource increases similar to the control group and in those with larger increases.

Overall, although resources were likely important for implementation and may explain part of the effects in some municipalities, the results in this section indicate that resource changes alone are not enough to explain our findings. The evidence is more consistent with a combination of deployment changes and the strategy adjustments enabled by local knowledge accumulation under permanent assignment to quadrants.

⁶⁷Information on police performance is only available in ENUSC round 2005. Using this round, we compare perceived police performance in municipalities that adopted *Plan Cuadrante* in 2005 or earlier and in municipalities adopting after 2005. Figure B.6 in Online Appendix B shows that the number of years of exposure to *Plan Cuadrante* is positively correlated with perceived police performance. While these exercises on perceived police performance should be interpreted with caution, we show in the same appendix that the timing of future adoption is not correlated with perceived police performance in 2005 for municipalities that had not yet adopted the program by 2005.

⁶⁸As with reported crime, this event study is restricted to 5 pre-treatment and 7 post-treatment periods for symmetry with the ENUSC-based analysis; Online Appendix Figure C.8 reports the results for arrests using the full set of pre-treatment and post-treatment periods.

Figure 6: *Plan Cuadrante* and Changes in Police Performance



Notes: Panel (a) reports the effects on perceived police presence. Panel (b) reports perceived police performance in 2005, comparing municipalities where *Plan Cuadrante* was already implemented before 2006 with municipalities where it was implemented after 2005. Panels (c), (d), and (e) report dynamic event-study estimates for arrests per 100,000 inhabitants, the ratio of arrests to reported crime, and trust in the police. The estimation follows Callaway and Sant'Anna (2021). The graphs display estimated coefficients and 95% confidence intervals. Standard errors are clustered at the municipality level using the wild bootstrapping procedure.

We also reproduce below the Online Appendix subsection that details the heterogeneity analysis and the robustness check controlling for police resources referenced above:

To make the implementation of *Plan Cuadrante* possible, police resources were augmented in some municipalities. To examine the extent to which the observed results might be driven by the increase in police resources rather than by the geographic specialization component of the reform, we conduct the following two empirical exercises.

Firstly, we examine the heterogeneity of *Plan Cuadrante*'s effects by the magnitude of resource increases in each municipality. Panels (a) and (b) of Figure 4 show that municipalities in the bottom 40% of the resource-increase distribution—represented by blue bars—experienced increases close to the average annual variation experienced by never treated or municipalities that had not yet adopted the program (green bar), while municipalities in the top 40% (red bars) experienced larger increases, with officers rising by around 22% and patrol vehicles by around 80%. If the effects of *Plan Cuadrante* were driven by resource increases alone, we would not expect to see any effect in municipalities with increases in police resources similar in magnitude to those observed in the control group. To conduct our heterogeneity analysis, we split the sample into municipalities with small and large resource increases and estimate specification (1) separately for each subsample. Yet panels (c) and (d) of Figure 4 show that the effects on most outcomes remain sizeable for most crime and crime perception outcomes even among municipalities with resource increases similar to the control group. While there might be a resource gradient for the effects on crime perception, the effects on twelve-month victimization and private investment in security, in municipalities with resource increases similar to those in the control group, are comparable in magnitude to the effects among municipalities with larger resource increases. The effect on the probability of complaining about police surveillance, arguably a proxy for police presence, was also similar for both groups.⁶⁹ More generally, we cannot reject the hypothesis that effects are equal between the two groups in all the examined outcomes. The heterogeneous effects on trust in the police, complaints about the lack of police officers, and arrests—the three outcomes not shown in Figure 4—are reported in Figure B.3 and are likewise similar across the two groups.

Finally, we re-estimate the main results of the study reported in Figure 2, controlling for the number of police officers and the number of patrol vehicles in the municipality.

The results of this analysis are reported in Table B.3. The magnitude of the estimated effects remains similar for most variables, although the statistical significance is reduced in some estimations. This likely reflects a mechanical feature of the Callaway and Sant'Anna (2021) estimation procedure: control variables are not incorporated parametrically, but are instead used to construct the comparison group via outcome regression or propensity-score weighting, so the resulting estimation uncertainty propagates to the ATT standard errors. Note, however, that these resource variables are bad controls, as they are affected by the adoption of *Plan Cuadrante*, so these results should be interpreted with caution.

We also reproduce below the Online Appendix subsection presenting two additional empirical analyses on the link between police resources and crime outside the scope of *Plan Cuadrante*, including the analysis of *Seguridad Ciudadana*:

⁶⁹We provide the heterogeneous effects of *Plan Cuadrante* on complaints about lack of police surveillance because information on this variable is available for the period 2003-2012. However, information on the share of households perceiving an increase in police presence, reporting that police passed by their house daily and never seeing police patrols near their house is only available in ENUSC rounds 2003, 2005 and 2006, and therefore there is not sufficient overlap with information on resources, available from 2005 for patrol vehicles and from 2006 for police officers.

This Online Appendix presents two additional empirical analyses examining the link between police resources and crime in Chile, outside the scope of *Plan Cuadrante*.

First, we assess the effects of *Seguridad Ciudadana*, a program that creates a security force that, while they do not have the competence to apprehend or carry weapons, assist Carabineros in their work, effectively increasing their human resources for tasks such as patrolling and other deterrence activities. Like *Plan Cuadrante*, the creation of these municipal security forces was staggered and started after 2012 in nearly all municipalities.⁷⁰ We hand-collected the year in which each municipality established its *Seguridad Ciudadana* program, requesting this information from all 346 Chilean municipalities through the national transparency portal (*Portal de Transparencia*) and compiling the responses they returned into an original dataset. Using administrative and ENUSC data available until 2017, the estimated effects of *Seguridad Ciudadana* program are presented in Figure B.4. We estimate the effects for all the outcomes examined in the *Plan Cuadrante* analysis except for trust in the police, since information on the latter variable is rarely collected in ENUSC rounds conducted after 2012. The results show that this program had little effects on every dimension of police effectiveness considered. This result is also consistent with the hypothesis that the large effects of *Plan Cuadrante* on police effectiveness are unlikely driven entirely by the change in police resources brought by the program.

Secondly, we use yearly information on the number of police officers and patrol vehicles at the municipality level facilitated by *Carabineros* for the period 2006-2012 to examine the statistical association between police resources and crime, controlling for year and municipality fixed effects. Specifically, we estimate the following regression:

$$Crime_{imt} = \delta_1 Police\ resources_{mt} + Municipality_m + Year_t + u_{imt} \quad (3)$$

where *Police resources* indicates either the number of police officers per 100,000 inhabitants in municipality m during year t or the number of patrol vehicles per 100,000 inhabitants in municipality m during year t . We focus on the 41 municipalities of Santiago surveyed in ENUSC for the period 2006-2012. The study period is delimited by the availability of data on police resources. We focus on the municipalities of Santiago because *Plan Cuadrante* was implemented in all of these municipalities by 2000. Expanding the analysis to other municipalities may confound the dynamic effects of *Plan Cuadrante* with the potential effects of subsequent increases in police resources. Between 2006 and 2012, these municipalities saw an average increase of 26% in police officers and 83% in patrol vehicles.

The results of the analyses are reported in Figure B.5. While the set of fixed effects account for time-invariant unobservable characteristics of the municipalities and the time shocks that affects all municipalities at the same time, the coefficients should be interpreted with caution due to potential endogeneity concerns. Overall, the results show little statistical association between police resources and the different outcomes examined, except for perceived crime. The lack of statistical association between police resources and the main outcomes is consistent with the increase of resources playing a limited role in explaining the effects of *Plan Cuadrante*. Figure B.5 reports these associations for two samples shown side by side: the 41 metropolitan

⁷⁰ Among the municipalities in the ENUSC survey, only Curicó introduced the *Seguridad Ciudadana* program before 2013, being 2012 the last year in which *Plan Cuadrante* was implemented for the first time in a municipality surveyed in ENUSC. Among the full set of 346 municipalities, only 10 of them introduced the program *Seguridad Ciudadana* before 2013, all of them except Curicó being small municipalities.

municipalities of Santiago (blue bars) and the broader set of municipalities that implemented *Plan Cuadrante* before 2006 together with the never treated municipalities surveyed in ENUSC (red bars). The results are similar across both samples.

Finally, we reproduce the subsection of Online Appendix B exploring which factors are associated with a larger increase in resources upon adoption of *Plan Cuadrante*:

As discussed in Section 2 and in Online Appendix B.4, the increase in police resources associated with the adoption of *Plan Cuadrante* varied substantially across municipalities. In principle, this increase should be related to each municipality’s pre-existing police resource deficit—measured in Equivalent Surveillance Units (UVEs)—which *Plan Cuadrante* was meant to help close. However, program evaluations do not clearly establish that this deficit is what determined the size of the resource increase each municipality received (Winchester et al., 2007; Salas P. et al., 2014).⁷¹ If the pre-existing deficit is not what determines the resource increase, what is it related to? Reassuringly, municipalities receiving larger versus smaller resource increases do not display differential pre-treatment trends in either the outcomes of interest or in police resources themselves.⁷² In this section we examine whether the magnitude of the resource increase experienced by a municipality upon adoption is correlated with its pre-treatment levels in crime, police resources, and other observable characteristics. Specifically, we regress the increase in resources at the time of adopting *Plan Cuadrante* on municipality characteristics measured before the program’s introduction, including crime and police resources in 2006, the first year for which information on police resources is available. Thus, we restrict this analysis to municipalities that adopted *Plan Cuadrante* from 2007 onward.

Table B.4 reports OLS regressions of the average annual increase in police officers and patrol vehicles per 100,000 inhabitants on pre-adoption municipality characteristics, separately for the administrative (SPD) sample at the municipality level (columns 1–2) and the ENUSC sample at the survey-respondent level (columns 3–4). All variables are standardized and standard errors are clustered by municipality, so coefficients are comparable in magnitude across regressors.

The results provide little support for the hypothesis that municipalities with worse pre-treatment crime outcomes received systematically larger resource increases. Pre-treatment reported crime is, if anything, negatively associated with the subsequent increase in both officers and vehicles in the administrative sample (columns 1–2); in the ENUSC sample, it is negatively associated with the increase in officers (column 3), although it is positively associated with the increase in vehicles (column 4). Pre-treatment victimization is, if anything, negatively associated with both increases in the ENUSC sample (columns 3–4). Pre-treatment arrests, crime perception, trust in the police, and private investment in security measures are not significantly associated with the resource increase. The increase in resources is instead more strongly predicted by structural municipality characteristics: less populous municipalities experienced larger increases in three of the four specifications, and municipalities with higher poverty rates predicted a larger increase in vehicles in the administrative sample (column 2), although this association

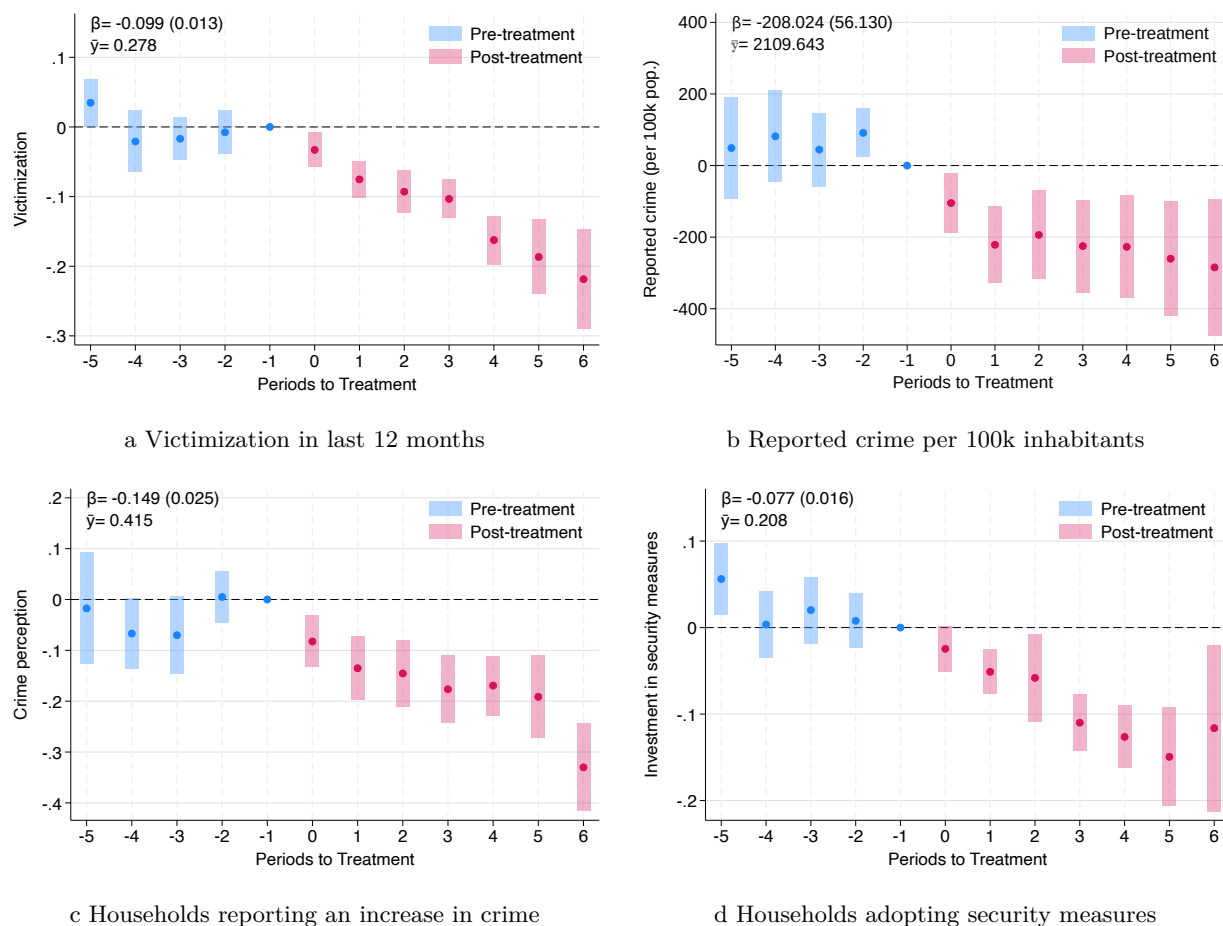
⁷¹Although we requested information on the demand and supply of UVEs underlying this allocation, this information was not provided by *Carabineros*.

⁷²We test for the joint significance of the pre-treatment coefficients between the two groups, separately for the split based on the increase in police officers and the split based on the increase in patrol vehicles. The resulting p-values are, respectively, 0.823 and 0.048 for victimization, 0.059 and 0.379 for crime perception, 0.820 and 0.157 for investment in security measures, and 0.213 and 0.419 for reported crime, as well as 0.616 and 0.444 for police officers and 0.405 and 0.157 for patrol vehicles.

is not significant for officers in that sample (column 1) or in the ENUSC sample (columns 3–4).

Other figures and tables referenced above:

Figure 2: Effects of *Plan Cuadrante* on Crime Outcomes and Security Perceptions



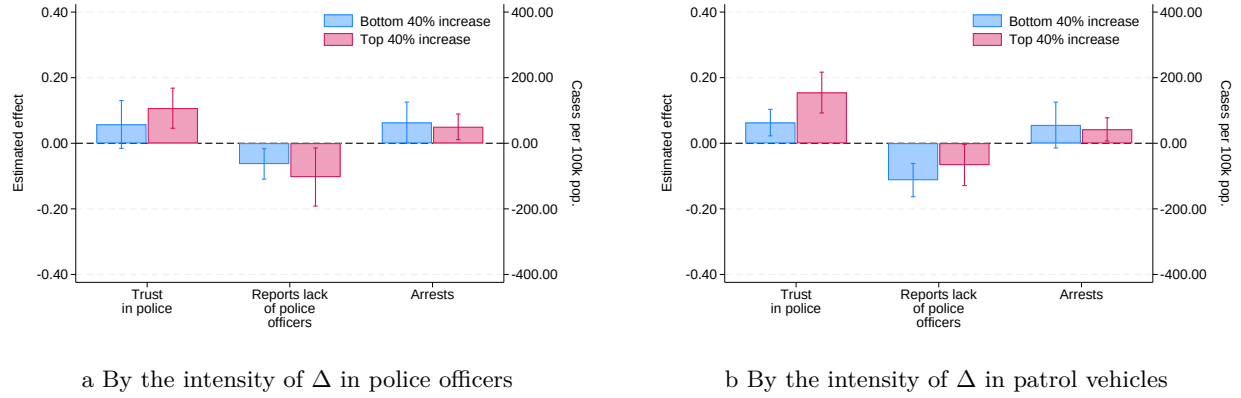
Notes: This figure shows the dynamic effects of the *Plan Cuadrante* on different measures of police effectiveness. The estimation is conducted using the procedure developed in Callaway and Sant'Anna (2021) for difference-in-differences with staggered treatment adoption. The dots in the figure represent the estimated coefficients and the bars behind them 95% confidence intervals. Standard errors are clustered at the municipality level using the wild bootstrapping procedure. The pooled effect and the mean outcome before the introduction of the treatment is also presented in each panel. Panel (a) reports the effect of *Plan Cuadrante* on the probability of household victimization in the last 12 months using the ENUSC survey. Panel (b) reports the effect on the number of crimes reported to the police in the municipality per 100,000 inhabitants using administrative data. Panel (c) reports the effect on the probability of perceiving an increase in crime using the ENUSC survey. Panel (d) reports the effect on the probability of adopting security measures in the last twelve months using the ENUSC survey. Panel (a) includes 93,041 observations; panel (b), 4,231; panel (c), 89,578; and panel (d), 93,041 observations.

Table A.1: Implementation of *Plan Cuadrante* by Year

Year	N	Municipalities
1998–2000	44	Metropolitan Region of Santiago (44 municipalities)
2002	10	Chiguayante, Concepción, Hualpén, Quilpué, San Antonio, San Pedro de la Paz, Talcahuano, Valparaíso, Villa Alemana, Viña del Mar
2003	2	Padre Las Casas, Temuco
2004	2	Antofagasta, Copiapó
2005	7	Alto Hospicio, Arica, Calama, Curicó, Iquique, Linares, Talca
2006	7	Coquimbo, La Serena, Los Andes, Ovalle, Rancagua, San Felipe, San Fernando
2007	7	Chillán, Chillán Viejo, Los Ángeles, Osorno, Puerto Montt, Punta Arenas, Valdivia
2008	9	Ancud, Castro, Constitución, Coronel, Lota, Peñaflor, Quillota, Rengo, Villarrica
2009	12	Angol, Aysén, Calera, Concón, Coyhaique, La Unión, Padre Hurtado, Parral, Penco, San Carlos, Tomé, Vallenar
2011	17	Arauco, Cañete, Cartagena, Cauquenes, Curanilahue, La Ligua, Limache, Molina, Nueva Imperial, Pucón, Puerto Varas, Río Bueno, San Clemente, San Javier, San Vicente, Santa Cruz, Victoria
2012	20	Cabrero, Calbuco, Casablanca, Chimbarongo, Collipulli, Curacaví, El Monte, Graneros, Illapel, Isla de Maipo, Lautaro, Machalí, Mostazal, Mulchén, Nacimiento, Natales, Panguipulli, Quellón, Quintero, Tocopilla
2013	13	Bulnes, Laja, Las Cabras, Lebu, Llaillay, Loncoche, Longaví, Los Álamos, Maule, Requínoa, Teno, Vicuña, Yumbel
Total	150	

Note: N is the number of municipalities that adopted *Plan Cuadrante* in each year. Data shared by Carabineros de Chile.

Figure B.3: Effects of *Plan Cuadrante* by Resource Increase: Trust, Complaints, and Arrests



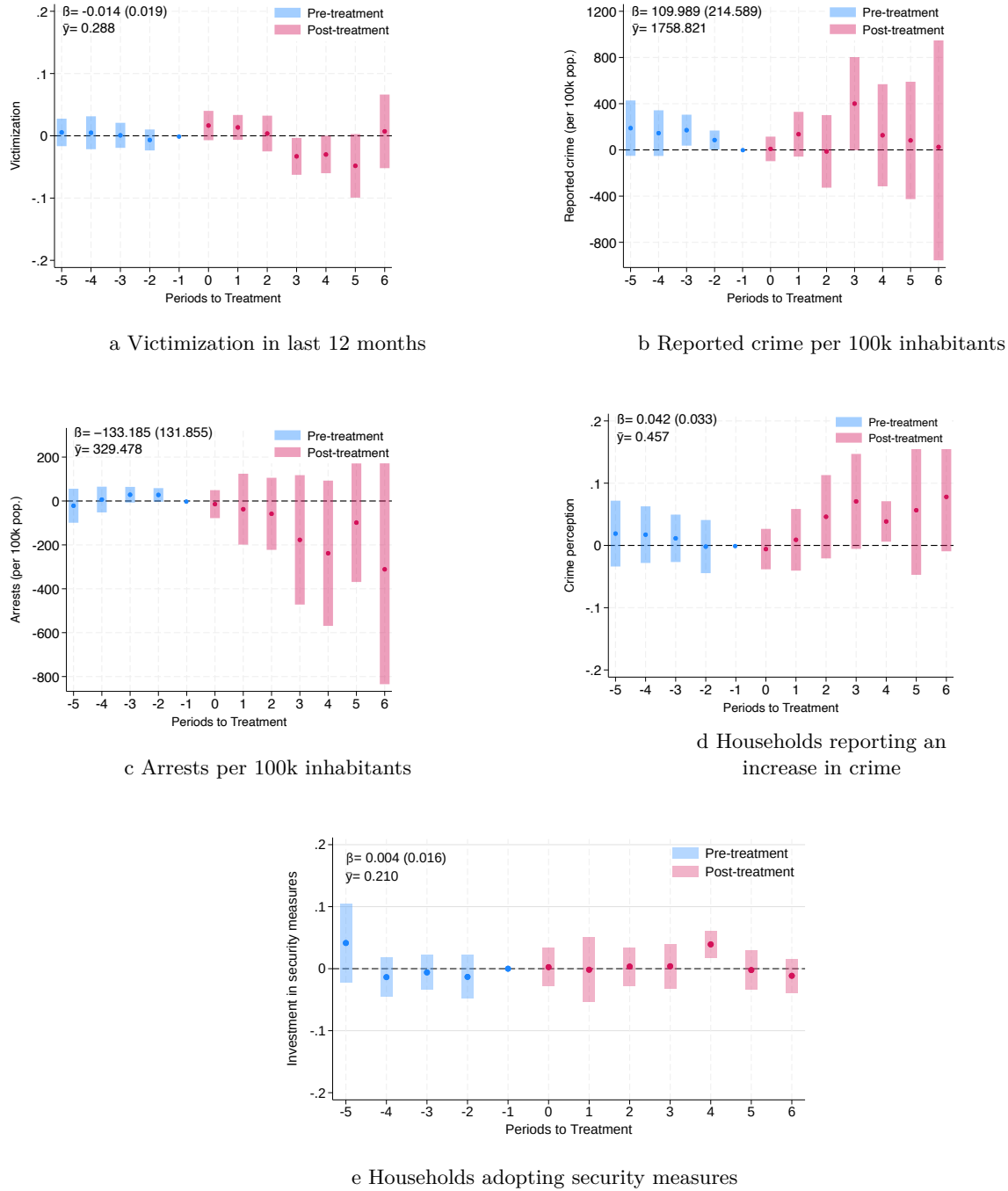
Notes: This figure complements Figure 4 by reporting, for the three outcomes not shown there, the effect of *Plan Cuadrante* estimated separately for municipalities in the bottom 40% (blue) and top 40% (red) of the resource-increase distribution. Panel (a) splits municipalities by the increase in police officers and panel (b) by the increase in patrol vehicles. In each panel, arrests (administrative data, in cases per 100,000 inhabitants) are read on the right axis; the other two outcomes, from the ENUSC survey (trust in the police, and reports about the lack of police officers), are read on the left axis. Effects are estimated using the difference-in-differences estimator for staggered treatment adoption developed in Callaway and Sant'Anna (2021); bars are point estimates and whiskers 95% confidence intervals, with standard errors clustered at the municipality level using the wild bootstrapping procedure.

Table B.3: Effects of *Plan Cuadrante* Controlling for Police Resources

	(1)	(2)	(3)	(4)	(5)
<i>Panel A. Victimization</i>					
ATT	-0.099*** (0.013)	-0.077*** (0.013)	-0.086** (0.043)	-0.094*** (0.028)	-0.087* (0.053)
N	93,041	53,973	53,553	72,017	53,553
<i>Panel B. Crime perception</i>					
ATT	-0.149*** (0.025)	-0.140*** (0.032)	-0.112** (0.057)	-0.082** (0.033)	-0.111 (0.106)
N	89,578	52,160	51,746	69,434	51,746
<i>Panel C. Investment in security measures</i>					
ATT	-0.077*** (0.016)	-0.066*** (0.019)	-0.109*** (0.041)	-0.081*** (0.019)	-0.168*** (0.064)
N	93,041	53,973	53,553	72,017	53,553
<i>Panel D. Reported crime (per 100k pop.)</i>					
ATT	-208.024*** (56.130)	-157.010*** (56.224)	29.230 (147.261)	-261.439 (170.954)	-245.555 (165.221)
N	4,231	2,963	1,911	2,248	1,911
Sample	Full	Resource yrs	Resource yrs	Resource yrs	Resource yrs
Police officers (per 100k pop.)			✓		✓
Patrol vehicles (per 100k pop.)				✓	✓

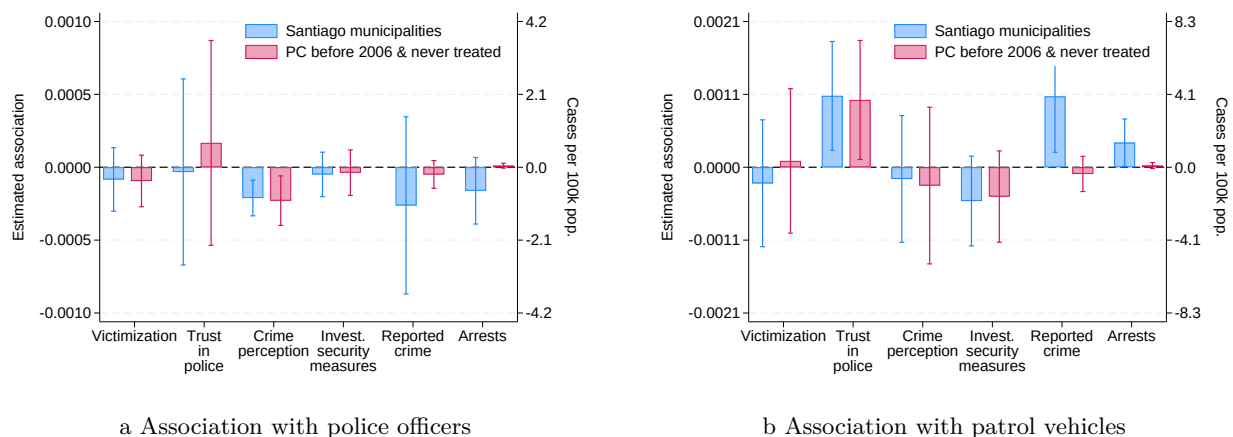
Note: ATT estimates of the effect of *Plan Cuadrante* on each outcome. Column (1) is the baseline (full estimation sample, no controls) and reproduces the headline estimate of the corresponding estimator. Column (2) restricts the estimation to the years in which police-resource data are available (from 2006 onward). No control variables are used in this estimation. Columns (3)–(5) add time-varying controls for police resources per 100,000 inhabitants: officers only; patrol vehicles only; and officers and patrol vehicles jointly (entered as separate regressors). Patrol vehicles are measured, per 100,000 inhabitants, as all patrol automobiles (radiopatrulla, auto patrullero, auto policial, automovil, station wagon, jeep and camioneta) plus motorcycles. Standard errors in parentheses; row N reports the estimation-sample size. Estimation follows Callaway and Sant’Anna (2021); standard errors clustered at the municipality level using wild bootstrapping procedures. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Figure B.4: Effects of *Seguridad Ciudadana* on Crime Outcomes and Security Perceptions



Notes: This figure shows the dynamic effects of the *Seguridad Ciudadana* program on different measures of crime and security perceptions. The estimation is conducted using the procedure developed in Callaway and Sant'Anna (2021) for difference-in-differences with staggered treatment adoption. The dots in the figure represent the estimated coefficients and the bars behind them 95% confidence intervals. Standard errors are clustered at the municipality level using wild bootstrapping. The pooled effect and the mean outcome before the introduction of the treatment for the outcome is also presented in each panel. Panel (a) reports the effect of *Seguridad Ciudadana* on the probability of household victimization in the last 12 months using the ENUSC survey. Panel (b) reports the effect on the number of crimes reported to the police in the municipality per 100,000 inhabitants using administrative data. Panel (c) reports the effect on the number of arrests in the municipality per 100,000 inhabitants using administrative data. Panel (d) reports the effect on the probability of perceiving an increase in crime using the ENUSC survey. Panel (e) reports the effect on the probability of adopting security measures in the last twelve months using the ENUSC survey. Panel (a) includes 210,102 observations; panel (b), 2,234; panel (c), 1,908; panel (d), 203,078; and panel (e), 74,316 observations.

Figure B.5: Association between Police Resources and Police Effectiveness (No *Plan Cuadrante* Variation)



Notes: The figure shows the estimated within-municipality association between police resources (per 100,000 inhabitants) and the main outcomes, in municipalities where *Plan Cuadrante* does not vary over the sample period. Estimates come from OLS regressions of each outcome on the resource measure with municipality and year fixed effects; standard errors are clustered at the municipality level. Blue bars use the 41 metropolitan municipalities of Santiago, all of which adopted *Plan Cuadrante* in 2000; red bars use the broader set of municipalities that adopted *Plan Cuadrante* before 2006 together with the never-treated municipalities surveyed in ENUSC (2006–2012). Reported crime and arrests (administrative data, in cases per 100,000 inhabitants) are read on the right axis; all other outcomes, from the ENUSC survey (victimization, trust in the police, crime perception, and investment in security measures), are read on the left axis. Panel (a) reports associations with the number of police officers; panel (b) with the number of patrol vehicles. Bars are point estimates with 95% confidence intervals.

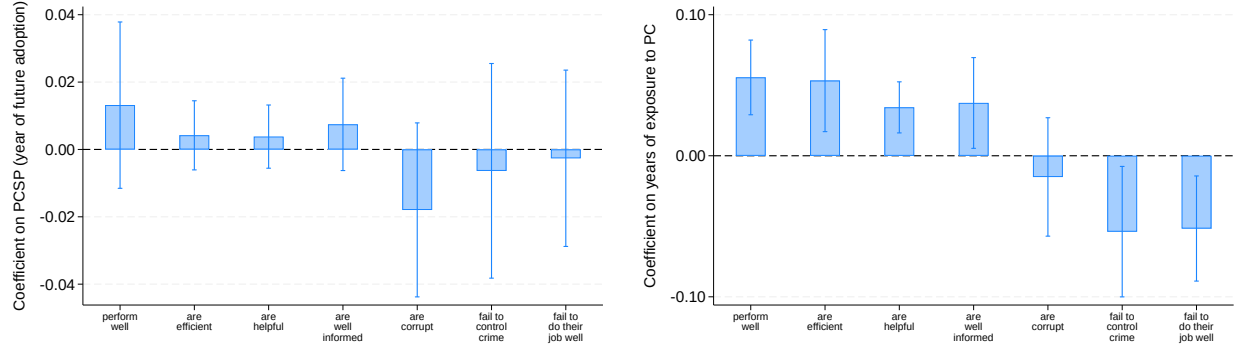
Table B.4: Predictors of the Increase in Police Resources under *Plan Cuadrante*

	Administrative sample		ENUSC sample	
	Police officers (1)	Patrol vehicles (2)	Police officers (3)	Patrol vehicles (4)
Dep. var: avg. annual increase in resources (share of pre-adoption level)				
Reported crime (per 100k pop.)	−0.203** (0.099)	−0.199** (0.097)	−0.319* (0.182)	0.272* (0.137)
Arrests (per 100k pop.)	0.099 (0.136)	−0.041 (0.129)	0.003 (0.186)	0.297 (0.223)
Victimization			−0.017* (0.010)	−0.025* (0.013)
Crime perception			0.005 (0.017)	0.008 (0.014)
Trust in the police			0.024 (0.018)	0.027 (0.019)
Investment in security measures			0.004 (0.010)	−0.003 (0.009)
Police officers (per 100k pop.)	−0.207 (0.155)	0.155 (0.129)	−0.888*** (0.154)	0.571*** (0.136)
Patrol vehicles (per 100k pop.)	−0.129 (0.129)	−0.139 (0.126)	0.537** (0.212)	−0.507*** (0.169)
Log population	−0.529*** (0.145)	−0.203* (0.112)	−0.193 (0.189)	−0.626** (0.269)
Poverty rate (2004)	0.170 (0.113)	0.308** (0.126)	0.027 (0.108)	−0.017 (0.108)
Observations	65	65	8,781	8,781
R-squared	0.342	0.282	0.614	0.545

Note: OLS regressions of the average annual increase in police officers (columns 1 and 3) and patrol vehicles (columns 2 and 4) per 100,000 inhabitants, from the year before adoption of *Plan Cuadrante* to 2012. As in panels (a) and (b) of Figure 4, the increase is expressed as a share of the resources available the year before adoption. Columns 1–2 use the administrative sample at the municipality level; columns 3–4 use ENUSC survey respondents interviewed before 2007 (survey outcomes at the respondent level, other regressors as pre-2006 municipality aggregates). All variables are standardized. Standard errors clustered by municipality in parentheses.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

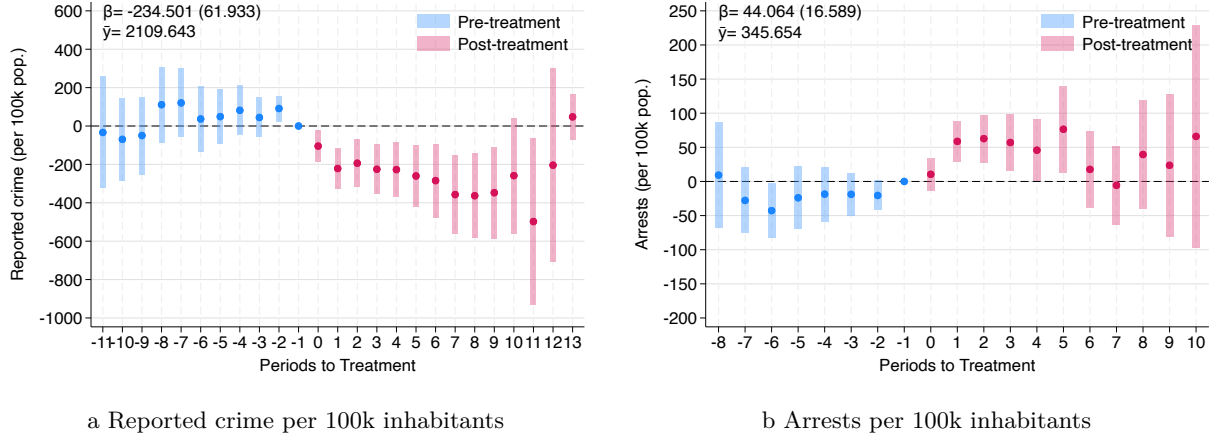
Figure B.6: Timing of Adoption of *Plan Cuadrante* and Perceived Police Performance



a Year of adoption and police performance before adoption b Years exposed to *Plan Cuadrante* (dose response)

Notes: The figure considers seven measures of police performance: the share of respondents who perceive police performance as good, and who perceive the police to be efficient, helpful, well informed, corrupt, failing to control crime, and failing to do their job well. Using the sample of municipalities that adopted *Plan Cuadrante* after 2005, panel (a) examines whether the future timing of adoption of *Plan Cuadrante* is correlated with perceived police performance in 2005, before the introduction of the program. Panel (b) reestimates the same comparison as Figure 4, replacing the binary adoption indicator with the municipality's number of years of exposure to *Plan Cuadrante* as of 2005. In both panels, each bar plots the coefficient from a separate regression of one performance measure. The graphs display the estimated coefficients and 95% confidence intervals.

Figure C.8: Effects of *Plan Cuadrante* on Reported Crime and Arrests Using the Full Set of Post-treatment and Pre-treatment Periods



Notes: This figure shows the dynamic effects of *Plan Cuadrante* on the number of crimes reported to the police in the municipality per 100,000 inhabitants (panel a) and on the number of arrests in the municipality per 100,000 inhabitants (panel b), using administrative data. The estimation is conducted using the procedure developed in Callaway and Sant'Anna (2021) for difference-in-differences with staggered treatment adoption. The dots in the figure represent the estimated coefficients and the bars behind them 95% confidence intervals. Standard errors are clustered at the municipality level using the wild bootstrapping procedure. The pooled effect and the mean outcome before the introduction of the treatment are also presented in each panel. Unlike Figure 2 in the main text, we do not cap the event study to the same 5 pre-treatment and 6 post-treatment periods used for symmetry with the ENUSC-based analysis. Instead, we exploit the full set of post-treatment and pre-treatment periods. Panel (a) includes 4,371 observations and panel (b) 3,372 observations.

Comment R2.2

Data: Chile has around 350 municipalities, yet the main analysis uses survey data for only 46 urban municipalities after excluding “always treated” municipalities. This restricted sample raises questions about external validity and potential biases: To what extent might this selection affect the results? Are these 46 municipalities spatially clustered or connected in ways that could bias comparisons? Could omitted municipalities differ systematically in crime dynamics or socio-economic characteristics? All these issues deserve some discussion in the paper.

Response to R2.2:

Thanks for the opportunity to clarify this point.

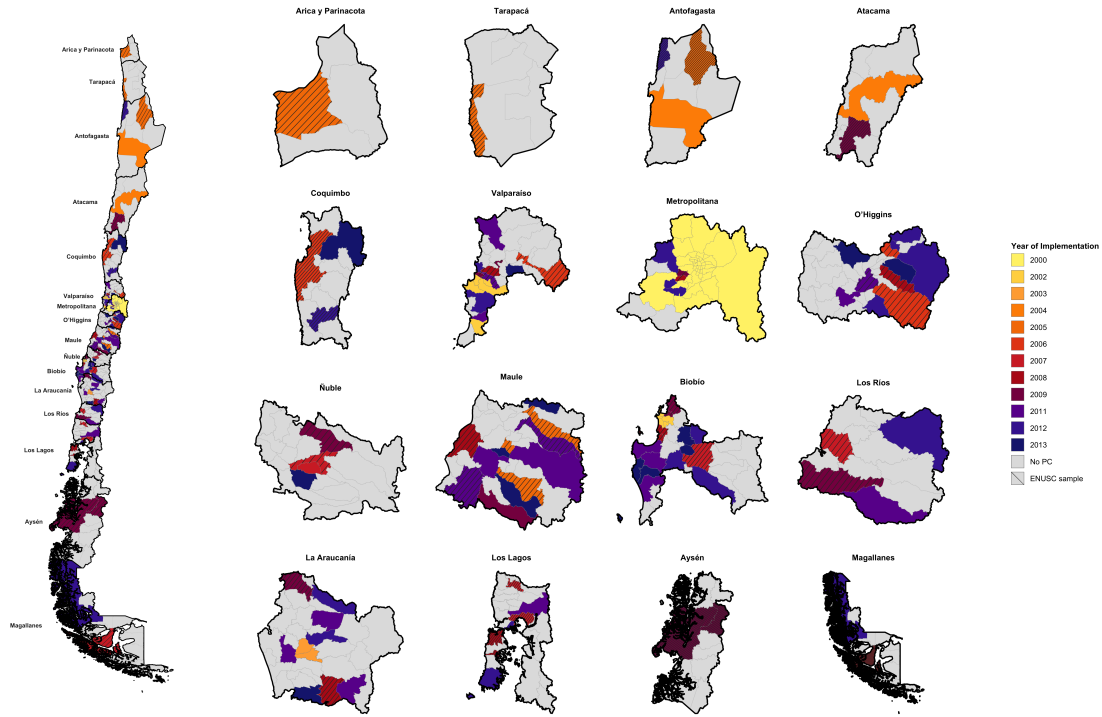
Plan Cuadrante is a policing program that targeted urban municipalities only. Specifically, it was designed to be introduced only in municipalities where at least 70% of the population was urban, although this threshold was not strictly applied to all municipalities incorporated in later waves. Between 1998 and 2013, when the last municipality received *Plan Cuadrante*, the program was introduced in 150 municipalities, out of a total of 346 municipalities in Chile, accounting for roughly 90% of the Chilean population.

We use two main data sources for outcome variables. First, we use the ENUSC survey. This survey is conducted only in 101 urban municipalities and includes information on household victimization, trust in the police, crime perception, and household investments in security. While these 101 municipalities are not a random draw of Chilean municipalities, they account for 87% of the Chilean population. Because the ENUSC was first conducted in 2003 and there was no ENUSC survey on 2004, the Callaway and Sant’Anna estimator mechanically excludes always-treated municipalities, i.e., those that adopted *Plan Cuadrante* before the start of the survey. This leads to the exclusion of 54 municipalities that introduced *Plan Cuadrante* before 2005. While this may seem like a large number, the excluded municipalities are highly spatially clustered, corresponding primarily to the municipalities of the metropolitan city of Santiago and a few additional clustered locations in the coast. The analytical sample in the analysis of the ENUSC outcomes is the remaining 46 treated municipalities and 1 never treated municipality, which account for approximately 27% of the Chilean population and are spatially distributed across the territory. These municipalities adopted *Plan Cuadrante* between 2005 and 2012.

Second, we use administrative information on reported crime and arrests at the municipality level, available on a yearly basis for the universe of Chilean municipalities since 2002 for reported crime and since 2005 for arrests. This expands substantially the number of municipalities in the analysis: 292 municipalities for reported crime, including never treated municipalities and municipalities that introduced *Plan Cuadrante* after 2002, and 281 municipalities for arrests, including never treated municipalities and municipalities that introduced *Plan Cuadrante* after 2005.

Regarding the external validity of our findings, the treated municipalities in our analytical samples are not spatially clustered or isolated. As shown in the map above, the 46 treated municipalities in the analytical sample using the ENUSC data are distributed across Chile from north to south, covering a wide range of geographic and socioeconomic contexts, and account for approximately 27% of Chile’s population. The treated municipalities in the administrative data sample provide even broader national coverage, together accounting for over half of Chile’s population. The consistency of results across the samples—which differ substantially in size, geographic scope, and outcome

Figure B.1: Geographic Distribution of *Plan Cuadrante*



Notes: This figure shows the spatial rollout of *Plan Cuadrante* across Chilean municipalities. The national map (left) is displayed alongside the 16 regional maps (right) for closer inspection. The color of each treated municipality indicates its year of program adoption, ranging from 2000 (lightest) to 2013 (darkest), following the legend. Municipalities shown in light grey were never treated under PCSP. Diagonal hatching identifies the 46 treated municipalities in the analytical sample used to estimate the effects on ENUSC outcomes.

measures—already provides substantial reassurance that our findings are not specific to a narrow group of municipalities.

Further evidence comes from the heterogeneity analysis reported in Section 4.3 of the updated manuscript and reproduced below. We find no significant differences in the effects of *Plan Cuadrante* across municipalities with different population densities, poverty rates, or household socioeconomic status (Figure 3). This result speaks directly to the concern that the particular composition of our analytical samples might drive the estimates, and suggests instead that the effects of the program are broadly consistent across the range of urban Chilean municipalities.

The table below characterizes the treated municipalities in our samples relative to all Chilean municipalities (Column 1) and all *Plan Cuadrante* municipalities (Column 2). On several dimensions they are broadly similar: the poverty rate of the 46 treated municipalities in the ENUSC sample (21.7%) and the 96 treated municipalities in the administrative data sample (24.0%) are both close to the national average of 23.7%, and the crime rates of the treated ENUSC municipalities closely track the average across all *Plan Cuadrante* municipalities (2,202 vs. 2,132 per 100,000 inhabitants). They do differ on other dimensions—particularly crime rates relative to all Chilean municipalities and population density. The differences have natural explanations. Higher crime rates are expected because *Plan Cuadrante* exclusively targeted urban municipalities, where crime is more prevalent. The lower-than-average population density in our samples (215 and 135 inhabitants per km², vs. a national mean of 870) reflects the fact that the most densely populated *Plan*

Cuadrante municipalities—those of the Santiago metropolitan area—were the earliest adopters of the program and are therefore not included in our analytical samples.

Table B.1: Baseline characteristics by subsample

	(1) All municipalities	PC municipalities				
		(2) All PC municipalities	(3) Always- treated	(4) Treated ENUSC	(5) Treated admin. data	(6) Never- treated
Reported crime (per 100k pop.)	1,650.23	2,132.29	2,437.25	2,202.06	1,976.59	1,282.47
Arrests (per 100k pop.)	267.62	442.08	606.01	459.05	358.11	134.11
Poverty rate (%)	23.73	20.50	14.88	21.73	23.96	26.82
Pop. density (pop./km ²)	870.28	1,904.14	4,593.32	215.52	134.50	27.02
Population	57,857.22	117,975.43	187,558.74	113,079.74	81,532.84	11,612.44
Municipalities	346	150	58	46	96	196

Note: Reported crime, poverty rate, population density, and population are measured as of 2003; arrests are measured as of 2005, the first year for which arrest data are available. Column (1) includes all municipalities in Chile. Column (2) includes all PC municipalities. Column (3) restricts to municipalities that adopted *Plan Cuadrante* before 2005, the always-treated in our sample, which are therefore excluded from our estimation samples (as depicted in Figure 1, panel (a)). Column (4) restricts to treated municipalities in the analytical sample using the ENUSC data. Column (5) restricts to treated municipalities included in the analytical sample using the administrative data. Column (6) includes never-treated municipalities.

Below we reproduce the updated data subsection of the manuscript:

Data

We combine administrative and survey data from *Carabineros*, the Undersecretary for Crime Prevention, and the National Institute of Statistics (INE).

Carabineros provided information on *Plan Cuadrante* implementation, which we use to define treatment status across municipalities over time. The National Institute of Statistics granted access to the National Survey on Security and Crime (ENUSC), a repeated cross-section first conducted in 2003 and administered annually between October and December since 2005. The survey covers more than 25,000 households across 101 urban municipalities, collecting information on crime victimization, perceptions, trust in police, and household prevention measures over the previous 12 months.⁷³

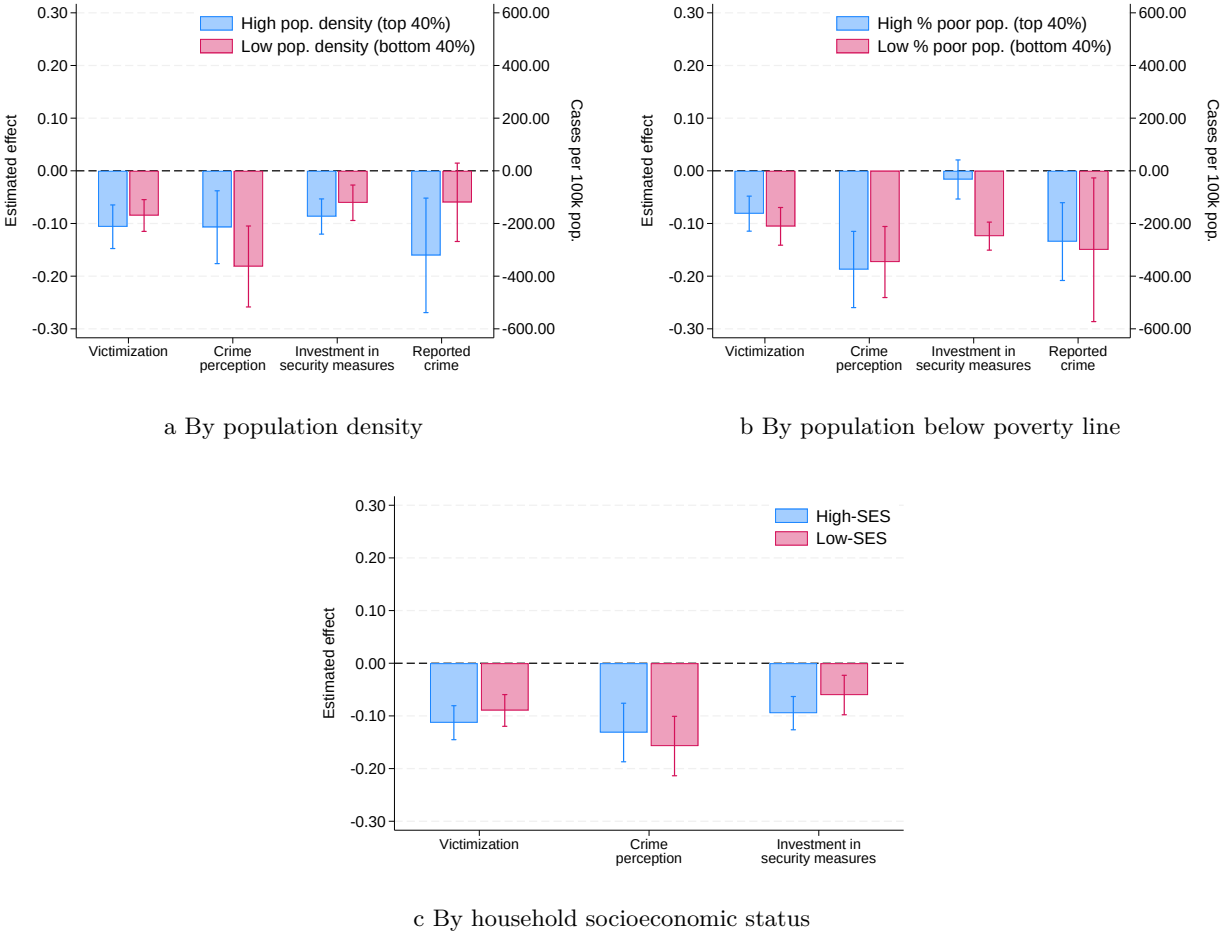
Our empirical approach excludes always-treated municipalities (Callaway and Sant’Anna, 2021). Given ENUSC’s first implementation in 2003 and absence in 2004, our main sample includes households from 46 urban municipalities that adopted *Plan Cuadrante* between 2005 and 2012 plus one never-treated municipality.⁷⁴ The treated municipalities in this analytical sample account for approximately 27% of Chile’s population.

The Undersecretary for Crime Prevention provided data on the number of reported crimes (2002–2016) and arrests (2005–2016) for all municipalities. This comprehensive coverage allows us to expand the analytical sample: for reported crime, we include all municipalities adopting *Plan Cuadrante* after 2002 together with never-treated municipalities, for a total of

⁷³The survey is administered to a randomly selected household member who must: a) live in the household, b) be 15 years or older, c) not be a domestic worker, and d) not have any disability preventing survey comprehension.

⁷⁴This sample excludes municipalities adopting *Plan Cuadrante* in 2013 as they are not covered by ENUSC. Also, there is only one never treated municipality covered by ENUSC.

Figure 3: Effects of *Plan Cuadrante* by Population Density, Poverty, and Socioeconomic Status



Notes: This figure reports the heterogeneous effects of *Plan Cuadrante* by population density (panel a), the proportion of poor people in the municipality (panel b), and household socioeconomic status (panel c). Each panel reports the effect on victimization, defined as the probability of household victimization in the last 12 months; crime perception, defined as the probability of perceiving an increase in crime; investment in security measures, defined as the probability of adopting security measures in the last twelve months; and reported crime, defined as the number of crimes reported to the police in the municipality per 100,000 inhabitants. Reported crime (administrative data, in cases per 100,000 inhabitants) is read on the right axis; all other outcomes, from the ENUSC survey (victimization, crime perception, and investment in security measures), are read on the left axis. Household socioeconomic status is only observed in the ENUSC, so panel (c) does not include reported crime. In panels (a) and (b), the bars compare municipalities in the two bottom quintiles (bottom 40%) and in the two top quintiles (top 40%) of the population density and poverty distributions, respectively. The average population density is 458 inhabitants per square kilometer in high-density municipalities and 29 in low-density municipalities; the average poverty rate is 28% in high-poverty municipalities and 14% in low-poverty municipalities. In panel (c), low-SES households include lower-class and lower-middle-class households, while high-SES households include upper-middle and upper-class households. The estimation is conducted using the difference-in-differences estimator for staggered treatment adoption developed in Callaway and Sant'Anna (2021). The graphs display the estimated coefficients and 95% confidence intervals. Standard errors are clustered at the municipality level using the wild bootstrapping procedure.

292 municipalities; for arrests, the corresponding sample includes all post-2005 adopters and never-treated municipalities, for a total of 281 municipalities. The treated municipalities in these analytical samples account for over half of Chile's population.

The study period differs across the two data sources, as each covers a different time span and a different set of municipalities. For the ENUSC-based analysis, it is restricted to 2003–2012. The Callaway and Sant’Anna (2021) estimator uses not-yet-treated and never-treated municipalities as the comparison group. Since only one municipality in the survey sample is never treated, extending the analysis beyond 2012—the last year in which a municipality surveyed in ENUSC adopts the program—would leave only that single never-treated unit as the comparison group for subsequent periods, making the estimates unreliable. The administrative data, by contrast, cover a large number of never-treated municipalities that can serve as controls throughout the full data period, allowing us to exploit the complete administrative coverage—2002–2016 for reported crime and 2005–2016 for arrests—and to include all municipalities adopting the program from 2003 to 2013, when the last municipality joins *Plan Cuadrante*.

Together, the treated municipalities in our two analytical samples represent substantial shares of the Chilean population—approximately 27% in the ENUSC-based analysis and over half in the administrative data analysis—and are geographically distributed across the country rather than clustered in a single region. Tables A.2 and A.3 in Online Appendix A and Table B.1 and Figure B.1 in Online Appendix B provide a detailed characterization of the treated municipalities in each analytical sample relative to the average Chilean municipality.

Below we also reproduce the new subsection of Appendix B of the updated manuscript characterizing the treated municipalities in our analytical samples:

Characterization of the Analytical Samples

The treated municipalities in our analytical samples are geographically distributed across the country. As shown in Figure B.1, the 46 treated municipalities in the ENUSC-based sample span Chile from north to south, covering a wide range of geographic and socioeconomic contexts, and together account for approximately 27% of Chile’s population. The 96 treated municipalities in the administrative data sample provide even broader national coverage, together accounting for over half of Chile’s population.

Table B.1 characterizes the treated municipalities in each analytical sample relative to all Chilean municipalities (Column 1), all *Plan Cuadrante* municipalities (Column 2), the always-treated municipalities excluded from our estimation samples (Column 3), and never-treated municipalities (Column 6). Columns (4) and (5) report characteristics for the treated municipalities in the ENUSC and administrative data analytical samples respectively. On several dimensions our samples broadly resemble the overall Chilean population: the poverty rate of the 46 treated municipalities in the ENUSC sample (21.7%) and the 96 treated municipalities in the administrative data sample (24.0%) are both close to the national average of 23.7%, and the crime rates of the treated ENUSC municipalities closely track the average across all *Plan Cuadrante* municipalities (2,202 vs. 2,132 per 100,000 inhabitants). They do differ on other dimensions—particularly crime rates relative to all Chilean municipalities and population density—but both differences have natural explanations. Higher crime rates are expected because *Plan Cuadrante* exclusively targeted urban municipalities, where crime is more prevalent. The lower-than-average population density in our samples (215 and 135 inhabitants per km², vs. a national mean of 870) reflects the fact that the most densely populated *Plan Cuadrante* municipalities—those of the Santiago metropolitan area—were the earliest adopters of the program and are therefore excluded from our analytical samples as always-treated.

(Figure B.1 and Table B.1 are reproduced above.)

Other tables referenced above:

Table A.2: Summary Statistics – ENUSC sample

	Mean	SD	Min	Max	N
<i>Household</i>					
Socioeconomic status	3.58	0.73	1	5	99,298
Victimization	0.26	0.44	0	1	99,298
Crime perception (neighborhood)	0.38	0.49	0	1	95,598
Crime perception (municipality)	0.66	0.48	0	1	96,794
Crime perception (country)	0.79	0.41	0	1	98,525
Trust in the police	0.44	0.50	0	1	51,445
Investment in security measures	0.21	0.40	0	1	99,298
<i>Plan Cuadrante</i> (year)	2007	1.85	2005	2012	97,847
<i>Household member interviewed</i>					
Female	0.55	0.50	0	1	99,298
Age	43.87	17.95	15	99	99,298
Primary school attendance	0.98	0.14	0	1	99,298
Secondary school attendance	0.72	0.45	0	1	99,298
University attendance	0.20	0.40	0	1	99,298

Note: This table presents summary statistics from the ENUSC sample, covering the 47 municipalities used in the survey-based analysis (46 that adopted *Plan Cuadrante* between 2005 and 2012 plus one never-treated municipality). Socioeconomic status is a 1 to 5 measure produced by ENUSC surveys and based on self-reported socioeconomic strata, being 1 the highest ranked and 5 the lowest. Crime perception is equal to 1 if the interviewed person reports that crime has increased in the last 12 months. The *Plan Cuadrante* (year) row is computed over treated municipalities only.

Table A.3: Summary Statistics – Administrative municipality level sample

	Mean	SD	Min	Max	N
<i>Panel A. Reported-crime sample (292 municipalities)</i>					
Reported crime (per 100k pop.)	1,732.39	1,036.68	0.00	11,434.40	4,371
Arrests (per 100k pop.)	332.07	271.75	0.00	1,712.30	3,504
Poverty rate (%)	25.74	9.72	5.65	58.33	3,720
Pop. density (pop./km ²)	63.51	140.47	0.09	1,109.58	4,065
<i>Plan Cuadrante</i> (year)	2010	2.82	2003	2013	1,440
<i>Panel B. Arrests sample (281 municipalities)</i>					
Reported crime (per 100k pop.)	1,684.74	1,014.67	0.00	11,434.40	4,209
Arrests (per 100k pop.)	313.21	251.23	0.00	1,712.30	3,372
Poverty rate (%)	26.01	9.72	5.65	58.33	3,570
Pop. density (pop./km ²)	57.90	127.30	0.09	1,109.58	3,900
<i>Plan Cuadrante</i> (year)	2010	2.26	2006	2013	1,275

Note: The table reports municipality-level summary statistics for the two administrative estimation samples used in the paper. Panel A covers the 292 municipalities in the reported-crime sample (municipalities adopting *Plan Cuadrante* from 2003 onward plus never-treated); Panel B the 281 municipalities in the arrests sample (adopters from 2006 onward plus never-treated — municipalities adopting in 2005 are excluded because arrest data begin that year, leaving no pre-treatment period). Reported crime and arrests are measured per 100,000 inhabitants over 2002–2016 and 2005–2016, respectively. Minimum reported crime and arrests of zero correspond to tiny, remote never-treated comunas (e.g., Juan Fernández, Cabo de Hornos). The *Plan Cuadrante* (year) row is computed over treated municipalities only. Data from the Undersecretary for Crime Prevention.

Comment R2.3

Methods and Results: My main concern relates to SUTVA violations due to spillovers.

Control municipalities may be indirectly affected by the implementation of “Plan Cuadrante” in treated municipalities, which could bias the ATT estimates. Two channels seem particularly relevant: a) Crime displacement (criminal activity might shift from treated to nearby untreated municipalities) and b) Resource displacement (personnel, vehicles, or budget may have been diverted from control to treated municipalities).

Unfortunately, the robustness check used by the authors to claim “no spillovers” to neighboring untreated municipalities is only briefly mentioned in Section 3.2. Having read the appendix looking for details, I still have two concerns about their exercise. First, given that the main ENUSC sample includes only 46 municipalities, it is unclear which municipalities are considered “neighboring municipalities” in this test (i.e., how many neighbor municipalities do you have among the 46 municipalities considered?). Second, identifying spatial spillovers generally requires more sophisticated designs such as two-stage randomization as in Crépon et al. (2013) in the context of labor markets, or Blattman et al. (2021) in the context of crime. Thus, it is unclear to what extent the current exercise can capture spillovers. A revised version of the paper should include a more serious discussion of this issue in the main body of the paper.

Response to R2.3:

Thank you very much for the feedback around spillovers, which we agree was crucial and did not receive the necessary attention in the previous version of the manuscript.

Spillovers from treated municipalities to control municipalities can bias the estimates upwards for two reasons. First, *Plan Cuadrante* might displace criminals from committing crime in treatment municipalities to committing crime in control municipalities. Second, the program could in principle relocate police resources from nearby control municipalities to treated ones. In the revised manuscript, we address both channels directly. We first use administrative data on reported crime and arrests, which are available since 2002 for crime and since 2005 for arrests for all Chilean municipalities, rather than being restricted to the 47 municipalities surveyed in ENUSC. These administrative samples include 292 municipalities for reported crime and 281 for arrests, out of a total of 346 municipalities.⁷⁵ The results, presented in the appendix and summarized in the new spillovers section of the main text, show no evidence that *Plan Cuadrante* displaced crime or arrests to nearby untreated municipalities.

We then expand the analysis to police resources, again finding no meaningful spillovers. This is consistent with the institutional functioning of Carabineros. Transfers of uniformed officers are difficult outside the 10-year service period in each location, and the program documents indicate that the resource increases associated with *Plan Cuadrante* were driven mainly by new hirings rather than by reallocating officers or vehicles from control to treatment areas. This is also consistent with the fact that police resources increased substantially in both treatment and control municipalities during the 2000s and 2010s.

We also examine the 46 ENUSC municipalities separately. Even in that smaller sample, where the potential for spillovers is harder to assess, we find no evidence that early adopters displaced crime or resources onto later adopters before those municipalities themselves joined the program.

⁷⁵The municipalities excluded from these samples are the always-treated municipalities, which adopted *Plan Cuadrante* before the administrative data begin.

Following your suggestion, we further conduct two additional exercises. First, following Crépon et al. (2013), we estimate the effect of a higher share of neighboring municipalities adopting *Plan Cuadrante* on crime, arrests, and police resources in control municipalities, finding no significant effects on any of these outcomes. Second, following Blattman et al. (2021a), we implement a randomization inference exercise in which treatment status and adoption dates are randomly assigned 1,000 times, constructing placebo distributions of treatment and spillover effects. The true treatment effect on crime falls in the bottom 1% of the placebo distribution, while the estimated spillover effects fall within the 40th–45th percentile, providing no evidence of meaningful spillovers on crime, arrests, or resources onto neighboring untreated municipalities.

Finally, the broad administrative coverage allows us to estimate spillovers credibly using the full set of administrative outcomes, while the ENUSC outcomes are more limited because only 46 adopting municipalities and 1 never-treated municipality are observed. Within ENUSC, only 10 municipalities share a border with another municipality surveyed in ENUSC, and only 6 share a border with either a never-treated municipality or a treated municipality that adopted the program in a different year. We therefore rely on the administrative exercises for the main spillover evidence and use the ENUSC subsample as a complementary check.

We are providing below the new section added to the manuscript summarizing the results of the spillover analyses and the Appendix providing the details of the spillover analyses:

Beyond parallel trends, a causal interpretation of our estimates requires the absence of spillover effects from treated to control municipalities, since such spillovers could bias treatment effects upward. This concern is relevant in our setting. Previous work on deployment-based policing reforms, such as hot-spot policing, shows that crime may be displaced to nearby areas (Blattman et al., 2021b). Spillovers could also arise through the reallocation of police resources from control to treated municipalities.

In our setting, spillovers are less likely for several reasons. First, as discussed in Section 2, *Plan Cuadrante* introduced beat policing in Chile by reorganizing deployment across the full municipal territory rather than concentrating resources in specific hot spots. Second, Chilean municipalities are large—on average 101,000 inhabitants and 2,000 square kilometers (Gobierno de Chile, 2024)—and criminals typically do not travel such long distances to commit crimes, which makes displacement to other municipalities unlikely (Kirchmaier et al., 2023). Third, and more directly related to police resources, police regulations assign officers to *comisaría*s—broader operational units that usually cover an entire municipality—for fixed periods, which limits rapid reallocation of officers across jurisdictions.⁷⁶

We nonetheless assess spillovers empirically through three complementary exercises. Online Appendix C reports the full results. First, we estimate the effect of *Plan Cuadrante* on police-reported crime and police resources in untreated municipalities whose neighbors adopt the program. Online Appendix Figure C.12 summarizes these estimates and shows no evidence of spatial spillovers.

Second, following Crépon et al. (2013), we test whether higher exposure to *Plan Cuadrante* among neighboring municipalities affects outcomes in control municipalities. Online Appendix Figures C.13 and C.14—which define neighbor exposure as the share and the number of treated

⁷⁶See the Transfer Manual for Carabineros de Chile Personnel available at https://www.carabineros.cl/transparencia/og/pdf/OG_2707_13112019.pdf

neighbors, respectively—show no meaningful effects across the outcomes we study, again suggesting the absence of spatial spillovers.

Third, following Blattman et al. (2021b), we implement a randomization-inference exercise in which treatment status and adoption dates are randomly reassigned 1,000 times. Online Appendix Figure C.15 shows that estimated spillover effects lie near the center of the placebo distribution—between the 40th and 45th percentiles—which provides further evidence against spatial spillovers. By contrast, the estimated effects for treated municipalities lie in the bottom first percentile of the placebo distribution, consistent with *Plan Cuadrante* significantly reducing crime in treated municipalities.

The exercises above evaluate spatial spillovers using administrative records for the extended municipality sample. We complement this evidence with two additional analyses focused on ENUSC municipalities. First, we re-estimate the administrative-outcome spillover analysis—for reported crime, arrests, police officers, and patrol vehicles—restricting the sample to neighbors of the 46 ENUSC municipalities (Online Appendix Figure C.16). Second, we test for spillovers directly in ENUSC outcomes. Because ENUSC coverage is limited—46 treated municipalities and 1 never-treated municipality, with only 10 sharing a border and 6 of those treated in different years—we estimate spillover effects from early-adopting ENUSC municipalities onto late-adopting ENUSC neighbors before the latter adopt the program (Online Appendix Figure C.17). Both exercises yield reassuringly null effects.

Taken together, these results indicate that spatial spillovers are unlikely to threaten identification in our setting.

This Online Appendix details the specifications behind the spillover tests summarized in Section 4.2.

The baseline test estimates the effect of *Plan Cuadrante* on reported crime, arrests, police officers, and patrol vehicles in neighboring untreated municipalities,

$$Outcome_{mt} = \delta_1 \text{Plan Cuadrante near } municip_{.mt} + Municipality_m + Year_t + u_{mt} \quad (4)$$

where the sample is restricted to municipalities that never adopted the program, each is assigned a treatment date equal to the earliest adoption among its treated neighbors, and δ_1 is estimated dynamically with the staggered difference-in-differences estimator of Callaway and Sant’Anna (2021). Because these administrative outcomes are observed for every municipality, we estimate the specification on the full national sample of never treated municipalities (Online Appendix Figure C.12).

For the intensity-based test in the spirit of Crépon et al. (2013), we replace the binary neighbor-treatment indicator with a continuous measure of exposure and restrict the sample to never-treated municipalities,

$$Outcome_{mt} = \delta_1 \text{Share neighbors with Plan Cuadrante}_{mt} + Municipality_m + Year_t + u_{mt} \quad (5)$$

Because the treatment is continuous, we estimate it with the heterogeneity-robust estimator of Chaisemartin and D’Haultfœuille (2020), measuring exposure both as the share of treated neighbors (Online Appendix Figure C.13) and as their number (Online Appendix Figure C.14).

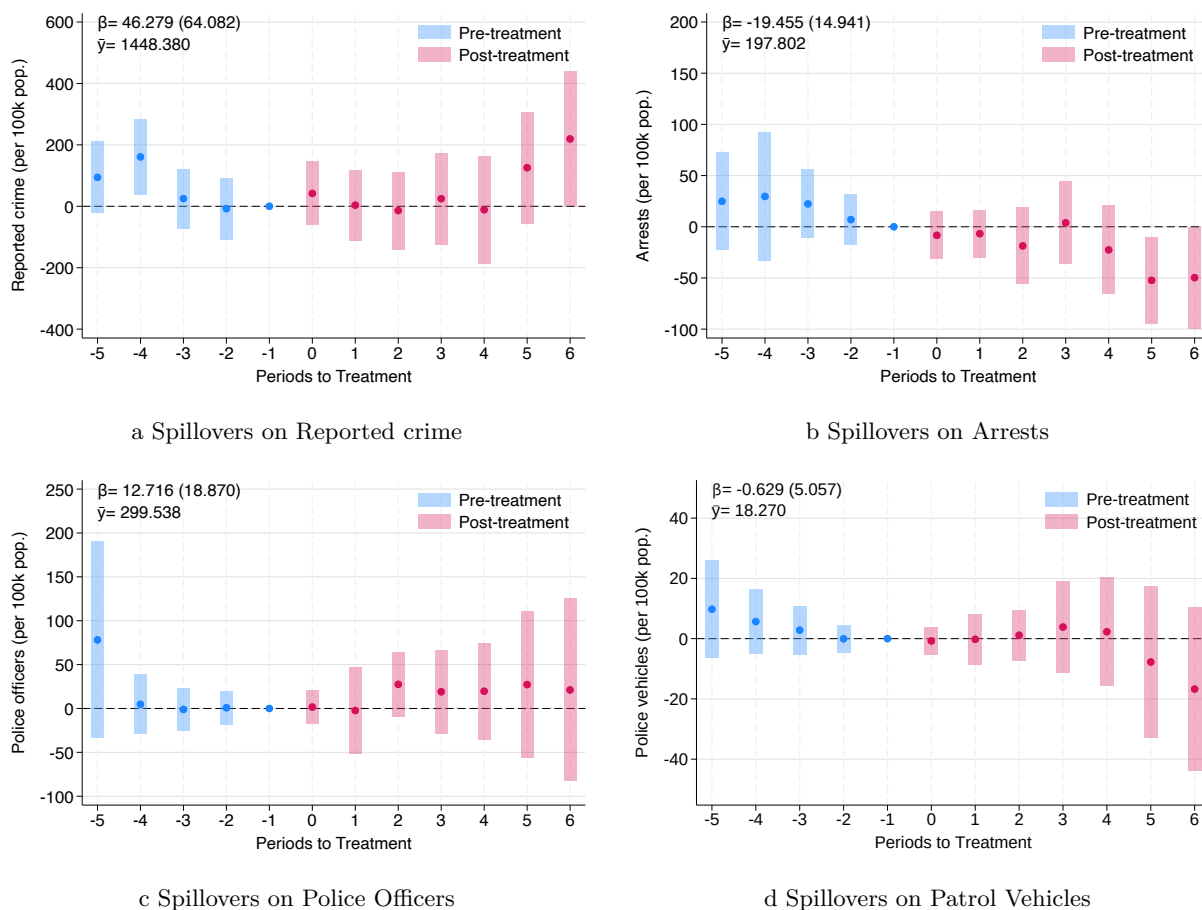
For the randomization-inference test, we reassign adoption dates and never-treated status across municipalities 1,000 times and re-estimate the direct and spillover effects under the sharp

null, yielding the placebo distributions in Online Appendix Figure C.15. For this exercise we rely on the estimator of Chaisemartin and D’Haultfoeuille (2020) rather than that of Callaway and Sant’Anna (2021): re-estimating the latter over a thousand permutations would require prohibitive computational power and running time.

Finally, we re-estimate the baseline administrative specification on the municipalities neighboring the 46 ENUSC municipalities (Online Appendix Figure C.16); and, for the outcomes measured only in ENUSC, we estimate the spillover of early-adopting ENUSC municipalities on their late-adopting ENUSC neighbors before the latter’s own adoption (Online Appendix Figure C.17). The six focal late-adopters, each shown with its earliest-adopting ENUSC neighbor and that neighbor’s adoption year, are Molina (Curicó, 2005), Tocopilla (Iquique, 2005), San Carlos (Chillán, 2007), Padre Hurtado (Peñaflor, 2008), Calera (Quillota, 2008), and Limache (Quillota, 2008). Across all of these specifications, the estimated spillover effects are small and statistically indistinguishable from zero.

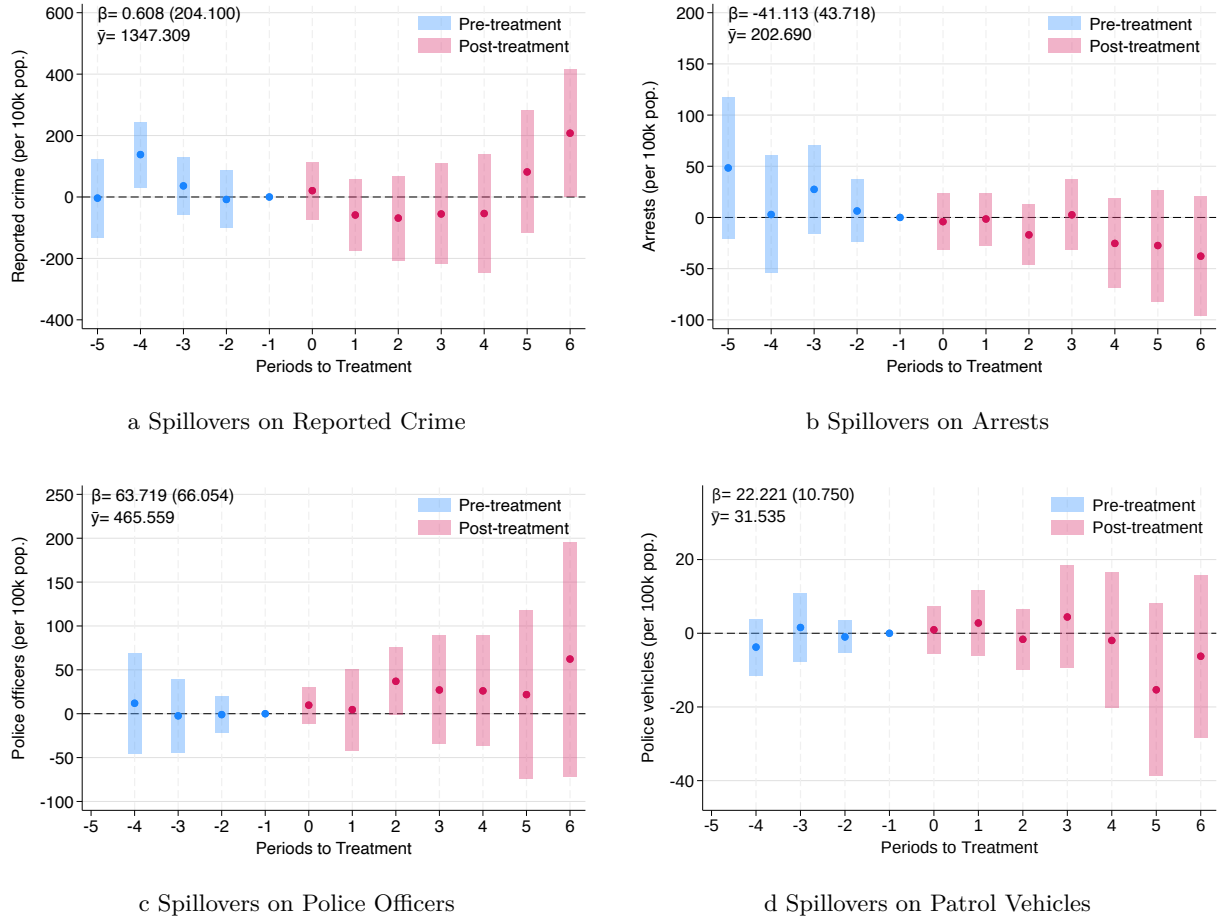
Other figures and tables referenced above:

Figure C.12: Geographic Spillovers of *Plan Cuadrante* on Neighboring Untreated Municipalities



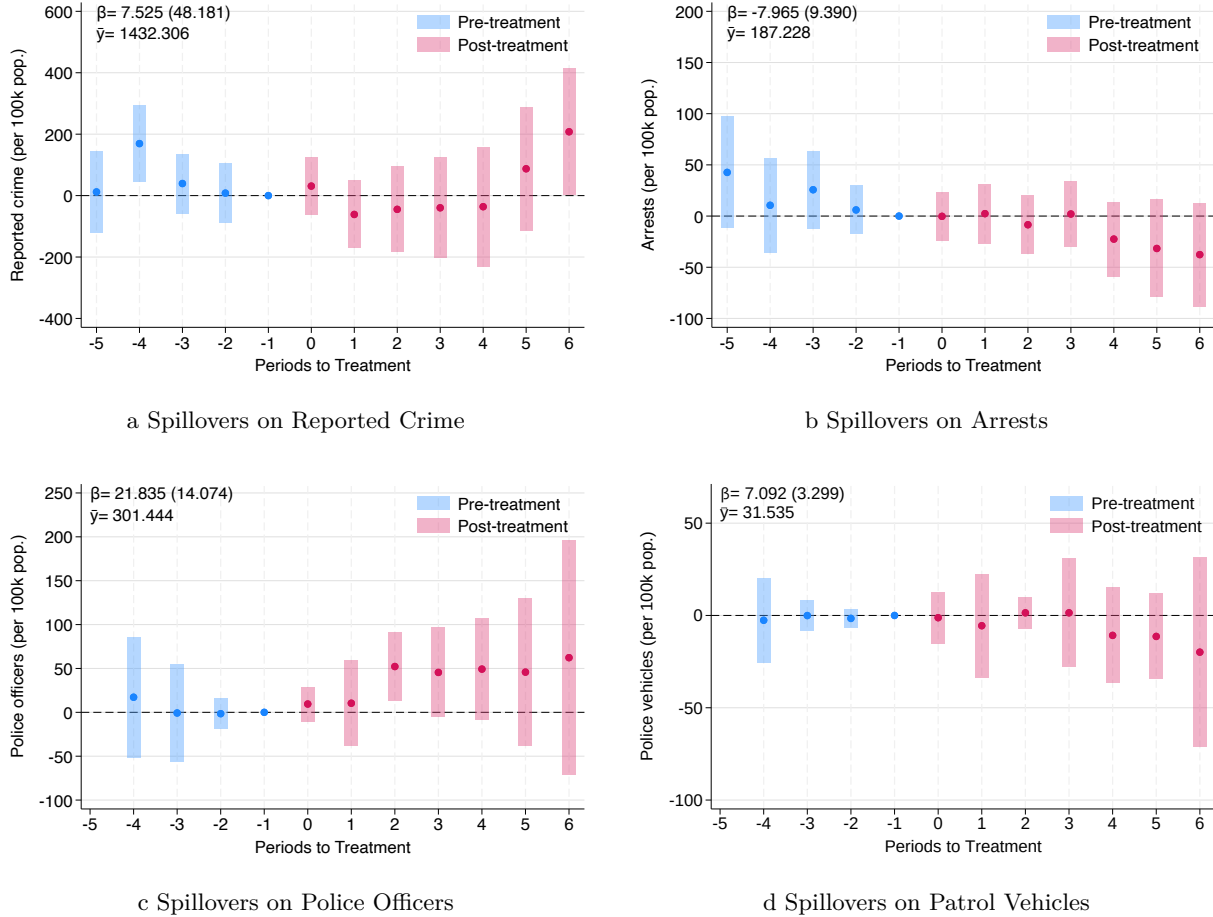
Notes: This figure reports the spillover effects of *Plan Cuadrante* on the number of crimes reported to the police in the municipality per 100,000 inhabitants (panel a), the number of arrests in the municipality per 100,000 inhabitants (panel b), the number of police officers per 100,000 inhabitants (panel c), and the number of patrol vehicles per 100,000 inhabitants (panel d). Panels (a)–(d) include 2,859, 2,166, 1,144, and 1,942 observations respectively. Specifically, the figure shows the dynamic effects of the *Plan Cuadrante* for neighbor municipalities which never introduced the program. The estimation is conducted using the procedure developed in Callaway and Sant’Anna (2021) for difference-in-differences with staggered treatment adoption, using as the treatment introduction date the earliest date of a neighboring municipality. The dots in the figure represent the estimated coefficients and the bars behind them 95% confidence intervals. Standard errors are clustered at the municipality level using the wild bootstrapping procedure. The pooled effect and the mean outcome before the introduction of the treatment for every outcome is also presented in each panel.

Figure C.13: Geographic Spillovers of *Plan Cuadrante* by Share of Treated Neighbors



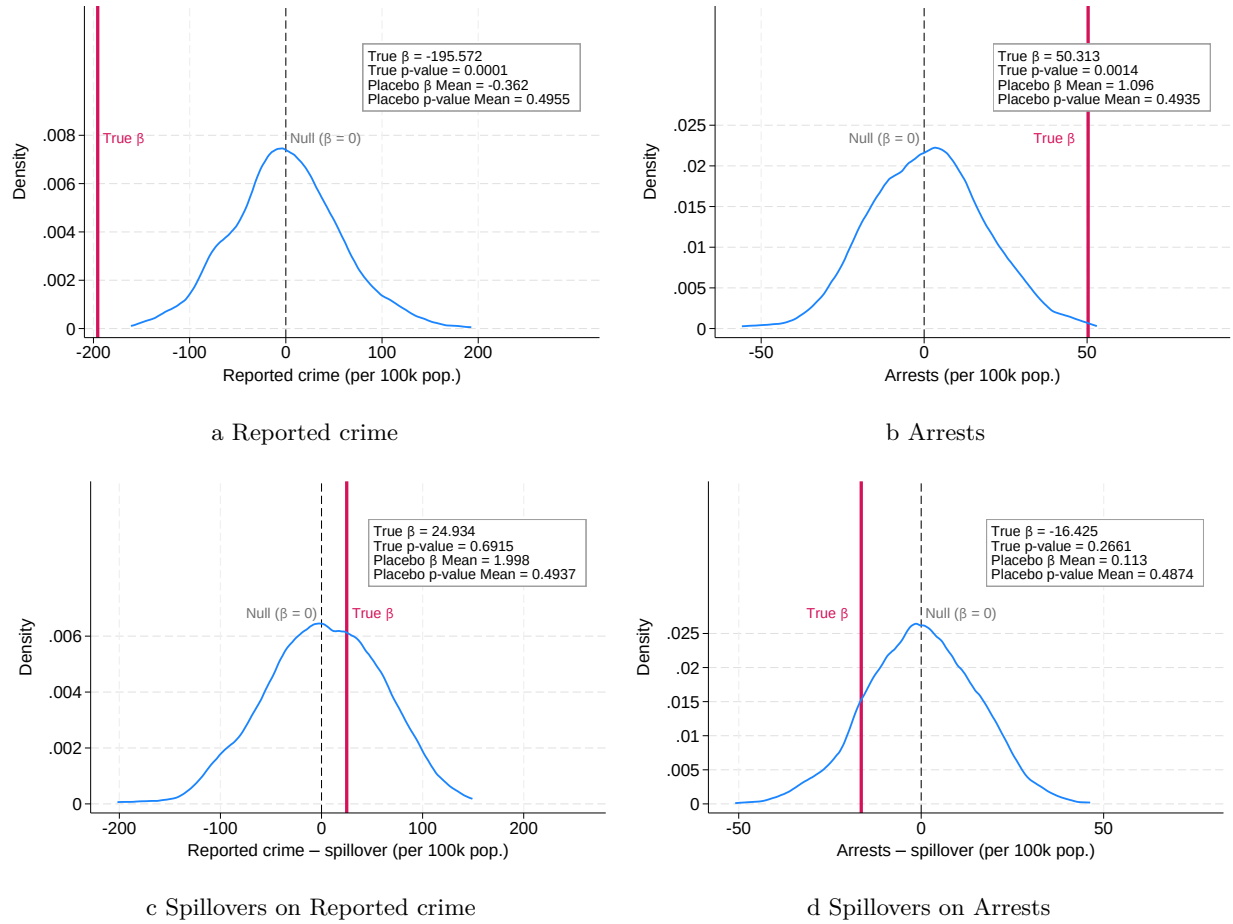
Notes: This figure reports the geographic spillover effects of *Plan Cuadrante* on four outcomes for never-treated municipalities: reported crime per 100,000 inhabitants (panel a), arrests per 100,000 inhabitants (panel b), the number of police officers per 100,000 inhabitants (panel c), and the number of patrol vehicles per 100,000 inhabitants (panel d). Unlike the dummy spillover specification, the treatment variable is now continuous and equals the share of neighboring municipalities that have implemented *Plan Cuadrante* by year t , following the spillover-intensity approach of Crépon et al. (2013). The sample is restricted to municipalities that never adopted *Plan Cuadrante*. Estimation is performed using the heterogeneity-robust estimator of Chaisemartin and D'Haultfœuille (2020), which accommodates a continuous, time-varying treatment intensity in a staggered design. The dots represent the estimated dynamic effects and the bars behind them denote 95% confidence intervals. Standard errors are clustered at the municipality level. The pooled effect and the mean of the outcome among municipalities with no treated neighbors are reported in each panel.

Figure C.14: Geographic Spillovers of *Plan Cuadrante* by Number of Treated Neighbors



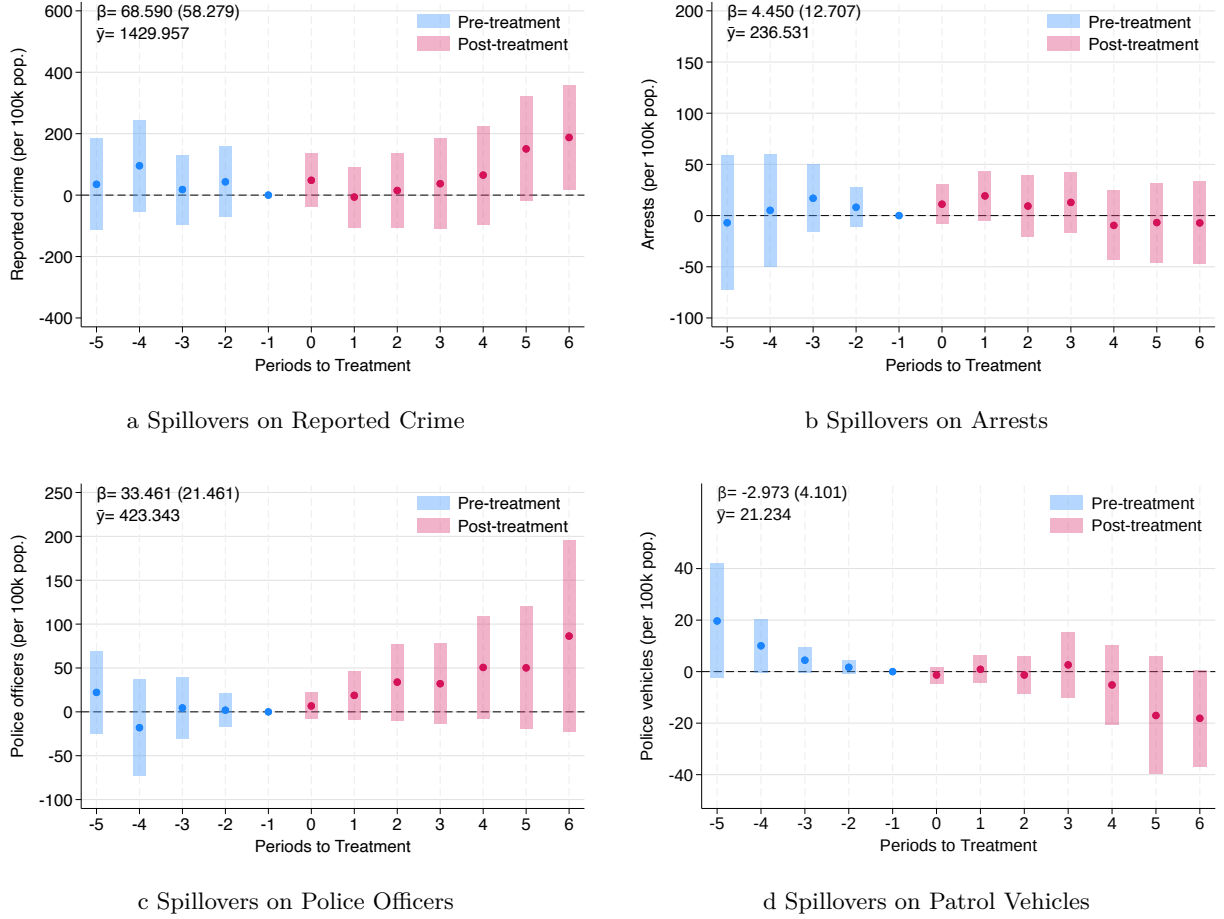
Notes: This figure reports the geographic spillover effects of *Plan Cuadrante* on four outcomes for never-treated municipalities: reported crime per 100,000 inhabitants (panel a), arrests per 100,000 inhabitants (panel b), the number of police officers per 100,000 inhabitants (panel c), and the number of patrol vehicles per 100,000 inhabitants (panel d). Unlike the dummy spillover specification, the treatment variable is now continuous and equals the number of neighboring municipalities that have implemented *Plan Cuadrante* by year t , following the spillover-intensity approach of Crépon et al. (2013). The sample is restricted to municipalities that never adopted *Plan Cuadrante*. Estimation is performed using the heterogeneity-robust estimator of Chaisemartin and D'Haultfœuille (2020), which accommodates a continuous, time-varying treatment intensity in a staggered design. The dots represent the estimated dynamic effects and the bars behind them denote 95% confidence intervals. Standard errors are clustered at the municipality level. The pooled effect and the mean of the outcome among municipalities with no treated neighbors are reported in each panel.

Figure C.15: Randomization Inference for the Direct and Spillover Effects of *Plan Cuadrante*



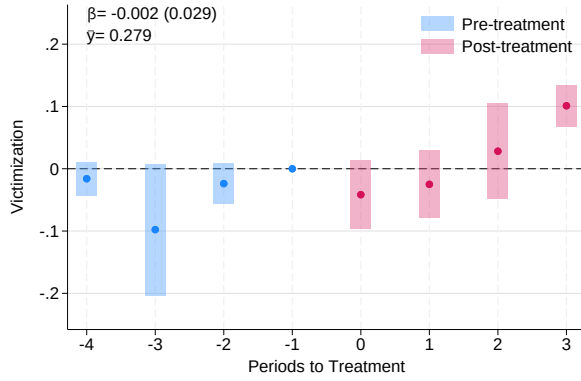
Notes: This figure presents randomization inference results under the sharp null of no treatment effect, using the heterogeneity-robust estimator of Chaisemartin and D'Haultfœuille (2020). Each curve is the kernel density of 1,000 placebo ATT estimates obtained by jointly permuting *Plan Cuadrante* adoption dates and never-treated status across municipalities (drawing from the empirical distribution of (date, status) pairs, so the marginal counts of treated and never-treated municipalities are preserved in every permutation). In each permutation the variable PCSP is shuffled across all 346 municipalities and a new time-varying treatment indicator is computed before re-estimating the dynamic effects. The solid colored vertical line marks the point estimate from the true rollout; the dashed black line marks the null expectation $\beta = 0$. The boxes report the true point estimate, its asymptotic two-sided p -value, and the mean of the placebo β and placebo p -value distributions.

Figure C.16: Geographic Spillovers of *Plan Cuadrante* for Neighbors of the ENUSC Municipalities (Administrative Outcomes)

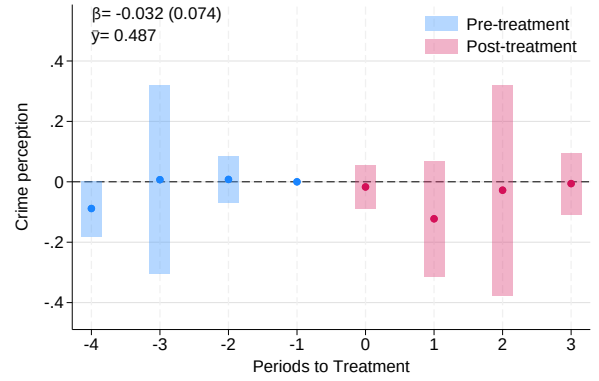


Notes: This figure reports the spillover effects of *Plan Cuadrante* on the number of crimes reported to the police in the municipality per 100,000 inhabitants (panel a), on the number of arrests in the municipality per 100,000 inhabitants (panel b), on the number of police officers (panel c), and on the number of patrol vehicles (panel d). Specifically, the figure shows the dynamic effects of *Plan Cuadrante* for neighboring never-treated municipalities to the 46 municipalities included in the ENUSC sample. The estimation is conducted using the procedure developed in Callaway and Sant'Anna (2021) for difference-in-differences with staggered treatment adoption, using as the treatment introduction date the earliest date of a neighboring municipality. The dots in the figure represent the estimated coefficients and the bars behind them 95% confidence intervals. Standard errors are clustered at the municipality level using the wild bootstrap procedure. The pooled effect and the mean outcome before the introduction of the treatment for each outcome are also presented in each panel.

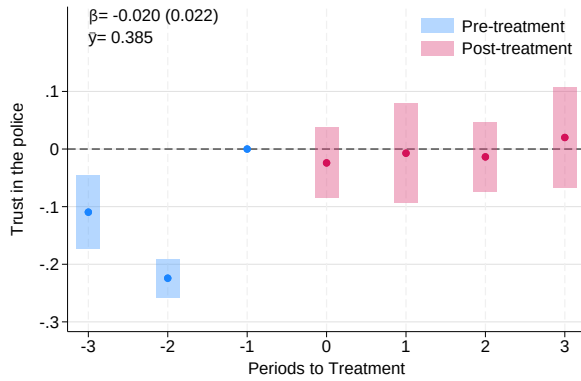
Figure C.17: Geographic Spillovers of *Plan Cuadrante* on ENUSC Survey Outcomes (Early- to Late-Adopting Municipalities)



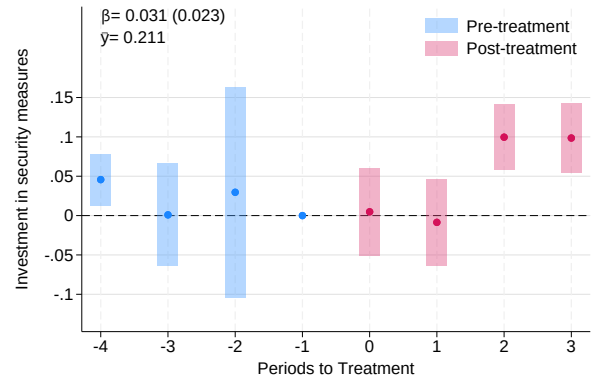
a Spillovers on Victimization in last 12 months



b Spillovers on Households reporting an increase in crime



c Spillovers on Trust in the police



d Spillovers on Households adopting security measures

Notes: This figure presents the geographic spillover effects of *Plan Cuadrante* from early-adopting ENUSC municipalities on their late-adopting ENUSC neighbours, using the ENUSC sample. Panel (a) reports the effect of *Plan Cuadrante* on the probability of household victimization in the last 12-months using the ENUSC survey. Panel (b) reports the effect on crime perception using the ENUSC survey. Panel (c) reports the effect on trust in the police using the ENUSC survey. Panel (d) reports the results on the probability of purchasing private security measures in the last twelve months using the ENUSC survey. The focal treated sample consists of six late-adopting ENUSC municipalities—Molina, Tocopilla, San Carlos, Padre Hurtado, Calera, and Limache—each assigned a spillover-onset date equal to the year its earliest-adopting ENUSC neighbour received *Plan Cuadrante*: Curicó (2005) for Molina, Iquique (2005) for Tocopilla, Chillán (2007) for San Carlos, Peñaflor (2008) for Padre Hurtado, and Quillota (2008) for both Calera and Limache. Estimation is performed using the difference-in-differences estimator for staggered treatment adoption of Callaway and Sant’Anna (2021). The dots represent the estimated dynamic effects and the bars behind them denote 95% confidence intervals. Standard errors are clustered at the municipality level. The pooled effect and the mean of the outcome among municipalities with no treated neighbors are reported in each panel.

Comment R2.4

Relatedly, criminals may easily cross municipal boundaries in Chile. Can you report the share of offenders apprehended in municipality X who committed crimes elsewhere? How should the reader think about this?

Response to R2.4:

Thanks for the comment. If criminals cross municipal boundaries to commit crime, it is reasonable to expect that the implementation of *Plan Cuadrante*, by increasing police effectiveness and deterrence, may have displaced some criminal activity from treated to untreated municipalities, biasing upward the estimated effects of the program.

Understanding the extent of that displacement is therefore important. We do not observe offenders' full travel patterns in Chile, so we cannot measure this directly. However, the institutional setting and the new spillover exercises both point to limited displacement. Chilean municipalities are large administrative units with an average area of about 2,000 square kilometers, making cross-boundary displacement costly. More importantly, in the revised manuscript we directly estimate spillover effects on crime, arrests, and police resources in neighboring untreated municipalities, and in the ENUSC subsample, and we find no evidence of meaningful spillovers in either direction. These analyses are described in detail in the response to the previous comment.

Comment R2.5

Interpretation of the Results: The authors argue that the effects on crime are driven primarily by geographic specialization, rather than additional resources, and leverage variation in the amount of additional resources across municipalities to provide evidence for their claim. However, it is unclear from the paper what drives the variation in additional resources. If municipalities experiencing larger increases in crime are those getting larger increases in resources, then this margin of exposure to the "resource channel" is endogenous. The paper may benefit from explaining what determines the variation in resources across municipalities and whether it correlates with trends in crime. As mentioned above, it may also help to provide more details on "Plan Cuadrante" such as whether it was accompanied by public information campaigns or media coverage, which could plausibly affect perceptions of security (one of their key outcomes) independently of actual crime rates (see Ajzenman et al. 2023 on this mechanism).

Response to R2.5:

Thanks for the comment. Let us start by clarifying our main claim: we do not argue that the additional resources brought by *Plan Cuadrante* have no effect on crime. Our key finding in the heterogeneity analysis by resource increases is that even municipalities that experienced no increase in resources relative to the control group show sizeable effects on nearly every outcome considered (see Figure 2). In this context, where resource increases may be endogenous, we cannot rule out a role for resources in explaining the effects of *Plan Cuadrante* in municipalities that received large increases, nor can we rule out that resources were important for the successful implementation of the beat policing component in those municipalities. However, the finding that municipalities with

no additional resources relative to the control group still experience sizeable effects provides clear evidence that the beat policing component of the program plays an important independent role, which is our main point. Consistent with this, and in line with the revised manuscript, we now emphasize that these effects remain sizeable in municipalities with resource increases similar to those in control municipalities. We also find that subsequent increases in police resources outside *Plan Cuadrante* have very limited effects on victimization. First, conditional on municipality and year fixed effects, changes in resources are not associated with reductions in crime. Second, we estimate the effect of a national program that increased police resources without any organizational component and find null effects on crime. Finally, the main estimates are robust to the inclusion of police resources as control variables, which, while by definition bad controls as they are themselves affected by *Plan Cuadrante*, do not meaningfully alter the coefficients.

On what determined the size of the resource increases: During the 2000s and 2010s, the Chilean government substantially increased the overall number of Carabineros and other police resources such as patrol vehicles, and this increase benefited both treated and control municipalities. On top of this general trend, *Plan Cuadrante* aimed to provide treated municipalities with sufficient additional resources to successfully implement the program. Panels (c) and (d) of Figure 4 show that, on average, *Plan Cuadrante* substantially increased police resources relative to the control group. Panels (a) and (b) of Figure 4 show, however, that this increase masks substantial heterogeneity across treated municipalities: while municipalities in the bottom 40% of the resource-increase distribution experienced increases very similar to those observed in control municipalities, municipalities in the top 40% experienced substantially larger increases, with officers rising annually by around 22% and patrol vehicles by around 80% upon joining *Plan Cuadrante*. So, what drives the variation in additional resources across treated municipalities? In principle, the goal was for *Plan Cuadrante* to increase resources enough to cover each municipality’s pre-existing resource deficit, measured in Equivalent Surveillance Units (UEs). However, program documents make clear that budgetary constraints meant that all municipalities joined *Plan Cuadrante* without fully closing this gap (Salas P. et al., 2014; Winchester et al., 2007), and program evaluations do not clearly establish that the size of this remaining deficit is what determined the resource increase each municipality received. We asked Carabineros for access to the official demand and supply figures that were supposed to guide this allocation, but this information was not provided. Importantly, we find that municipalities receiving larger versus smaller resource increases—whether the split is based on the increase in police officers or in patrol vehicles—do not display differential pre-treatment trends in either the outcomes of interest or in police resources themselves.⁷⁷ We also explore the correlates of the resource increase in Online Appendix B, where we show that pre-treatment outcomes levels are not predictive of the size of the resource increase, while structural municipality characteristics such as population and poverty are more predictive.

Regarding the possibility that the effects on crime perception are driven by accompanying public information campaigns or media coverage, we believe this is unlikely for two reasons. First, *Plan Cuadrante* was an internal operational reform of Carabineros with no formal public communication component in its design. While the program received some media attention when it was broadly rolled out across Santiago in 2000, we find no evidence of targeted information campaigns at the municipality level when individual municipalities subsequently joined the program, and the staggered rollout over more than a decade makes a sustained media salience mechanism implausible.

⁷⁷The p-values for the joint test of equal pre-trends between the two groups are, for the officers-based split and the patrol-vehicles-based split respectively: 0.823 and 0.048 for victimization, 0.059 and 0.379 for crime perception, 0.820 and 0.157 for investment in security measures, 0.213 and 0.419 for reported crime, 0.616 and 0.444 for police officers, and 0.405 and 0.157 for patrol vehicles.

Second, and more importantly, if the improvement in crime perception were driven by media coverage or information campaigns, one would expect the effects to be concentrated shortly after adoption and to fade over time. Instead, the dynamic estimates show that the effects on crime perception grow and persist over time, a pattern more consistent with a genuine improvement in security than with a transitory salience effect of the kind documented in Ajzenman et al. (2023).

All these implementation details are provided in Section 2 in the updated manuscript, which expands substantially the section describing the implementation of *Plan Cuadrante*:

Carabineros, Chile’s national police force, was founded in 1927. Through most of our study period, Carabineros operated under the Ministry of Defense, which oversaw both military and police functions until domestic security responsibilities were transferred to the Ministry of Interior in 2011.⁷⁸

Carabineros’ mission, functions, and structure are established by Law 18.961 and the Regulation on Carabineros’ Organization (Reglamento de Carabineros) (Biblioteca del Congreso Nacional de Chile, 1990; Ministerio de Defensa Nacional, República de Chile, 1989). Under this framework, Carabineros is a professional, disciplined, and hierarchical police institution whose mission is to preserve public order and security with a central emphasis on preventive policing. The General Director heads the strategic level, responsible for defining doctrine and institutional strategy, while the Subdirector coordinates implementation. The Directorate of Order and Security administers police functions at the tactical level.

Territorially, Carabineros is organized into zones that oversee *prefecturas*. Zones and *prefecturas* are primarily management layers. Zones map roughly one-to-one to Chile’s regional divisions, with two exceptions: the Metropolitan Region of Santiago—the most populated in the country—and La Araucanía—which has recently experienced significant social conflict—are each divided into two zones. Each zone is subdivided into *prefecturas*—54 nationwide—which supervise 227 *comisaría*s, the main operational units at the tactical level. These *comisaría*s oversee smaller units—*subcomisaría*s, *tenencias*, *retenes*, *puestos*, and *avanzadas*—that operate within their assigned jurisdictions.

In practice, the *comisaría* is the basic unit of routine frontline policing and, in most municipalities, serves as the territorial anchor of police work. Officers conduct patrol, surveillance, prevention, and first response at this level. In most municipalities, a single *comisaría* covers the entire municipal territory, though larger municipalities may have multiple ones. Santiago, for instance, has four. All municipalities that adopted *Plan Cuadrante* during our study period had a *comisaría*. With the exception of 9 municipalities, each had a single *comisaría* covering the full municipal territory. On average, a *comisaría* in our sample covered 1,608 square kilometers and served 57,642 inhabitants.

Plan Cuadrante was introduced within this institutional framework without altering Carabineros’ hierarchical structure or its strategic and tactical levels. The reform emerged in a context of rising victimization and insecurity, with the explicit goal of reversing these trends (Ministerio del Interior y Seguridad Pública, 2014). Its central innovation was a shift toward a beat policing model. In the criminology literature, beat policing refers to assigning officers to fixed territorial areas so they can accumulate local knowledge, strengthen ties with residents, and respond more effectively to local crime patterns (Cordner, 1995; Skogan and Hartnett,

⁷⁸Currently, Carabineros reports to the Ministry of Public Security, created in 2024 to focus exclusively on domestic security.

1997). In practice, the catchment area of police forces in municipalities adopting *Plan Cuadrante* was partitioned into smaller, fixed geographic units (quadrants), to which officers were permanently assigned. Rather than building new infrastructure, the reform redefined the territorial responsibility of officers in already existing police stations (Winchester et al., 2007).⁷⁹ This represented a clear break from the pre-reform system, in which officers rotated across broad operational territories, very often spanning entire municipalities.

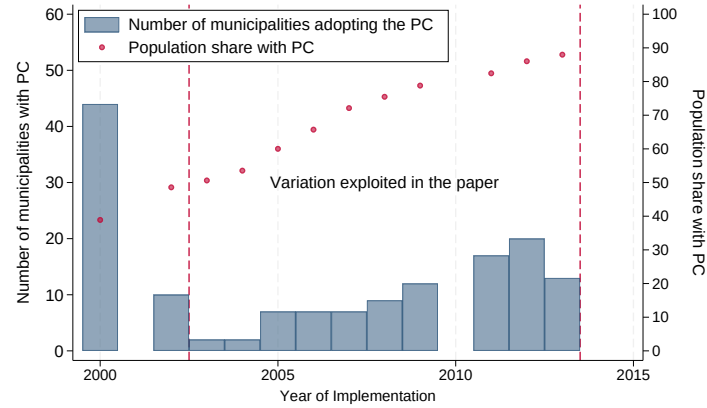
Carabineros began piloting *Plan Cuadrante* in 1998. The program expanded gradually through 2013, as shown in Panel (a) of Figure 1, reaching 150 urban municipalities and covering roughly 90% of Chile’s population.⁸⁰ Online Appendix A provides further details about the rollout of the program. The rollout was designed to follow a municipality ranking based on a weighted index of victimization, police resource deficits, drug prevalence, unemployment, and demand for police services, with annual entry determined by available budget.⁸¹ In practice, however, Budget Office evaluations show that adoption diverged from this ranking for undocumented reasons, so the rule operated more as a loose guideline than as a binding assignment mechanism (Salas P. et al., 2014).

⁷⁹In 2016, Carabineros launched the *Plan Cuadrante 2.0* program in those municipalities served by *Plan Cuadrante*. While this reform introduced additional reforms aimed at integrating the community into the provision of security and building capacity to gather and analyze crime data in police stations, this paper’s study period ends before these improvements were introduced.

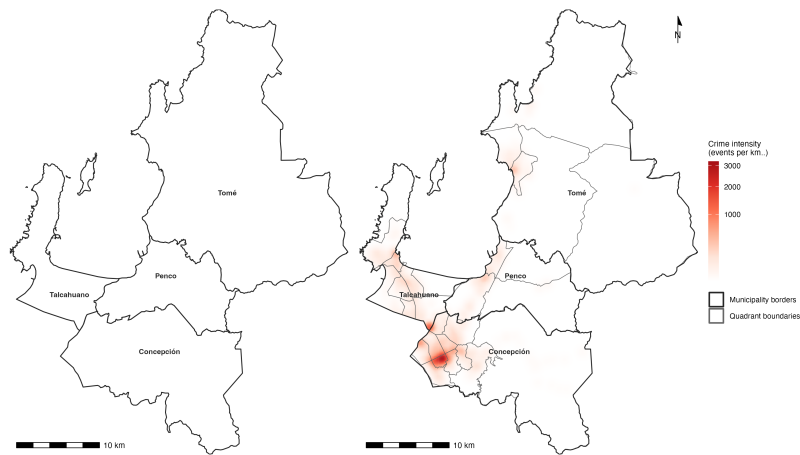
⁸⁰Further details about the expansion of *Plan Cuadrante* are provided in Online Appendix A and Table A.1.

⁸¹The weights assigned to each component changed on a yearly basis (Winchester et al., 2007; Salas P. et al., 2014).

Figure 1: Implementation of *Plan Cuadrante*



a Adoption of *Plan Cuadrante*



b *Plan Cuadrante* in the Concepción area

Notes: Panel (a) shows the number of municipalities adopting the *Plan Cuadrante* each year (bars), while the circles above them illustrate the cumulative share of the Chilean population living in a municipality with the *Plan Cuadrante* already in place. Panel (b) depicts the metropolitan area of Concepción, the second-largest in Chile: the left map shows its four municipalities—the original police catchment areas—and the right map overlays the Carabineros quadrants together with a kernel-density estimate of all types of robberies and thefts recorded by the police (both reported crimes and arrests). It illustrates how *Plan Cuadrante* subdivided each catchment area (which corresponds to the territory of a municipality) into smaller quadrants, as well as the spatial concentration of crime within them.

Conceptually, police effectiveness can be viewed as a function of three inputs: total resources, territorial deployment, and policing strategy. Although in some municipalities police resources were increased significantly to operationalize the beat policing model (Winchester et al., 2007; Salas P. et al., 2014), the program primarily transformed the latter two elements. The creation of quadrants was accompanied by permanent geographic assignment of officers to specific areas. This change was operationally critical. By making officers responsible for their quadrant, it encouraged accumulation of local knowledge and stronger community ties, allowing them to adapt patrols and prevention activities to local crime patterns. Importantly, this was not driven by program-specific operational directive changes. Throughout the period we study, a single national set of operational guidelines applied to officers in both treated and control municipalities (Carabineros de Chile, 2018; Winchester et al., 2007). Thus, while the average effects we document on crime and crime perception likely reflect combined changes in the three elements we discuss—i.e., resources, deployment, and strategy—the deployment change itself was essential to operationalizing the reform. We discuss the drivers of the effects we document in Section 4.4.

Quadrant size and shape were determined by local police authorities based on factors such as road networks, geographic features, and available resources. In principle, resource allocation across quadrants followed demand for police services, measured in Equivalent Surveillance Units (UEVs); in practice, the station commander retained discretionary authority, as was also the case prior to *Plan Cuadrante* (Salas P. et al., 2014). Municipalities implementing *Plan Cuadrante* were subdivided into an average of seven quadrants, each typically staffed by approximately three officers per shift. On average, each quadrant covered 185 square kilometers and served 13,466 inhabitants. Panel (b) of Figure 1 illustrates this structure, showing the 29 quadrants created across the four municipalities of the Concepción metropolitan region—Chile’s second-largest metropolitan area and the largest in our sample. Quadrants represented approximately 14% of the original *comisaría* territory. Officers continued to operate from the same police station but were permanently assigned to a specific sub-area within its jurisdiction. While Carabineros’ hierarchical structure limits frontline officer decision-making authority, each quadrant was assigned a Quadrant Delegate (*Delegado de Cuadrante*), a non-commissioned officer, responsible for immediate operational decisions within the quadrant.

Plan Cuadrante may affect public safety through two complementary channels: deterrence and police effectiveness, understood as the capacity to respond to and resolve crimes. The changes induced by *Plan Cuadrante*—permanently assigning officers to well-defined neighborhoods and, in some municipalities, increasing available resources—can increase both police visibility and local knowledge. As officers repeatedly patrol the same area, they become familiar with its streets, residents, and routines, which can raise the perceived probability of identification and apprehension and reduce the expected returns to crime. At the same time, this deeper knowledge of local territory and communities can improve police response and case resolution once crimes occur. The reform therefore likely affects both dimensions. Consistent with this interpretation, in Section 4 we document declines in victimization and in reported crime—patterns consistent with deterrence—together with increases in arrests, consistent with improved police effectiveness.

Although beat policing shares features with other policing approaches, such as hot-spot policing and community-based policing, which have been widely studied, it has distinctive features that make it worth studying as an independent approach. Hot-spot policing concentrates resources in high-crime areas and does not focus on building local knowledge and community

ties. It is typically implemented through targeted interventions at specific locations rather than systematic territorial organization, and the same officers are not necessarily sent repeatedly to the same areas. Evidence on hot-spot policing is mixed, with some studies showing crime reductions in targeted areas but also potential displacement effects (Braga et al., 2019). *Plan Cuadrante*, by contrast, subdivided all urban municipalities into quadrants with permanent officer assignment. As Panel (b) of Figure 1 illustrates, some quadrants contained crime hot-spots, but many did not. The reform thus addressed entire urban territories rather than specific high-crime areas, potentially mitigating displacement effects associated with hot-spot policing.

Community-based policing emphasizes building relationships with residents and addressing underlying social issues, and typically involves active community participation in defining strategy and priorities—in some cases, even assuming surveillance and prevention responsibilities. While increasing trust in the police and strengthening community ties were among the declared goals of *Plan Cuadrante*, and may have resulted from its territorial reorganization, the community was not formally involved in the provision of security.⁸²

In recent decades, police departments worldwide have transitioned toward beat policing reforms (National Academies of Sciences, Engineering, and Medicine, 2018; Ministerio del Interior y Seguridad Pública, 2014), yet the effectiveness of this model has not been rigorously assessed. Although the beat policing approach introduced through *Plan Cuadrante* shares features with programs previously studied in the economic literature, its distinctive characteristics—comprehensive territorial reorganization and systematic local officer specialization—make it a valuable case for studying how deployment and policing strategies not yet examined in the economics literature affect crime and crime perception.

We also include below Section 4.4, which summarizes the analyses disentangling the role of resources from that of the beat policing component discussed above:

To reduce crime and improve security perceptions, police forces need to strengthen both prevention and police effectiveness. Both margins can be shaped by resources, territorial deployment, and policing strategy. This section examines which components of *Plan Cuadrante* are most likely to drive the effects documented in Section 4.

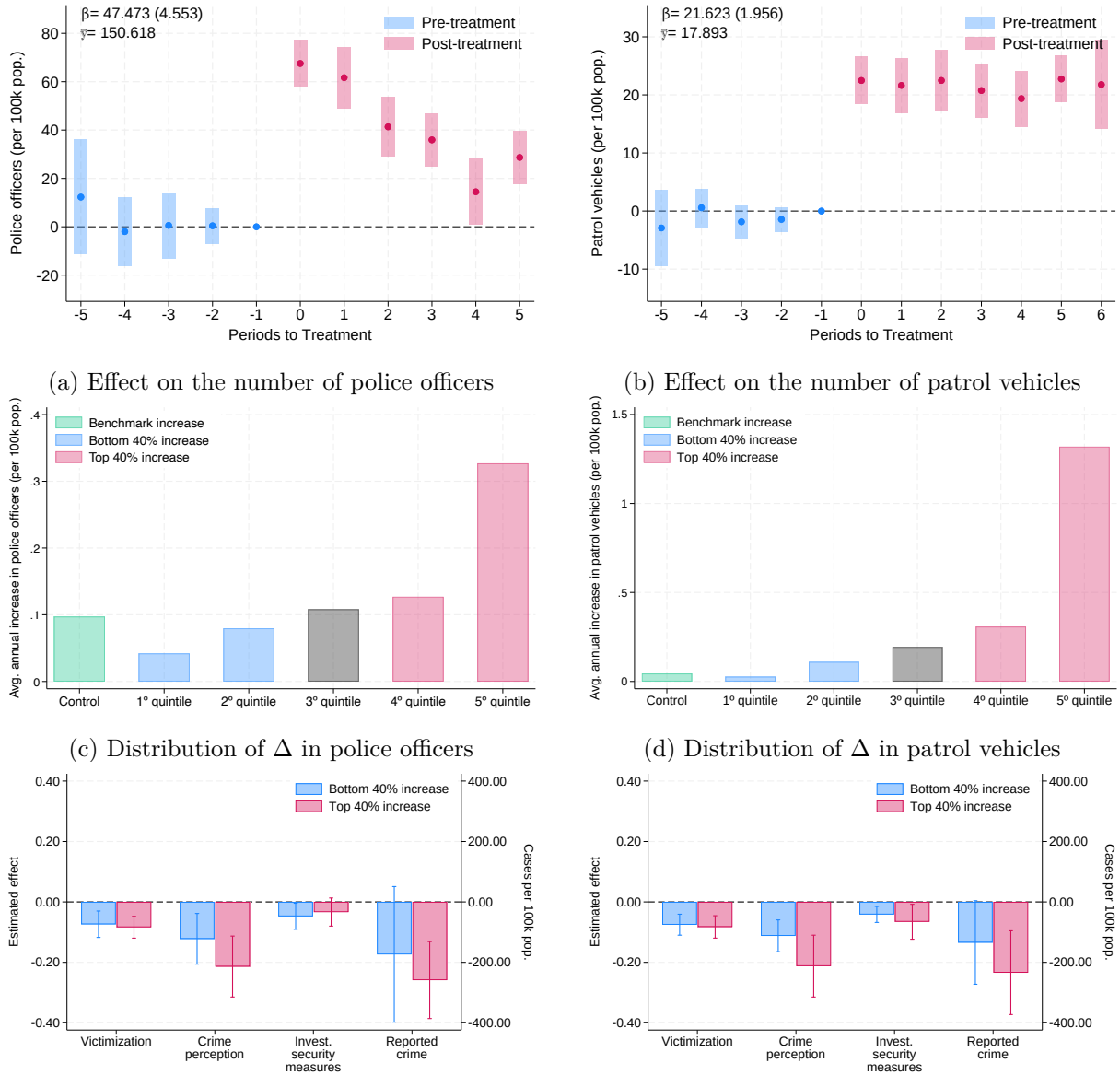
The central objective of *Plan Cuadrante* was to introduce a beat policing model by reorganizing deployment. Police operational territories that had typically matched municipal boundaries were subdivided into quadrants, and officers were permanently assigned to these quadrants. During our study period, national operational guidelines remained the same in adopting and non-adopting municipalities. Still, this new deployment likely affected policing strategy at a more operational level, because it allowed officers to adapt patrol routes, schedules, and prevention practices using the local knowledge that the program sought to build.

Although changes in resources were not the core of the reform, implementation contemplated increases in municipalities that lacked the personnel and patrol capacity required to move to

⁸²The *Plan Cuadrante*’s original design contemplated a formal community integration model (MICC) with a Community Affairs Office (*Oficina de Asuntos Comunitarios*) and a dedicated officer in each police station. However, this model was not implemented during our study period. Independent program evaluations from the Budget Office document that these Community Affairs Offices did not exist and had no dedicated personnel, budget, or structured activities in the period we study (Winchester et al., 2007; Salas P. et al., 2014). The community integration model was piloted in four police stations during 2012–2013, none of which are in our analytical sample. Financing was then discontinued in early 2014 and only restored and expanded after our study period ends.

this model. During the period we study, Chile was expanding public spending on security, and police resources increased in most municipalities regardless of whether or when they adopted *Plan Cuadrante*. On average, however, municipalities adopting the program experienced larger increases in officers and patrol vehicles, as shown in panels (a) and (b) of Figure 4. Since increases in police resources have been shown to reduce crime in some contexts (Evans and Owens, 2007), this pattern raises a central question for interpretation, namely whether our estimates reflect resource expansion or the deployment and strategy changes introduced by *Plan Cuadrante*. We address this question through a set of exercises that exploit heterogeneity in resource increases across municipalities and over time.

Figure 4: Plan Cuadrante and Changes in Police Resources



(e) Effects by the intensity of Δ in police officers (f) Effects by the intensity of Δ in patrol vehicles

Notes: Panels (a) and (b) show the effect of *Plan Cuadrante* on police officers and patrol vehicles per 100,000 inhabitants. Panels (c) and (d) show the distribution of annual changes in officers and patrol vehicles across municipalities. Panels (e) and (f) report effects by the intensity of resource changes, comparing municipalities in the bottom 40% and top 40% of the distribution. The estimation follows Callaway and Sant'Anna (2021). The graphs display estimated coefficients and 95% confidence intervals. Standard errors are clustered at the municipality level using wild bootstrapping.

Panels (c) and (d) of Figure 4 show that the average increase in police resources masks substantial heterogeneity. Municipalities in the bottom 40% of the resource-change distribution (blue bars) experienced increases close to the average annual variation observed among never treated and not-yet-treated municipalities (green bar). By contrast, municipalities in the top 40% (red bars) experienced much larger increases, with officers rising by around 22% and patrol vehicles

by around 80%. If the effects of *Plan Cuadrante* were driven by resource increases alone, we would not expect effects in municipalities whose resource increases were similar to those of the control group. Yet panels (e) and (f) of Figure 4 show sizeable effects for most crime and crime-perception outcomes even in that low-increase group. While there may be a resource gradient for some crime-perception outcomes, the effects on twelve-month victimization and private investment in security among municipalities with control-like resource increases are close in magnitude to those in municipalities with larger increases. Reassuringly, municipalities in the top and bottom 40% of the resource-increase distribution do not display differential pre-treatment trends in outcomes or police resources, which suggests that this heterogeneity is not driven by differential pre-existing trends.⁸³

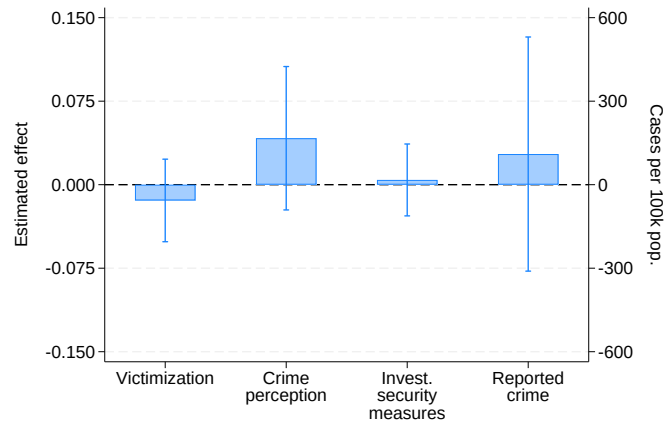
Although we cannot fully separate the beat policing component from the resource component of the reform—resources may have been important to enable effective implementation of the program, or may explain part of the effects in some municipalities—the fact that municipalities with resource increases similar to those experienced by control municipalities still display sizeable effects indicates that the effects of *Plan Cuadrante* are not driven entirely by resources.⁸⁴⁸⁵

⁸³We test for the joint significance of the pre-treatment coefficients between the two groups, separately for the split based on the increase in police officers and the split based on the increase in patrol vehicles. The resulting p-values are, respectively, 0.823 and 0.048 for victimization, 0.059 and 0.379 for crime perception, 0.820 and 0.157 for investment in security measures, and 0.213 and 0.419 for reported crime, as well as 0.616 and 0.444 for police officers and 0.405 and 0.157 for patrol vehicles.

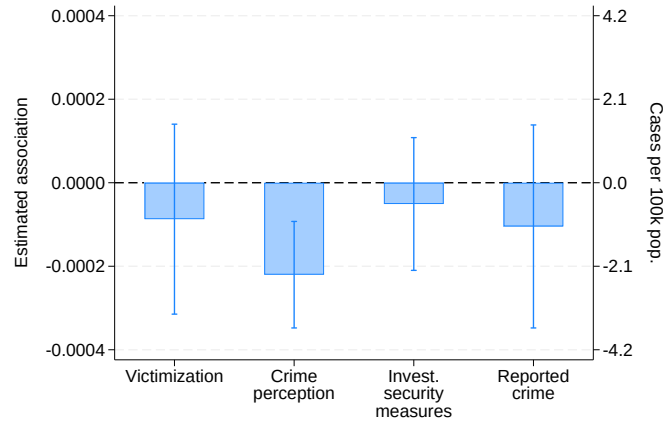
⁸⁴We also re-estimate the results in Figure 2 but controlling for police resources. Online Appendix Table B.3 shows that the magnitude of the estimated effects remains similar for most variables, although the statistical significance is reduced in some estimations. This likely reflects a mechanical feature of the Callaway and Sant’Anna (2021) estimation procedure: control variables are not incorporated parametrically, but are instead used to construct the comparison group via outcome regression or propensity-score weighting, so the resulting estimation uncertainty propagates to the ATT standard errors. Note, in addition, that police resources are a bad control—i.e., at least part of their variation is a result of the adoption of *Plan Cuadrante*—so these results should be interpreted cautiously.

⁸⁵Table B.4 in Online Appendix B examines which factors are associated with a larger increase in police resources upon the adoption of *Plan Cuadrante*.

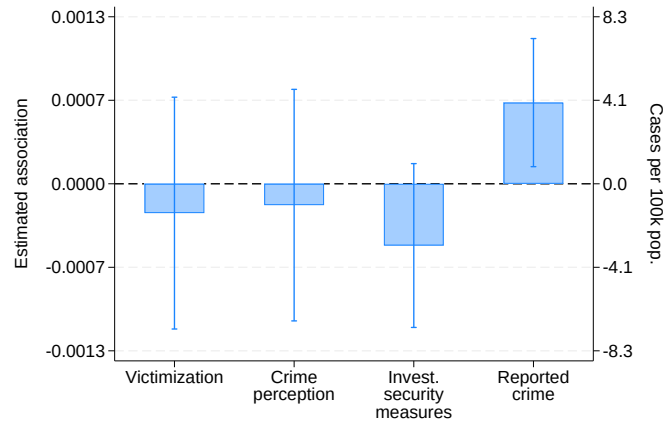
Figure 5: Police Resources and Public Safety



a Effects of *Seguridad Ciudadana*



b Association between police officers and effectiveness



c Association between patrol vehicles and effectiveness

Notes: Panel (a) reports the pooled effect of *Seguridad Ciudadana*, a program that expanded local security personnel to support patrolling and deterrence but did not replicate the deployment-and-strategy features of *Plan Cuadrante*. Panels (b) and (c) report within-municipality associations between police resources and police-effectiveness outcomes among the 41 municipalities of the Santiago Metropolitan Region, all of which had adopted *Plan Cuadrante* by 2000. The regressions include municipality and year fixed effects. Standard errors are clustered at the municipality level using the wild bootstrapping procedure.

To further investigate whether resource increases alone can explain our results, we assess the effects of *Seguridad Ciudadana*, a program that expanded local security personnel to support patrolling and deterrence, but did not replicate the core deployment-and-strategy features of *Plan Cuadrante*. Using a difference-in-differences specification following Callaway and Sant’Anna (2021), panel (a) of Figure 5 shows no significant effects of these resource changes on police effectiveness. Further details about this exercise and the corresponding event-study estimates are reported in Online Appendix B and Online Appendix Figure B.4.

Finally, we study whether variation in police resources in municipalities that had already adopted *Plan Cuadrante* improves public safety and security perceptions. Using ENUSC data and information on resources for 2006–2012 in the 41 municipalities of the Santiago Metropolitan Region—all of which had adopted *Plan Cuadrante* by 2000—we exploit variation in officers and patrol vehicles that is unrelated to program adoption. Panels (b) and (c) of Figure 5 show the association between police resources and our main outcomes, conditional on municipality and year fixed effects. With one exception—the association between police officers and crime perception—none of these associations is statistically significant at conventional confidence levels.⁸⁶

Taken together, these results suggest that resource increases alone do not account for the effects of *Plan Cuadrante*. We next turn to evidence that speaks more directly to the beat-policing components of the reform.

Panel (a) of Figure 4 shows that, following the adoption of *Plan Cuadrante*, households are much more likely to report increased police presence in their neighborhoods. The share of households perceiving increased police presence rises by 35 percentage points (156%), the share reporting that police pass by their house daily rises by 22 percentage points (59%), the share reporting never seeing police patrols near their house falls by 10 percentage points (57%), and complaints about lack of police surveillance fall by around 5 percentage points (20%). This evidence is consistent with both larger resources and the new deployment model. Yet the latter estimate—the only police-presence outcome for which we can separately estimate effects in municipalities with small and large resource increases—is similar in magnitude across the two groups, which suggests that the increase in police presence is not explained only by larger resource increases (see Online Appendix Figure B.3).⁸⁷ More importantly, the broader pattern of results suggests that *Plan Cuadrante* did more than simply increase patrolling intensity. Existing work shows that police presence can reduce crime while officers are present, but that its overall effects may be attenuated when crime is displaced across places or time windows (Mastrobuoni, 2019; Blanes-i Vidal and Mastrobuoni, 2018). Our results point to something broader. As discussed in Section 4, we find actual declines in victimization and reported crime, together with no evidence of spillovers to neighboring municipalities. This pattern is more consistent with officers accumulating local knowledge and adapting prevention strategies to local conditions under stable territorial assignment than with a purely mechanical increase in patrol intensity.

Panel (b) of Figure 4 reinforces this interpretation. Households in early-adopting municipalities

⁸⁶Online Appendix Figure B.5 reports the same exercise and also shows similar results for the broader set of municipalities that adopted *Plan Cuadrante* before 2006 and never-treated municipalities (red bars); see Online Appendix B for further details.

⁸⁷We cannot estimate heterogeneous effects for other police-presence outcomes, such as the reporting of police passing by or the share of households perceiving an increase in police presence, because those variables are available only in ENUSC rounds 2003, 2005, and 2006, while resource data start in 2006.

report better police performance across multiple dimensions. They are 8.1 percentage points (16%) more likely to rate police performance as good or very good, 7.6 percentage points (11%) more likely to describe police as efficient, 5 percentage points (6%) more likely to report that police are helpful, and 4.9 percentage points (7%) more likely to report that police are well informed. They are also 3.9 percentage points (10%) less likely to perceive police as corrupt, 7.9 percentage points (21%) less likely to believe police fail to control crime, and 7.6 percentage points (22%) less likely to report that police do not perform their duties effectively.⁸⁸

The administrative evidence points in the same direction. Panel (c) of Figure 4 shows that arrests per 100,000 inhabitants increase by 48 (14%) in our main estimates, even as crime declines overall.⁸⁹ This effect is also significant among municipalities with resource increases similar to those in the control group (see Online Appendix Figure B.3). Consistent with this pattern, panel (d) of Figure 4 shows that the arrest rate per reported crime also increases by 1.7 percentage points (10%), suggesting that a higher share of crimes reported to the police are being solved. Because these gains appear even where resources did not increase more than in control municipalities, they are difficult to reconcile with a pure resource story and are more naturally interpreted as improvements in police effectiveness under the deployment and strategy changes introduced by *Plan Cuadrante*. The corresponding event-study estimates for arrests, the arrest rate per reported crime, and trust in the police in Figure 4 confirm that these changes emerge only after adoption.

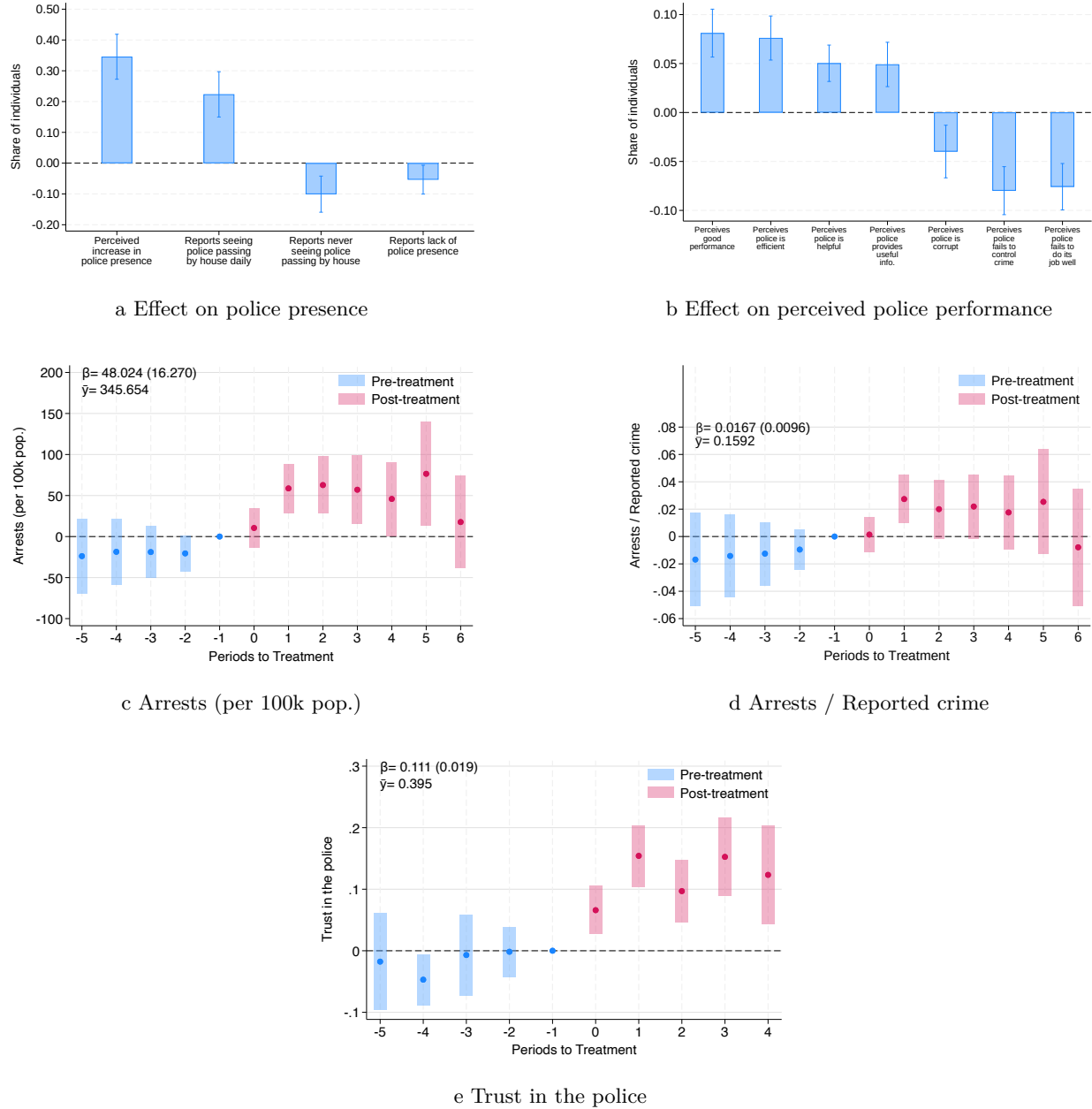
Finally, panel (e) of Figure 4 shows a sharp increase in trust in the police in our main specification (Section 4), consistent with one of the objectives of beat policing, namely bringing officers closer to the community and allowing them to accumulate local knowledge through repeated interaction. Online Appendix Figure B.3 shows that this effect remains sizeable both in municipalities that experienced resource increases similar to the control group and in those with larger increases.

Overall, although resources were likely important for implementation and may explain part of the effects in some municipalities, the results in this section indicate that resource changes alone are not enough to explain our findings. The evidence is more consistent with a combination of deployment changes and the strategy adjustments enabled by local knowledge accumulation under permanent assignment to quadrants.

⁸⁸Information on police performance is only available in ENUSC round 2005. Using this round, we compare perceived police performance in municipalities that adopted *Plan Cuadrante* in 2005 or earlier and in municipalities adopting after 2005. Figure B.6 in Online Appendix B shows that the number of years of exposure to *Plan Cuadrante* is positively correlated with perceived police performance. While these exercises on perceived police performance should be interpreted with caution, we show in the same appendix that the timing of future adoption is not correlated with perceived police performance in 2005 for municipalities that had not yet adopted the program by 2005.

⁸⁹As with reported crime, this event study is restricted to 5 pre-treatment and 7 post-treatment periods for symmetry with the ENUSC-based analysis; Online Appendix Figure C.8 reports the results for arrests using the full set of pre-treatment and post-treatment periods.

Figure 6: *Plan Cuadrante* and Changes in Police Performance



Notes: Panel (a) reports the effects on perceived police presence. Panel (b) reports perceived police performance in 2005, comparing municipalities where *Plan Cuadrante* was already implemented before 2006 with municipalities where it was implemented after 2005. Panels (c), (d), and (e) report dynamic event-study estimates for arrests per 100,000 inhabitants, the ratio of arrests to reported crime, and trust in the police. The estimation follows Callaway and Sant'Anna (2021). The graphs display estimated coefficients and 95% confidence intervals. Standard errors are clustered at the municipality level using the wild bootstrapping procedure.

We also added a new Online Appendix B subsection, “Factors Associated with the Increase in Police Resources,” regressing the increase in resources at the time of adoption on pre-treatment municipality characteristics. Because the UVE demand-and-supply files were not provided by *Carabineros*, we cannot infer the exact administrative rule used to assign the additional resources:

As discussed in Section 2 and in Online Appendix B.4, the increase in police resources associated with the adoption of *Plan Cuadrante* varied substantially across municipalities. In principle, this increase should be related to each municipality’s pre-existing police resource deficit—measured in Equivalent Surveillance Units (UEs)—which *Plan Cuadrante* was meant to help close. However, program evaluations do not clearly establish that this deficit is what determined the size of the resource increase each municipality received (Winchester et al., 2007; Salas P. et al., 2014).⁹⁰ If the pre-existing deficit is not what determines the resource increase, what is it related to? Reassuringly, municipalities receiving larger versus smaller resource increases do not display differential pre-treatment trends in either the outcomes of interest or in police resources themselves.⁹¹ In this section we examine whether the magnitude of the resource increase experienced by a municipality upon adoption is correlated with its pre-treatment levels in crime, police resources, and other observable characteristics. Specifically, we regress the increase in resources at the time of adopting *Plan Cuadrante* on municipality characteristics measured before the program’s introduction, including crime and police resources in 2006, the first year for which information on police resources is available. Thus, we restrict this analysis to municipalities that adopted *Plan Cuadrante* from 2007 onward.

Table B.4 reports OLS regressions of the average annual increase in police officers and patrol vehicles per 100,000 inhabitants on pre-adoption municipality characteristics, separately for the administrative (SPD) sample at the municipality level (columns 1–2) and the ENUSC sample at the survey-respondent level (columns 3–4). All variables are standardized and standard errors are clustered by municipality, so coefficients are comparable in magnitude across regressors.

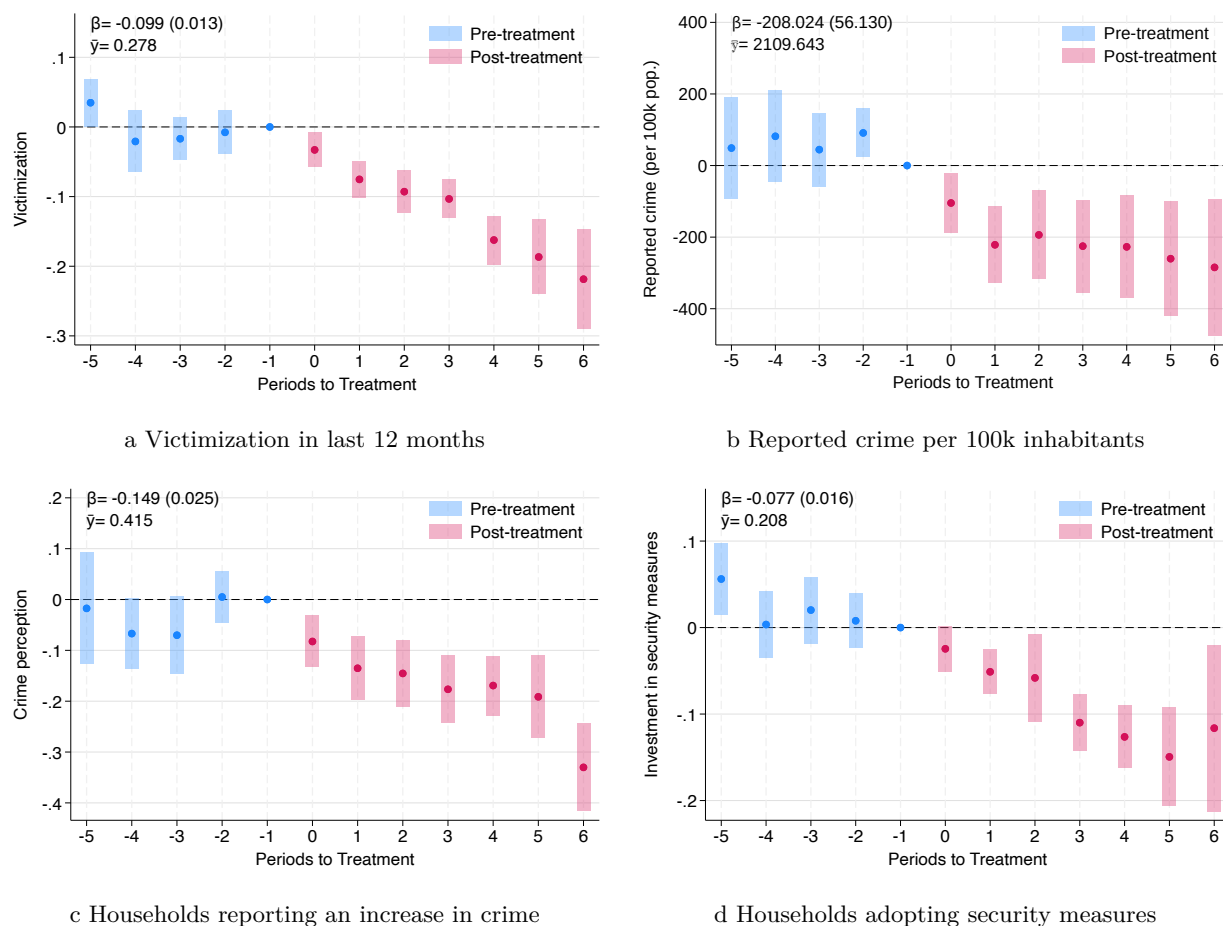
The results provide little support for the hypothesis that municipalities with worse pre-treatment crime outcomes received systematically larger resource increases. Pre-treatment reported crime is, if anything, negatively associated with the subsequent increase in both officers and vehicles in the administrative sample (columns 1–2); in the ENUSC sample, it is negatively associated with the increase in officers (column 3), although it is positively associated with the increase in vehicles (column 4). Pre-treatment victimization is, if anything, negatively associated with both increases in the ENUSC sample (columns 3–4). Pre-treatment arrests, crime perception, trust in the police, and private investment in security measures are not significantly associated with the resource increase. The increase in resources is instead more strongly predicted by structural municipality characteristics: less populous municipalities experienced larger increases in three of the four specifications, and municipalities with higher poverty rates predicted a larger increase in vehicles in the administrative sample (column 2), although this association is not significant for officers in that sample (column 1) or in the ENUSC sample (columns 3–4).

⁹⁰Although we requested information on the demand and supply of UEs underlying this allocation, this information was not provided by *Carabineros*.

⁹¹We test for the joint significance of the pre-treatment coefficients between the two groups, separately for the split based on the increase in police officers and the split based on the increase in patrol vehicles. The resulting p-values are, respectively, 0.823 and 0.048 for victimization, 0.059 and 0.379 for crime perception, 0.820 and 0.157 for investment in security measures, and 0.213 and 0.419 for reported crime, as well as 0.616 and 0.444 for police officers and 0.405 and 0.157 for patrol vehicles.

Other figures and tables referenced above:

Figure 2: Effects of *Plan Cuadrante* on Crime Outcomes and Security Perceptions



Notes: This figure shows the dynamic effects of the *Plan Cuadrante* on different measures of police effectiveness. The estimation is conducted using the procedure developed in Callaway and Sant'Anna (2021) for difference-in-differences with staggered treatment adoption. The dots in the figure represent the estimated coefficients and the bars behind them 95% confidence intervals. Standard errors are clustered at the municipality level using the wild bootstrapping procedure. The pooled effect and the mean outcome before the introduction of the treatment is also presented in each panel. Panel (a) reports the effect of *Plan Cuadrante* on the probability of household victimization in the last 12 months using the ENUSC survey. Panel (b) reports the effect on the number of crimes reported to the police in the municipality per 100,000 inhabitants using administrative data. Panel (c) reports the effect on the probability of perceiving an increase in crime using the ENUSC survey. Panel (d) reports the effect on the probability of adopting security measures in the last twelve months using the ENUSC survey. Panel (a) includes 93,041 observations; panel (b), 4,231; panel (c), 89,578; and panel (d), 93,041 observations.

Table B.4: Predictors of the Increase in Police Resources under *Plan Cuadrante*

	Administrative sample		ENUSC sample	
	Police officers (1)	Patrol vehicles (2)	Police officers (3)	Patrol vehicles (4)
Dep. var: avg. annual increase in resources (share of pre-adoption level)				
Reported crime (per 100k pop.)	−0.203** (0.099)	−0.199** (0.097)	−0.319* (0.182)	0.272* (0.137)
Arrests (per 100k pop.)	0.099 (0.136)	−0.041 (0.129)	0.003 (0.186)	0.297 (0.223)
Victimization			−0.017* (0.010)	−0.025* (0.013)
Crime perception			0.005 (0.017)	0.008 (0.014)
Trust in the police			0.024 (0.018)	0.027 (0.019)
Investment in security measures			0.004 (0.010)	−0.003 (0.009)
Police officers (per 100k pop.)	−0.207 (0.155)	0.155 (0.129)	−0.888*** (0.154)	0.571*** (0.136)
Patrol vehicles (per 100k pop.)	−0.129 (0.129)	−0.139 (0.126)	0.537** (0.212)	−0.507*** (0.169)
Log population	−0.529*** (0.145)	−0.203* (0.112)	−0.193 (0.189)	−0.626** (0.269)
Poverty rate (2004)	0.170 (0.113)	0.308** (0.126)	0.027 (0.108)	−0.017 (0.108)
Observations	65	65	8,781	8,781
R-squared	0.342	0.282	0.614	0.545

Note: OLS regressions of the average annual increase in police officers (columns 1 and 3) and patrol vehicles (columns 2 and 4) per 100,000 inhabitants, from the year before adoption of *Plan Cuadrante* to 2012. As in panels (a) and (b) of Figure 4, the increase is expressed as a share of the resources available the year before adoption. Columns 1–2 use the administrative sample at the municipality level; columns 3–4 use ENUSC survey respondents interviewed before 2007 (survey outcomes at the respondent level, other regressors as pre-2006 municipality aggregates). All variables are standardized. Standard errors clustered by municipality in parentheses.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Comment R2.6

Since the authors emphasize local knowledge as the key mechanism, the paper would also benefit from a deeper integration with the economics literature on local governance and institutional design. In particular, Ostrom (1973) provides relevant conceptual grounding for specialization and local monitoring in the context of policing, yet her work and related contributions are not cited.

Response to R2.6:

We thank the reviewer for this suggestion. We were indeed unaware of her research program on metropolitan policing in the 1970s (Ostrom et al., 1973, 1978), which provides indeed a relevant conceptual foundation for our work. Comparing police forces in Indianapolis, the authors find that areas covered by smaller and less centralized police forces outperform areas covered by larger ones. The authors argue that a key driver of the difference might be that officers in smaller departments operate in smaller catchment areas developing deeper local knowledge and citizens engage more actively in monitoring their neighborhoods, which anticipates the mechanism we propose. *Plan Cuadrante* achieves a similar effect not by fragmenting police departments but by subdividing their operational territories, permanently assigning officers to smaller areas within the same institutional structure rather than rotating them across larger areas. In the revised manuscript, we discuss this connection, situating our contribution within this broader literature on the relationship between police organizational scale and effectiveness.

Specifically, we have added the following passage to the introduction when discussing our contribution to the literature:

Second, our results speak to a broader literature showing that specific features of police organization matter for police effectiveness and public security outcomes... Earlier work also found that areas covered by smaller, more localized police departments outperform, in terms of security, those covered by larger centralized ones, hypothesizing that this might be related, among other factors, to closer citizen-police engagement (Ostrom et al., 1973, 1978).

Comment R2.7

Some additional comments.

1. The authors should provide more details on the timing of adoption among municipalities. What criteria or circumstances led to a staggered adoption?

Response to R2.7:

Thank you very much. In the updated version of the manuscript this is discussed in Section 2 and Appendix A. Table A.1 in Appendix A provides information on the year of introduction of the program in each treated municipality.

The selection of municipalities into *Plan Cuadrante* is described in the program documents (Winchester et al., 2007; Salas P. et al., 2014). Eligibility was restricted to municipalities where at least 70% of the population is urban, which corresponds to 150 out of 346 Chilean municipalities. Among

eligible municipalities, adoption was intended to follow a weighted index (with changing weights) combining five characteristics: victimization rate, police resource deficit, drug prevalence, unemployment rate, and demand for police services measured in Equivalent Surveillance Units (UEs). The staggered rollout reflected annual budget constraints: each year, the available budget determined how many municipalities could join the program, with the ranking intended to determine which ones would be incorporated first. However, program evaluations document that the actual pattern of adoption diverged substantially from this ranking, with the rule operating as a guideline rather than a binding assignment mechanism (Salas P. et al., 2014). For example, all municipalities in the metropolitan area of Santiago were incorporated in the initial years of the program (before our study period begins), regardless of their ranking. By 2013, all 150 eligible municipalities had been incorporated into the program. We requested information on the ranking index and its inputs but were not granted access to this information.

The validity of our identification strategy rests primarily on the parallel trends assumption, which would be threatened if adoption timing were correlated with pre-existing trends in outcomes. The event study estimates presented in Figure 2 directly address this concern: they show no evidence of differential pre-treatment trends in terms of most outcomes examined, supporting the key identification condition of our difference-in-differences design.

Below we reproduce the relevant passages from Section 2 and Appendix A of the updated manuscript, which discuss the selection criteria and timing of adoption in detail:

From Section 2 of the updated manuscript:

Carabineros began piloting *Plan Cuadrante* in 1998. The program expanded gradually through 2013, as shown in Panel (a) of Figure 1, reaching 150 urban municipalities and covering roughly 90% of Chile’s population.⁹² Online Appendix A provides further details about the rollout of the program. The rollout was designed to follow a municipality ranking based on a weighted index of victimization, police resource deficits, drug prevalence, unemployment, and demand for police services, with annual entry determined by available budget.⁹³ In practice, however, Budget Office evaluations show that adoption diverged from this ranking for undocumented reasons, so the rule operated more as a loose guideline than as a binding assignment mechanism (Salas P. et al., 2014).

From Appendix A of the updated manuscript:

Implementation

Plan Cuadrante was launched in 1998, initially as a pilot program in a few municipalities in the southern part of the city of Santiago. This initial phase aimed to test the plan’s implementability. The six-month pilot revealed promising outcomes, which led to its broader implementation despite the need for additional resources and personnel.

By 2000, the *Plan Cuadrante* had expanded to 44 municipalities in Santiago, covering approximately 96.1% of the city’s population. This expansion was based on expert recommendations and the division of urban areas into manageable patrol zones. The plan was then implemented gradually throughout urban municipalities in Chile. The staggered expansion of the *Plan Cuadrante* beyond Santiago was, in principle, meant to follow a prioritization based on a

⁹²Further details about the expansion of *Plan Cuadrante* are provided in Online Appendix A and Table A.1.

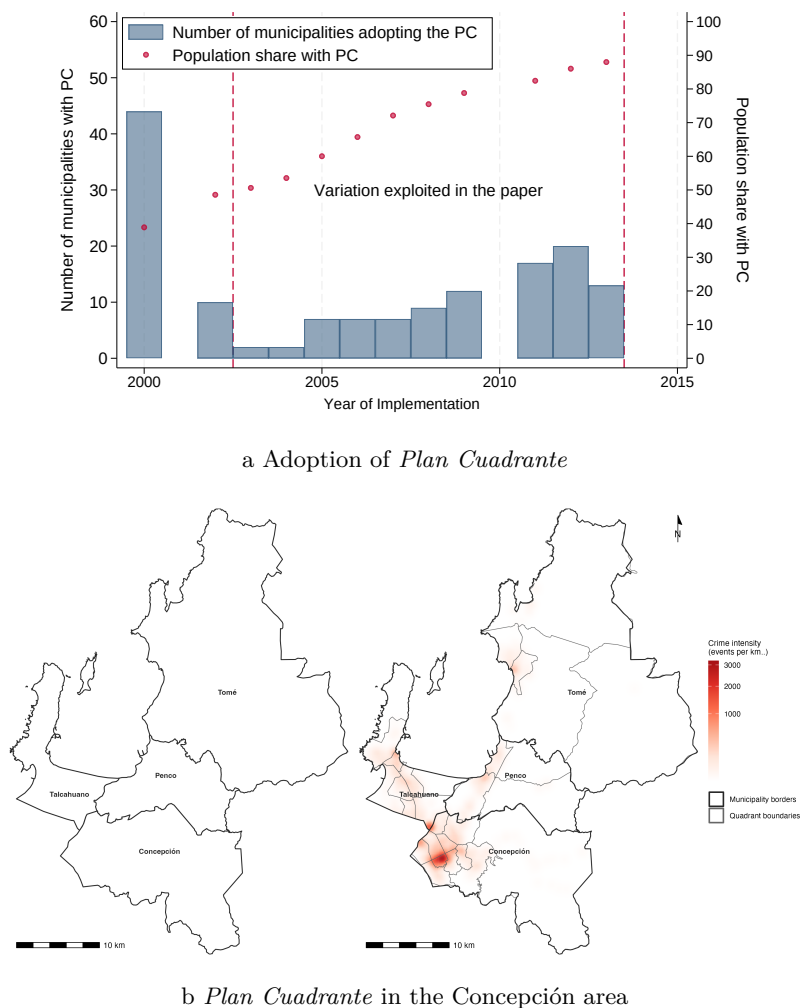
⁹³The weights assigned to each component changed on a yearly basis (Winchester et al., 2007; Salas P. et al., 2014).

weighted index of victimization, police resource deficits, drug prevalence, unemployment, and demand for police services, with annual entry determined by available budget. The weights assigned to each component changed on a yearly basis and, in practice, Budget Office evaluations show that adoption diverged from this ranking for undocumented reasons, so the rule operated more as a loose guideline than as a binding assignment mechanism (Salas P. et al., 2014).

The plan saw significant growth between 2002 and 2006. In 2002, it extended beyond Santiago into the V and VIII Regions, adding ten new municipalities. The expansion continued from 2003 to 2006, with 18 additional municipalities incorporated into the plan. By 2006, the *Plan Cuadrante* covered a total of 72 municipalities across Chile. Between 2007 and 2009, the plan expanded to include 28 more municipalities, reaching the 100 municipalities covered by the *Plan Cuadrante*. The *Plan Cuadrante* faced significant challenges in 2010 due to the earthquake that struck Chile. The disaster necessitated a reallocation of resources to support recovery efforts, temporarily halting the plan's expansion. Despite this setback, the *Plan Cuadrante* resumed its implementation in 2011. Between 2011 and 2013, the plan was expanded to include an additional 50 municipalities, bringing the total to 150 across Chile. Tables A.2 and A.3 present summary statistics at the household and at the municipality level of the municipalities implementing *Plan Cuadrante* in the period that we study.

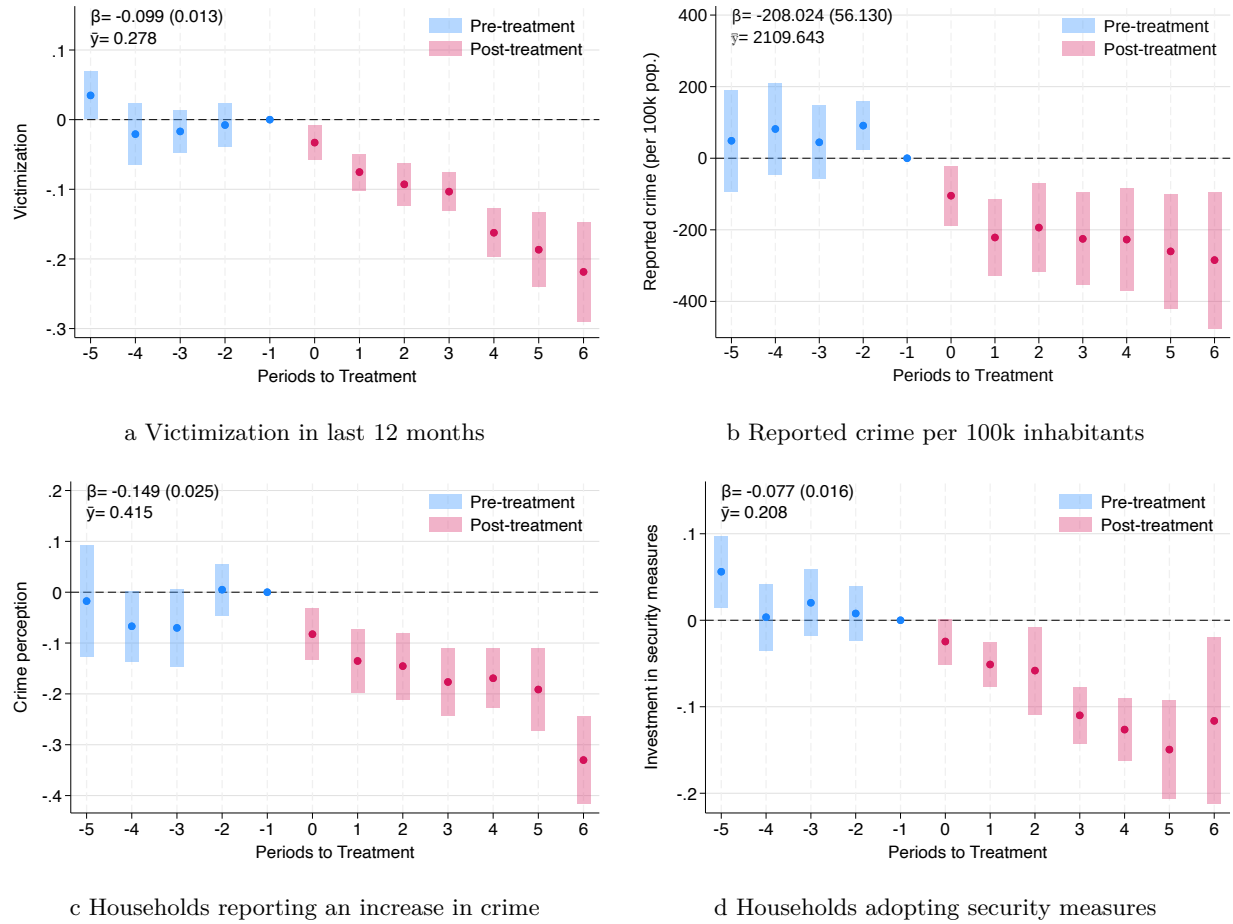
Other figures and tables referenced above:

Figure 1: Implementation of *Plan Cuadrante*



Notes: Panel (a) shows the number of municipalities adopting the *Plan Cuadrante* each year (bars), while the circles above them illustrate the cumulative share of the Chilean population living in a municipality with the *Plan Cuadrante* already in place. Panel (b) depicts the metropolitan area of Concepción, the second-largest in Chile: the left map shows its four municipalities—the original police catchment areas—and the right map overlays the Carabineros quadrants together with a kernel-density estimate of all types of robberies and thefts recorded by the police (both reported crimes and arrests). It illustrates how *Plan Cuadrante* subdivided each catchment area (which corresponds to the territory of a municipality) into smaller quadrants, as well as the spatial concentration of crime within them.

Figure 2: Effects of *Plan Cuadrante* on Crime Outcomes and Security Perceptions



Notes: This figure shows the dynamic effects of the *Plan Cuadrante* on different measures of police effectiveness. The estimation is conducted using the procedure developed in Callaway and Sant'Anna (2021) for difference-in-differences with staggered treatment adoption. The dots in the figure represent the estimated coefficients and the bars behind them 95% confidence intervals. Standard errors are clustered at the municipality level using the wild bootstrapping procedure. The pooled effect and the mean outcome before the introduction of the treatment is also presented in each panel. Panel (a) reports the effect of *Plan Cuadrante* on the probability of household victimization in the last 12 months using the ENUSC survey. Panel (b) reports the effect on the number of crimes reported to the police in the municipality per 100,000 inhabitants using administrative data. Panel (c) reports the effect on the probability of perceiving an increase in crime using the ENUSC survey. Panel (d) reports the effect on the probability of adopting security measures in the last twelve months using the ENUSC survey. Panel (a) includes 93,041 observations; panel (b), 4,231; panel (c), 89,578; and panel (d), 93,041 observations.

Table A.1: Implementation of *Plan Cuadrante* by Year

Year	N	Municipalities
1998–2000	44	Metropolitan Region of Santiago (44 municipalities)
2002	10	Chiguayante, Concepción, Hualpén, Quilpué, San Antonio, San Pedro de la Paz, Talcahuano, Valparaíso, Villa Alemana, Viña del Mar
2003	2	Padre Las Casas, Temuco
2004	2	Antofagasta, Copiapó
2005	7	Alto Hospicio, Arica, Calama, Curicó, Iquique, Linares, Talca
2006	7	Coquimbo, La Serena, Los Andes, Ovalle, Rancagua, San Felipe, San Fernando
2007	7	Chillán, Chillán Viejo, Los Ángeles, Osorno, Puerto Montt, Punta Arenas, Valdivia
2008	9	Ancud, Castro, Constitución, Coronel, Lota, Peñaflor, Quillota, Rengo, Villarrica
2009	12	Angol, Aysén, Calera, Concón, Coyhaique, La Unión, Padre Hurtado, Parral, Penco, San Carlos, Tomé, Vallenar
2011	17	Arauco, Cañete, Cartagena, Cauquenes, Curanilahue, La Ligua, Limache, Molina, Nueva Imperial, Pucón, Puerto Varas, Río Bueno, San Clemente, San Javier, San Vicente, Santa Cruz, Victoria
2012	20	Cabrero, Calbuco, Casablanca, Chimbarongo, Collipulli, Curacaví, El Monte, Graneros, Illapel, Isla de Maipo, Lautaro, Machalí, Mostazal, Mulchén, Nacimiento, Natales, Panguipulli, Quellón, Quintero, Tocopilla
2013	13	Bulnes, Laja, Las Cabras, Lebu, Llaillay, Loncoche, Longaví, Los Álamos, Maule, Requínoa, Teno, Vicuña, Yumbel
Total	150	

Note: N is the number of municipalities that adopted *Plan Cuadrante* in each year. Data shared by Carabineros de Chile.

Table A.2: Summary Statistics – ENUSC sample

	Mean	SD	Min	Max	N
<i>Household</i>					
Socioeconomic status	3.58	0.73	1	5	99,298
Victimization	0.26	0.44	0	1	99,298
Crime perception (neighborhood)	0.38	0.49	0	1	95,598
Crime perception (municipality)	0.66	0.48	0	1	96,794
Crime perception (country)	0.79	0.41	0	1	98,525
Trust in the police	0.44	0.50	0	1	51,445
Investment in security measures	0.21	0.40	0	1	99,298
<i>Plan Cuadrante</i> (year)	2007	1.85	2005	2012	97,847
<i>Household member interviewed</i>					
Female	0.55	0.50	0	1	99,298
Age	43.87	17.95	15	99	99,298
Primary school attendance	0.98	0.14	0	1	99,298
Secondary school attendance	0.72	0.45	0	1	99,298
University attendance	0.20	0.40	0	1	99,298

Note: This table presents summary statistics from the ENUSC sample, covering the 47 municipalities used in the survey-based analysis (46 that adopted *Plan Cuadrante* between 2005 and 2012 plus one never-treated municipality). Socioeconomic status is a 1 to 5 measure produced by ENUSC surveys and based on self-reported socioeconomic strata, being 1 the highest ranked and 5 the lowest. Crime perception is equal to 1 if the interviewed person reports that crime has increased in the last 12 months. The *Plan Cuadrante* (year) row is computed over treated municipalities only.

Table A.3: Summary Statistics – Administrative municipality level sample

	Mean	SD	Min	Max	N
<i>Panel A. Reported-crime sample (292 municipalities)</i>					
Reported crime (per 100k pop.)	1,732.39	1,036.68	0.00	11,434.40	4,371
Arrests (per 100k pop.)	332.07	271.75	0.00	1,712.30	3,504
Poverty rate (%)	25.74	9.72	5.65	58.33	3,720
Pop. density (pop./km ²)	63.51	140.47	0.09	1,109.58	4,065
<i>Plan Cuadrante</i> (year)	2010	2.82	2003	2013	1,440
<i>Panel B. Arrests sample (281 municipalities)</i>					
Reported crime (per 100k pop.)	1,684.74	1,014.67	0.00	11,434.40	4,209
Arrests (per 100k pop.)	313.21	251.23	0.00	1,712.30	3,372
Poverty rate (%)	26.01	9.72	5.65	58.33	3,570
Pop. density (pop./km ²)	57.90	127.30	0.09	1,109.58	3,900
<i>Plan Cuadrante</i> (year)	2010	2.26	2006	2013	1,275

Note: The table reports municipality-level summary statistics for the two administrative estimation samples used in the paper. Panel A covers the 292 municipalities in the reported-crime sample (municipalities adopting *Plan Cuadrante* from 2003 onward plus never-treated); Panel B the 281 municipalities in the arrests sample (adopters from 2006 onward plus never-treated — municipalities adopting in 2005 are excluded because arrest data begin that year, leaving no pre-treatment period). Reported crime and arrests are measured per 100,000 inhabitants over 2002–2016 and 2005–2016, respectively. Minimum reported crime and arrests of zero correspond to tiny, remote never-treated comunas (e.g., Juan Fernández, Cabo de Hornos). The *Plan Cuadrante* (year) row is computed over treated municipalities only. Data from the Undersecretary for Crime Prevention.

Comment R2.8

2. The paper should provide additional information on the estimation sample: Are the event-study models estimated within a fixed window? What happened to observations outside the window (included but not reported or binned-up)?

Response to R2.8:

Thanks. Information on the estimation sample is now described in the data section and then provided in the footnote below each figure in the manuscript.

In the ENUSC data, the analytical sample includes the 47 municipalities surveyed in the ENUSC that were not treated in 2003 or earlier, the first year for which data are available. The sample period is restricted to 2003–2012 by two features of the data. First, the ENUSC is only available from 2003. Second, since all but one ENUSC municipality is eventually treated, the Callaway and Sant’Anna estimator cannot use already-treated municipalities as controls, and extending the sample beyond 2012—the last year in which an ENUSC municipality adopted the program—would leave only a single never-treated municipality as the comparison group. This results in an event study with a maximum of 5 pre-treatment and 7 post-treatment periods.⁹⁴ Note that since treatment adoption is staggered, not all municipalities contribute to the estimation of every pre- and post-treatment period. As a robustness check, we also estimate the main results using a balanced panel of municipalities, which is reported in Online Appendix C in the updated manuscript. We restrict the sample to municipalities that implemented *Plan Cuadrante* between 2006 and 2009, enabling the construction of a symmetric window from $t - 2$ to $t + 3$. The resulting estimates are very similar to those from the main specification.

The administrative data are available for the full set of Chilean municipalities—reported crimes from 2002 to 2016 and arrests from 2005 to 2016. Because reported crime is available from 2002, the analytical sample for this outcome includes 292 municipalities: all municipalities that adopted *Plan Cuadrante* from 2003 onward, together with all never-treated municipalities. Because arrests are only available from 2005, the corresponding analytical sample is restricted to 281 municipalities: all municipalities that adopted the program from 2005 onward, together with all never-treated municipalities. For symmetry with the ENUSC analysis, the main specification is nonetheless restricted to 5 pre-treatment and 7 post-treatment periods in both samples. We conduct here two robustness checks. First, we also estimate the effects using the full set of pre- and post-treatment periods available in each administrative sample and find consistent results. Second, we restrict the sample to municipalities that implemented *Plan Cuadrante* between 2006 and 2009 and construct a symmetric window from $t - 2$ to $t + 3$. Once again, the resulting estimates are very similar to those from the main specification.

Below we reproduce the relevant paragraphs from the Data subsection of the updated manuscript, describing the construction of both analytical samples:

We combine administrative and survey data from *Carabineros*, the Undersecretary for Crime Prevention, and the National Institute of Statistics (INE).

⁹⁴Some variables—such as trust in the police, police presence, or complaints about lack of police presence—are not available in every ENUSC wave, so the number of pre- and post-treatment periods may differ across event studies for these variables.

Carabineros provided information on *Plan Cuadrante* implementation, which we use to define treatment status across municipalities over time. The National Institute of Statistics granted access to the National Survey on Security and Crime (ENUSC), a repeated cross-section first conducted in 2003 and administered annually between October and December since 2005. The survey covers more than 25,000 households across 101 urban municipalities, collecting information on crime victimization, perceptions, trust in police, and household prevention measures over the previous 12 months.⁹⁵

Our empirical approach excludes always-treated municipalities (Callaway and Sant’Anna, 2021). Given ENUSC’s first implementation in 2003 and absence in 2004, our main sample includes households from 46 urban municipalities that adopted *Plan Cuadrante* between 2005 and 2012 plus one never-treated municipality.⁹⁶ The treated municipalities in this analytical sample account for approximately 27% of Chile’s population.

The Undersecretary for Crime Prevention provided data on the number of reported crimes (2002–2016) and arrests (2005–2016) for all municipalities. This comprehensive coverage allows us to expand the analytical sample: for reported crime, we include all municipalities adopting *Plan Cuadrante* after 2002 together with never-treated municipalities, for a total of 292 municipalities; for arrests, the corresponding sample includes all post-2005 adopters and never-treated municipalities, for a total of 281 municipalities. The treated municipalities in these analytical samples account for over half of Chile’s population.

The study period differs across the two data sources, as each covers a different time span and a different set of municipalities. For the ENUSC-based analysis, it is restricted to 2003–2012. The Callaway and Sant’Anna (2021) estimator uses not-yet-treated and never-treated municipalities as the comparison group. Since only one municipality in the survey sample is never treated, extending the analysis beyond 2012—the last year in which a municipality surveyed in ENUSC adopts the program—would leave only that single never-treated unit as the comparison group for subsequent periods, making the estimates unreliable. The administrative data, by contrast, cover a large number of never-treated municipalities that can serve as controls throughout the full data period, allowing us to exploit the complete administrative coverage—2002–2016 for reported crime and 2005–2016 for arrests—and to include all municipalities adopting the program from 2003 to 2013, when the last municipality joins *Plan Cuadrante*.

Together, the treated municipalities in our two analytical samples represent substantial shares of the Chilean population—approximately 27% in the ENUSC-based analysis and over half in the administrative data analysis—and are geographically distributed across the country rather than clustered in a single region. Tables A.2 and A.3 in Online Appendix A and Table B.1 and Figure B.1 in Online Appendix B provide a detailed characterization of the treated municipalities in each analytical sample relative to the average Chilean municipality.

Note also the following footnote to Section 4 in the main manuscript:

Online Appendix Figure C.8 reports the results for reported crime and arrests using the full set of post-treatment and pre-treatment periods, which are very similar to those presented here.

and the following subsection of Online Appendix C, “Effects of *Plan Cuadrante* Using the Full Set

⁹⁵The survey is administered to a randomly selected household member who must: a) live in the household, b) be 15 years or older, c) not be a domestic worker, and d) not have any disability preventing survey comprehension.

⁹⁶This sample excludes municipalities adopting *Plan Cuadrante* in 2013 as they are not covered by ENUSC. Also, there is only one never treated municipality covered by ENUSC.

of Post-treatment and Pre-treatment Periods”, which reports the first robustness check described above:

For symmetry with the ENUSC-based analysis, the event studies for reported crime and arrests reported in the main text are restricted to 5 pre-treatment and 7 post-treatment periods. Because the administrative data are available for the full set of Chilean municipalities and therefore include a much larger number of never-treated municipalities than the ENUSC sample, we can estimate these effects over a longer window without exhausting the pool of valid comparison units. Figure C.8 reports the effects of *Plan Cuadrante* on reported crime and arrests using the full set of post-treatment and pre-treatment periods rather than capping the event study to the 5 pre-treatment and 7 post-treatment periods used in the main specification. The results are very similar to those presented in Section 4.1.

as well as the following subsection of Online Appendix C, “Effects of *Plan Cuadrante* using a Balanced Panel”, which reports the second robustness check described above for both analytical samples:

Figure C.7 shows the effects of *Plan Cuadrante* on the main outcomes, but estimated using a balanced panel of municipalities. For the ENUSC-based outcomes, this panel includes only municipalities that implemented the *Plan Cuadrante* between 2006 and 2009, which allows the construction of a balanced panel from $t-2$ to $t+3$ (31 municipalities). For the administrative outcomes, the balanced panel comprises those municipalities observed over that same $t-2$ to $t+3$ window around their own adoption (94 for reported crime and 78 for arrests). The resulting estimates are very similar to those presented in the main document.

Other figures and tables referenced above:

Table A.2: Summary Statistics – ENUSC sample

	Mean	SD	Min	Max	N
<i>Household</i>					
Socioeconomic status	3.58	0.73	1	5	99,298
Victimization	0.26	0.44	0	1	99,298
Crime perception (neighborhood)	0.38	0.49	0	1	95,598
Crime perception (municipality)	0.66	0.48	0	1	96,794
Crime perception (country)	0.79	0.41	0	1	98,525
Trust in the police	0.44	0.50	0	1	51,445
Investment in security measures	0.21	0.40	0	1	99,298
<i>Plan Cuadrante</i> (year)	2007	1.85	2005	2012	97,847
<i>Household member interviewed</i>					
Female	0.55	0.50	0	1	99,298
Age	43.87	17.95	15	99	99,298
Primary school attendance	0.98	0.14	0	1	99,298
Secondary school attendance	0.72	0.45	0	1	99,298
University attendance	0.20	0.40	0	1	99,298

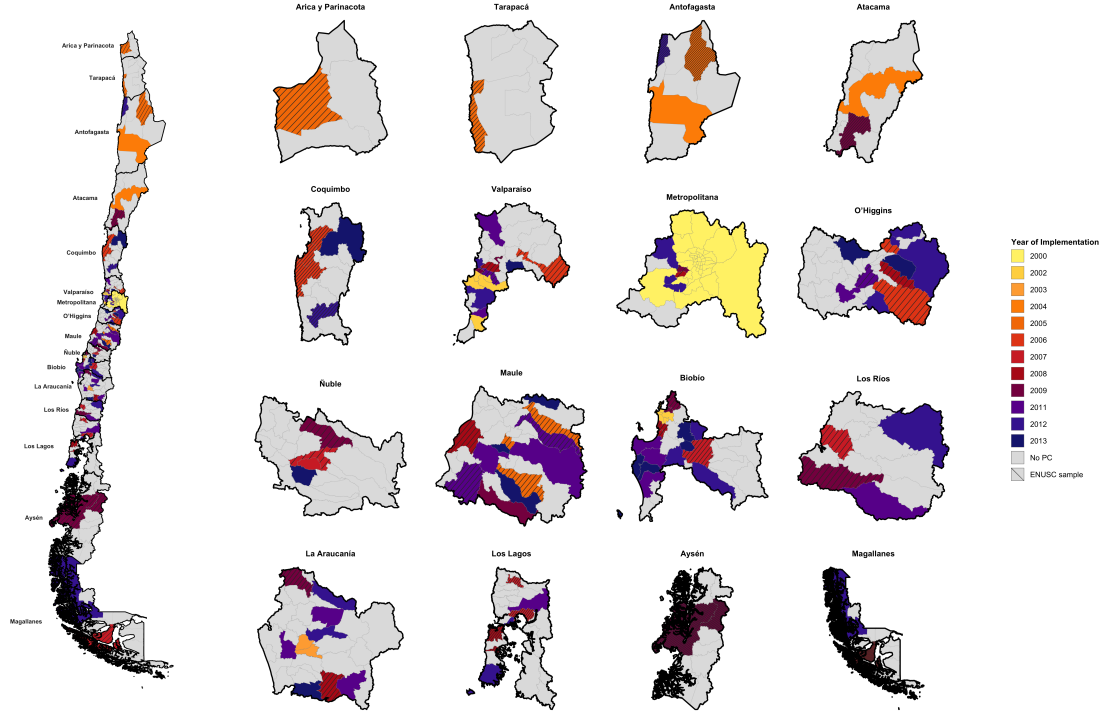
Note: This table presents summary statistics from the ENUSC sample, covering the 47 municipalities used in the survey-based analysis (46 that adopted *Plan Cuadrante* between 2005 and 2012 plus one never-treated municipality). Socioeconomic status is a 1 to 5 measure produced by ENUSC surveys and based on self-reported socioeconomic strata, being 1 the highest ranked and 5 the lowest. Crime perception is equal to 1 if the interviewed person reports that crime has increased in the last 12 months. The *Plan Cuadrante* (year) row is computed over treated municipalities only.

Table A.3: Summary Statistics – Administrative municipality level sample

	Mean	SD	Min	Max	N
<i>Panel A. Reported-crime sample (292 municipalities)</i>					
Reported crime (per 100k pop.)	1,732.39	1,036.68	0.00	11,434.40	4,371
Arrests (per 100k pop.)	332.07	271.75	0.00	1,712.30	3,504
Poverty rate (%)	25.74	9.72	5.65	58.33	3,720
Pop. density (pop./km ²)	63.51	140.47	0.09	1,109.58	4,065
<i>Plan Cuadrante</i> (year)	2010	2.82	2003	2013	1,440
<i>Panel B. Arrests sample (281 municipalities)</i>					
Reported crime (per 100k pop.)	1,684.74	1,014.67	0.00	11,434.40	4,209
Arrests (per 100k pop.)	313.21	251.23	0.00	1,712.30	3,372
Poverty rate (%)	26.01	9.72	5.65	58.33	3,570
Pop. density (pop./km ²)	57.90	127.30	0.09	1,109.58	3,900
<i>Plan Cuadrante</i> (year)	2010	2.26	2006	2013	1,275

Note: The table reports municipality-level summary statistics for the two administrative estimation samples used in the paper. Panel A covers the 292 municipalities in the reported-crime sample (municipalities adopting *Plan Cuadrante* from 2003 onward plus never-treated); Panel B the 281 municipalities in the arrests sample (adopters from 2006 onward plus never-treated — municipalities adopting in 2005 are excluded because arrest data begin that year, leaving no pre-treatment period). Reported crime and arrests are measured per 100,000 inhabitants over 2002–2016 and 2005–2016, respectively. Minimum reported crime and arrests of zero correspond to tiny, remote never-treated comunas (e.g., Juan Fernández, Cabo de Hornos). The *Plan Cuadrante* (year) row is computed over treated municipalities only. Data from the Undersecretary for Crime Prevention.

Figure B.1: Geographic Distribution of *Plan Cuadrante*



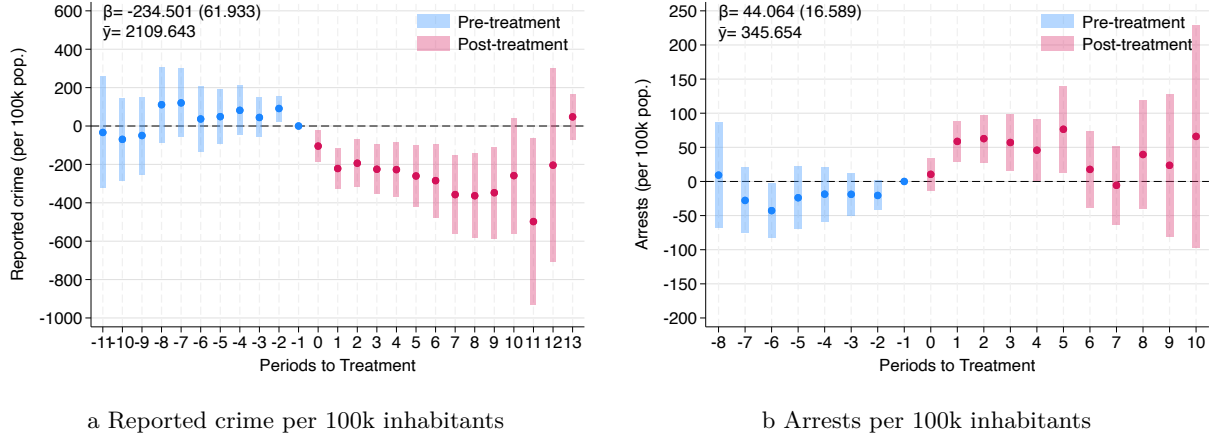
Notes: This figure shows the spatial rollout of *Plan Cuadrante* across Chilean municipalities. The national map (left) is displayed alongside the 16 regional maps (right) for closer inspection. The color of each treated municipality indicates its year of program adoption, ranging from 2000 (lightest) to 2013 (darkest), following the legend. Municipalities shown in light grey were never treated under PCSP. Diagonal hatching identifies the 46 treated municipalities in the analytical sample used to estimate the effects on ENUSC outcomes.

Table B.1: Baseline characteristics by subsample

	(1) All municipalities	PC municipalities				(6) Never- treated
		(2) All PC municipalities	(3) Always- treated	(4) Treated ENUSC	(5) Treated admin. data	
Reported crime (per 100k pop.)	1,650.23	2,132.29	2,437.25	2,202.06	1,976.59	1,282.47
Arrests (per 100k pop.)	267.62	442.08	606.01	459.05	358.11	134.11
Poverty rate (%)	23.73	20.50	14.88	21.73	23.96	26.82
Pop. density (pop./km ²)	870.28	1,904.14	4,593.32	215.52	134.50	27.02
Population	57,857.22	117,975.43	187,558.74	113,079.74	81,532.84	11,612.44
Municipalities	346	150	58	46	96	196

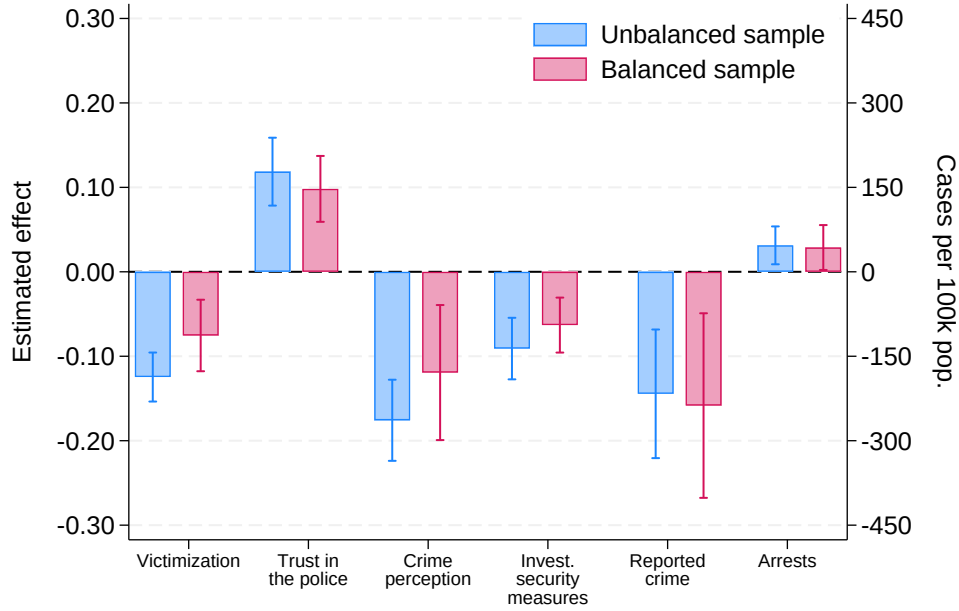
Note: Reported crime, poverty rate, population density, and population are measured as of 2003; arrests are measured as of 2005, the first year for which arrest data are available. Column (1) includes all municipalities in Chile. Column (2) includes all PC municipalities. Column (3) restricts to municipalities that adopted *Plan Cuadrante* before 2005, the always-treated in our sample, which are therefore excluded from our estimation samples (as depicted in Figure 1, panel (a)). Column (4) restricts to treated municipalities in the analytical sample using the ENUSC data. Column (5) restricts to treated municipalities included in the analytical sample using the administrative data. Column (6) includes never-treated municipalities.

Figure C.8: Effects of *Plan Cuadrante* on Reported Crime and Arrests Using the Full Set of Post-treatment and Pre-treatment Periods



Notes: This figure shows the dynamic effects of *Plan Cuadrante* on the number of crimes reported to the police in the municipality per 100,000 inhabitants (panel a) and on the number of arrests in the municipality per 100,000 inhabitants (panel b), using administrative data. The estimation is conducted using the procedure developed in Callaway and Sant'Anna (2021) for difference-in-differences with staggered treatment adoption. The dots in the figure represent the estimated coefficients and the bars behind them 95% confidence intervals. Standard errors are clustered at the municipality level using the wild bootstrapping procedure. The pooled effect and the mean outcome before the introduction of the treatment are also presented in each panel. Unlike Figure 2 in the main text, we do not cap the event study to the same 5 pre-treatment and 6 post-treatment periods used for symmetry with the ENUSC-based analysis. Instead, we exploit the full set of post-treatment and pre-treatment periods. Panel (a) includes 4,371 observations and panel (b) 3,372 observations.

Figure C.7: Effects of *Plan Cuadrante* Using a Balanced Panel of Municipalities



Notes: The figure reports the effects of *Plan Cuadrante* on different measures of police effectiveness using the main analytical sample (unbalanced, blue bars) and a balanced panel of municipalities observed over the full event-study window from $t-2$ to $t+3$ (red bars). For the ENUSC sample the balanced panel consists of the 31 municipalities that implemented *Plan Cuadrante* between 2006 and 2009; for the administrative municipality-level sample it consists of 94 municipalities for reported crime and 78 municipalities for arrests. The bars represent the average post-treatment effect estimated with the difference-in-differences estimator developed in Callaway and Sant'Anna (2021), and the whiskers the 95% confidence intervals; standard errors are clustered at the municipality level. Victimization is the probability of household victimization in the last 12 months; reported crime is the number of crimes reported to the police in the municipality per 100,000 inhabitants; arrests is the number of arrests per 100,000 inhabitants; trust in the police is the probability of reporting high trust in the police; crime perception is the probability of perceiving an increase in crime; and investment in security measures is the probability of adopting security measures in the last twelve months. Reported crime and arrests (administrative data) are read on the right axis; all other outcomes, from the ENUSC survey, are read on the left axis.

Comment R2.9

3. Although I appreciate the robustness check to the various difference-in-differences estimators presented in Appendix B, it would be more reassuring to see the event studies.

Response to R2.9:

Thank you very much for the suggestion. In the revised manuscript, we have expanded the “Alternative Estimation Methods” subsection of Appendix C to include event-study plots for all main outcomes under each alternative estimator. We have also added a sentence in the Empirical Strategy section noting that the event studies using these staggered difference-in-differences estimators confirm flat pre-adoption trends under all estimation methods.

We reproduce this sentence from the Empirical Strategy section below:

Online Appendix Figure C.2 shows robustness to alternative estimators, including those developed by Chaisemartin and D’Haultfoeulle (2020); Gardner (2022); Wooldridge (2023), as well as to the canonical two-way fixed effects (TWFE) estimator. The resulting estimates are remarkably similar across the alternative staggered-adoption estimators, except for the canonical TWFE, which shows attenuated effects, arguably driven by the negative weights problem discussed above (Goodman-Bacon, 2021). The corresponding event studies, reported in Online Appendix Figures C.3, C.4, C.5, and C.6, confirm that pre-adoption trends are flat under each alternative estimator.

We also reproduce below the expanded “Alternative Estimation Methods” subsection of Online Appendix C:

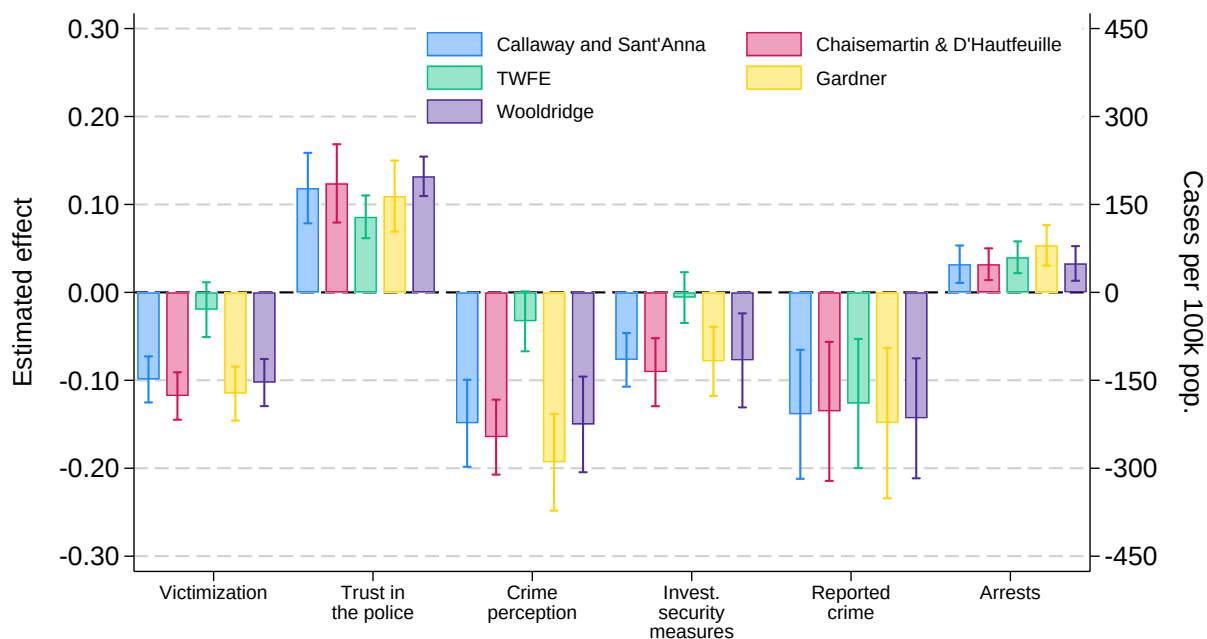
The results reported in Section 4.1 are estimated using the difference-in-differences estimator for staggered treatment adoption developed in Callaway and Sant’Anna (2021). In this subsection, we examine the robustness of the results to the use of alternative difference-in-difference estimators for staggered treatment adoption including those developed in Chaisemartin and D’Haultfoeulle (2020), Gardner (2022), Wooldridge (2023), and also to the canonical two-way fixed effects (TWFE).

The results of these analyses are reported in Figure C.2 and show similar effects of *Plan Cuadrante* in terms of both magnitude and statistical significance across most specifications. On the other hand, the canonical TWFE shows smaller effects, likely due to the negative weights problem highlighted by Goodman-Bacon (2021) when estimating difference-in-differences with staggered treatment adoption using TWFE.

Figures C.3 through C.6 report the corresponding event-study estimates for victimization, crime perception, investment in security measures, and reported crime under each alternative estimator. Pre-treatment coefficients are consistently small and statistically indistinguishable from zero across all estimators and outcomes, confirming that the parallel trends assumption holds regardless of the estimation method. Post-treatment effects remain negative and statistically significant across these outcomes, closely mirroring the results from the main specification in Figure 2.

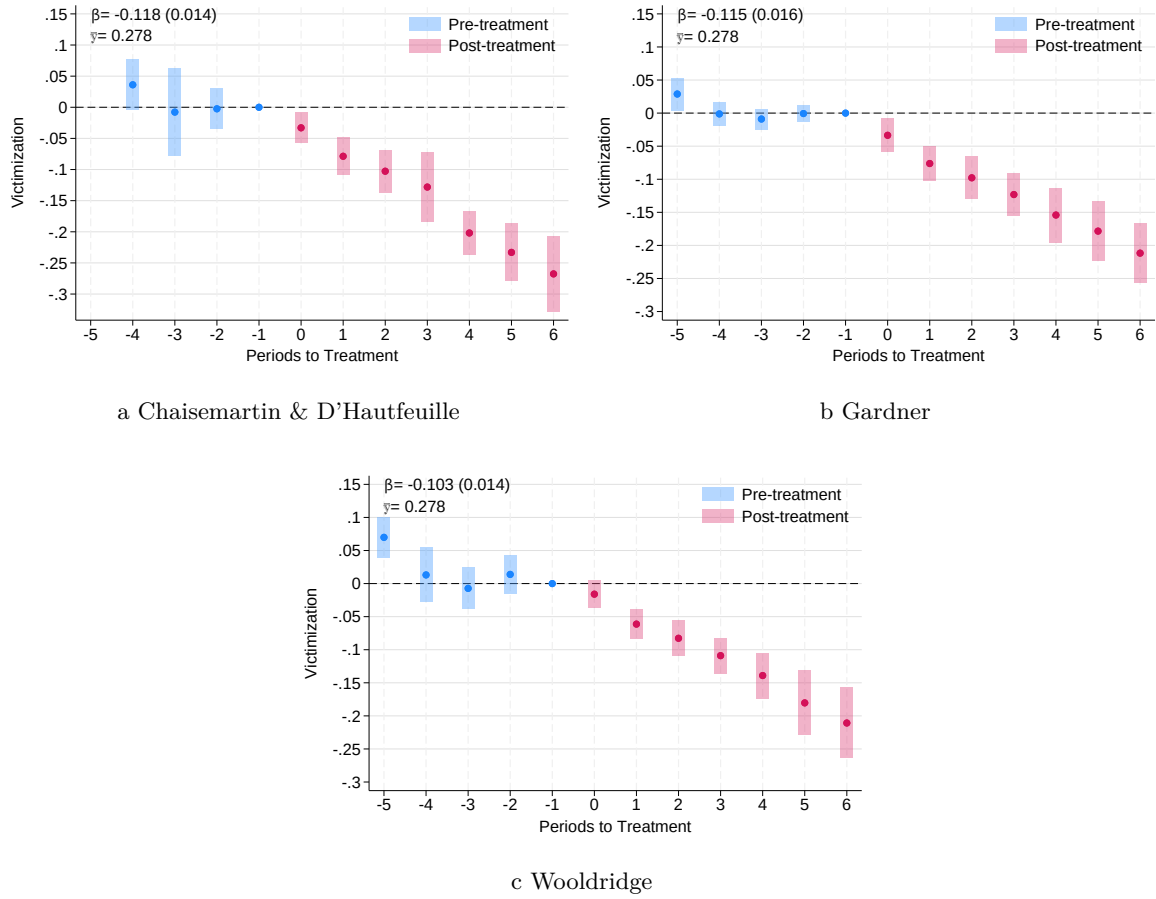
Other figures referenced above:

Figure C.2: Effects of *Plan Cuadrante* on Police Effectiveness Using Alternative Estimators



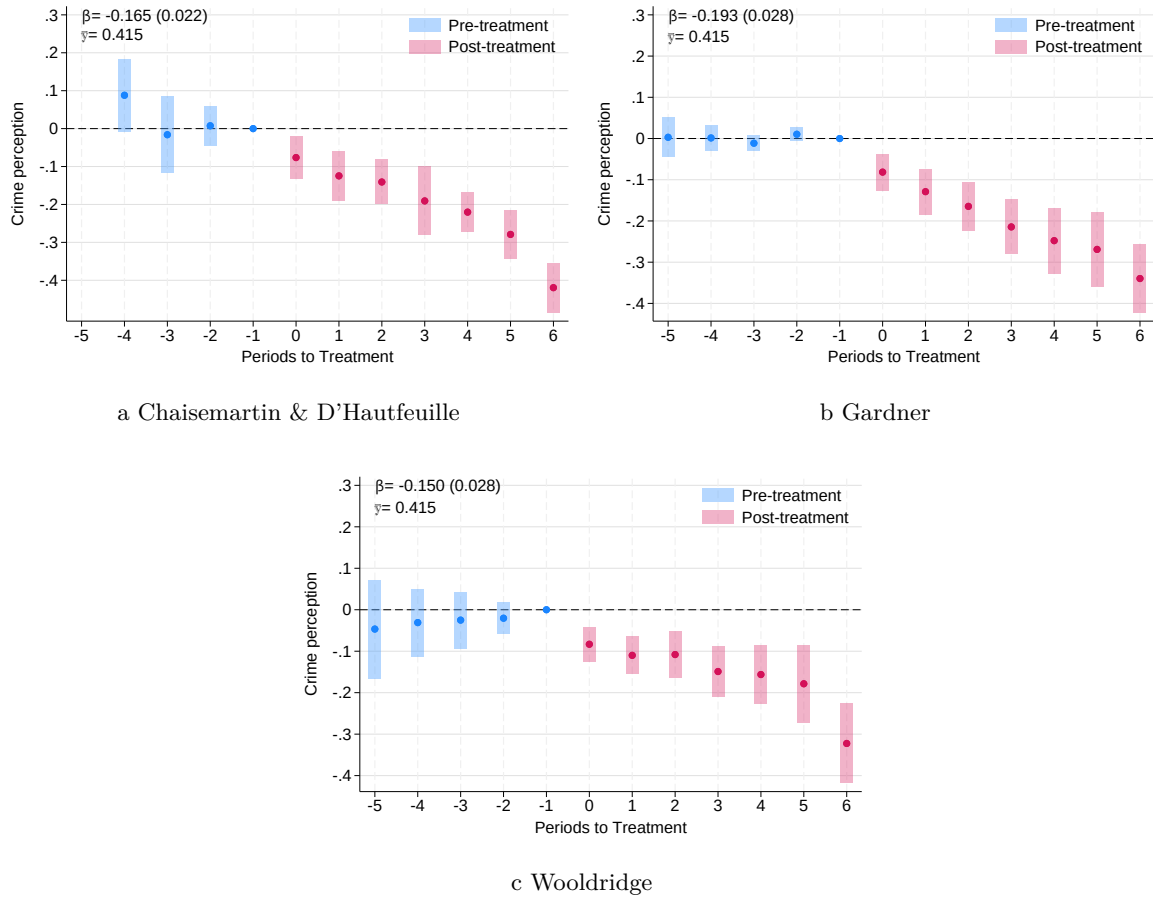
Notes: The figure reports the effects of *Plan Cuadrante* on different measures of police effectiveness using various difference-in-differences estimators, including the methods developed in Callaway and Sant'Anna (2021), Chaisemartin and D'Hautfeuille (2020), Gardner (2022), Wooldridge (2023), and the canonical two-way fixed effects (TWFE). Bars are the estimated post-treatment effects and the whiskers the 95% confidence intervals; standard errors are clustered at the municipality level. Victimization is the probability of household victimization in the last 12 months; reported crime is the number of crimes reported to the police in the municipality per 100,000 inhabitants; crime perception is the probability of perceiving an increase in crime; investment in security measures is the probability of adopting security measures in the last twelve months; trust in the police is the probability of reporting high trust in the police; and arrests is the number of arrests per 100,000 inhabitants. Reported crime and arrests (administrative data) are read on the right axis; all other outcomes, from the ENUSC survey, are read on the left axis.

Figure C.3: Effects of *Plan Cuadrante* on Victimization Using Alternative Estimators



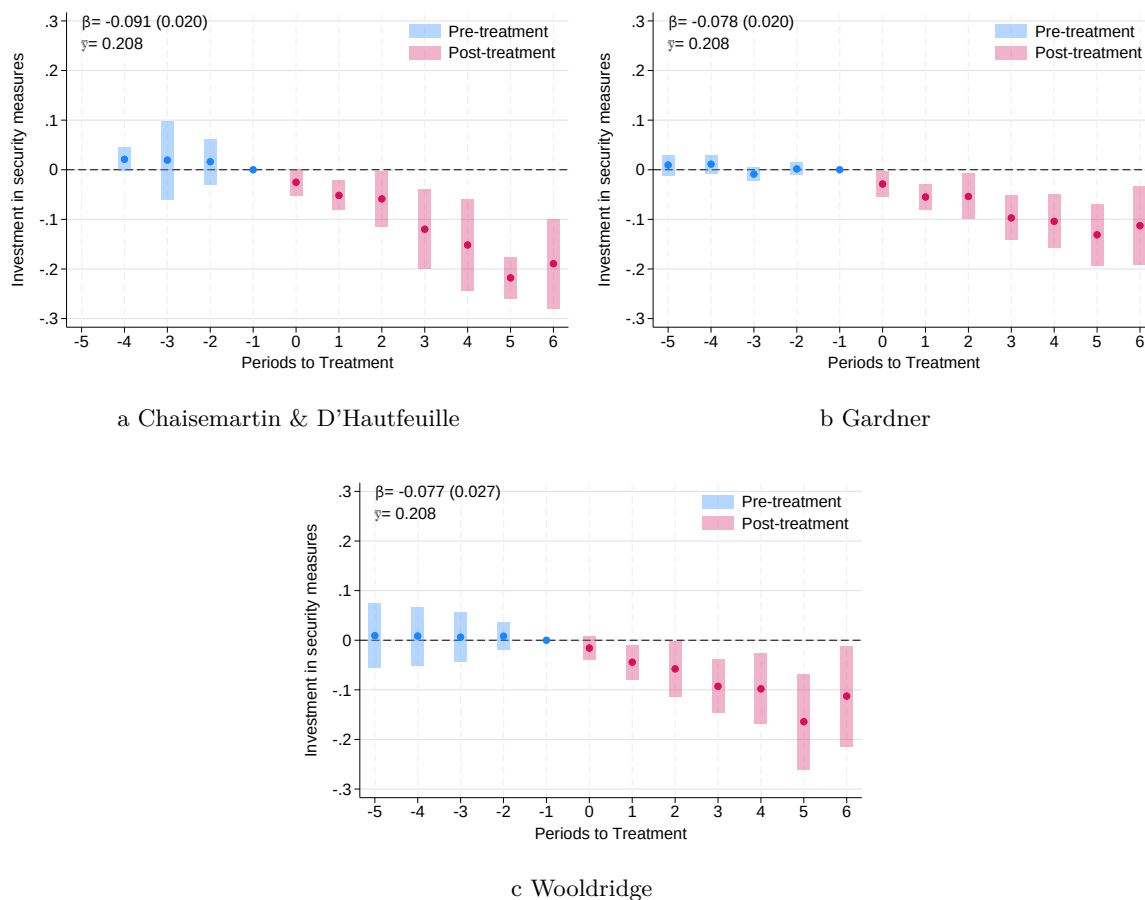
Notes: The figure reports event-study estimates of the effect of *Plan Cuadrante* on the probability of household victimization in the last 12 months using the ENUSC survey, estimated with three alternative difference-in-differences estimators robust to staggered adoption: Chaisemartin and D'Haultfœuille (2020) in panel (a), Gardner (2022) in panel (b), and Wooldridge (2023) in panel (c). Each panel plots the estimated coefficient at each event time relative to treatment along with 95% confidence intervals. Standard errors are clustered at the municipality level. The headline coefficient β corresponds to the average post-treatment effect, and $\bar{y}$ denotes the never-treated mean of the outcome over the sample period.

Figure C.4: Effects of *Plan Cuadrante* on Crime Perception Using Alternative Estimators



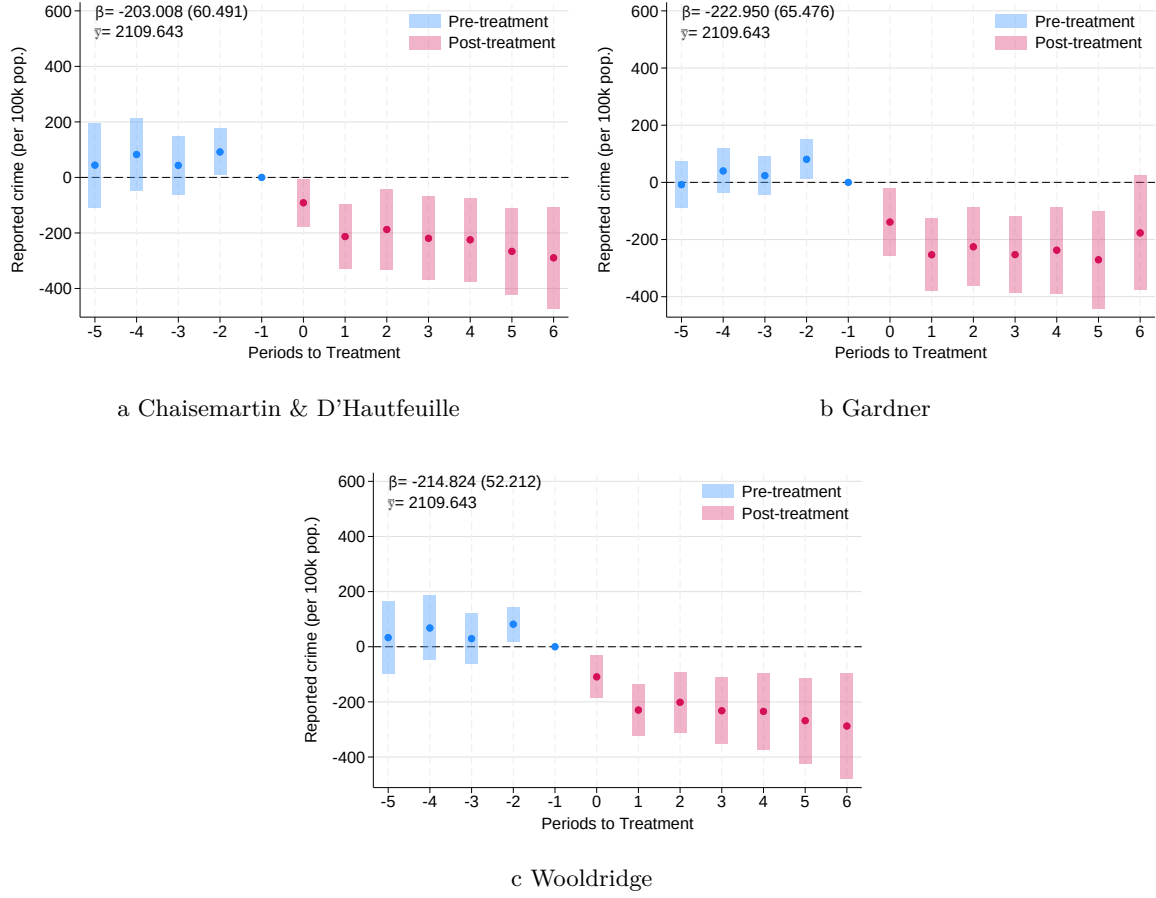
Notes: The figure reports event-study estimates of the effect of *Plan Cuadrante* on the probability of perceiving an increase in crime using the ENUSC survey, estimated with three alternative difference-in-differences estimators robust to staggered adoption: Chaisemartin and D'Hautfeuille (2020) in panel (a), Gardner (2022) in panel (b), and Wooldridge (2023) in panel (c). Each panel plots the estimated coefficient at each event time relative to treatment along with 95% confidence intervals. Standard errors are clustered at the municipality level. The headline coefficient β corresponds to the average post-treatment effect, and $\bar{y}$ denotes the never-treated mean of the outcome over the sample period.

Figure C.5: Effects of *Plan Cuadrante* on Investment in Security Measures Using Alternative Estimators



Notes: The figure reports event-study estimates of the effect of *Plan Cuadrante* on the probability of adopting security measures in the last twelve months using the ENUSC survey, estimated with three alternative difference-in-differences estimators robust to staggered adoption: Chaisemartin and D'Hautfeuille (2020) in panel (a), Gardner (2022) in panel (b), and Wooldridge (2023) in panel (c). Each panel plots the estimated coefficient at each event time relative to treatment along with 95% confidence intervals. Standard errors are clustered at the municipality level. The headline coefficient β corresponds to the average post-treatment effect, and $\bar{y}$ denotes the never-treated mean of the outcome over the sample period.

Figure C.6: Effects of *Plan Cuadrante* on Reported Crime Using Alternative Estimators



Notes: The figure reports event-study estimates of the effect of *Plan Cuadrante* on the number of crimes reported to the police in the municipality per 100,000 inhabitants using administrative data, estimated with three alternative difference-in-differences estimators robust to staggered adoption: Chaisemartin and D'Hautfeuille (2020) in panel (a), Gardner (2022) in panel (b), and Wooldridge (2023) in panel (c). Each panel plots the estimated coefficient at each event time relative to treatment along with 95% confidence intervals. Standard errors are clustered at the municipality level. The headline coefficient β is the equal-weighted average of the post-treatment effects from $t = 0$ to $t = 6$, and $\bar{y}$ denotes the never-treated mean of the outcome over the sample period.

Comment R2.10

4. Given the large number of related outcomes (e.g., Figure 2), the authors may want to consider adjustments for multiple hypothesis testing.

Response to R2.10:

We thank the reviewer for this suggestion.

We acknowledge that testing multiple outcomes simultaneously raises concerns about false discoveries. We note, however, that all six outcomes in our main analysis—including the four now presented in Figure 2 and the arrests and trust-in-police outcomes moved to the mechanisms section in the revised manuscript—are statistically significant at the 1% significance level. Under the null hypothesis of no effects, one would expect at most 1% of tested coefficients to be false positives by chance at the same level. The observed pattern of results is therefore difficult to reconcile with false discoveries driven by multiple testing. Nonetheless, to address this concern formally, we apply a multiple hypothesis testing correction to the p-values. While procedures such as Romano and Wolf (2005) would be the natural choice, their implementation is not computationally straightforward with the Callaway and Sant’Anna (2021) estimator. We therefore apply the Bonferroni correction, which can be applied directly to any set of estimated p-values regardless of the underlying estimator. The adjusted p-values, reported in Table C.2, confirm that the main results remain statistically significant after correction for every outcome considered.

In the revised manuscript, we have added the following footnote in Section 4.1:

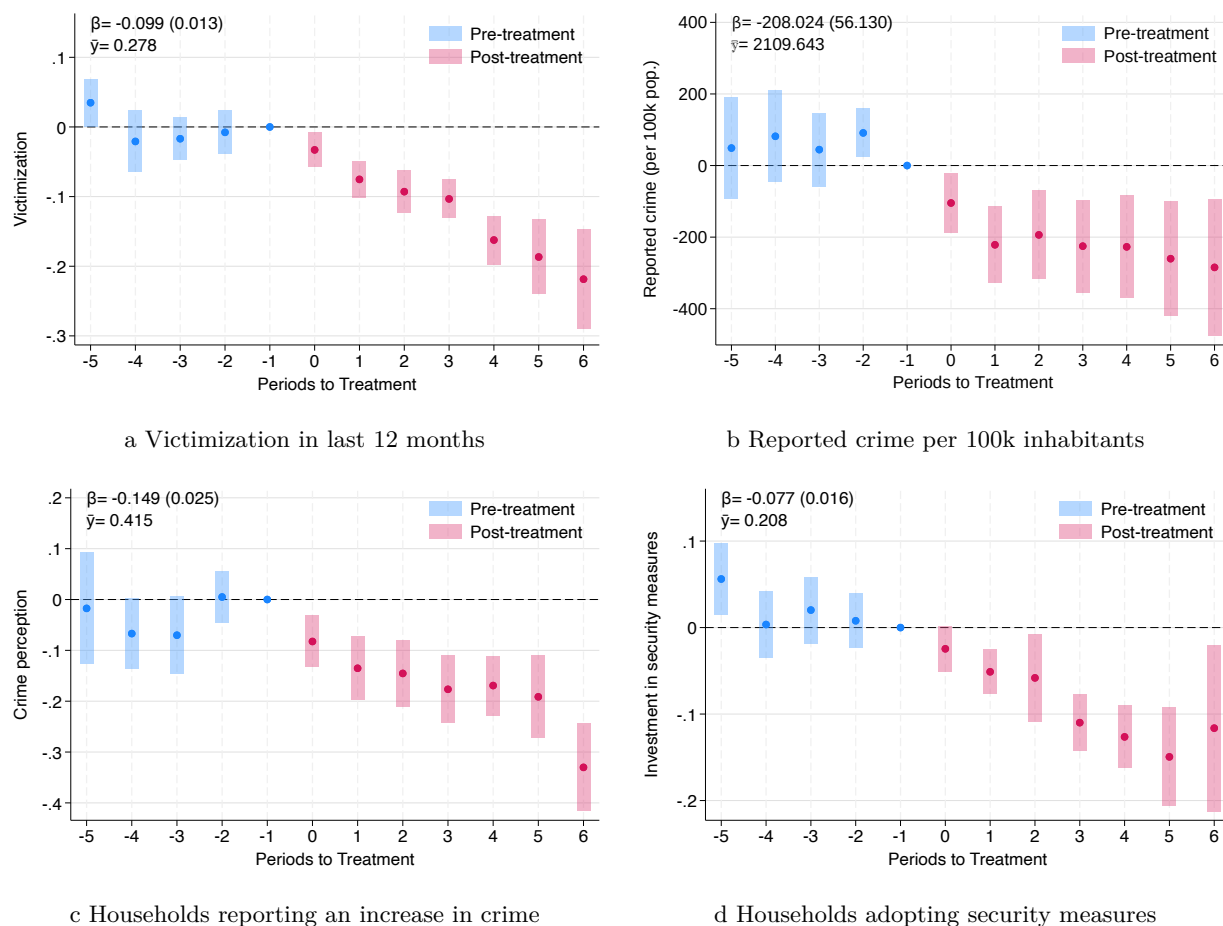
Table C.2 in Online Appendix C applies the Bonferroni correction across the main outcome estimates and confirms that these results are not a product of multiple hypothesis testing.

We also add the following subsection, “Multiple Hypothesis Testing,” to Online Appendix C reporting this exercise in detail:

Testing multiple outcomes simultaneously raises concerns about false discoveries. In our analyses, all six outcomes in our main analyses are statistically significant at the 1% significance level. Under the null hypothesis of no effects, one would expect at most 1% of tested coefficients to be false positives by chance at the same level. The observed pattern of results is therefore difficult to reconcile with false discoveries driven by multiple testing. Nonetheless, to address this concern formally, we apply a multiple hypothesis testing correction to the p-values. While procedures such as Romano and Wolf (2005) would be a natural choice, their implementation is not computationally straightforward with the Callaway and Sant’Anna (2021) estimator. We therefore apply the Bonferroni correction, which can be applied directly to any set of estimated p-values regardless of the underlying estimator. The adjusted p-values, reported in Table C.2, confirm that the main results remain statistically significant at conventional confidence levels after correction for every outcome considered.

Other figure and table referenced above:

Figure 2: Effects of *Plan Cuadrante* on Crime Outcomes and Security Perceptions



Notes: This figure shows the dynamic effects of the *Plan Cuadrante* on different measures of police effectiveness. The estimation is conducted using the procedure developed in Callaway and Sant'Anna (2021) for difference-in-differences with staggered treatment adoption. The dots in the figure represent the estimated coefficients and the bars behind them 95% confidence intervals. Standard errors are clustered at the municipality level using the wild bootstrapping procedure. The pooled effect and the mean outcome before the introduction of the treatment is also presented in each panel. Panel (a) reports the effect of *Plan Cuadrante* on the probability of household victimization in the last 12 months using the ENUSC survey. Panel (b) reports the effect on the number of crimes reported to the police in the municipality per 100,000 inhabitants using administrative data. Panel (c) reports the effect on the probability of perceiving an increase in crime using the ENUSC survey. Panel (d) reports the effect on the probability of adopting security measures in the last twelve months using the ENUSC survey. Panel (a) includes 93,041 observations; panel (b), 4,231; panel (c), 89,578; and panel (d), 93,041 observations.

Table C.2: Multiple Hypothesis Testing: Bonferroni Correction

Outcome	Original		Bonferroni adjusted	
	p-value		p-value	
Victimization	0.00000	***	0.00000	***
Reported crime (per 100k pop.)	0.00021	***	0.00126	***
Crime perception	0.00000	***	0.00000	***
Investment in security measures	0.00000	***	0.00000	***
Trust in the police	0.00000	***	0.00000	***
Arrests (per 100k pop.)	0.00316	***	0.01896	**

Notes: Bonferroni correction multiplies each p-value by the number of outcomes tested ($n = 6$). Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Comment R2.11

5. I would qualify the claims associated to Table B1 since column (2) shows an estimate twice as large as that in column (1), and column (4) does not allow to reject an ATT equal to zero.

Response to R2.11:

Thanks for the point. We agree that the description of the results of this table (now Table C.1, following the reorganization of the appendices in the revised manuscript) required qualification.

In the previous version of the paper, we wrote the following: “In this appendix, we estimate the effects of *Plan Cuadrante* on the number of crimes reported to the police and on the total number of arrests, using only the subsample of municipalities included in the analysis of the ENUSC outcomes. While the sample size is smaller, the estimates using this restricted analytical sample, reported in Table B1 and Figure B2, show similar patterns to those observed when using the full sample.”

While the effects on reported crime have the same sign and are statistically significant in both samples, the effects on reported crime are nearly twice as large in the ENUSC sample of municipalities. These municipalities are on average larger and experienced a higher incidence of crime, which may partly explain the larger estimated effect.

Regarding the effects on arrests, the estimates are not statistically significant in the ENUSC sample of municipalities, which is likely due to the smaller sample size. The point estimates are negative and of similar magnitude to those in the full sample, but the standard errors are larger, leading to a failure to reject the null hypothesis of no effect in this sample.

We have edited the text to reflect these points:

In this Online Appendix, we estimate the effects of *Plan Cuadrante* on the number of crimes reported to the police and on the total number of arrests, using only the subsample of municipalities included in the analysis of the ENUSC outcomes. The sample size in this analysis is smaller, and the results are reported in Figure C.1. While the effects for reported crime are larger in magnitude and the effects on arrests are not statistically significant in the ENUSC subsample, the estimates are qualitatively similar to those observed in Table C.1 with the full sample.

If the editor or the reviewer feel that this wording should be qualified further, we are happy to do so.

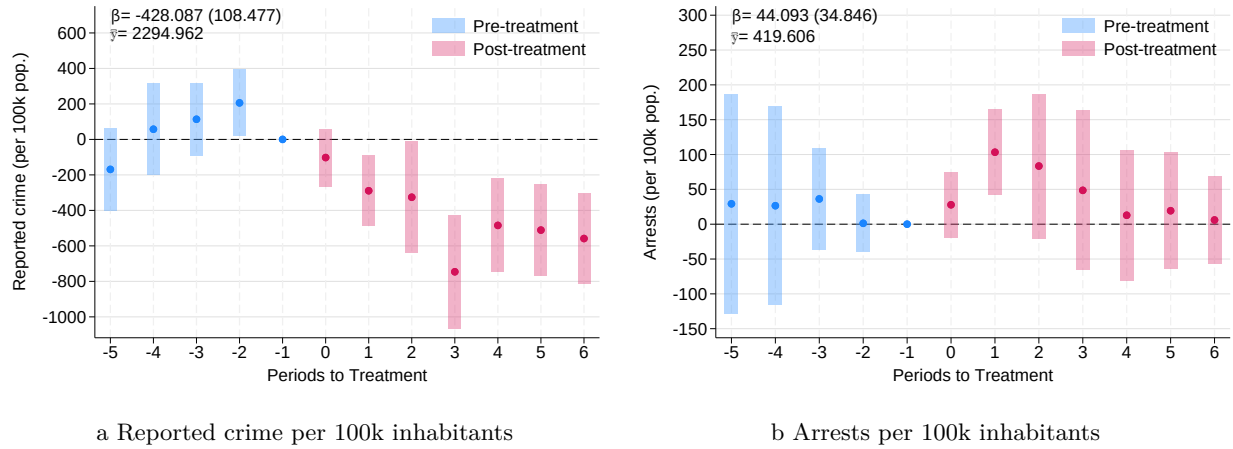
Other figure and table referenced above:

Table C.1: Effects of *Plan Cuadrante* on Reported Crime and Arrests: Full Sample and ENUSC Municipalities

Outcome	(1)	(2)	(3)	(4)
	Reported crime		Arrests	
	Sample Admin	Sample ENUSC	Sample Admin	Sample ENUSC
ATT	-208.02*** (56.13)	-428.09*** (108.48)	48.02*** (16.27)	44.09 (34.85)
N	4,231	592	3,293	405
Municipalities	292	47	281	40
Y mean	2,109.64	2,294.96	345.65	419.61

Note: This table shows the effects of *Plan Cuadrante* on the number of crimes reported to the police in the municipality per 100,000 inhabitants and on the number of arrests in the municipality per 100,000 inhabitants using administrative data from the Undersecretary for Crime Prevention. The table reports the results for two samples: the full sample of municipalities included in the administrative database that is used for the analysis of these outcomes and the subsample of these municipalities included in the analysis of ENUSC outcomes. The estimation is conducted using the procedure developed in Callaway and Sant'Anna (2021) for difference-in-differences with staggered treatment adoption. Standard errors are clustered at the municipality level using the wild bootstrapping procedure. ***p<0.01, **p<0.05, *p<0.1.

Figure C.1: Effects of *Plan Cuadrante* on Reported Crime and Arrests (Sample of Municipalities Included in the Analysis of ENUSC Outcomes)



Notes: This figure shows the dynamic effects of *Plan Cuadrante* on the number of crimes reported to the police per 100,000 inhabitants in the municipality in panel (a) and on the number of arrests per 100,000 inhabitants in the municipality in panel (b) using administrative data from the Undersecretary for Crime Prevention. The analysis is conducted using only the sample of municipalities included in the analysis of ENUSC outcomes. Panel (a) includes 592 observations, while panel (b) includes 405 observations. The estimation is conducted using the procedure developed in Callaway and Sant'Anna (2021) for difference-in-differences with staggered treatment adoption. The dots in the figure represent the estimated coefficients and the bars behind them 95% confidence intervals. Standard errors are clustered at the municipality level using wild bootstrapping. The pooled effect and the mean outcome before the introduction of the treatment is also presented in each panel.

Comment R2.12

Minor Comments:

1. There is an additional period (.) after footnote 9.

Response to R2.12:

Thanks. This is corrected in the updated version of the manuscript.

Comment R2.13

2. Add space before “Using a difference-in-differences specification...” on page 11.

Response to R2.13:

Thanks. This is now corrected in the updated manuscript.

Comment R2.14

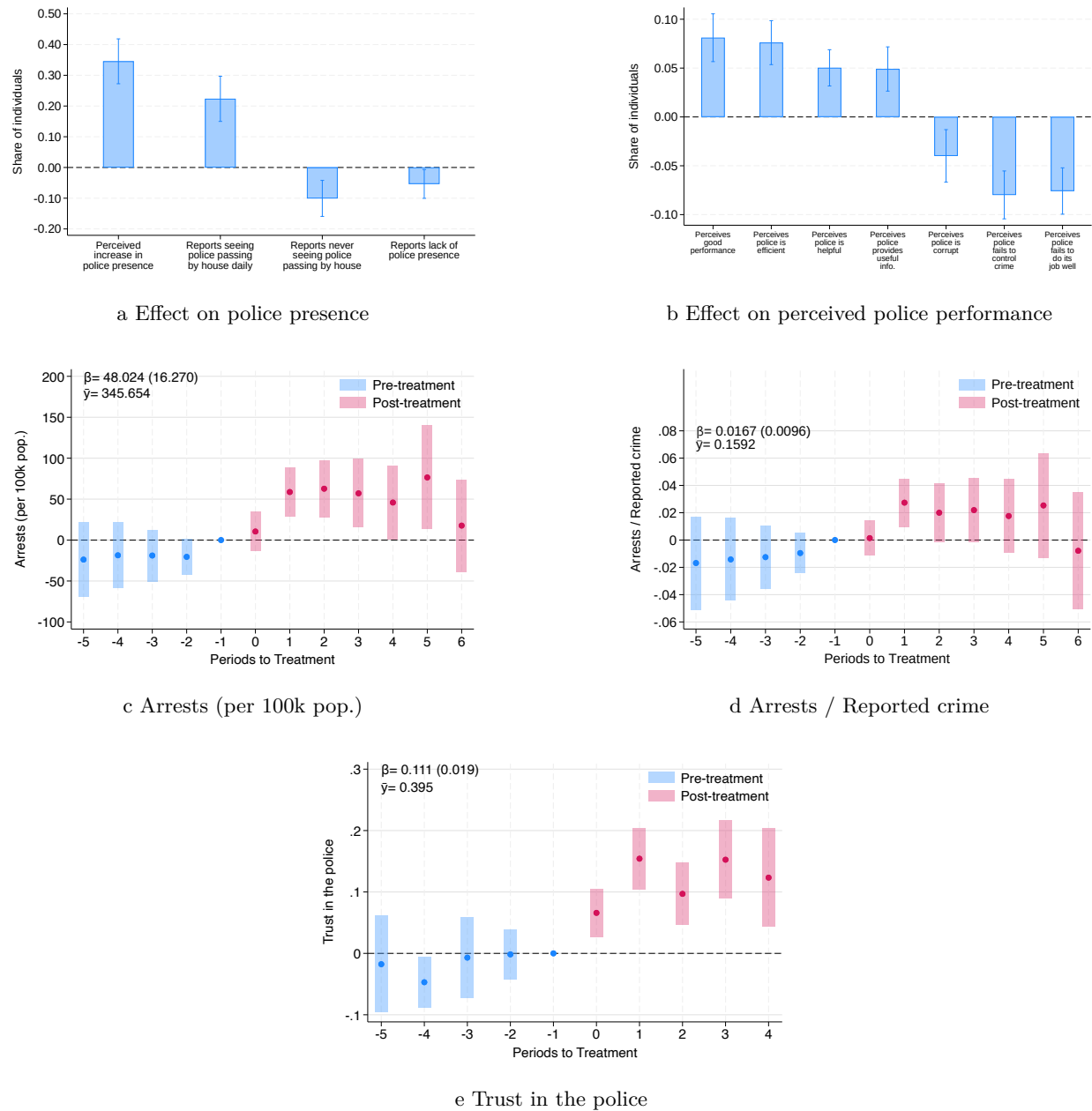
3. Notes on Figure 4 are missing the word “observations” after “This panel includes 8,495, 7,975, 8,326, 7,439, 6,621, 7,712, 7,889”

Response to R2.14:

Thanks. This is corrected in the updated manuscript.

Other figure referenced above:

Figure 4: *Plan Cuadrante* and Changes in Police Performance



Notes: Panel (a) reports the effects on perceived police presence. Panel (b) reports perceived police performance in 2005, comparing municipalities where *Plan Cuadrante* was already implemented before 2006 with municipalities where it was implemented after 2005. Panels (c), (d), and (e) report dynamic event-study estimates for arrests per 100,000 inhabitants, the ratio of arrests to reported crime, and trust in the police. The estimation follows Callaway and Sant'Anna (2021). The graphs display estimated coefficients and 95% confidence intervals. Standard errors are clustered at the municipality level using the wild bootstrapping procedure.

Comment R2.15

4. Sometimes the authors use the word “Police” and others they use the word “Carabineros” (e.g., Panel a of Figure 4), I suggest sticking to one.

Response to R2.15:

Thanks. In the updated version of the manuscript, we restrict the use of the term “Carabineros” only to the description of the institution.

References

- Ajzenman, N., P. Dominguez, and R. Undurraga (2023, October). Immigration, crime, and crime (mis)perceptions. *American Economic Journal: Applied Economics* 15(4), 142–76.
- Biblioteca del Congreso Nacional de Chile (1990). Ley orgánica constitucional de carabineros de chile (ley 18.961). Promulgada el 27 de febrero de 1990; publicada el 7 de marzo de 1990; última versión consolidada consultada en la Biblioteca del Congreso Nacional.
- Blanes-i Vidal, J. and G. Mastrobuoni (2018, Mar). Police patrols and crime. IZA Discussion Paper 11393, Institute of Labor Economics (IZA), Bonn.
- Blattman, C., D. P. Green, D. Ortega, and S. Tobón (2021a, 02). Place-Based Interventions at Scale: The Direct and Spillover Effects of Policing and City Services on Crime. *Journal of the European Economic Association* 19(4), 2022–2051.
- Blattman, C., D. P. Green, D. Ortega, and S. Tobón (2021b, 02). Place-Based Interventions at Scale: The Direct and Spillover Effects of Policing and City Services on Crime. *Journal of the European Economic Association* 19(4), 2022–2051.
- Braga, A. A., B. Turchan, A. V. Papachristos, and D. M. Hureau (2019). Hot spots policing of small geographic areas effects on crime. *Campbell Systematic Reviews* 15(3).
- Callaway, B. and P. H. Sant’Anna (2021). Difference-in-differences with multiple time periods. *Journal of Econometrics* 225(2), 200–230.

- Carabineros de Chile (2018). *Manual Operativo: Plan Cuadrante de Seguridad Preventiva 2.0*.
- Carabineros de Chile. Aprobado conforme a Orden General N° 1960 del 14 de septiembre de 2010, actualizado para el PCSP 2.0.
- Chaisemartin, C. and X. D’Haultfoeulle (2020, September). Two-way fixed effects estimators with heterogeneous treatment effects. *American Economic Review* 110(9), 2964–96.
- Cordner, G. W. (1995). Community policing: Elements and effects. *Police Forum* 5(3), 1–8.
- Crépon, B., E. Duflo, M. Gurgand, R. Rathelot, and P. Zamora (2013). Do labor market policies have displacement effects? Evidence from a clustered randomized experiment. *The Quarterly Journal of Economics* 128(2), 531–580.
- Evans, W. N. and E. G. Owens (2007). Cops and crime. *Journal of Public Economics* 91(1), 181–201.
- Gardner, J. (2022, July). Two-stage differences in differences. Papers 2207.05943, arXiv.org.
- Gobierno de Chile (2024). Subsecretaria de Desarrollo Regional y Administrativo. Sistema Nacional de información Municipal. <https://datos.sinim.gov.cl>. Consultado el 6 de agosto de 2024.
- Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of Econometrics* 225(2), 254–277.
- Kirchmaier, T., M. Langella, and A. Manning (2023). Commuting for crime. *The Economic Journal* 134(659), 1173–1198.
- Mastrobuoni, G. (2019). Police disruption and performance: Evidence from recurrent redeployments within a city. *Journal of Public Economics* 176, 18–31.
- Ministerio de Defensa Nacional, República de Chile (1989). Reglamento de organización de carabineros de Chile n° 1. Decreto N° 77, de 1 de marzo de 1989.
- Ministerio del Interior y Seguridad Pública (2014). Evaluación del Programa Plan Cuadrante de Seguridad Preventiva. Technical report.
- National Academies of Sciences, Engineering, and Medicine (2018). *Proactive Policing: Effects on Crime and Communities*. Washington, DC: The National Academies Press.

- Ostrom, E., R. B. Parks, and G. P. Whitaker (1973). Do we really want to consolidate urban police forces? A reappraisal of some old assertions. *Public Administration Review* 33(5), 423–432.
- Ostrom, E., R. B. Parks, and G. P. Whitaker (1978). *Patterns of Metropolitan Policing*. Cambridge, MA: Ballinger Publishing Company.
- Romano, J. P. and M. Wolf (2005). Stepwise multiple testing as formalized data snooping. *Econometrica* 73(4), 1237–1282.
- Salas P., R., J. Fernández L., and J. Muñoz V. (2014, July). Evaluación de programas gubernamentales: Informe final corregido. programa plan cuadrante de seguridad preventiva. Evaluación de programas gubernamentales, División de Control de Gestión, Dirección de Presupuestos, Ministerio de Hacienda, Gobierno de Chile. Ministerio Responsable: Ministerio del Interior y Seguridad Pública. Período de evaluación: 2010–2013.
- Skogan, W. G. and S. M. Hartnett (1997). *Community Policing, Chicago Style*. New York: Oxford University Press.
- Winchester, L., G. Zambrano, and J. Le-Bert (2007, June). Informe final de evaluación: Programa plan cuadrante de seguridad preventiva. Evaluación de programas gubernamentales, Dirección de Presupuestos, Ministerio de Hacienda, Gobierno de Chile. Ministerio Responsable: Ministerio de Defensa. Servicio Responsable: Carabineros de Chile. Período de evaluación: 2004–2006.
- Wooldridge, J. M. (2023). Simple approaches to nonlinear difference-in-differences with panel data. *The Econometrics Journal* 26(3), C31–C66.

Referee 3 Suggestions, Comments and Questions

Thank you very much for your previous feedback on our paper, which greatly strengthened it. We are glad that you consider we have addressed all your comments adequately.

Comment R3.1

The paper studies whether reorganizing police work around small, fixed geographic areas improves police effectiveness. The authors exploit the staggered rollout of Plan Cuadrante in Chile and estimate treatment effects using Callaway & Sant'Anna (2021). They combine household survey data with municipal administrative records on reported crimes and arrests. The paper reports sizeable and policy-relevant effects: a decline in 12-month household victimization, a decline in reported cases, higher arrests, increased trust in police, reduced perceived crime, and lower private security investment. The paper also provides analyses on mechanisms and resources, spillovers, and robustness.

I refereed this paper before for another journal, and I am happy to see that, overall, the paper is strong and much improved. It offers credible evidence that how police are organized matters, not only how many police jurisdictions deploy. I recommend some revisions focused on clarifications and a few additional robustness/interpretation checks.

Below, I reproduce the main points from my prior report and assess whether the current version addressed them.

1. Mechanism beyond resource expansion. I asked for clearer evidence that specialization—rather than more officers/vehicles—drives the results. The revised manuscript adds two complementary pieces of evidence: (i) early adopters report better police performance in 2005 cross-sections; and (ii) treatment raises perceived police presence and reduces complaints about lack of surveillance. The paper also presents three resource tests: dynamic changes in officers/vehicles after adoption, heterogeneity by actual and predicted resource increases, and no meaningful association between resources and outcomes where Plan Cuadrante had long been in place. The analysis of Seguridad Ciudadana (a later municipal security force program) shows no effects. Together, these additions make a convincing case that specialization—stable assignment to small areas that deepens local knowledge and community contact—explains the gains.

2. Parallel trends and pre-trends. I asked for a stronger pre-trends discussion. The event-study figures show flat pre-paths, and the text reports joint pre-treatment p-values for all primary outcomes, which supports the identifying assumption.

3. Administrative measurement and reporting behavior. I asked for a short discussion because higher trust can raise reporting. The manuscript notes that improved trust could bias administrative counts upward, so observed declines likely understate true crime reductions. It also shows consistent declines by crime type and a placebo with non-neighborhood crimes (sexual abuse) that does not change.

4. External validity. I asked for a short discussion of what features of Chile matter for generalization. The conclusion explicitly flags two contextual features—limited organized crime presence during the study window and relatively high police training—and argues how these shape transferability.

This is careful, policy-relevant work that convincingly shows that geographic specialization of police can reduce crime and improve public confidence. The revisions address virtually all of the concerns in my previous report, and the paper now contains a thorough and transparent robustness and mechanism section.

Response to R3.1:

Thank you for your thoughtful feedback and comments, which greatly enriched this paper.